

IPOÁ REDD+ PROJECT



Document Prepared By CARBONEXT

Project Title	IPOÁ REDD+ PROJECT
Version	1.9
Date of Issue	06 th October 2023
Project Location	Novo Aripuanã municipality, State of Amazonas, Brazil
Project Proponent(s)	- Carbonext Tecnologia em Soluções Ambientais Ltda.: redd.ipoa@carbonext.com.br + 55 (11) 3168 8521 - Silvio Antonio Balen: silviobalen@hotmail.com - Carina Marcia Klein Balen
Prepared By	Carbonext Tecnologia em Soluções Ambientais Ltda.
Validation Body	Ruby Canyon Environmental
Project Lifetime	17 March 2022 – 16 March 2052; 30–year total period
GHG Accounting Period	17 March 2022 – 16 March 2052; 30–year total period
History of CCB Status	Not applicable
Gold Level Criteria	Community: The Project meets the GL2 Gold Level – Exceptional Community Benefits according to the CCBS by bringing short and long-term benefits to low-income communities as described in section 4.5 of the present document.
Expected Verification Schedule	September 2022

Table of Contents

1	Summary of Project Benefits	9
1.1.	Unique Project Benefits.....	9
1.2.	Standardized Benefit Metrics	10
2	General	14
2.1.	Project Goals, Design and Long-Term Viability	14
2.1.1	Summary Description of the Project (G1.2)	14
2.1.2	Project Scale	17
2.1.3	Project Proponent (G1.1)	17
2.1.4	Other Entities Involved in the Project	18
2.1.5	Physical Parameters (G1.3)	18
2.1.6	Social Parameters (G1.3)	30
2.1.7	Project Zone Map (G1.4-7, G1.13, CM1.2, B1.2).....	35
2.1.8	Stakeholder Identification (G1.5)	36
2.1.9	Stakeholder Descriptions (G1.6, G1.13)	37
2.1.10	Sectoral Scope and Project Type	41
2.1.11	Project Activities and Theory of Change (G1.8).....	41
2.1.12	Sustainable Development.....	54
2.1.13	Implementation Schedule (G1.9)	66
2.1.14	Project Start Date	68
2.1.15	Benefits Assessment and Crediting Period (G1.9).....	69
2.1.16	Differences in Assessment/Project Crediting Periods (G1.9).....	70
2.1.17	Estimated GHG Emission Reductions or Removals	70
2.1.18	Risks to the Project (G1.10).....	71

2.1.19	Benefit Permanence (G1.11).....	73
2.1.20	Financial Sustainability (G1.12)	74
2.1.21	Grouped Projects	74
1)	Eligibility Criteria for Grouped Projects (G1.14)	74
2.1.22	Scalability Limits for the Grouped Projects (G1.15).....	75
2)	Risk Mitigation Approach for Grouped Projects (G1.15)	75
2.2.	Without-project Land Use Scenario and Additionality	76
2.2.1	Land Use Scenarios without the Project (G2.1)	76
2.2.2	Most-Likely Scenario Justification (G2.1)	76
2.2.3	Community and Biodiversity Additionality (G2.2)	77
2.2.4	Benefits to be used as Offsets (G2.2)	79
2.3.	Stakeholder Engagement.....	79
2.3.1	Stakeholder Access to Project Documents (G3.1)	79
2.3.2	Dissemination of Summary Project Documents (G3.1)	80
2.3.3	Informational Meetings with Stakeholders (G3.1)	80
2.3.4	Community Costs, Risks, and Benefits (G3.2)	83
2.3.5	Information to Stakeholders on Validation and Verification Process (G3.3)	87
2.3.6	Site Visit Information and Opportunities to Communicate with Auditor (G3.3).....	87
2.3.7	Stakeholder Consultations (G3.4)	87
2.3.8	Continued Consultation and Adaptive Management (G3.4).....	89
2.3.9	Stakeholder Consultation Channels (G3.5).....	90
2.3.10	Stakeholder Participation in Decision-Making and Implementation (G3.6).....	92
2.3.11	Anti-Discrimination Assurance (G3.7)	94
2.3.12	Feedback and Grievance Redress Procedure (G3.8).....	94

2.3.13	Accessibility of the Feedback and Grievance Redress Procedure (G3.8)	95
2.3.14	Worker Training (G3.9)	96
2.3.15	Community Employment Opportunities (G3.10)	96
2.3.16	Relevant Laws and Regulations Related to Worker's Rights (G3.11)	97
2.3.17	Occupational Safety Assessment (G3.12)	97
2.4.	Management Capacity	101
2.4.1	Project Governance Structures (G4.1)	101
2.4.2	Required Technical Skills (G4.2)	104
2.4.3	Management Team Experience (G4.2)	104
2.4.4	Project Management Partnerships/Team Development (G4.2)	112
2.4.5	Financial Health of Implementing Organization(s) (G4.3)	113
2.4.6	Avoidance of Corruption and Other Unethical Behavior (G4.3)	114
2.4.7	Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)	115
2.5.	Legal Status and Property Rights	115
2.5.1	Statutory and Customary Property Rights (G5.1)	115
2.5.2	Recognition of Property Rights (G5.1)	117
2.5.3	Free, Prior and Informed Consent (G5.2)	118
2.5.4	Property Rights Protection (G5.3)	119
2.5.5	Illegal Activity Identification (G5.4)	119
2.5.6	Ongoing Disputes (G5.5)	120
2.5.7	National and Local Laws (G5.6)	121
2.5.8	Approvals (G5.7)	126
2.5.9	Project Ownership (G5.8)	127
2.5.10	Management of Double Counting Risk (G5.9)	127

2.5.11	Emissions Trading Programs and Other Binding Limits	127
2.5.12	Other Forms of Environmental Credit	127
2.5.13	Participation under Other GHG Programs	128
2.5.14	Projects Rejected by Other GHG Programs	128
2.5.15	Double Counting (G5.9)	128
3	Climate	128
3.1.	Application of Methodology	128
3.1.1	Title and Reference of Methodology	128
3.1.2	Applicability of Methodology	130
3.1.3	Project Boundary	142
3.1.4	Baseline Scenario	154
3.1.5	Additionality	169
3.1.6	Methodology Deviations	184
3.2.	Quantification of GHG Emission Reductions and Removals	185
3.2.1	Baseline Emissions	185
3.2.2	Project Emissions	255
3.2.3	Leakage	268
3.2.4	Net GHG Emission Reductions and Removals	300
3.3.	Monitoring	304
3.3.1	Data and Parameters Available at Validation	304
3.3.2	Data and Parameters Monitored	318
3.3.3	Monitoring Plan	342
3.3.4	Dissemination of Monitoring Plan and Results (CL4.2)	353
3.4.	Optional Criterion: Climate Change Adaptation Benefits	354
3.4.1	Regional Climate Change Scenarios (GL1.1)	354

3.4.2	Climate Change Impacts (GL1.2).....	354
3.4.3	Measures Needed and Designed for Adaptation (GL1.3).....	354
4	Community.....	354
4.1.	Without-Project Community Scenario	354
4.1.1	Descriptions of Communities at Project Start (CM1.1).....	354
4.1.2	Interactions between Communities and Community Groups (CM1.1)	369
4.1.3	High Conservation Values (CM1.2).....	369
4.1.4	Without-Project Scenario: Community (CM1.3).....	373
4.2.	Net Positive Community Impacts	374
4.2.1	Expected Community Impacts (CM2.1)	374
4.2.2	Negative Community Impact Mitigation (CM2.2)	381
4.2.3	Net Positive Community Well-Being (CM2.3, GL1.4).....	382
4.2.4	High Conservation Values Protected (CM2.4)	387
4.3.	Other Stakeholder Impacts	389
4.3.1	Impacts on Other Stakeholders (CM3.1).....	389
4.3.2	Mitigation of Negative Impacts on Other Stakeholders (CM3.2)	390
4.3.3	Net Impacts on Other Stakeholders (CM3.3)	390
4.4.	Community Impact Monitoring	391
4.4.1	Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)	391
4.4.2	Monitoring Plan Dissemination (CM4.3)	400
4.5.	Optional Criterion: Exceptional Community Benefits	401
4.5.1	Exceptional Community Criteria (GL2.1)	401
4.5.2	Short-term and Long-term Community Benefits (GL2.2)	402
4.5.3	Community Participation Risks (GL2.3)	403
4.5.4	Marginalized and/or Vulnerable Community Groups (GL2.4)	403

4.5.5	Net Impacts on Women (GL2.5)	405
4.5.6	Benefit Sharing Mechanisms (GL2.6)	406
4.5.7	Benefits, Costs, and Risks Communication (GL2.7)	409
4.5.8	Governance and Implementation Structures (GL2.8)	409
4.5.9	Smallholders/Community Members Capacity Development (GL2.9)	410
5	Biodiversity	411
5.1.	Without-Project Biodiversity Scenario	411
5.1.1	Existing Conditions (B1.1)	411
5.1.2	High Conservation Values (B1.2)	420
5.1.3	Without-project Scenario: Biodiversity (B1.3)	422
5.2.	Net Positive Biodiversity Impacts	422
5.2.1	Expected Biodiversity Changes (B2.1)	422
5.2.2	Mitigation Measures (B2.3)	424
5.2.3	Net Positive Biodiversity Impacts (B2.2, GL1.4)	426
5.2.4	High Conservation Values Protected (B2.4)	428
5.2.5	Species Used (B2.5)	428
5.2.6	Invasive Species (B2.5)	428
5.2.7	Impacts of Non-native Species (B2.6)	428
5.2.8	GMO Exclusion (B2.7)	428
5.2.9	Inputs Justification (B2.8)	428
5.2.10	Waste Products (B2.9)	428
5.3.	Offsite Biodiversity Impacts	429
5.3.1	Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2)	429
5.3.2	Net Offsite Biodiversity Benefits (B3.3)	429
5.4.	Biodiversity Impact Monitoring	430

5.4.1	Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)	430
5.4.2	Biodiversity Monitoring Plan Dissemination (B4.3)	431
5.5.	Optional Criterion: Exceptional Biodiversity Benefits	431
5.5.1	High Biodiversity Conservation Priority Status (GL3.1)	431
5.5.2	Trigger Species Population Trends (GL3.2, GL3.3)	431
6	Appendices	432
	Appendix 1: Project Activities	432
	Appendix 2: Theory of Change	436
	Appendix 3: Total VCUs	445
7	Bibliographic references	447

1 SUMMARY OF PROJECT BENEFITS

1.1. Unique Project Benefits

Outcome or Impact Estimated by the End of Project Lifetime	Section Reference
1) Increase in community members with access to the internet and electricity, broadening their life opportunities.	2.1.11
2) Improved health and safety at work for community members	2.1.11

1.2. Standardized Benefit Metrics

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
GHG emission reductions or removals	Net estimated emission removals in the project area, measured against the without-project scenario	N/A	-
	Net estimated emission reductions in the project area, measured against the without-project scenario	2,522,057 tCO ₂	3.2.4
Forest ¹ cover	For REDD ² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	4,200.43 ha	2.2
	For ARR ³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario	N/A	-
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices are expected to occur as a result of project activities, measured against the without-project scenario	N/A	-
	Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured against the without-project scenario	N/A	-
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	94 community members	4.1.1

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (*VCS Program Definitions*)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (*VCS Program Definitions*)

³ Afforestation, reforestation, and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (*VCS Program Definitions*)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (*VCS Program Definitions*)

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	23 women	4.1.1
Employment	Total number of people expected to be employed in project activities, ⁵ expressed as number of full-time employees ⁶	Up to 8	2.1.11 2.3.15
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	Up to 4 predicted	2.1.11 2.3.15
Livelihoods	Total number of people expected to have improved livelihoods ⁷ or income generated as a result of project activities	Up to 8	2.1.11 2.3.15
	Number of women expected to have improved livelihoods or income generated as a result of project activities	Up to 4 predicted	2.1.11 2.3.15
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	94 community members	2.1.11 4.1.1
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	23 women	2.1.11 4.1.1

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. *The Sustainable Livelihood Approach to Poverty Reduction*. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	35 children	4.1.1
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	16 girls	4.1.1
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	N/A	-
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	N/A	-
Well-being	Total number of community members whose well-being ⁸ is expected to improve as a result of project activities	94 community members	2.1.11 4.1.1 4.2.1
	Number of women whose well-being is expected to improve as a result of project activities	23 women	2.1.11 4.1.1 4.2.1

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g., Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation, ⁹ measured against the without-project scenario	4,200.43 ha	5.1.3
	Expected number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities, ¹¹ measured against the without-project scenario	4 ¹²	5.1.1

⁹ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

¹² Number predicted by literature (see section 5.1.1), to be confirmed by biodiversity contractor

2 GENERAL

2.1. Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (G1.2)

The primary objective of the IPOÁ REDD+ Project is to reduce the emission of 2,522,057 tCO₂e (average of 84,069 tCO₂e/year), corresponding to 4,200.43 hectares of avoided deforestation over the 30-year project lifetime, or an average of 140.01 hectares a per year.

The Project Area is of 17,329.73 hectares, operating within private areas located in the municipality of Novo Aripuanã. The latter ranks 15th for highest deforestation of the municipalities of the Amazon biome: According to PRODES, between 2016 and 2021, an average of 160km² were deforested per year.¹³

The IPOÁ REDD+ project is located at approximately 13km from the Juma Sustainable Development Reserve (RDS) and 25km from the Acari National Park^{14,15}. It is aligned with the broader mission of these National Conservation Units, providing synergies through a buffer zone supporting the combatting of illegal deforestation.

The project's benefits for communities, climate and biodiversity are an excellent case in point for the application of CCB Standards, which will serve for their integral quantification and monitoring. The definition of the project's activities and strategies was carried out in a participatory and consultative fashion with stakeholders.

The latter aim to provide positive outcomes and impacts for its participants in a deep, continuous, and long-term fashion. Activities ranging from education and health service reforms, to installation of communications capabilities, to annual cultural and leisure activities are planned to initiate from the project's year 1 see section 2.1.11 and 0). As of year 2, a variety of activity fronts will initiate, including and training to create a community association, hiring and training forest monitors, and biodiversity study and conservation, contributing to all three pillars of the CCB – climate, community, and biodiversity – together with the continuous forest monitoring and conservation actions which are central to the REDD+ project's core conservation aims.

¹³ PRODES, 2021. http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/increments

¹⁴ SEMA - AM, 2022. http://meioambiente.am.gov.br/wp-content/uploads/2018/03/Decreto-26.010_2006-Cria-RDS-do-Juma.pdf (Acesso em 14/03/2022).

¹⁵ ICMBio. https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/unidade-de-conservacao/unidades-de-biomas/amazonia/lista-de-ucs/parna-do-acari/arquivos/dcom_decreto_criacao_parna_do_acari.pdf (Acesso em 14/03/2022).

The grouped model adopted by this project keeps open the possibility of its replication in neighbouring areas. The central idea is to multiply the preserved areas in the surrounding regions by adopting sustainable practices, converting the region into a model for sustainable development allied with REDD+.

A summary of the project's expected climate, community and biodiversity benefits follows below, with full details presented in sections 2.1.11 and Appendix 2, among others:

Community Benefits:

- Healthcare: tele-medical capabilities for the Basic Health Facility enabled by telecommunication infrastructure, and funding for continuous medical supplies and water ambulance operation and maintenance.
- Education: additional teacher training and enhanced lessons enabled by installed telecommunication infrastructure, and support for basic and additional teaching supplies, promoting quality education services.
- Communication and electrification: Installation of antennas and solar panels enabling transformative communication capabilities, which is of central importance to many aspects of the project including health, education, and forest conservation.
- Employment: Facilitating creation of a community association and hiring and training of community forest monitors activities will strengthen community governance and enable communities to demand essential public services, including health and education.
- Occupational health and safety: Training and qualification courses on health and safety for workers and provision of Personal Protective Equipment are planned.
- Training: Forest conservation, and sustainable and improved agricultural techniques training are planned for communities.

Biodiversity Benefits:

Flora and fauna inventories in the Project Zone, followed by biodiversity research, monitoring and conservation initiatives, are foreseen, aiming to have targeted impacts on increased biodiversity knowledge and conservation, promoting the habitat and survival of different animal and plant species, including threatened and endangered species.

Climate benefits:

The project's climate activities aim to conserve forest, through: remote monitoring via Carbonext's CarbonEye system; fire brigade training; and targeted deforestation prevention and mitigation actions by the field team.

The key implementation dates of and milestones each of the activity scopes in the three pillars (community, biodiversity and climate) are presented below, for the thirty years of the project lifetime.

Activity	Implementation schedule	Milestone
Pillar: Social		
Construction and maintenance of basic health units	Annual starting 2024	Start construction of Basic Health Facility by the end of 2024
Health transport program	Every 3 years	Articulate with public and private partners as of 2022, implement by 2025
School renovation	Every 2 years	Supply equipment for the school by second semester 2023
Teacher support and training	Annual	Implement activities in the field with specialist partner company by 2° semester 2024
Training and workshop: Occupational health and safety	Annual	Train community members in the second half of 2023
Personal protective equipment for worker health and safety	Annual	Deliver PPE in the second half of 2023
Installation and maintenance of technology and infrastructure for communication, internet access and electricity	Annual	Install antennas and satellites by the 2nd semester of 2023
Hiring and training forest monitors	Annual starting 2024	Hire monitors by 2024
Courses on sustainable practices and conservation	Annual starting 2024	Train community monitors by 2024
HCVs - social: resource harvesting areas, cultural areas, etc.	Every 2 years	Carry out participatory mapping by 2023
Pillar: Biodiversity		
Fauna inventory	Every 5 years	Inventory by 2025
Flora inventory	Every 5 years	Inventory by 2025
Environmental awareness and education actions	Every 5 years	Start activities in 2025
Research, monitoring or conservation program for target species	Every 5 years	Start activities in 2025

Survey of areas of high conservation value	Every 5 years	Start activities in 2025
Remote monitoring	Annual	Initiate remote monitoring in 2022 Draw up internal and external deforestation occurrence procedure by 2022
Pillar: Climate		
Fire brigade training (in accordance with the PIF)	Annual	Start activities in 2024
Deforestation prevention and mitigation actions	Annual	Start activities in 2022
Other		
Relationship-building actions with communities, such as festive, cultural, and sports activities.	Every two years	Carry out 1 activity every two years
Surplus budget may be allocated to unforeseen costs (e.g. maintenance) or to existing or new activity scopes as required	Every two years	Carry out 1 activity every two years

The predicted impacts of this are to avoid a total of 4,200.43 ha of deforestation, corresponding to avoided net emissions of 2,522,057 tCO₂e over the Project's 30-year lifetime. This is sub-divided into Avoided Planned Deforestation of 3,547.95 ha, corresponding to avoided net emissions of 2,361,764 tCO₂e, and Avoided Unplanned Deforestation of a predicted 652.48 ha, avoiding 160,293 tCO₂e of net emissions.

2.1.2 Project Scale

Project Scale	
Project	x
Large project	

2.1.3 Project Proponent (G1.1)

Organization name	Carbonext Tecnologia em Soluções Ambientais Ltda.
Contact person	Janaína Dallan

Title	Project Developer
Address	R. Gomes de Carvalho, 1510 – 19º andar – Vila Olímpia, São Paulo – SP, Brazil, ZIP 04547-005
Telephone	+ 55 (11) 3168-8521
Email	redd.ipoa@carbonext.com.br

Organization name	Silvio Antonio Balen
Contact person	Silvio Antonio Balen
Title	Project Proponent (Landowner)
Address	Rua Pedro Ivo nº 3959, Chopinzinho – PR – Brasil
Telephone	+ 55 (46) 9903-0076
Email	silviobalen@hotmail.com

Organization name	Carina Marcia Klein Balen
Contact person	Silvio Antonio Balen
Title	Project Proponent (Landowner)
Address	Rua Pedro Ivo nº 3959, Chopinzinho – PR – Brasil
Telephone	+ 55 (46) 9903-0076
Email	silviobalen@hotmail.com

2.1.4 Other Entities Involved in the Project

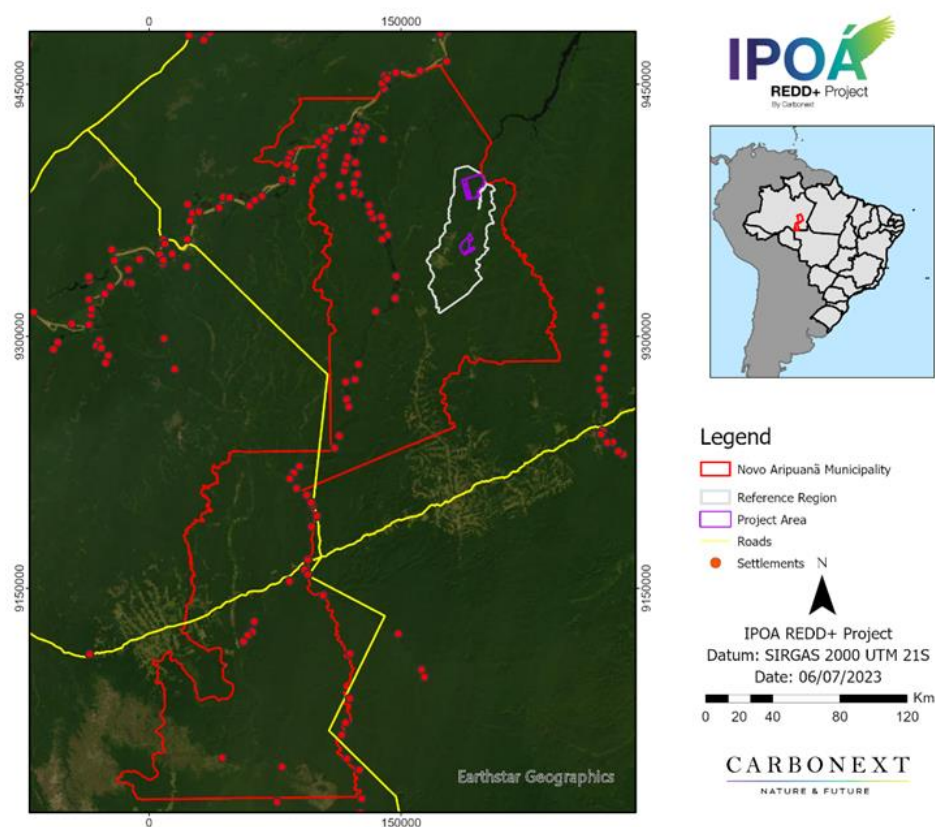
This item does not apply to the present project.

2.1.5 Physical Parameters (G1.3)

Location of the project

The project is located in Northern Brazil's Amazon region, in the state of Amazonas, municipality of Novo Aripuanã, as shown in the figure below. Novo Aripuanã falls within the Mesoregion of the South Amazon and Microregion of the Madeira watershed¹⁶.

The most common way to reach the municipality is by water, on large boats or speedboats that leave the state capital, Manaus, crossing parts of the Amazon River and Madeira River. The duration of travel is 24 hours in larger boats or 10 hours in speedboats, as approximately 450 kilometers separate the two cities. However, there is also the option of travelling by seaplane from Manaus, which takes under two hours. The main bodies of water in the vicinity of the project area are the Sucunduri River and the Acari River.



¹⁶ <https://portalamazonia.com/amazonia-az/letra-n/novo-aripuana>

Figure 1. Location of the Project Area and Reference Region in Novo Aripuanã Municipality, Amazonas State.

The IPOÁ REDD+ Project Area is of 17,329.73 ha in area and is composed of two properties: the Maanaim property, geodetic coordinates: 05°31'04,33" S 59°46'00,53" W.

And the Campos property, geodetic coordinates: 05°50'15,93" S 59°48'26,53" W.

In terms of applied methodology, the Project Areas are subdivided as follows:

- Maanaim property Avoided Unplanned Deforestation (AUD) area: 9,914.10 ha.
- Maanaim property Avoided Planned Deforestation (APD) area: 2,666.48 ha.
- Campos property Avoided Unplanned Deforestation (AUD) area: 3,867.68 ha.
- Campos property Avoided Planned Deforestation (APD) area: 881.47 ha.

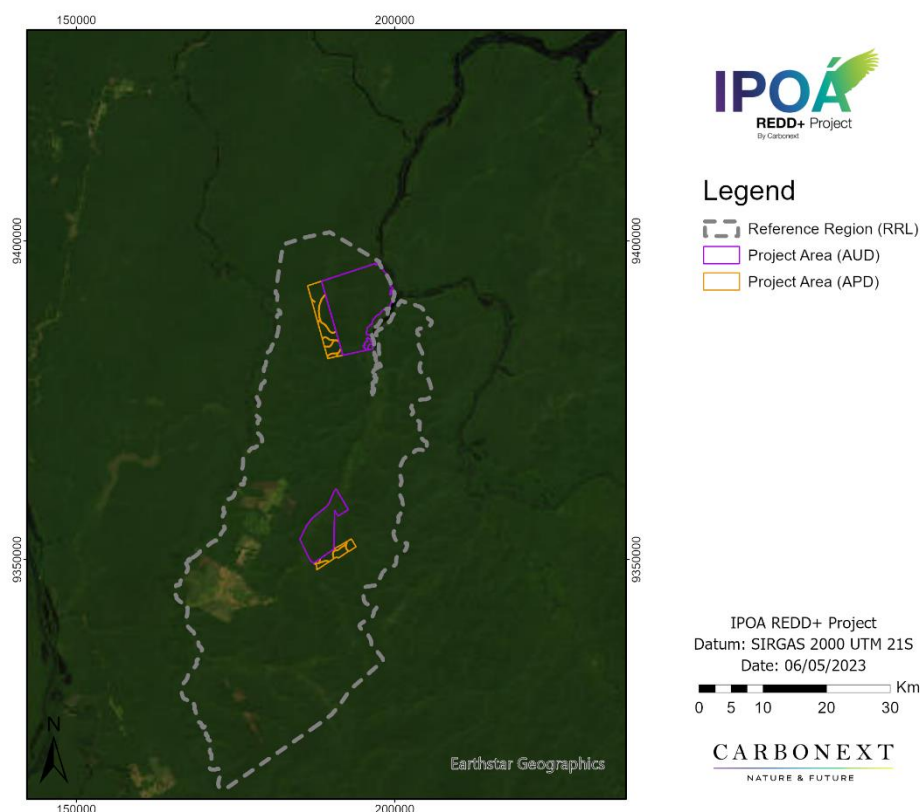


Figure 2. Ipoá REDD+ Project PA and Reference Region

Topography

The municipality of Novo Aripuanã is located in lowland areas, of which the topography is flat and slightly undulated.¹⁷

The highest elevation within the Reference Region is of 134 meters, while in the Project Area it is of 101 meters, with Maanaim's area being flatter, while Campos's displays greater variation.¹⁸

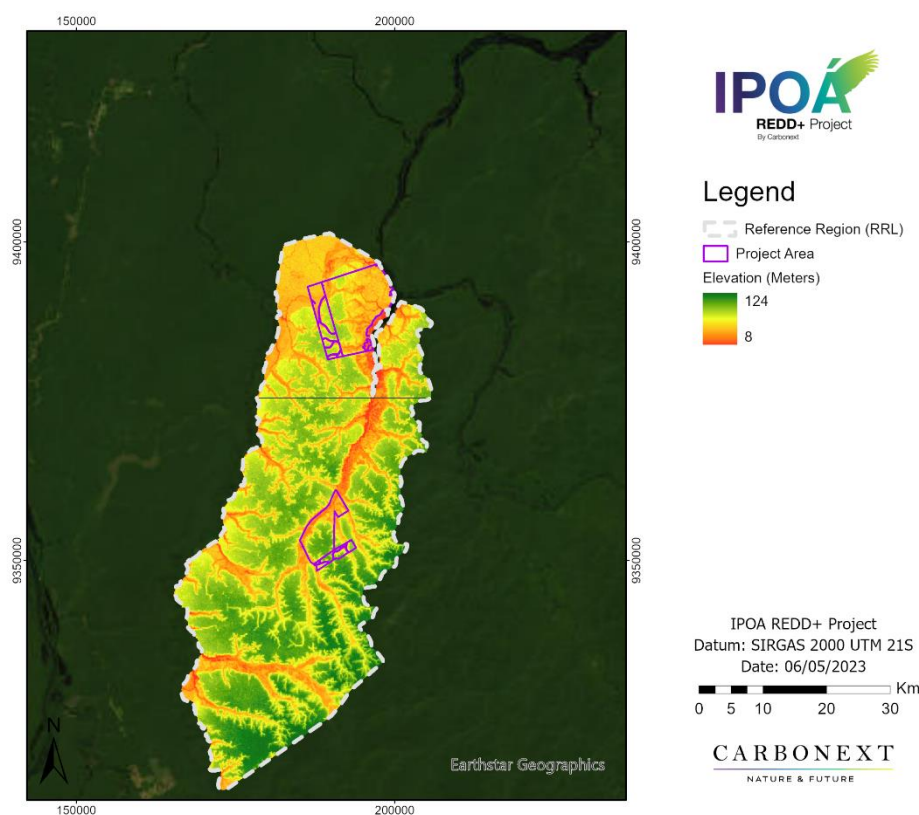


Figure 3. Elevation of the Reference Region and Project Area

¹⁷<http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSIC%3%87%20O%20FLOR%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20C%81REA%20COM%20EXPLORA%20O%20FLORESTAL%20NO%20MUNIC%20DE%20PIO>

¹⁸Mapa Índice TOPODATA (webmapit.com.br)

The Project Area and Reference Region have a slope percentage lower than or equal to 15 in 97% and 95% of their areas, respectively. Thus, the slope of >15% is similar between areas PA, RRL, RRD and LB, equivalent to more than 94% in each area (Table 1 - Stakeholder Identification.).

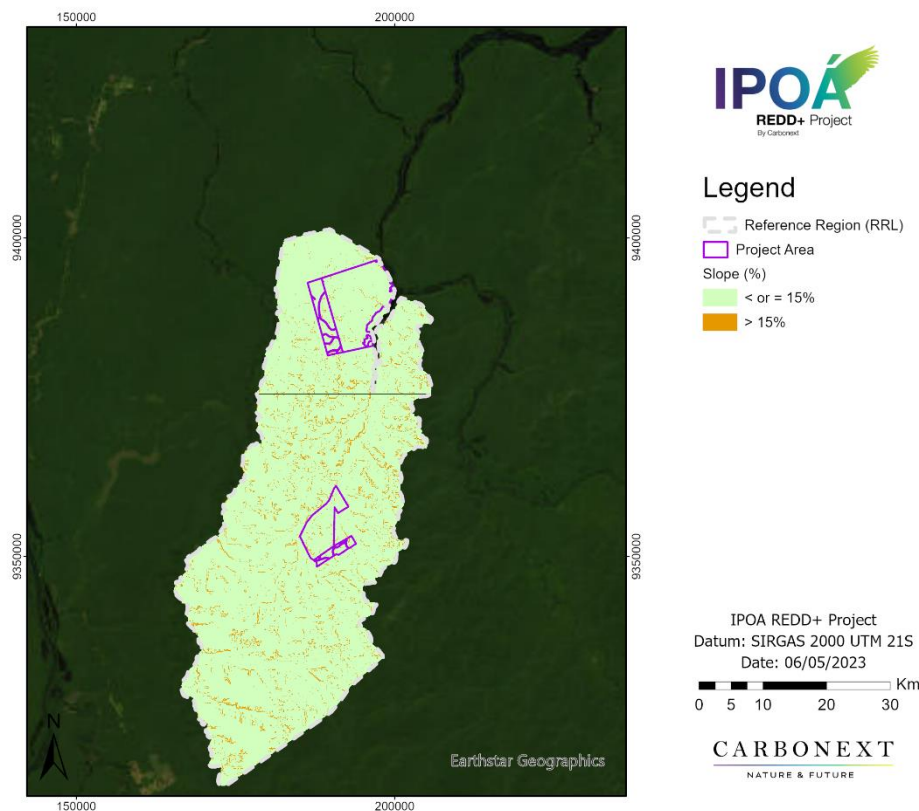


Figure 4. Slope of Reference Region and Project Area

Soil

According to the Brazilian Soil Classification System, from the Brazilian Agricultural Research Corporation (EMBRAPA, 2026) ¹⁹, the Region of Reference comprises the following soil types: Haplic Tb Distrophic Gleisols, Yellow Distrophic Ferrasols, and Red-Yellow Distrophic Ferrasols, as shown in the figure below:

¹⁹http://geoinfo.cnps.embrapa.br/layers/geonode%3Abrasil_solos_5m_20201104

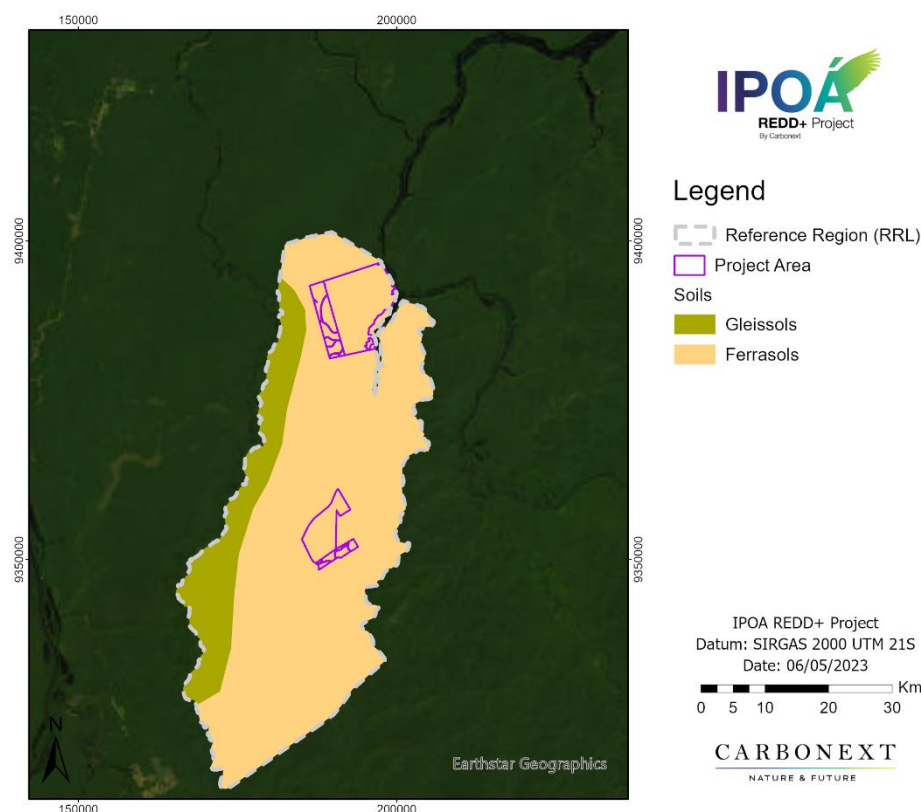


Figure 5. Soil types in the Reference Region and Project Area

Yellow and Red- Yellow Ferrasols are derived from clayey or sandy-clayey sedimentary materials from the Barriers formation in the coastal region of Brazil, or in the low plateaus of the Amazon region related to the Alter-do-Chão Formation, which may also occur outside these environments when they meet the color requirements defined by the SiBCS.²⁰ The most common texture is clayey or very clayey. The yellow and red color is uniform in depth, as is the clay content. Another field aspect refers to the high cohesion of the structural aggregates. They have good physical conditions of moisture retention and permeability. The gleissols are periodically or permanently flooded soils, and present the predominant gray color due to drainage, also for being in an area of low relief. Those classified as dystrophic are those with low fertility²¹.

²⁰ SiBCS. Sistema Brasileiro de Classificação de Solos. Brasília: Embrapa -SPI; Rio de Janeiro: Embrapa Solos, 2006.

²¹https://www.agencia.cnptia.embrapa.br/gestor/solos_tropicais/arvore/CONT000fzyjaywi02wx5ok0q43a0r58asu5l.html.

The soil types are similar between the PA, RRL, RRD, and LB (see Figure 5). The Ferrasols class represents more than 83% of all these areas.

Climate

The climate of Amazon region is classified as equatorial, characterized by high temperatures and air humidity, besides being constantly subjected to the oscillations of the rivers²². Within the equatorial, according to Köppen-Geiger climatic classification the predominant type of climate in Novo Aripuanã and in the Reference Region is type Af (tropical humid climate), with maximum temperatures of 37.5°C, minimum of 16°C and an average temperature of 26.7°C.

The relative humidity is quite high in the municipality, where it presents minimum and maximum factors of 85 and 95%. The high rainfall of the region is one of the characteristic factors, with an average variation of 2250 to 2750 mm in the year, and rainfall rates tend to be high all year round. The rains generally start in October, with the rainiest four months between January and April, and the driest between June and September.²³

A small part of the Reference Region is classed as Am climate type (humid or sub-humid tropical), according to Köppen-Guiger²⁴, a transition between the Af and Aw climatic type. It is characterized by an average temperature of the coldest month that is always above 18°C, with a short dry season that is counter-balanced by high precipitation totals.

²² <http://eventoscopq.mackenzie.br/index.php/jornada/xvijornada/paper/view/2159/1295>.

²³ <http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSIC%3%87%20O%20FLOR%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20C%81REA%20COM%20EXPLORA%20O%20FLORESTAL%20NO%20MUNIC%8dPIO%20DE%20>

²⁴ https://ri.ufmt.br/bitstream/1/2160/1/TESE_2016_Jos%20a%20Maur%20adicio%20da%20Cunha.pdf

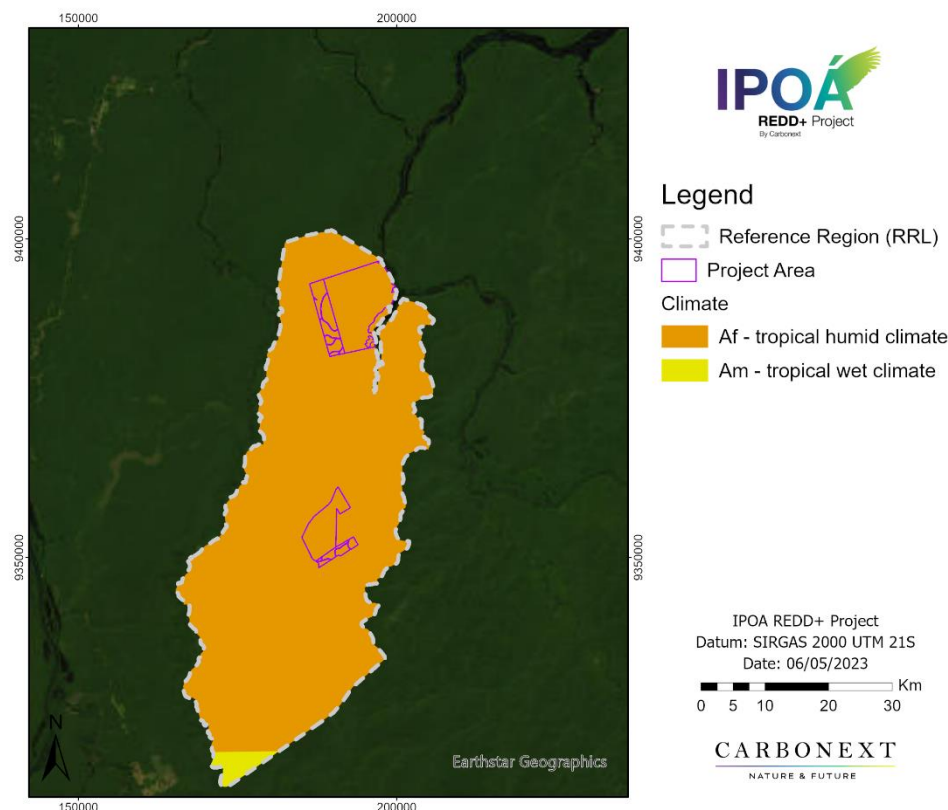


Figure 6. Climate types in Reference Region according to Köppen 's climate classification.

Hydrology

The river system of the municipality of Novo Aripuanã is that of the Madeira River Basin and its tributary the Aripuanã River²⁵. The high-water period is from December to May, with the maximum level in the month of April. In the Novo Aripuanã region, floodplains are full from June to November, and the lowest waters are in the month of September, and the influence of the floods in the Amazon River in the May to July period²⁶.

²⁵<http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSIC%3%87%20%20FLORESTAL%20NO%20MUNIC%3%8dPIO%20DE%20.pdf>

²⁶<http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSIC%3%87%20%20FLORESTAL%20NO%20MUNIC%3%8dPIO%20DE%20.pdf>

Water bodies close to the project area are important for river transport. Access to the Maanaim property is via the Sucunduri River and to reach the Campos property, the path is via the Capinarana Stream. The Project Area and entire Reference Region are located in are in Lower Madeira Hydrographic Microregion²⁷.

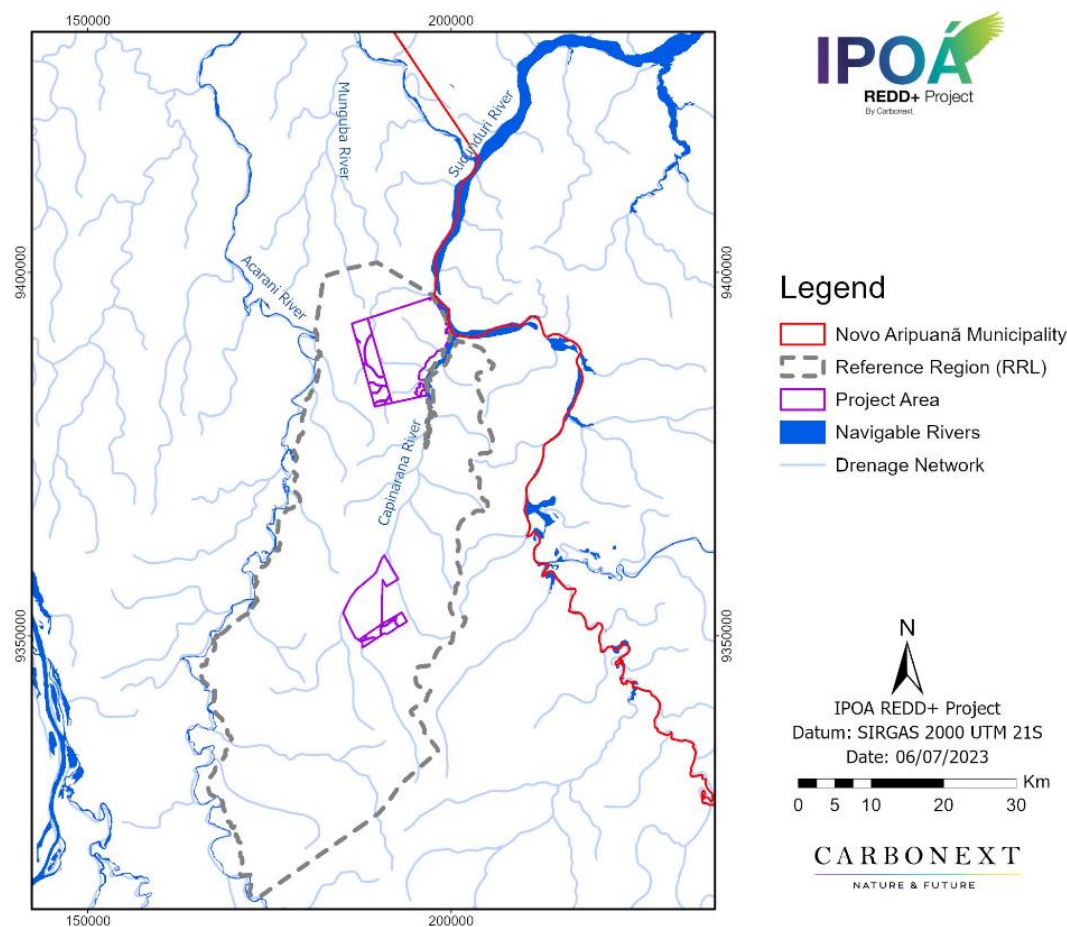


Figure 7. Drainage network and main rivers in Project Area and Reference Region.

²⁷https://geoftp.ibge.gov.br/informacoes_ambientais/estudos_ambientais/bacias_e_divisoes_hidrograficas_do_brasil/2021/Divisao_Hidrografica_Nacional_DHN250/mapas/mapa_das_divisoes_hidrograficas_do_brasil_2021.pdf

Vegetation

The vegetation generally present in the municipality of Novo Aripuanã (AM) is classed as Dense Tropical Rainforest, with emergent species and a group of palm trees, which compete for light in the upper tree stratum²⁸. This forest type is defined by the presence of phanerophytes, woody lianas, and epiphytes in abundance, and is subdivided into five formations that vary according to topography.

In the RRD Reference Region, the predominant vegetation type is Submontane Dense Tropical Rainforest (87%). The vegetation types are similar between PA, RRL, RRD, and LB, with a predominance of Submontane Dense Tropical Rainforest (Dse), representing 73% of the PA, 87% of the RRL, 86% of the RRD and 78% of the LB.

²⁸<http://www.bdta.ufra.edu.br/jspui/bitstream/123456789/839/1/TCC%20COMPOSIC%3%87%83O%20FLOR%3%8dSTICA%20E%20FITOSSOCIOLOGIA%20DE%20UMA%20%81REA%20COM%20EXPLORA%3%87%83O%20FLORESTAL%20NO%20MUNIC%3%8dPIO%20DE%20.pdf>

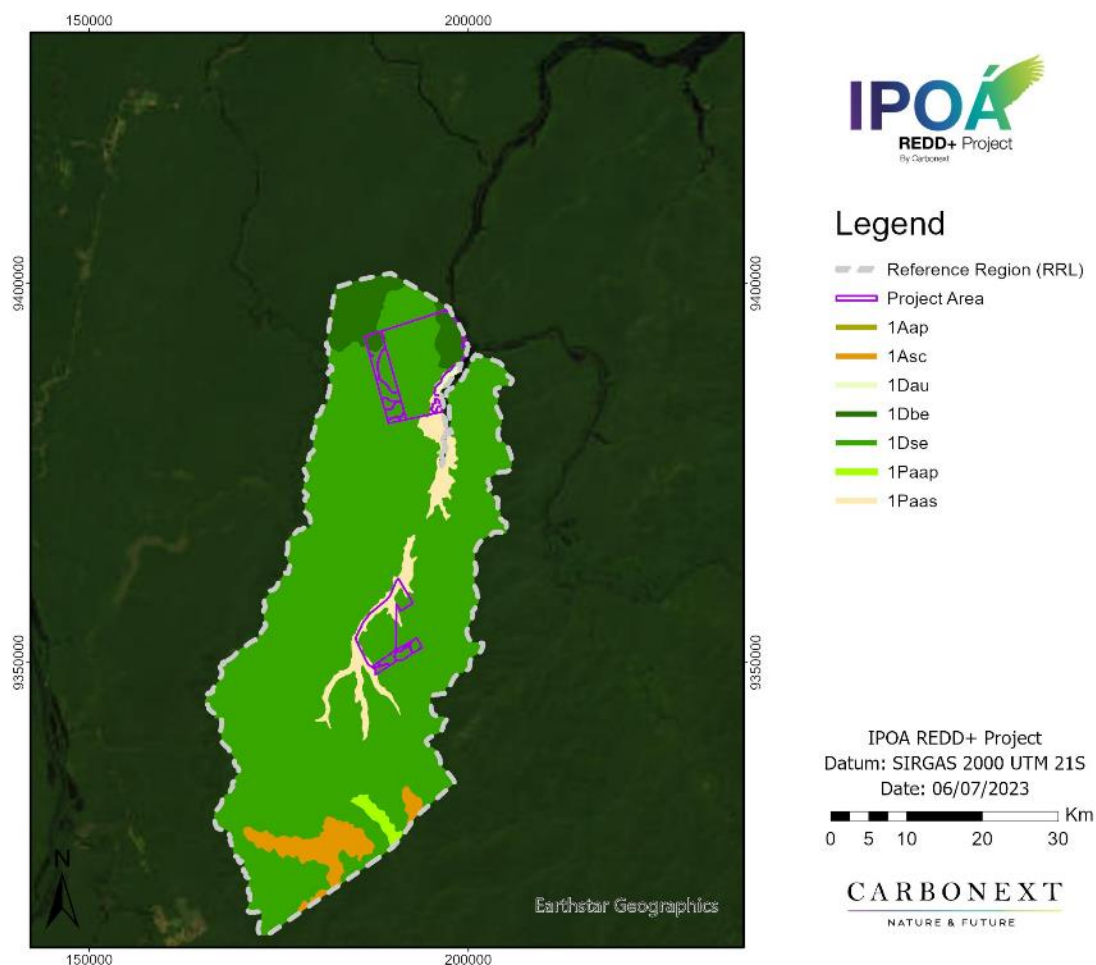


Figure 8. Vegetation types in Reference Region (Source: IBGE, 2021)

²⁹Three types of Dense Tropical Rainforest are found in the Project Area: a) Alluvial; b) Lowland; c) Submontane; and d) Pioneer formation. Sub-types can be defined with open or uniform canopy.³⁰

²⁹ Legend key: **Aap** – Alluvial Open Tropical Rainforest; **Asc** – Submontane Open Tropical Rainforest; **Dau** – Alluvial Dense Tropical Rainforest; **Dbe** – Lowland Dense Tropical Rainforest; **Dse** – Submontane Dense Tropical Rainforest; **Paap** – Pioneer Formation with fluvial influence with palm trees; **Paas** – Pioneer Formation with fluvial influence without palm trees

³⁰ http://redd.mma.gov.br/images/gttredd/daltonvaleriano_definicaofloresta.pdf

In the AUD Project Area, the predominant vegetation type is Submontane Dense Tropical Rainforest (73.28%), followed by Lowland Dense Tropical Rainforest (16.31%), and finally Alluvial Dense Tropical Rainforest (9.09%).

In the APD Project Area, the predominant vegetation type is Submontane Dense Tropical Rainforest (87.19%), followed by Lowland Dense Tropical Rainforest (12.57%), and finally Pioneer Formation with Fluvial Influence without Palm Trees (0.24%).

Some notes on the forest types follow below:

a) Alluvial: does not display marked topographic variation, being a riverine or riparian forest that occurs along watercourses. It consists of fast-growing phanerophytes, generally with smooth bark, conical trunk, and sometimes with tabular roots. It is a formation with many palm trees in the primary stratum and the forest understory. It contains many woody and herbaceous lianas, besides a large number of epiphytes and few parasites.³¹

b) Lowland: found between 4° N latitude and 16° S latitude, from 5 m to 100 m above sea-level; from 16° S to 24° S latitude; from 24° S latitude to 32° S latitude from 5 m to 30 m: It is a forest typology that in general occupies the coastal plains, capped by plio-pleistocene tablelands of the Barreiras Group.³²

c) Submontane: occurs between 4° N and 16° S latitude from 100 m to 600 m; from 16° S to 24° S latitude from 50 m to 500 m; from 24° S to 32° S latitude from 30 m to 400 m. Its main features are high phanerophytes, some exceeding 50 m in Amazonia and rarely 30 m in other parts of the country. The forest understory is composed of naturally regenerating seedlings, few nanophanerophytes and chamaephytes, in addition to the presence of small palm trees and herbaceous lianas in larger numbers.³³ The main species that characterize the emerging Submontane Dense Ombrophylous Forests are: *Dinizia excelsa* Ducke (angelimpedra), *Bertholletia excelsa* (chestnut tree) and *Cedrelinga cateniformis* (Ducke), *Ducke* (cedrorana). The uniform stratum is characterized by *Manilkara* spp. (maçarandubas), *Protium* spp. (breus) and *Pouteria* spp. (abius)³⁴.

³¹ https://ambientes.ambientebrasil.com.br/natural/regioes_fitoeologicas/regioes_fitoeologicas_-_floresta_ombrofila_densa.html

³² https://ambientes.ambientebrasil.com.br/natural/regioes_fitoeologicas/regioes_fitoeologicas_-_floresta_ombrofila_densa.html

³³ https://ambientes.ambientebrasil.com.br/natural/regioes_fitoeologicas/regioes_fitoeologicas_-_floresta_ombrofila_densa.html

³⁴ <http://scielo.iec.gov.br/pdf/bmpegn/v4n1/v4n1a02.pdf>

c) Pioneer formations: are the vegetation types that are first to grow on previously bare soils, caused by natural processes³⁵. Plant communities of floodplains reflect the effects of river floods in rainy seasons, or floodplains³⁶, thus this type of vegetation is found adjacent to small and medium-sized rivers in western Amazonia. Alluvial communities are characterized by the presence of herbaceous and shrub species, and it is common for them to include arboreal species, sometimes with palm trees present.



Figure 9. Dense Tropical Rainforest schematic profile. 1 - Altomontane; 2 - Montane; 3 - Submontane; 4- Lowland; 5- Alluvial. Source: Veloso, Rangel Filho and Lima (1991).

2.1.6 Social Parameters (G1.3)

The Madeira Microregion and deforestation factors

The microregion Madeira is in the mesoregion of South Amazon. It has an area of 221,036.579 km², which represents 14.07% of the total area of the State (AM). The region is home to 165,663 inhabitants (IBGE, 2010) with a population density of 0.75 hab./km², and is composed of 5 municipalities: Novo Aripuanã, Apuí, Borba, Humaitá, and Manicoré.

Located in the southeastern portion of Amazonas, this region has different transportation logistics than the rest of the state and is strategically important in containing the advance of deforestation. The region is endowed with federal and state highways that constitute the main means of transportation, although the region also presents an extensive water network, only the Madeira River is widely navigable. The channel

³⁵ <https://snif.florestal.gov.br/pt-br/conhecendo-sobre-florestas/168-tipologias-florestais>

³⁶ Manual técnico da vegetação brasileira / Fundação Instituto Brasileiro de Geografia e Estatística, Departamento de Recursos Naturais e Estudos Ambientais -Rio de Janeiro: IBGE, 1991. <<https://biblioteca.ibge.gov.br/visualizacao/livros/liv23267.pdf>>. Accessed 16/09/2022.

of the Madeira River, in the southeastern region of Amazonas, is cut by the Transamazon Highway (BR-230), which passes through of these municipalities³⁷.

The occupation of the Madeira Microregion was more intense in the 1960s with the construction of the BR-230 highway. The central axis was the occupation of the Amazon with the increase of the exploitation of natural and agricultural resources through the construction of roads aiming at exportation (MAHAR 1978; FEARNESIDE, 1986). In recent years, the Madeira micro-region has suffered from deforestation for the expansion of farming and cattle raising activities, mainly due to the advance of the Arc of deforestation towards the Amazon³⁸.

In 2014, Novo Aripuanã, Manicoré, and Apuí occupied the top positions in the deforestation ranking in southeastern Amazonas³⁹. Novo Aripuanã saw a jump in deforestation in mid-2019⁴⁰. The margins of the BR-230 and AM-174 roads, which connect Apuí to Novo Aripuanã, are formed almost entirely by a landscape of deforested areas and pastures. The AM-174 road is one of the largest drivers of deforestation in southern Amazonas⁴¹. As soon as lands becomes unproductive, new areas of pasture are opened up. The dense forest, which was once the basis for wild-harvesting activities, is being quickly replaced through the practices of slash-and-burn livestock, involving deforestation to create pastureland, burning, and the production of cattle⁴².

According to the Mapbiomas database, in 2020 about 3.21% of the total area of the municipality of Novo Aripuanã had already been converted to pasture, which is the predominant land-use in the region⁴³.

³⁷ <https://1library.org/article/microrregi%C3%A3o-madeira-no-estado-do-amazonas.y816jjwz>

³⁸ https://repositorio.unb.br/bitstream/10482/15216/1/2013_KatiaVianaCavalcante.pdf

³⁹ <https://www.revistaterceiramargem.com/index.php/terceiramargem/article/view/318/225>

⁴⁰ <https://www.redalyc.org/journal/5999/599963212009/599963212009.pdf>

⁴¹ <https://www.sciencedirect.com/science/article/abs/pii/S0921800918304130>

⁴² <https://www.redalyc.org/journal/5999/599963212009/599963212009.pdf>

⁴³ <https://mapbiomas.org/download>

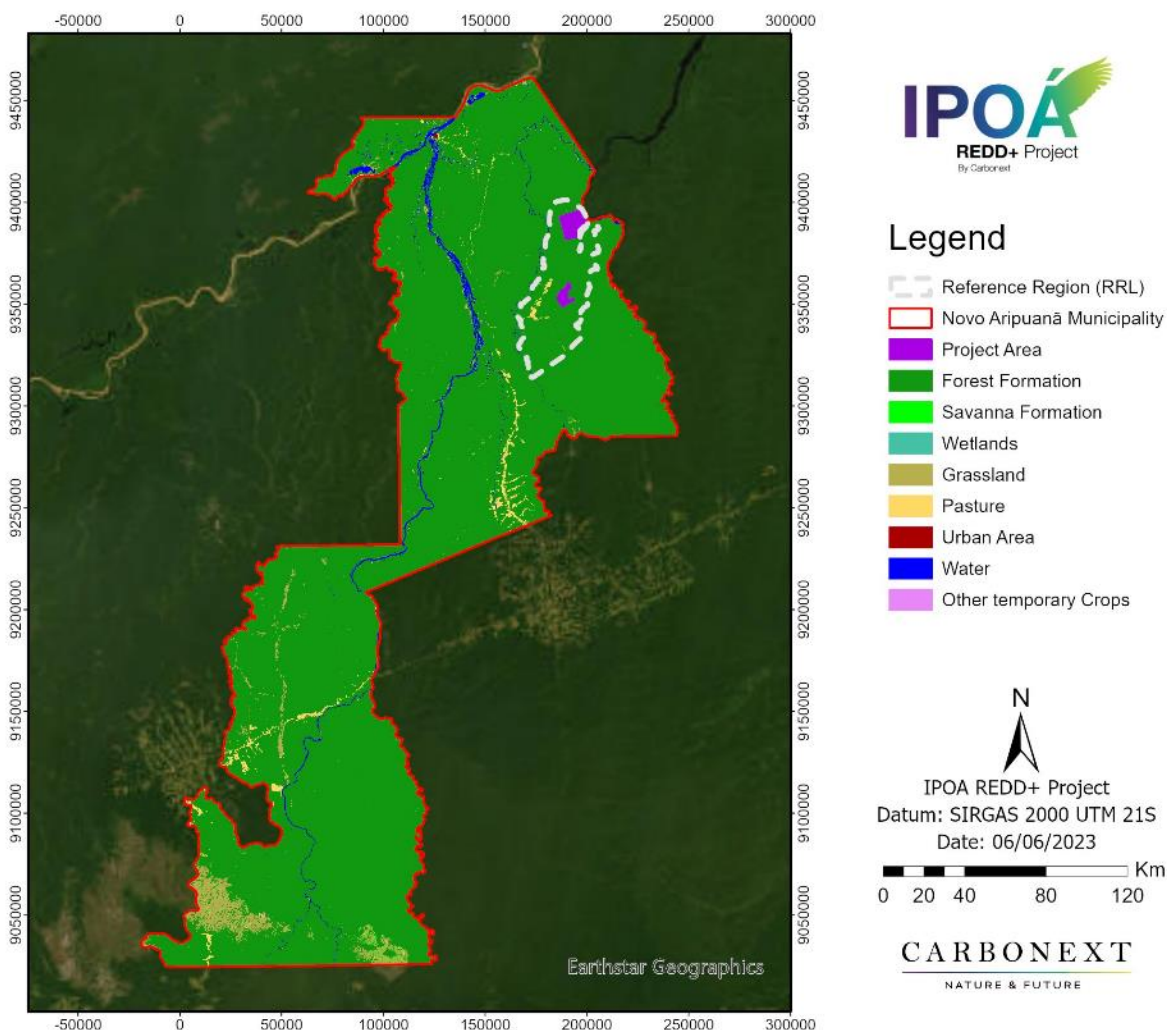


Figure 10. Land use, year 2020, in the municipality of Novo Aripuanã, AM, Brazil.

Municipality of Novo Aripuanã

The Novo Aripuanã municipality is made up of territory separated from Borba and Manicoré, so its history is closely linked to those municipalities. Its territory was inhabited by the indigenous ethnic groups Toras, Barés, Muras, Urupás, Araras Indians, among others (IBGE, 2012).

In 2021, its population was 26,443 people with population density reaching 0.52 per km² (IBGE; 2021). Most of the population in this municipality is recognized as mixed race (“pardo”) (80%), followed by white (12,4%) and black (5,5%) races. About 63% of the population lives in the urban area, while the remaining 37% is in the rural area. Regarding to the distribution of inhabitants by gender, 53% are men and 47% are women (IBGE,2010).



Figure 11. Municipality of Novo Aripuanã, AM, Brazil⁴⁴.

Health

The municipality of Novo Aripuanã has seven health care facilities, all of which are public facilities that are part of SUS (Brazilian Unified Health System), with approximately 3.26 units for 10,000 inhabitants. Hospitalizations happen mainly due to problems in the reproductive system (IBGE,2009;2010).

The basic sanitation service is provided only for water supply and is carried out exclusively by a state government entity. Its situation is classified as +inadequate for the more than 6,000 households served (IBGE, 2017).

In 2020, the infant mortality rate in the municipality reached 8.13 deaths per thousand live births, a considerable drop aiming at the previous year's rate that reached 35.17 (IBGE, 2020).

⁴⁴ <https://ibge.gov.br/cidades-e-estados/am/novo-aripuanã.html>

Education

In terms of quality of education, the municipality has the largest number of schools focused on elementary and middle school with 95% concentrated in this segment and the remaining 5% for high school (IBGE, 2020). In educational level, most of the population falls into no education or incomplete elementary school (71,8%). Only about 0,2% of the population has completed higher education (IBGE, 2010).

The Basic Education Development Index (IDEB) for elementary education in the city was 3.9, in which the city ranks 35th in the state and performs below the state average of 4.5 (IBGE, 2019).

Land-use and economic activities

Agriculture, livestock, services, and industry are the main economic activities in Novo Aripuanã, where the municipality has a Municipal Human Development Index (HDI) of 0.554. The economy is concentrated in agricultural and livestock establishments where they are managed by individual producers (71,4%) with collective title deed (93,4%) and the service sector activities are mainly linked to public administration, agriculture, and education (IBGE, 2010; 2017). The per capita GDP of the municipality is R\$ 8,263.52 (IBGE, 2019).

Land use is divided into four segments: farming, pasture, and forest areas. Farming areas are subdivided into two groups of uses: temporary (44,9%) and permanent (55,1%). While pasture areas are subdivided into three conditions: natural (43,7%), good condition (52,6%) or poor condition (3,7%). The most significant agricultural species for the region are cassava, watermelon, and banana, while in the forestry segment the highlight is açaí. Of the livestock activities, cattle and chicken raising are the most significant (IBGE, 2017).

The forest areas present in the municipality are classified as natural areas (38%) or areas designated for permanent conservation or legal reserve (62%). At the same time, there are 3 local establishments with areas of agroforestry systems with forest species for crops and grazing by animals (IBGE, 2017).

Relevant historical conditions

The history of Novo Aripuanã dates back to the 17th century, where the first recorded expeditions aiming to explore the Madeira River were in early 1617, with an accentuation from 1637 onwards. Among these expeditions was that of Pedro Teixeira.

The village of Borba was founded in 1728, being the first to be established in the Upper Madeira region. With the creation of the village, the economy of the region began to be boosted, with the cultivation of

tobacco and coffee, and these products began to be exported to Belém. During all this time, the territory that currently makes up the municipality of Novo Aripuanã belonged to the village of Borba.

2.1.7 Project Zone Map (G1.4-7, G1.13, CM1.2, B1.2)

The IPOÁ Project Zone was delimited by the Project Area and a supplementary area where additional project activities will be undertaken. Thus, the Project Zone includes the communities of Vila Batista, Boa Esperança and Vale da Benção, identified in Section 2.1.9, whose population will benefit from the project activities.

There are no provisions for new communities to be included in the project. The communities that are part of the area are defined based on the eligibility criteria found in section 2.1.21: 1) Eligibility Criteria for Grouped Projects, and any communities to be included in future would have to meet these requirements. However, the other communities identified during the PD development phase, which have not been included, do not meet these requirements.

The high conservation value (HCV) areas (identified in Sections 4.1.3 and 5.1.2) area are delimited by the Project Area, with the exception of the HCV 5 and HCV 6 areas which are located in the communities listed above and indicated on the map.

No negative offsite biodiversity impacts (Section 5.3) or negative climate impacts resulting from the implementation of the project's activities are expected. There are also no direct impacts to other stakeholders from project activities (Section 4.3).

The areas of the Project Zone that are outside the property of the proponent are not in private areas registered in the Land Management System (SIGEF) maintained by the National Institute for Colonization and Agrarian Reform (INCRA), the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country. The areas are also not located in Federal Conservation Units or Indigenous Territories, as described in Section 2.5.

The Project Zone map below was developed in compliance with CCB Standards requirements.

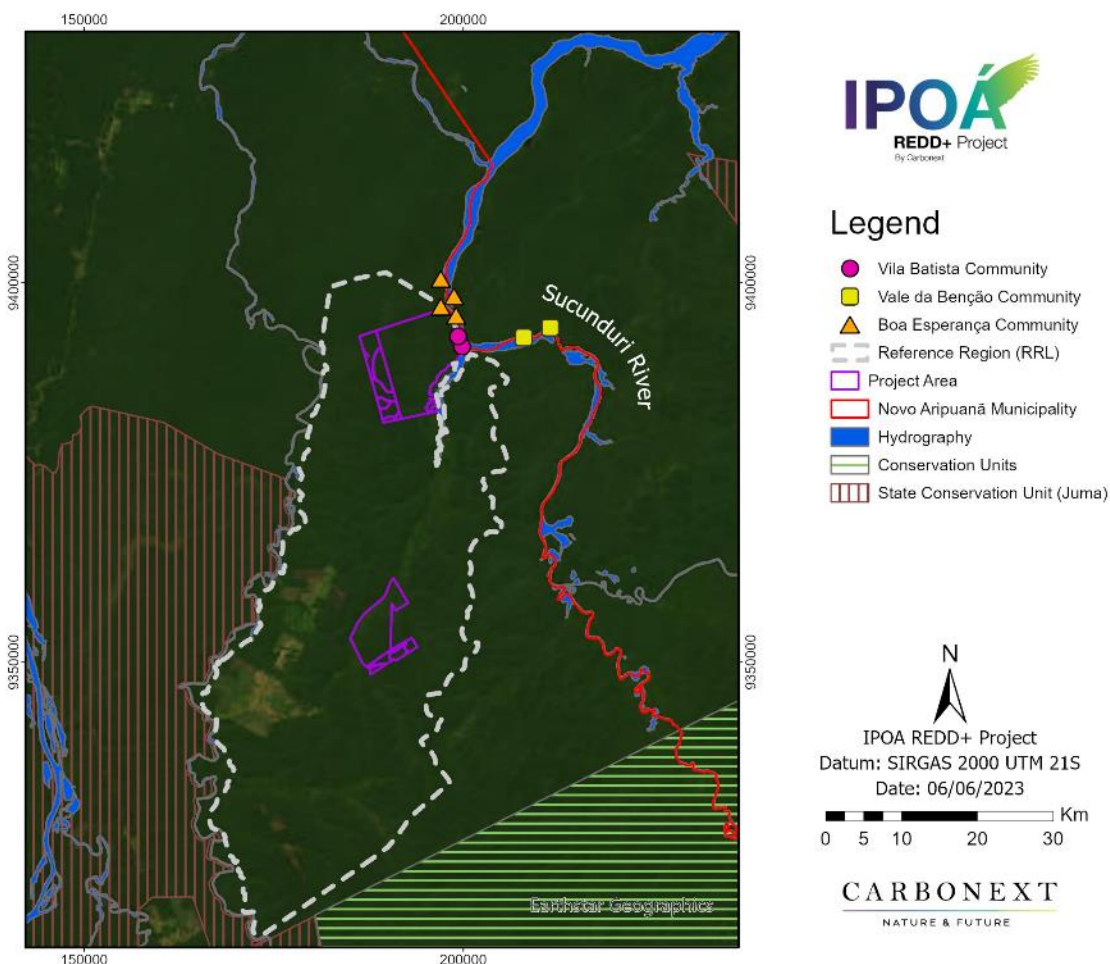


Figure 12. Project Zone: Project Area and communities

2.1.8 Stakeholder Identification (G1.5)

Once the Project Area and Reference Region had been identified, research regarding stakeholders in and around the Project Area, as well as the surrounding municipality, was initiated.

Stakeholder identification was initially based on secondary data, followed by interviews conducted in the field in March, April and October, 2022. The first step in identifying them was via remote sensing software, overlaying the Project Area with various mapping databases, in order to verify possible overlaps with officially demarcated or non-demarcated areas, and the boundaries between areas. Then, following the recommendations regarding non-permanence risk analysis (AFOLU Non-Permanence Risk Tool V4.0), a

20 km radius was drawn around the Project Area, which was analyzed for existing infrastructure indicating possible human occupation of the area.

In this way, focal groups that might be affected directly or indirectly, in a positive or negative sense, by the project were identified. The objective was to contact a variety of stakeholder organizations, groups and individuals; to inform them of the key aspects of the project, and invite those located within the Project Zone (PZ) to begin to interact with the project, and ultimately participate in the design of its activities.

Via satellite images and local visits, a community called Vila Batista (1km from the PA) was identified, which was visited in order to understand the profile of the beneficiary families and start the process of collecting information and community needs for the creation of activity plans. Also, the communities of Boa Esperança (1 km from PA) and Vale da Benção (8 km from PA) were identified and visited to diagnose the needs of the communities through interviews. The geographical limits of the Project Zone were defined according to the locations of the communities Vila Batista, Boa Esperança and Vale da Benção, as shown in section 2.1.7.

The process for stakeholder identification varied according to their characteristics, as presented in the table below. In the next section, the stakeholders identified through the mapping of the Reference Region and the Project Zone, as well as interview data collected by Carbonext's social team, are described.

Table 1 - Stakeholder Identification.

Type	Description
I	People within the Project Zone directly or indirectly affected by the carbon project and their representatives or advocates.
II	Other affected stakeholders outside the project area.
III	Companies/institutions operating in the project area that may be affected.
IV	Local policy makers and local authority representatives.
V	Any regional authorities.
VI	National government institutions or National Focal Point.

2.1.9 Stakeholder Descriptions (G1.6, G1.13)

Institutional Consultation

The stakeholders were defined as introduced in 2.1.8. Thus, the public consultation with the institutions and associations of interest was carried out. To this end, e-mails with a summary of the project were sent to the institutions listed below:

Table 2. Description of Institutional Stakeholders – State and Municipal Bodies.

Stakeholder	Description and Relevance to the Project	Average distance from PA (km)
Novo Aripuanã Local Government	Responsible for public management, promoting sustainable, social, and economic development	80
Municipal Secretariat of Social Assistance of Novo Aripuanã	Implements the municipality's social assistance policy, aimed at meeting the social interests and aspirations of the population in situations of social risk	80
Municipal Secretariat of Environment of Novo Aripuanã	Responsible for carrying out activities related to environmental management, promoting environmental education, regulation, control, regularization, protection, conservation, and recovery of natural resources	80
Municipal Secretary of Hab. Int. Social and Land Regularization of Novo Aripuanã	Responsible for the formulation, planning, coordination, and articulation of social interest housing and urban development policies, with the objective of promoting universal access to housing.	80
Municipal Secretariat of Education of Novo Aripuanã	Promote, implement, and maintain the education for the municipal population according to the Law of Directives and Bases of National Education (Lei nº 9.394/1996).	80
Municipal Secretariat of Agriculture and Supply in Novo Aripuanã	Promote rural development based on the search for alternatives to priority problems and on local potentialities	80
Municipal Secretariat of Health in Novo Aripuanã	Health unit that provides health care services in the locality of the Novo Aripuanã - AM district of the city.	96
Borba Local Government	Responsible for public management, promoting sustainable, social, and economic development.	120

Stakeholder	Description and Relevance to the Project	Average distance from PA (km)
Borba City Councilor	Politician key for contact and networking with key local government secretariats and other players.	120
Sucunduri Lodge	Popular hotel and tourism business in the project region.	20
IDESAM - Institute for Conservation and Sustainable Development of the Amazon	Non-profit organization. Works to promote the valuation and sustainable management of natural resources in the Amazon, seeking alternatives for environmental conservation, social development, and the mitigation of climate change.	268
IDAM - Institute of Agricultural Development and Sustainable Forestry of the State of Amazonas	Responsible for providing Technical Assistance and Extension services to farmers and educational producers in the State through participatory processes that ensure sustainability, citizenship, and improvement in the quality of life.	260
ICMBio - CR1	Management of the National Flona Aripuanã	80
ICMBio: Humaitá Integrated Management Center	Management of the Acari National Park management	426
ICMBio: Integrated Management Center	Management of the Pau-Rosa National Forest	426
FAS - Amazonas Sustainable Foundation	Acting in the Juma and Rio Madeira RDS	264
ISA - Instituto Socioambiental	Non-Governmental Organization (NGO) that supports the Setemã and Coatá-Laranjal indigenous territories	257
ODESPI - Organization of Development and Economic Sustainability for Indigenous Peoples	Associations for the defense of social rights: acting in causes of a social nature, such as the defense of human rights, defense of the environment, defense of ethnic minorities, etc.	266

Stakeholder	Description and Relevance to the Project	Average distance from PA (km)
General Council of the Sateré-Mawé Tribe – CGTSM	Organizational and support groups of the referred indigenous communities.	457
Union of Sateré Mawé and Munduruku Indigenous Peoples		457
Association of Sateré Mawé Indigenous Women		464
Mura Indigenous Organization of the Municipality of Novo Aripuanã and Borba		494
Association of Residents and Friends of the Juma Reserve	Organizational and support groups of the Juma RDS	80
SEMA/AM - Secretary of Environment of the State of Amazonas	Manager of the Juma RDS and manager of the Madeira River RDS	263
DEMUC - Department of Climate Change and Conservation Management	Responsible for UC Management in the State of AM, such as the Juma and Madeira River RDS.	263
Federal University of Amazonas (UFAM)	Educational institution. Carries out research in the field of forestry engineering and sustainability.	261
Amazonas State University (UEA)		262

Community Public Consultation

The public consultations with the communities directly involved (those within the Project Zone) were carried out face-to-face, through the exhibition of promotional material (see Figure 19 - Banner IPOÁ REDD+ Project.) and meetings. As mentioned above, the communities involved are described below:

- Vila Batista Community

- Boa Esperança Community
- Vale da Benção Community

Widely publicized informational meetings will be held with communities and other stakeholders to explain project activities and respond to stakeholder questions and concerns related to the project. During the course of the project, other stakeholders may be raised and consulted in the future as well. More details about community meetings and public consultation are described in section 2.3.7.

The criteria for inclusion of communities in the project are described in 2.1.21 Grouped Projects topic 1) Eligibility Criteria for Grouped Projects, define that all the communities included in the Reference Region will be welcomed and benefited by the actions of the project.

No communities are currently foreseen to join the IPOÁ REDD+ project because, although some have been identified nearby (Bom Jardim and Munguba, both in the district of Borba), they are outside the Reference Region, and therefore ineligible according to the aforementioned criteria.

2.1.10 Sectoral Scope and Project Type

The project corresponds to the VCS Scope VM0007

Sectoral Scope: 14 - Agriculture, Forestry, Land Use

Project Category: Avoided Unplanned Deforestation (AUD project activity) and Avoided Planned Deforestation (APD project activity).

This is a grouped project.

2.1.11 Project Activities and Theory of Change (G1.8)

The Theory of Change (ToC) is the causal model that maps out the logical sequence of how a given project activity will achieve various results, defined as, from short- to long-term: output, outcome, and impact, respectively⁴⁵. Ipoá's ToC is divided into four sectoral pillars – Social, Biodiversity, Climate, and Further

⁴⁵ SBIA_Part_1.pdf (verra.org).

leisure, cultural and sporting activities – which are sub-divided into a total of fourteen specific activity scopes. Please refer to the full Theory of Change framework presented in section 0 of the present PD.

The Ipoá REDD Project's activities were defined through a fully participatory process of a community workshop, described further below, allowing the communities control and agency over the project. The activities are implemented through management documents known as Operational Procedures (OPs), also presented fully in this section.

Using the available information about the pre-project social and environmental conditions in the Project Zone and surrounding areas, collected through *in loco* social and environmental diagnoses, the main necessities and wishes of the communities were identified. Subsequently, these were analyzed and the project activities were outlined by Carbonext's social and technical teams.

The participatory process with the communities involved a workshop to prioritize the activities, using a prioritization matrix. In this way, the residents identified their own main demands and necessities, in a participatory fashion, enabling the creation of short-, medium- and long-term solutions. The results of the community vote on project activities are presented in the table below. The data and evidence on this process was provided at audit.



Figure 13 - Prioritization of activities.

Table 3. Results of community vote on project activities to be implemented.

Categories	Votes
Internet	10
Medical care (more frequent)	9
Solar energy	8
Brushcutter/ hoe equipment	6
Health care	6
Improved transportation	5

School refurbishment	4
Water pump motor	4
School supplies	4
Medical center	3
Handicrafts	3
Toiletries for women	3
City Hall course to work in catering and cleaning	2
Road transportation fro the Arara community	2
Ground transportation - car	2
School transport "catraia"	1
Irrigation in the summer	1
Internet course	1
Provide trained and responsible professionals	0
Playground for children	0
Health agent - medicines	0

The current diagnosis of project activities can be improved throughout the lifetime of the project, through a continuous adaptive management methodology so that the activities of the Theory of Change can always be as robust and pertinent to the reality and objectives of the project as possible.

Carbonext's management documents are primarily composed of the overarching FIP – Forest Integrity Plan – and various Operational Procedures (OPs). They are as follows:

1. OP for community conflict resolution
2. OP for stakeholder communication
3. OP for irregular or illegal community activity (Annex III of FIP)
4. OP for community health and safety at work
5. OP for definition of alternative land-uses (Annex IV of FIP)
6. FIP – Forest Integrity Plan

7. Responsibility Agreement and Workshop
8. OP for deforestation events (Annex II of FIP)

The applicability of the OPs to each of the project's activities is summarized in the table below. They govern the management and risk mitigation of various aspects of the Theory of Change, in the ways described below.

Table 3. Applicability of the Operational Procedures to each of the project's activities

Operating Procedures/ Activities		Healthcare	Education	Occupational health and safety	Communication and Electrification	Employment	Training	Biodiversity ⁴⁶	Fire brigade training	Deforestation prevention and mitigation actions	Cultural, leisure and sporting
1.	OP for community conflict resolution										
2.	OP for stakeholder communication										
3.	OP for irregular or illegal community activity										
4.	OP for community health and safety at work										
5.	OP for definition of alternative land-uses										
6.	FIP – Forest Integrity Plan										
7.	Responsibility Agreement and Workshop										

⁴⁶ Activities to be defined by a specialist

[illegible]

Pillars

Social

Biodiversity

Climate

Cultural, leisure
and sports



It is noted that the full presentation of project activities and the corresponding Theory of Change (ToC) are found in 0 and 0 of the present document, respectively. However, a brief outline of the activities, their outcomes and impacts follows below. They are treated according to three main pillars: Social, Biodiversity, and Climate Action, complemented by other activities.

SOCIAL PILLAR

Healthcare

The healthcare activities of the IPOÁ REDD+ project consist of two main fronts:

- i. A permanent Basic Health Facility (BHF), run by the health secretariat and Community Health Agent (CHA)⁴⁷, will be supported to receive communication and electrification facilities, enabling tele-medicine services by a specialized service provider, and support for continuous medical supplies will be provided.
- ii. Emergency services support: costs will be covered for continuous maintenance and operation of water ambulance (“ambulancha”) services, which provides emergency transport for critical conditions. The water ambulance was provided initially by government entities but has been non-operational due to a lack of funding for maintenance and operational inputs.

It is important to note that both i and ii above involve facilitation by Carbonext of community-government relations, to improve situations which have historically negatively impacted local health services.

Carbonext’s Responsibility Agreement and Workshop is applicable to the healthcare activities, as they provide the framework to make collaborative definitions regarding responsibilities and legal liabilities involved, between community stakeholders, project proponents, private service providers, and public sector actors. Furthermore, Carbonext’s Operational Procedure for Community Conflict Resolution applies because the installation of benefits in any given community may cause tension with another. All Operating Procedures (OPs) and other management tools were made available at audit.

It is important to note that the OP for stakeholder communication applies to all project activities, mainly because both of the following are essential to all activities: (i) the definition of project activities prior to their

⁴⁷ *Agente Comunitário de Saúde (ACS)* in Portuguese

implementation; (ii) the communication of results to communities after their achievement. Because the communication OP is considered to apply to all activities and will not be cited after each one.

Education

The project's education activities also consist of two main fronts:

- i. School infrastructure and equipment: consisting of the project installing electricity and telecommunication equipment in schools, and purchase and transport of educational material to support what the public sector supplies.
This component involves facilitation by Carbonext of community-government relations, to improve situations which have historically negatively impacted local education services.
- ii. Teacher training and support primarily through provision of further teacher training through specialized programs.

Expected impacts are to have a school-age population not only with an educational service brought into line with the national curriculum, but also with educational levels measurably above the regional average.

Carbonext's Responsibility Agreement and Workshop is also applicable to the education activities, as they provide the framework to make collaborative definitions regarding responsibilities and legal liabilities involved, between community stakeholders, project proponents, private service providers, and public sector actors. Furthermore, Carbonext's Operating Procedure for Community Conflict Resolution applies, because the installation of benefits in any given community may cause tension with another.

Occupational health and safety

This activity also has bipartite approach, as follows:

- i. Organization of community workshops on occupational health and safety.
- ii. Distribution of personal protective equipment (PPE) distribution for worker health and safety.

These activities will contribute to improvement of work conditions for the community.

Carbonext's Operating Procedure for Health and Safety at work is applicable to the present aspect of the ToC.

Communication and Electrification

The objective is for electricity, and mobile internet communication technology to be installed in community residences. The primary items of technology at the time of writing are solar panels and antennas to improve 3G mobile phone and internet signal, making it usable for the first time. The technology applied may evolve over the project lifetime, however the objectives remain as such.

The outcomes and impacts will be to increase access to modern telecommunication for the communities. This will create effective communication and dissemination channels, bringing the Project Zone (PZ) population into contact with the wider world, including providing access to the concepts, methodologies and news regarding REDD+ and the IPOÁ project itself. This vital information will enable the communities to participate autonomously in the project they live within.

Carbonext's Responsibility Agreement and Workshop apply to communication and electrification activities, as they provide the framework to make collaborative definitions regarding responsibilities and legal liabilities involved, between community stakeholders, project proponents, private service providers, and public sector actors. Furthermore, the Operating Procedure for Community Conflict Resolution is relevant, because the installation of benefits in any given community may cause tension with another, among other considerations.

Employment

Carbonext will support the creation of a community organization, most likely an association, which will involve a specialized consultant for its constitution and relevant community training. The association will of course serve for communities to carry out commercial activities such as trading non-timber forest products (NTFPs), artisanal work, or others of their choosing. However it will also be instrumental in strengthening community governance, enabling the community to demand and obtain essential public services in the areas of health and education, for example. These activities will be facilitated and intermediated by Carbonext where it is beneficial to do so.

Forest monitors will be identified and trained from the community, to conduct surveillance rounds and send deforestation send reports and alerts to the Project Proponent (PP), applying various protocols. The social outcomes and impacts (see also Appendix 2) of this will be to increase PZ residents in employment opportunities in the sustainability sector, increase professional communication between the communities and the PP, and ultimately reduce poverty.

Carbonext's management tool of the Forest Integrity Plan (FIP) and the deforestation event OP, which define, for example, procedures and safety measures regarding the forest monitors's operations in the case of a deforestation invasion, is applicable to the present aspect of the ToC. Furthermore, the OP for definition of alternative land-uses, defining areas of community land-use and their governing policies and monitoring procedures, is relevant. Finally, the OP for irregular or illegal community activity is necessary to govern procedures regarding deforestation carried out by the communities themselves.

Training

Community members will receive annual training courses falling broadly into the areas of sustainability and conservation. These will range from sustainable agricultural production, to wild-harvesting, to avoiding deforestation, to ecotourism, to give just a few examples. The outcomes and impacts will be to increase knowledge of these sustainability themes within the communities, and to provide employment opportunities in sustainability and conservation for them, ultimately this will contribute to the creation of a new bioeconomy in the region.

Carbonext's OP regarding health and safety at work is relevant to training, as they are the activities which will build capacity in the communities for new professional activities and manages their corresponding risks.

BIODIVERSITY PILLAR

Fauna and Flora Inventories and Conservation Initiatives

Fauna and flora inventories will be carried out within the PZ at the expected frequency of every five years, in accordance with guidelines given to the PP from specialized service providers.

Based on such inventories and the specialized advice of service providers, development of conservation initiatives will be enabled.

The outcome of this will be both to monitor biodiversity indicators such as diversity, population size and health, threat levels or indicator species such as flagship, trigger or endemic. The impacts will be to improve knowledge and conservation status within the PZ.

Regarding Operational procedures to be applied to all the project's biodiversity activities, these are to be defined by the specialist consultant, following the fauna and flora inventories.

Biodiversity Research, Monitoring, or conservation activities*Biodiversity Research, Monitoring, or conservation activities*

This category of activity includes specific Biodiversity Research, Monitoring, or conservation activities which are to be determined by biodiversity consultant following the fauna and flora inventories, as well as environmental education activities, described below.

Environmental Education

Aiming to involve the community and community agents in awareness-raising and environmental education (regarding trafficking, wild-harvesting, hunting, wildlife capture, and impact of domestic animals, for example) and encouraging locals to practice species conservation and to help reduce the impacts of external pressures.

High Conservation Values

A survey of High Conservation Value (HCV) areas and species will be conducted within the PZ. The HCV concept encompasses both high biological and ecosystem value, as well as sites and species considered of importance by the communities themselves, as further described in section 4.1.3 and 5.1.2 below. Similarly, to general fauna and flora, the outcomes relate to creation of study and conservation programs for HCVs in the PZ, with positive impacts predicted on knowledge of HCVs and their very biodiversity and conservation indicators.

CLIMATE PILLAR

The climate pillar consists of the three activity categories below.

Remote Monitoring and Deforestation Prevention and Mitigation Actions

Carbonext's dedicated SIG team have developed an automated weekly monitoring system called CarbonEye, which has a strong track record in detecting invasion and deforestation events. This system is applied throughout the project lifetime as of its initiation.

Fire brigade training (according to PIF)

Community fire brigade training is foreseen within the project's climate pillar, to build community firefighting capacity. The applicable OPs are described in the category below.

Deforestation prevention and mitigation actions

In conjunction with CarbonEye, deforestation prevention and mitigation actions on the ground will be implemented when necessary, consisting of activities such as forest monitor patrols, installation of guard base infrastructure, purchase of vehicles, as well as collaboration with public authorities and police. The combined outcome of these actions is reduction in deforestation and invasion events within the PZ.

For both Fire Brigade training and deforestation prevention and mitigation actions, the Protocols relating to forest integrity, area invasion, risk mitigation, and their relevant management processes are relevant, as all such activities involve significant responsibility and risk.

Firstly, the Forest Integrity Plan (FIP) and the deforestation event OP are important here for defining procedures and safety measures regarding the forest monitors' operations, including in case of deforestation invasions.

The Responsibility Agreement and Workshop are important to define, in collaboration with communities, the responsibilities and legal liabilities of the community stakeholders, project proponents, and any further private or public sector actors involved.

The OP for definition and mapping of various land-uses by the communities is important for forest monitors to assess and track any new land-use changes, and the relevant governing policies and monitoring procedures. The OP for irregular or illegal community activity is, furthermore, necessary to govern procedures regarding deforestation carried out by the communities themselves. Finally, the OP regarding health and safety at work is relevant to these activities of the climate pillar, as they involve significant risk. All the relevant OPs cited were made available at audit.

FURTHER LEISURE, CULTURAL AND SPORTING ACTIVITIES PILLAR

It is planned that activities falling within the cultural, leisure and sporting categories will be implemented on an annual basis by the project. These activities, which are not included in the three main strategic pillars presented above, nonetheless have important outcomes, including improved access to access to leisure, sporting and cultural activities for the communities, and relationship-building both within communities and – perhaps especially relevant – between neighbouring communities, and between the communities and the PPs. Ultimately, the intended impacts are of improved quality of life for the PZ's communities.

Jurisdictional and Nested REDD+ (JNR) Programmes

Neither the Brazilian nation nor Amazonas State has an active JNR programme at the time of writing. This is despite significant advances in various frameworks, including the CONAREDD+ (National REDD+ Commission, or *Comissão Nacional Para REDD+*), within which the Brazilian national strategy for REDD+ operates. Under the CONAREDD scheme, Amazonas State announced the issuance of credits for sale at COP-27.

Carbonext is actively liaising with the Federal and State leadership and monitoring developments in the relevant organs, including the Verra JNR programme, as well as participating in the COP meetings in order to ensure that the IPOÁ REDD+ Project, and others, is integrated into any JNR framework at the earliest possible opportunity.

2.1.12 Sustainable Development

Priority Areas for Conservation

Priority areas for conservation, sustainable use and benefit sharing of biodiversity ⁴⁸ are a public policy instrument aimed at decision making for planning and implementing appropriate measures for the conservation, restoration, and sustainable use of ecosystems, and are based on the methodology of Systematic Conservation Planning (SCP)⁴⁹. In this process, spatial information is collected and processed on the occurrence of species and ecosystems, costs and opportunities for conservation. It is an ongoing process of searching for subsidies and validating the results, which results in the construction of a map of the areas and the definition of priority actions for biodiversity conservation in all the major biomes and in the Coastal and Marine Zone, in addition to a database with information about the areas.

The rules for the identification of such Priority Areas and Actions were formally instituted by the Decree n° 5.092/2004⁵⁰ by the Ministry of the Environment (MMA), where the areas to be instituted will be considered for the purposes of research and inventory of biodiversity, use, recovery of degraded areas and of overexploited or endangered species, among others.

⁴⁸ <https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias>

⁴⁹ <https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias/metodologia-1>

⁵⁰ https://www.planalto.gov.br/ccivil_03/_Ato2004-2006/2004/Decreto/D5092.htm

The IPOÁ REDD+ project area is within two priority areas for conservation, sustainable use and benefit sharing of biodiversity, which are: AMZ-359 and AMZ-669. The area sheets present in the 2nd update ⁵¹ of the priority areas for conservation in the Amazon, presents the biological importance, the priority level, recommended actions (main and secondary), biodiversity conservation targets, opportunities for the area, general characteristics and conflicting activities existing in the area.

Such information and data are fundamental and validate the importance of the REDD+ project in the region. The IPOÁ REDD+ Project is being developed to promote actions that provide significant and dignified climate, community and biodiversity benefits in an integrated and sustainable manner, cooperating with the state to preserve these areas with the expansion of tools and investments in forest conservation and biodiversity, the inspection and control of illegal activities, integrated and participatory management, the recovery of degraded areas and the restoration of ecosystem services⁵².

⁵¹<https://www.gov.br/mma/pt-br/assuntos/servicosambientais/ecossistemas-1/conservacao-1/areas-prioritarias/2a-atualizacao-das-areas-prioritarias-para-conservacao-da-biodiversidade-2018>. Published at 09/12/2021.

⁵²https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias/arquivos/fichas_amz_.pdf

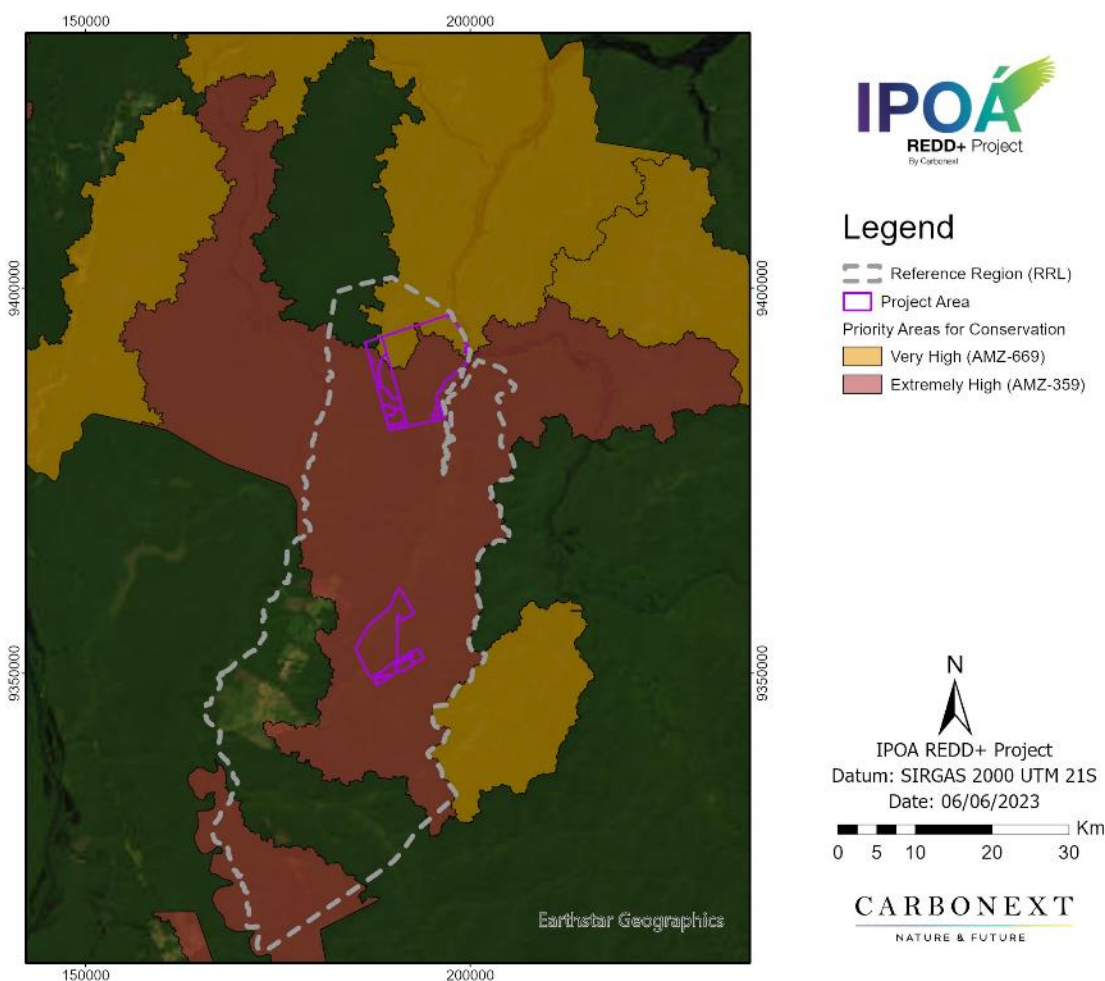


Figure 14. Priority Areas for Conservation.

Surrounding Conservation Areas

The IPOÁ REDD+ project is located at approximately 13km from the Juma Sustainable Development Reserve (RDS) and 25km from the Acari National Park. The Juma RDS was created through State Decree AM nº 26.010 of July 3, 2006, with the objective of preserving nature, ensuring the necessary conditions and means for reproduction and improvement of the ways and quality of life and exploitation of the natural resources of traditional populations, to value, conserve and improve the knowledge and techniques of environmental management developed by traditional populations, among others. The Acari National Park was created on May 11, 2016, with the objective of protecting the biological diversity of part of the Acari,

Camaiú, Sucunduri, Abacaxis rivers and their tributaries, their natural landscapes and associated abiotic values, in addition to guaranteeing the sustainability of the ecosystem services, contribute to the environmental stability of the region where it is located and provide the development of recreation activities in contact with nature and ecological tourism ^{53,54}.

The IPOÁ REDD+ project is aligned with the broader mission of these national Conservation Units, and will provide a buffer zone, supporting the combatting of illegal deforestation.

Territorial Management in Brazil

The State of Amazonas presents the Legal Amazon Ecological-Economic Macro-Zoning (MacroZEE) established by Decree N° 7.378/2010 ⁵⁵. The MacroZEE⁵⁶ maintains a two-way relationship with the main initiatives that are already transforming the Amazon and have strong political and social legitimacy, generally referenced in the Sustainable Amazon Plan (PAS), such as, for example, the Action Plan for Prevention and Control of Deforestation in the Amazon (PPCDAm) carries out mitigation actions and ongoing actions to reduce deforestation. The IPOÁ REDD+ project is in line with the main premises of the PPCDAm⁵⁷. To achieve this goal, it will be necessary to join governmental initiatives with independent actions, such as those proposed under this project, but it is not mandatory for the IPOÁ project to report to the Brazilian government on the actions taken on the national SDGs.

Other public policies are part of the PAS, such as Regional Development Policies (PNDR), Defense Policies (PND), the National Plan on Climate Change (PNMC). In addition, the state offerings the Territories of Citizenship Program, the Regional Development Plans, such as the Marajó and Xingu Plans, the Legal Amazon Land Regularization Program (Terra Legal), the Public Forest Management Law (federal law n° 11.284/2006), the Community and Family Forest Management Program (federal decree n° 6.874/2009) and the Regional Development Plan for the Amazon (PRDA).

⁵³ SEMA - AM, 2022. http://meioambiente.am.gov.br/wp-content/uploads/2018/03/Decreto-26.010_2006-Cria-RDS-do-Juma.pdf (Acesso em 14/03/2022).

⁵⁴ ICMBio. https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/unidade-de-conservacao/unidades-de-biomas/amazonia/lista-de-ucs/parna-do-acari/arquivos/dcom_decreto_criacao_parna_do_acari.pdf (Acesso em 14/03/2022).

⁵⁵ Decreto n° 7.378_2_de 01/12/2010 (abce.org.br).

⁵⁶ Referências (mma.gov.br).

⁵⁷ PPCDAm (mma.gov.br). Acessado 12/05/2022.

Recently the state of Amazonas adhered the Race to Zero campaign and reduce to the Under Coalition (Decree N° 44.16/2021), Amazonas assumed the commitment to achieve carbon neutrality by 2050 and reach its goals by half by 2030. In another opportunity to integrate international climate agreements, in 2021 Amazonas became eligible for Coal LEAF (Lowering Emissions by Accelerating Forest Finance).

Without the economy of the state of Amazonas, the "race to zero" campaign, including a goal of 2050 such as making the state a low-carbon line of financing, the IPOÁ REDD+ project will contribute to this, through its forest conservation results. To meet the neutralization goals of the government of Amazonas, committing to assume as goals of reducing deforestation and expanding a preserved area⁵⁸, an implantation of the conservation aid in the region of the goal and in the forest of the region.

Brazil's Nationally Determined Contribution (NDC)

At the international level, the IPOÁ REDD+ Project is in line and contributes with the environmental priorities set by the Brazilian Federal Administration at COP 14 in December 2008, where a 70% deforestation by the year 2018 was declared, and then further targets of zero illegal deforestation by 2030, and offsetting greenhouse gas emissions from illegal sources of vegetation removal were declared. The latter are elements of the Brazilian NDC, which the country intends to adopt more firmly in the framework of the Paris Climate Accords (COP-21)⁵⁹. Recently Brazil made official at COP 26, in November 2021, that the country's goal is to eliminate deforestation by 2028, increase to 50% the goal of reducing greenhouse gas emissions by 2030, and reduce by 30% its methane emissions in the atmosphere⁶⁰. At this COP, the Glasgow Climate Pact was signed, which reinforced the parties' commitments to keep warming below the 1.5°C limit ⁶¹. Another important decision taken was the authorization of the global carbon market, which means that countries will be able to exchange carbon credits among themselves, favoring Brazil because of the Amazon Forest (IDESAM, 2021). To achieve these goals, an essential component will be to ally governmental initiatives with independent actions such as the one proposed under this project.

Sustainable Development Goals

⁵⁸Source Agência Manaus: <https://informemanaus.com/2021/governo-do-amazonas-adere-a-campanha-global-race-to-zero-para-zerar-emissoes-ate-2050/>

⁵⁹ Source MMA: <http://redd.mma.gov.br/pt/redd-e-a-indc-brasileira>.

⁶⁰ Source FGV: <https://portal.fgv.br/artigos/quem-e-brasil-cop-26>. Access 12/05/2022.

⁶¹ Source IDESAM (2021): <https://idesam.org/opinio/cop26-entre-erros-e-acertos/>.

The SDGs - Sustainable Development Goals are a campaign of the UN, the United Nations Organization, to promote positive changes in the world of the future. The SDGs seek to promote urgent measures to ensure the 17 goals proposed in this UN sustainable agenda by conserving natural resources, combating climate change, and adopting more sustainable production and consumption practices. The SDGs are central to Brazil's 2030 Agenda and poverty can be the main mission for all internal or external initiatives, programs, projects, and activities, including REDD+ projects.

In 2015, the conclusion of the negotiations on the 2030 Agenda culminated in an ambitious result for Brazil, which was the proposal of 17 Sustainable Development Goals; 231 Sustainable Development Global Indicators to monitor and measure progress in the implementation of the SDGs; and 169 targets, representing the central axis of the 2030 Agenda, which will guide actions in the three dimensions of sustainable development⁶². Brazil has highlighted this as a great opportunity to eradicate poverty that may be within the mandate of this new Agenda.

In 2017, the Brazilian government's first "Voluntary National Reviews" focusing on institutional frameworks were put in place to incorporate the Sustainable Development Goals into public policy. In terms of conservation and sustainable use of natural resources, environmental sustainability is not only based on reducing damage to the environment, but also on the efficient management of ecosystem services that favor human development through increasing economic opportunities as well as social and ecological resilience.

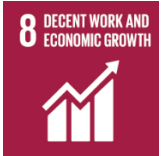
After considering the national sustainable development targets and indicators, the IPOÁ REDD+ Project aligned its activities in such a way that, at the local level, progress is being made towards achieving the targets of each of the following SDGs of the 2030 Agenda⁶³: Health and well-being, safe working environment, gender equality, climate action, among others. These SDGs that the project seeks to target with its activities are presented in the table below. The table below (Goals of sustainable development considered by the project.) points out the objectives contemplated by the project activities and are linked to the Theory Of Change presented in section 2.1.11 and attached to 0.

⁶² ODS – Metas Nacionais dos Objetivos de Desenvolvimento Sustentável (portalods.com.br). Access 04/05/2022.

⁶³ Pacto Global.

Table 4. Goals of sustainable development considered by the project.

Sustainable Development Goal	Goal	Indicator type
	1.2 - By 2030, at least halve the proportion of men, women and children of all people living in poverty, in all its dimensions, according to national definitions.	Poverty
	3.8 - Achieve universal health coverage, including financial risk protection, access to quality essential health services, and access to safe, effective, quality, and affordable essential medicines and vaccines for all.	Health
	4.a - Build and improve physical education facilities that are appropriate for children and sensitive to disability and gender, and that provide safe, non-violent, inclusive, and effective learning environments for all.	Education
	5.b - Increase the use of basic technologies, in particular information and communication technologies, to promote women's empowerment.	Equality

Sustainable Development Goal	Goal	Indicator type
 7 AFFORDABLE AND CLEAN ENERGY	7.1 - By 2030, ensure universal, reliable, modern, and affordable access to energy services.	Clean and Affordable Energy
 8 DECENT WORK AND ECONOMIC GROWTH	8.8 - Protect labor rights and promote safe and secure work environments for all workers, including migrant workers, in particular migrant women, and people in precarious employment.	Economic and Health
 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.1 - Develop quality, reliable, sustainable, and resilient infrastructure, including regional and transboundary infrastructure, for the project of economic development and human well-being, with a focus on equitable access and affordable prices for all	Infrastructure
 10 REDUCED INEQUALITIES	10.2 - By 2030, empower and promote the social, economic, and political inclusion of all, regardless of age, gender, disability, race, ethnicity, origin, religion, economic or other status	Equality

Sustainable Development Goal	Goal	Indicator type
	11.4 - Strengthen efforts to protect and safeguard the world's cultural and natural heritage	Communities
	13.2 Integrate climate change measures into national policies, strategies, and planning.	Environment
	15.2 - By 2020, Promote the implementation of sustainable management of all types of forests, stop deforestation, degraded forests and restore, increase isolation, afforestation and reforestation globally.	Environment
	16.b - Promote and enforce non-discriminatory laws and policies for sustainable development	Justice

Sustainable Development Goal	Goal	Indicator type
	17.8 - Fully operationalize the Technology Bank and the Science, Technology and Innovation Capacity Building Facility for Least Developed Countries by 2017, and increase the use of enabling technologies, in particular information and communication technologies.	Economic and social

In addition to the international laws and regulations already mentioned, the present project is also in accordance with: Convention on the International Trade in Endangered Species of Wild Flora and Fauna, 1973; Cartagena Protocol on Biosafety to the Convention on Biological Diversity, 2000; International Tropical Timber Agreement, 1994; United Nations Convention on Biological Diversity, 1992; United Nations Framework Convention on Climate Change, 1992; United Nations Declaration of the Rights of indigenous Peoples, 2007; The Kyoto Protocol to the Convention on Climate Change, 1997.

Cancun Safeguards

In 2010 in Cancun, Mexico, the parties to the United Nations Framework Convention on Climate Change (UNFCCC) agreed to seven safeguards for implementing activities that reduce emissions from deforestation and forest degradation and contribute to conservation, sustainable management of forests, and enhancement of forest carbon stocks (REDD+)⁶⁴. These safeguards address transparency, stakeholder participation, protection of biodiversity and ecosystem services, and respect for the rights of indigenous and local communities. These clauses have also been adapted by Brazil, as reported by the Ministry of Environment (MMA)⁶⁵:

*"When undertaking [REDD+] activities the following safeguards should be promoted and supported: (a) That actions complement or are consistent with the objectives of national forest programmers and relevant international conventions and agreements; (b) Transparent and effective national forest governance structures, taking into account national legislation and sovereignty; (c) Respect for the knowledge and rights of indigenous peoples and members of local communities, by taking into account relevant international obligations, national circumstances and laws, and noting that the United Nations General Assembly has adopted the United Nations Declaration on the Rights of Indigenous Peoples; (d) The full and effective participation of relevant stakeholders, in particular indigenous peoples and local communities, in the [REDD+] actions ; (e) That actions are consistent with the conservation of natural forests and biological diversity, ensuring that the [REDD+] actions are not used for the conversion of natural forests, but are instead used to incentivize the protection and conservation of natural forests and their ecosystem services, and to enhance other social and environmental benefits; (f) Actions to address the risks of reversals; (g) Actions to reduce displacement of emissions."*⁶⁶

⁶⁴ Source: "The Cancun Agreements: Outcome of the work of the Ad Hoc Working Group on Long-term Cooperative Action under the Convention", FCCC/CP/2010/7/Add.1. <https://unfccc.int/documents/6527>. Access: 05/09/2022.

⁶⁵ Source: http://redd.mma.gov.br/images/publicacoes/salvaguardas_1sumario.pdf

⁶⁶ Source: UNFCCC Decision 1/CP.16, Annex I, paragraph 2, available at: <https://www.un-redd.org/glossary/cancun-safeguards>

The present project is developed in accordance with the Climate, Community and Biodiversity (CCB) standards that are aligned and help us to demonstrate that the IPOÁ REDD+ project meets the guards for REDD+ activities in all aspects. The Table 5. The project's relationship to the Cancun safeguards. below illustrates the project's relationship to Cancun safeguards.

Table 5. The project's relationship to the Cancun safeguards.

Cancun safeguards	Project IPOÁ REDD+
A	Section: 2.5.7 <i>National and Local Laws (G5.6)</i>
B	Sections: 2.4 <i>Project Governance Structures (G4.1)</i> ; 2.5.3 <i>Free, Prior and Informed Consent (G5.2)</i> ; 2.5.7 <i>National and Local Laws (G5.6)</i> .
C	Sections: 2.5 <i>Statutory and Customary Property Rights (G5.1)</i> ; 2.5.2 <i>Recognition of Property Rights (G5.1)</i> ; 2.5.3 <i>Free, Prior and Informed Consent (G5.2)</i> ; 2.5.4 <i>Property Rights Protection (G5.3)</i>
D	Sections: 2.3 <i>Stakeholder Access to Project Documents (G3.1)</i> ; 2.3.2 <i>Dissemination of Summary Project Documents (G3.1)</i> ; 2.3.3 <i>Informational Meetings with Stakeholders (G3.1)</i> ; 2.3.4 <i>Community Costs, Risks, and Benefits (G3.2)</i> ; 2.3.5 <i>Information to Stakeholders on Validation and Verification Process (G3.3)</i> ; 2.3.6 <i>Site Visit Information and Opportunities to Communicate with Auditor (G3.3)</i> ; 2.3.7 <i>Stakeholder Consultations (G3.4)</i> ; 2.3.8 <i>Continued Consultation and Adaptive Management (G3.4)</i> ; 2.3.9 <i>Stakeholder Consultation Channels (G3.5)</i> ; 2.3.10 <i>Stakeholder Participation in Decision-Making and Implementation (G3.6)</i> and 2.5.3 <i>Free, Prior and Informed Consent (G5.2)</i> ; 2.5.4 <i>Property Rights Protection (G5.3)</i> .
E	Sections: 5.1.1 <i>Existing conditions (B1.1)</i> ; 5.1.2 <i>High Conservation Values (B1.2)</i> ; 5.1.3 <i>Without-project Scenario: Biodiversity (B1.3)</i> and 4.1.1 <i>Descriptions of Communities at Project Start (CM1.1)</i> ; 4.1.2 <i>Interactions between Communities and Community Groups (CM1.1)</i> ; 4.1.3 <i>High Conservation Values (CM1.2)</i> ; 4.1.4 <i>Without-Project Scenario: Community (CM1.3)</i> .
F	Sections: 2.1.18 <i>Risks to the Project (G1.10)</i> and 2.1.19 <i>Benefit Permanence (G1.11)</i> .
G	Sections: 3.1.3 <i>Project Boundary</i> ; 3.2.3 <i>Leakage</i> ; 4.3 <i>Impacts on Other Stakeholders (CM3.1)</i> ; 4.3.2 <i>Mitigation of Negative Impacts on Other Stakeholders (CM3.2)</i> ; 4.3.3 <i>Net</i>

	<i>Impacts on Other Stakeholders (CM3.3); 5.3 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2); 5.3.2 Net Offsite Biodiversity Benefits (B3.3).</i>
--	--

2.1.13 Implementation Schedule (G1.9)

Over the project lifetime, from the development to implementation, certain important dates and milestones are foreseen. These include introductory meeting dates, start and end dates of each project activity, GHG monitoring events, project socio-environmental activities (distinguishing partnership with public and private sectors), etc. The table below presents important milestones already achieved or foreseen over the first four years of the project, as others are still being defined.

Table 6. Implementation Schedule – first four years

Year	2021				2022				2023				2024			
Activity/ Trimester	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project start date					Proj. Start Date											
Project development with community stakeholders																
Project activity with private sector																
Biodiversity activity																
Deforestation prevention field activity																
Validation or verification visit																

To described in more detail the activities depicted above, the following activities have already occurred or are planned:

- Dec 2021: Signing the contract to begin the project;
- Mar 2022: Social mapping of the PZ and surrounding communities;
- Mar – Apr 2022: Research visit to the Project Zone to make first contact, present the project to communities and their leaders, and conduct social research and first contacts through Carbonext's social team and the local field team;
- Jun 2022: Remote Sensing Monitoring via CarbonEye is conducted continuously throughout the project lifetime, which led to the first deforestation alerts in June 2022,
- July 2022: Following this detection, deforestation prevention activities including installing signage in the field, and meeting with local organs were conducted as of July 2022;
- Oct 22 – Feb 23: Implementation of project activities as described in 2.1.11, including public and private partnership development, and installation of communication technology in the field have occurred in October 2022 and January – February 2023;
- Sept 23: data gathering for MR2, continuation of public and private sector activities initiated above, and biodiversity inventory, and subsequent research and conservation actions are predicted as of September 2023;
- 2nd trimester 24: The site visit for MR2 verification is predicted in the quarter following the completion of the project's second year;
- 3rd trimester 24: field visit to continue project activities and gather data in preparation for MR3.

Further actions to be repeated throughout the project lifetime, including the project activities described in the Theory of Change (see section 2.1.11) and routine GHG project activities, are presented in the table below.

Table 7. Implementation Schedule – repeated throughout project lifetime.

Milestone	Date	Strategic lines	Milestone(s) in the project's development and implementation
1	Before every verification	Social/ Biodiversity/ Climate	Monitoring site visit to: (i) gather field data prior to MR (ii) update community, biodiversity, and climate indicators
2	Annual minimum at	Social/ Biodiversity/ Climate	Activity implementation and development site visit regarding project actions in the social, biodiversity and climate areas
3	Continuous	Climate	Remote monitoring and deployment of deforestation event protocol in the field following any occurrences

4	Before every verification	GHG Project	Monitoring of deforestation and quantification of emission reductions
4	Every 5 years	Biodiversity	Fauna, and flora inventory
5	Annual	GHG Project	Verification audit site visits

2.1.14 Project Start Date

The project start date of the Ipoá REDD+ project is March 17, 2022. Explanations per component follow below.

AUD component:

The activities that led to the generation of GHG emission reductions began on March 17, 2022.

The project proponent (landowner) acquired the Campos property on August 21, 2019, and the Maanaim property on October 13, 2019, with the objective, as stated in the purchase and sale agreements, of carbon credit activities in the component of the legal reserve, being 80% of forests on the property (corresponding to the AUD component), and timber production activities, which includes a component of deforestation, to maximize timber production capacity, in the remaining 20% of the property (corresponding to the APD component).

In December 2021, the landowner and Carbonext (Project Developer) signed an agreement for the development and implementation of the REDD+ project.

After the recognition and analysis step, Carbonext started the implementation of project activities in the field on March 17, 2022. This event is considered as the Project Start Date.

On this occasion, an experienced social technician from Carbonext, accompanied by a local consultant, visited the Project Area, carrying out public consultations with the communities in the Project Zone, presenting the principles and benefits generated by the REDD+ project. The presentation lecture given to local inhabitants is considered the first activity to generate reductions in the project's GHG emissions, as it made communities aware of preserving the forest and encouraged them to monitor possible illegal deforestation.

APD component:

Regarding APD activities, May 3rd, 2022 was the date that the project owner submitted a request to the official organ IPAAM, to obtain permission to deforest the legal proportion (the aforementioned 20%) of the Maanaim and Campos properties in the Project Area (PA). The documents required and submitted to IPAAM for the request were made available at audit.

The planned deforestation license was considered necessary by the project owner as a form of backup plan, in case the REDD+ activities are unsuccessful, in which case he would opt for the regional business-as-usual of deforestation with timber extraction, and possible installation of permanent agricultural activity, following deforestation. This business-as-usual activity is also what the proponents personal background is in, and would provide the financial returns he requires, in the scenario of non-success of the present REDD+ project. To summarize, the proprietor's stated plan, since the purchase of the land, was: REDD carbon activities as the primary plan, with planned deforestation combined with timber production activities, as the backup alternative.

Regarding planned deforestation documentation, the VM0007 methodology, VMD0006-BL-PL_v1.3 module, states that: "if government approval is required for deforestation to occur, the intention to deforest within the project area must be demonstrated by evidence:

- Recent approval from relevant government department (local to national) for conversion of forest to an alternative land use; or
- Documentation that a request for approval has been filed with the relevant government department for permission to deforest and convert to alternative land use;"

In accordance with these methodology requirements, the filed deforestation request mentioned above is the documentation corresponding to the proof and start of the project's APD activities.

2.1.15 Benefits Assessment and Crediting Period (G1.9)

The crediting period is from March 17, 2022, to March 16, 2052, therefore comprising a duration of 30 years.

2.1.16 Differences in Assessment/Project Crediting Periods (G1.9)

There is no difference between the GHG emission accounting period and the community and biodiversity benefits assessment periods.

2.1.17 Estimated GHG Emission Reductions or Removals

The estimated total net GHG emission reductions or removals by the Project Activity over 30 years is 2,522,057 tCO₂e. This represents an average of 84,069 tCO₂e per year, summing all types of avoided deforestation (AUD and APD).

Table 8. Total ex ante estimated actual net carbon stock changes and emissions of non-CO₂ gases in the project area.

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
2022	246,706
2023	463,954
2024	483,341
2025	498,194
2026	164,407
2027	85,314
2028	79,726
2029	56,353
2030	56,353
2031	56,322
2032	61,237
2033	47,115
2034	53,940
2035	10,868
2036	6,476
2037	18,598
2038	7,310

2039	20,051
2040	41,527
2041	14,952
2042	7,102
2043	6,607
2044	3,511
2045	8,805
2046	2,688
2047	2,773
2048	2,131
2049	1,703
2050	11,861
2051	825
2052	1,307
Total estimated ERs	2,522,057
Total number of crediting years	30
Average annual ERs	84,069

2.1.18 Risks to the Project (G1.10)

Some natural and human-induced risks are expected to occur to the climate, community, and biodiversity benefits during the project lifetime. They each have corresponding measures taken to mitigate the risk, as presented in the table below (Potential project risks on community, biodiversity, and climate aspects. This item is to be revised at each monitoring period along the project lifetime, in accordance with renewed assessments of each factor.

Further risks to the project were analysed using the guidelines of the Non-Permanence Risk Tool, version 4.0, and the results are presented in the IPOÁ Non-Permanence Risk Report. And it is noted that section 2.3.4 below presents project risks focused on community aspects, and the Adaptive Management Plan (section 2.3.8) identifies, assesses, and creates a mitigation plan to address them.

Table 9 - Potential project risks on community, biodiversity, and climate aspects.

Identify Risk	Potential impact of risk on community, biodiversity, and climate	Actions needed and designed to mitigate the risk
Disputes over access/use rights (or overlapping rights)	Disputes may negatively impact communities. Or indeed compromise the continuity of the project itself. and consequently, the benefits to the climate, biodiversity and communities may be compromised.	External disputes: Application of invasion protocol (provided at audit): Deployment of on-the-ground field team or government organs, as applicable. Mediation of potential disputes and judicial monitoring of possible claims. Internal disputes (PP – community or intra community): conversation, explanation and adjustment of project benefits, or judicial support for possible claims. ⁶⁷
Changes in policies and laws governing external actors	Changes can compromise the continuity of the project and consequently the benefits to the climate, biodiversity, and communities.	Legal monitoring and adaptation to new rules

⁶⁷ See section 2.3.4 for further details on Operational Procedures applicable to mitigate risks to Community from land and deforestation issues.

Increase of fires	Carbon emissions, biodiversity loss and risks for the communities.	Anthropogenic forest burning is common in the Amazon biome, as landowners use fire to clear their land⁶⁸. Although project field and GIS analyses assessed this risk as low in the PZ, the possibility of fire by anthropogenic, natural or an interaction between the two will continue to be monitored, weekly by the CarbonEye system, in conjunction with forest monitors and community reports.
Deforestation/Degradation	Deforestation or degradation of forest cover.	Remote satellite monitoring (Carbonext's CarbonEye System) to identify and act on deforestation; on-site visits and community environmental education to combat degradation; measures for filing a report of occurrences and/or recovery of the area if necessary

2.1.19 Benefit Permanence (G1.11)

The community activities are designed to transform local economies into activities based on the low carbon economy that become perennial and financially attractive in the long term.

⁶⁸ <https://g1.globo.com/natureza/noticia/2019/08/23/queimadas-e-desmatamento-estao-relacionados-na-amazonia-entenda.ghml>

The project's priority will ensure the substitution of pastureland use, the main cause of deforestation in the region, with alternative activities with low environmental impact, in order to maintain the benefits to the climate, communities and biodiversity.

Such activities are all designed to be managed in the future by the communities themselves, without the need for external interventions. This item may be revised considering the manifestations of stakeholders in the public consultation in progress.

2.1.20 Financial Sustainability (G1.12)

There are no economic activities being carried out by the proponent in the Project Area. The project implementation annual costs are estimated at R\$ 296,620.05 throughout the project.

As of 2023, costs are expected to be covered by the revenues obtained from the sales of carbon credits. The Project Proponent have sufficient cash funds to keep the project until breakeven. This financial analysis may change due to needs identified in the ongoing public consultation.

2.1.21 Grouped Projects

This is a grouped project, and it was designed as an Avoided Unplanned Deforestation and Avoided Planned Deforestation, applying VM0007 methodology, Version 1.6, dated 08 September 2020.

In this sense, this project activity is designed to include more than one “project activity instance”, such as the accession of new communities or landowners to the project across its lifetime. Thus, this grouped project is designed to allow the expansion of the project activity, subsequent to project validation.

1) Eligibility Criteria for Grouped Projects (G1.14)

For this grouped project, the following set of eligibility criteria for the inclusion of new project activity instances has been defined, which is applicable for VM0007 REDD+ AUD activities, and the geographic area demarcated by the Reference Region:

This grouped project has one clearly defined geographic area within which project activity instances may be developed, which is defined using geodesic polygons, corresponding to the current Reference Region. The determination of baseline scenario and demonstration of additionality were based upon the initial project activity instances (two farms, Maanaim and Campos), that are presented in this CCB-PD for validation. For inclusion of new geographic areas, it will be demonstrated that such areas are subject to the same baseline scenario and rationale for the demonstration of additionality as the geographic area that does include initial project activity instances.

A single baseline scenario is determined for the entire designated geographic area (Reference Region), in accordance with VM0007 methodology. The additionality of the initial project activity instances was demonstrated for each designated geographic area, in accordance with the methodology applied to the project. All factors relevant to the determination of the baseline scenario or demonstration of additionality (i.e., common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; historical deforestation rates) were assessed across the grouped project geographic area and respective Reference Region.

The project proponent has defined the capacity limit for this project activity in terms of the Reference Region, with possible exceptions as defined below.

2.1.22 Scalability Limits for the Grouped Projects (G1.15)

The project scalability is delimited by the geographic area, in this project defined as the Reference Region. The geographical and jurisdictional limits for this expansion are predicted to be within the municipalities of Novo Aripuanã, however this may change at the monitoring events. If, in an exceptional case, a new instance is located outside the Reference Region, it will be guaranteed that all limiting criteria respect the same conditions of similarity of historical deforestation rates as applied to the initial project instances, within the Reference Region. Also, the eligibility criteria described in the section above shall be met. For the addition of new instances to the project, the interested institutions must present a financial plan as well as an implementation schedule to guarantee the permanence of the project and the fulfillment of its objectives in accordance with the CCB standard.

2) Risk Mitigation Approach for Grouped Projects (G1.15)

The risks associated with the non-continuity of benefits are reduced since the project activities proposed for the new instances will be designed and led in partnership with the owners of these new instances, who are

directly involved in the management and administration of the areas. Carbonext's technical team, which has the technical capacity to determine the project strategies and activities, in addition to offering technical support to the landowners for the correct execution of the activities and monitoring will develop this plan in case the project expansion occurs. Thus, Carbonext's continuous technical and scientific monitoring generates a support framework for monitoring, reporting and verification issues, which ensures the quality of the work and a continuous process of adjustment and improvement.

2.2. Without-project Land Use Scenario and Additionality

2.2.1 Land Use Scenarios without the Project (G2.1)

The present project activity has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal, or destruction. Instead, this project only foresees the reduction of emissions that would occur in the Project Area.

The baseline scenario is the same as the conditions existing prior to the project initiation. The description of scenarios is detailed in section 3.1.4.

2.2.2 Most-Likely Scenario Justification (G2.1)

The most-likely scenario to occur is planned and unplanned deforestation followed by pasture with cattle ranching.

Forest land is expected to be converted to non-forest land in the baseline scenario, under two mechanisms, or components of the baseline: Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD). The landowners cannot afford efforts and costs to keep long term vigilance of the project boundaries to avoid unplanned land use change from uncontrolled trespassing and deforestation, without carbon credit income.

A study conducted by Soares (2022) in the neighboring municipality of Apuí identified that the municipality of Novo Aripuanã is a new pole of expansion of the agricultural frontier, following the dynamics of the search for cheap land, investment in deforestation and agricultural production.⁶⁹

According with Map Biomas⁷⁰, pasture is the most common alternative land use in the municipality of Novo Aripuanã. In 2010, pastures covered only 1.25% of the municipal territory. However, in 2020 this percentage rose to 3.21%, totalling 132,082.99 hectares. This value represents 99,41% of non-natural land uses in Novo Aripuanã.

The increasing trend was also observed in the number of cattle identified in the municipality of Novo Aripuanã by the agricultural census. In 2010, the number of heads was 13,989, while in 2020 this amount increased to 42.100⁷¹, an increase of 300%.

The percentage of the total area of pasture in Reference Region increased from 0.57% in 2010 to 1.67%, More information is provided in Section 3.1.4.

2.2.3 Community and Biodiversity Additionality (G2.2)

As mentioned in item 0, the most likely land-use scenario in the absence of the project is the expansion of pasture resulting in forest deforestation for the creation of cattle ranching. According to the Forest Code (Law 12.651/2012), 80% of private properties in the Legal Amazon are defined as Legal Reserve, as described in:

“Article 16. The forests and other types of native vegetation, excepting those located in Areas of Permanent Preservation, as well as those not subject to the politics of restricted use or subject to specific legislation, are susceptible to suppression, as long as a portion of vegetation is preserved, as Legal Reserve, at a minimum:

I – eighty percent (80%), in rural estates located in forest zones located in the Legal Amazon.”

⁶⁹ https://www.teses.usp.br/teses/disponiveis/91/91131/tde-11022022-160839/publico/Pedro_Gandolfo_Soares_versao_revisada.pdf

⁷⁰ <https://mapbiomas.org/>

⁷¹ <https://sidra.ibge.gov.br/tabela/3939>

Thus, the landowner has the right to convert forest to other land-uses on 20% of his property, excluding the permanent conservation areas. In the case of Manaaim and Campos ranches, in the absence of the IPOÁ REDD+ Project, the owner had demonstrated the intention to use this part of the property available for cattle ranching, removing the characteristics of the local landscape and negatively impacting local biodiversity.

Despite the 80% - 20% division stipulated in the Forest Code, the project considers the whole area as potentially vulnerable to unplanned deforestation, since – due to its largely illegal nature – unplanned deforestation could occur anywhere in the Project Area, not only in the area available for Legal Reserve.

This can be contrasted with APD – Avoided Planned Deforestation – which can only be applied conserved above and beyond the 80% stipulated as Legal Reserve areas.

The removal of the forest for the implementation of ranching activities besides resulting in GHG emissions and destruction of the vegetation present, removes the habitat of several faunal populations. In addition to the space destined for cattle cultivation, in a future with ranching activities on the farms it would still be necessary to remove the forest in the direction of the Sucunduri River for the flow of the beef cattle, resulting in greater impacts to local biodiversity.

The creation of the project REDD + IPOÁ serves as an aid to the protection of biodiversity of the JUMA RDS, a neighboring conservation unit located 13km from the PA, serving as a refuge for wildlife as well as a temporary or permanent habitat for animal populations (such as birds and bats), including endemic and threatened ones, that may share the same environment. The presence of the REDD+ IPOÁ project also allows the survey and monitoring of flora and fauna species, including those at different levels of threat and endemism, enabling a better understanding of the region's scenario for conservationist decisions.

As for the social and environmental aspects, in the absence of the project, the communities of Vale da Benção, Boa Esperança and Vila Batista would continue without infrastructure support for communication, and the IPOÁ REDD+ Project is dynamizing efforts to bring internet to the region. Furthermore, planned socio-environmental actions regarding aspects of health improvement, work safety, connectivity and accessibility, educational program and training (better described in section 4.4) will only be possible due to the presence of the IPOÁ REDD+ Project. Without the project's activities, it is expected that communities will continue to have precariousness in meeting their basic needs.

Considering that the project region is at risk of deforestation due to the cited motives above, the IPOÁ REDD+ Project climate, community and biodiversity benefits of conservation would not occur without the project's activities. The benefits include the following:

- a. Protection and conservation of 4,200.43 ha.
- b. Creation of jobs position, directly or indirectly.
- c. Increase of life quality and access to internet, health care and education in the communities.
- d. Monitoring of fauna and flora species, including endangered ones.
- e. Technical capacity building in different topics.

2.2.4 Benefits to be used as Offsets (G2.2)

This item is not applicable, no community and biodiversity benefits will be used as offsets.

2.3. Stakeholder Engagement

2.3.1 Stakeholder Access to Project Documents (G3.1)

Project documents, such as project description documentation and monitoring reports, will be available through the VERRA website⁷² and will be made public and interested parties will be invited to review them and submit their comments. In addition, project stakeholders will be invited to participate in all public consultation periods and will receive a project summary.

All communities will have access to project information and documents pertinent to them in the local language. These documents will be given to them in an accessible and inclusive language so that they can be understood, and we will make them public in regular meetings with the communities, showing the project's progress and other questions.

Finally, we will provide a short and very illustrative copy of the Project Description that will be presented after the Validation period and registration in VERRA; of the Monitoring Reports after each verification period; and of the validation and verification processes by providing timely information on the auditor's visit, the location of the visits, the date, and how this process will occur (details in section 2.3.5). Thus, facilitating communication between the parties.

⁷² <https://registry.verra.org/>

2.3.2 Dissemination of Summary Project Documents (G3.1)

The project summary documentation, including the minimum points for the overall project profile described in G1.1-9⁷³, has been disseminated by email to all project stakeholders and will be actively disseminated to the communities with distribution of the summary copy to each of them in the local language (Portuguese). The summaries of the Monitoring Reports will be disseminated in the same way after each verification period.

In addition, the summaries of the project documents will be publicly available through the VERRA website.

During the technical team's visits to the Project Area, the project proposal is presented to communities and any questions that arise are answered in the field. Copies of the project summary in Portuguese will be made available in the project area and distributed to community leaders for interested people, in addition to explanatory folders with the main points of the project that are distributed to community members.

2.3.3 Informational Meetings with Stakeholders (G3.1)

Between March 14 and 21, 2022, a technical social analyst from Carbonext conducted an initial diagnosis in the vicinity of the IPOÁ project area, traveling along the Acari and Suncunduri Rivers and the Arara and Capinarana Igarapés. This trip was important to make the first contacts with the communities and the detailed description of each family present in this journey.

On March 16, 2022, in the community of Boa Esperança, an informative meeting about the IPOÁ REDD+ project was held with the residents. The social analyst who was responsible for the presentation of the project used clear communication to notify about the main points of the project and clarified the doubts raised by the inhabitants. To illustrate the project area, a banner (Figure 19 - Banner IPOÁ REDD+ Project.) was used with a short description of the project and the contact email of Carbonext, in addition, the WhatsApp contact of the social analyst was presented to the community leader who has a cell phone. On this occasion the project was also presented to the leaders of Vila Batista and Vale da Benção, where both signed the consent form of the project.

⁷³ https://verra.org/wp-content/uploads/2016/05/SBIA_Part_1.pdf



Figure 15 - Visit to the leaders of the communities of Vale da Benção and Vila Batista.



Figure 16 - Presentation of the project to the Boa Esperança community.

Between April 23 and 25, 2022 we conducted a new visit to the project communities, and on this occasion socio-economic questionnaires were applied to families in all communities, articulation and public consultation was carried out with the residents present in the three communities involved in the IPOÁ REDD+ project: Vila Batista, Boa Esperança and Vale da Benção. During the visit, 44 people were interviewed in 6 family nuclei distributed in the three communities.

The informative meeting held on April 25th in the community of Vila Batista involved 24 participants from the three communities. This meeting was fundamental for the residents to understand more about the project, for those who were not present at the previous month's visit to get to know the project, answer the questions that the residents had, and understand and write down the demands they have. In none of the meetings so far, we have had negative answers or opinions about the project's development in the region.

The means of dissemination of the April meeting was done with the articulation executed the day before by the social analyst and the technical project analyst of Carbonext, who visited all the houses that would correspond to one of the three communities involved (Project Zone) and that were identified in March 2022, informing one by one about the meeting, in addition to the distribution of flyers for all residents presents.



Figure 17 - Invitation flyer for community consultation meeting.

2.3.4 Community Costs, Risks, and Benefits (G3.2)

During the community consultations, information about the project was provided to the communities of Vila Batista, Vale da Benção and Boa Esperança, where visual material was used to demonstrate the demarcation information and the Project Zone. The explanation was delivered to the communities by a social specialist, being communicated in an accessible way and transparent, where the community members could understand the purpose of the project, and participate in identification of its potential costs, risks, and benefits.

Despite the premises that the Project Developer envisages, for example, no financial costs to the communities, the equitable distribution and access to project benefits to all, and that the project activities aim to promote ecosystem services, nonetheless the identification of such risks and benefits in a transparent and participatory aspect is essential to the project design.

It is noted that the table below focuses the risks already identified in 2.1.18 on the community scope.

Table 10 – Potential Community risks

Identify Risk	Potential impact of risk on community benefits	Actions needed and designed to mitigate the risk
External: Disputes and changes relating to land rights or access/use rights	Displacement of community by potential invaders. Loss of potential property rights for community members. Potential negative legal consequences for community members carrying out illegal deforestation	Forest cover and land-use monitoring and control. Judicial support for PPs and communities regarding potential legal claims. Carbonext has Operational Procedures (OPs), which differentiate invasive deforestation events from illegal community activity. They are addressed by the OP for Deforestation Events and the OP for Illegal Community Activity, respectively. The procedures adopted by each are very different because, for example, while the first involves the filing of complaints with responsible government and police organs, the second involves establishment of conversation with communities and reinforcement of areas

Identify Risk	Potential impact of risk on community benefits	Actions needed and designed to mitigate the risk
		for alternative land-uses, as first courses of action. ⁷⁴
Internal: Disputes and changes in land rights or access/use rights	Increase in tensions between communities and the PP, due to issues such as understanding of benefit-sharing. Increase in tensions between communities of the PZ, due to, e.g. movement of populations, as a result of, for example, project activities benefitting one community or another.	Education and explanation of project resource allocation, benefit-sharing, and other policies. Design of project activities promoting, as much as possible, access to all and equitable benefits Mediation of potential disputes and judicial support of possible claims.
Changes in policies and laws governing external actors	Various changes in community circumstances resulting from legal changes: land tenure; resource use rights; access rights; carbon credit rights; obligations to conserve; financial penalties for legal violations, etc.	Legal monitoring and adaptation to new rules carried out by project developer legal team.
Natural risks including fire; plagues; ecosystem imbalances, etc.	Various impacts resulting from: fauna depletion; loss of sustenance from hunting and gathering; alterations in weather patterns and phenomena.	Anthropogenic forest burning is common in the Amazon biome, as landowners use fire to "clean" their land ⁷⁵ .

⁷⁴ These mitigation measures are understood to be in line with Verra safeguards 3.18.5: Verra safeguards, 3.18.5: "The project proponent shall identify the risks for local stakeholders to participate in the project, including project design and consultation. Risks may include trade-offs with food security, land loss, loss of yields, and climate change adaptation. The project shall be designed and implemented to avoid trade-offs and mitigate the identified risks to local stakeholders".

⁷⁵ <https://g1.globo.com/natureza/noticia/2019/08/23/queimadas-e-desmatamento-estao-relacionados-na-amazonia-entenda.ghtml>

Identify Risk	Potential impact of risk on community benefits	Actions needed and designed to mitigate the risk
		Although project field and GIS analyses assessed this risk as low in the PZ, the possibility of fire by anthropogenic, natural or an interaction between the two will continue to be monitored, weekly by the CarbonEye system, in conjunction with forest monitors and community reports. Regarding the latter, the awareness of residents in the communities about the risks of fires and the procedure of communication to Carbonext or responsible agencies if this occurs is important.
Deforestation/Degradation	Deforestation or degradation of forest cover. Increase in deforestation due to movement of populations between communities of the PZ as a result of project activities.	Automatic satellite monitoring (Carbonext's CarbonEye System) to identify and act on deforestation; on-site visits and community environmental education to combat degradation; measures for filing a report of occurrences and/or recovery of the area if necessary.

Regarding the table above, it should be noted that the approach to land rights situation or access/use rights is designed to actively facilitate possible community claims, which have reasonable claims to legitimacy, and but are not fully recognized, legally. Any claims to land rights which are fully legally recognized will be honored with the corresponding proportion of carbon credit sales proceeds.

2.3.5 Information to Stakeholders on Validation and Verification Process (G3.3)

The means of communication with the community stakeholders in the pre-validation and verification visits was project agent visit supported by presentation including visual cues. At this time, the audit process was explained to the community, including the return audit visits for initial validation and verification, followed by annual verification visits.

The communication methods above were designed to be accessible intelligible to the target audience, delivered in Portuguese in person, in relatable language.

2.3.6 Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

As of the project agent's validation visit to this remote region, all audit field visits will be communicated along with invitations to stakeholders to comment on the project documents in the online public consultation period. This is achieved, initially, through WhatsApp message contact with certain community members who have this capability, who are requested to pass the message on. After the project's telecommunication initiatives are operational, the message will be broadcast to all community households.

The site visits are informed with at least a month's notice, so that stakeholders can plan to be available. As the project developer is in the process of building the relationship with to the community, this section will progress along the subsequent Monitoring Reports.

2.3.7 Stakeholder Consultations (G3.4)

Consultations

The communities involved in the project were invited to participate fully and effectively in the decision-making process and in the implementation of the project throughout its lifetime, with regular meetings held to disseminate information.

In the meetings and public consultations held in the communities of Boa Esperança and Vila Batista the main information about the IPOÁ REDD+ project was transmitted, and the residents were encouraged to comment, suggest and participate in the development of the project. So far, there were no negative

comments about the existence of the project and its development, in general, all comments from residents were positive for the project development.

With the first meeting, in March in the community of Boa Esperança, we had the consent of all the communities involved, through the signatures of the leaders of the three communities: Carlinhos from Vila Batista, Magno from Boa Esperança, and Bráulio from Vale da Benção. Although the communities do not have any kind of association or documents that recognize these leaders, in the second visit to the communities, questionnaires were applied and interviews were carried out with all the residents present, and all of them recognized these figures as representatives of the communities.

In the second visit there was also the articulation with the communities involved in the project, to invite the largest number of residents present to attend the meeting and be aware of the project. In this meeting, held in April in the community of Vila Batista, the residents were encouraged to ask questions, suggest improvements, and participate in the decision-making process in the choice of activities that will be implemented in the region. The meeting had (according to the meeting's attendance list) 24 residents present.

Project Designer

The strategy was to incorporate the communities in the decision-making processes of the IPOÁ REDD+ Project, and this occurred during meetings and public consultations held in the communities of Vila Batista and Boa Esperança, where the communities involved were informed about the project and had full and effective participation in the process.

Based on the data collected through the social and environmental diagnosis carried out in loco about the social and environmental conditions in the project area, the main demands of the communities were identified. Subsequently, these demands were analyzed by the technical team and the project activities were outlined considering climatic, community and biodiversity factors.

In a participatory process with the communities, an activity prioritization workshop was held (see section 2.1.11), where in a clear and inclusive way the residents themselves draw their main demands and select the central demands for the creation of short-, medium- and long-term solutions, thus contributing to the design of the project activities. From the technical perspective and from the perspective of the communities

themselves, the main strategies and activities needed to meet these demands were mapped out within the context of the IPOÁ Project.

With the design of the communities' main activities of interest, the project team started a long process of structuring and planning these activities raised by the communities, in order to execute and benefit from them. And this process of incorporating the communities in the decision-making processes will be part of the IPOÁ REDD+ project during its thirty-year duration, so this section will progress in relation to the subsequent monitoring reports, and new methodologies and tools may be used to guarantee the effective participation of the communities.

2.3.8 Continued Consultation and Adaptive Management (G3.4)

The project understands as fundamental the existence of continuous communication channels between the project proponents, the stakeholders, and the communities involved. To guarantee that communication and consultations are permanent throughout the existence of the project, a communication plan was created for the IPOÁ REDD+ Project, but for this plan to be applied, it was essential for the communities to have a fixed means of communication, which was implemented in 2022. Below, have the table with the project's communication plan.

Table 11 - Communication plan.

Communication Action	Focus	Parties	Frequency	Public	Evidence
Informative Meeting about the Project	Inform residents and stakeholders about the project. And consult with the communities about the implementation of the project	In person	Annual	Community residents and stakeholders who want to participate	Attendance list List of suggestions
Consultation Meeting	Consult with the communities about the project implementation	In person	Annually	Community residents	Attendance list List of suggestions

Communication Action	Focus	Parties	Frequency	Public	Evidence
Request for Information	Soliciting community leaders for information about community and forest situations	Telephone	Monthly	To community leaders	Questionnaires
Notifications	Notify the community about technical visits, auditor visits, etc.	Telephone, and occasionally in person	Whenever necessary	To community leaders	-
Submission of Project Summary	Inform identified stakeholders by e-mail about public consultation periods and verifications	E-mail	Annually	Community residents and stakeholders who want to participate	Evidence of shipments and receipts

Depending on the type of concerns and demands raised by stakeholders and communities, these will be included in the adaptive management process and will be taken into consideration in the development of project activities and operation. As we are in the middle of a community outreach process, this section will advance on subsequent monitoring reports.

2.3.9 Stakeholder Consultation Channels (G3.5)

In the community visits of the IPOÁ REDD+ project, consultations were conducted directly with communities and other stakeholders or their legitimate representatives. They take place in a location easily accessible to the community and the project was presented in full to the participants by social experts. For stakeholders, the project was presented by email, with the project summary sent to them with the appropriate level of information, and they were invited to comment on the project.

A Carbonext has a communication channel by WhatsApp and a 0800 where any staff will have direct access to our social team for complaints, suggestions, denunciations, etc. Even though, the residents at the time of the beginning of the Project do not have a telephone signal in the communities, the residents so far tend to contact the project team by phone when they are in the closest municipalities.



Figure 18 - IPOÁ REDD+ Project Consultation in April 2022.

A banner containing information about the project was available during the consultation, containing a map with the project area and the main information about the project, such as location, name of the farms, expected benefits, and project contact. During the meeting, the community had an open space to ask questions and give suggestions about the project.



Figure 19 - Banner IPOÁ REDD+ Project.

2.3.10 Stakeholder Participation in Decision-Making and Implementation (G3.6)

To ensure the effective participation of communities in decision making, as well as the implementation of necessary actions, it is essential to identify all communities and other stakeholders in the project. From this,

they are invited to participate in the decision-making process, as well as to be informed about any activity related to the implementation of the project.

The field visits were important to understand the local context and to propose activities that might be of interest to the community. Participatory workshops were held with the goal of understanding the residents' main wishes, dreams, desires, and needs.

It is essential to adopt viable communication channels that are effectively used by the communities and other stakeholders, allowing a continuous flow of information and transparency about the actions and allow community participation throughout the project lifetime.

The figure below presents the governance structure and parties involved in the decision-making and implementation processes.

Regarding the schedule for each phase:

- Decision-making is concentrated in the project development phase (2022 up to validation), and although the evaluation of its effectiveness is according to annual results, significant changes are predicted only every 6 years (at each baseline reevaluation event).
- Implementation is conducted on an annual basis, in accordance with each MR.

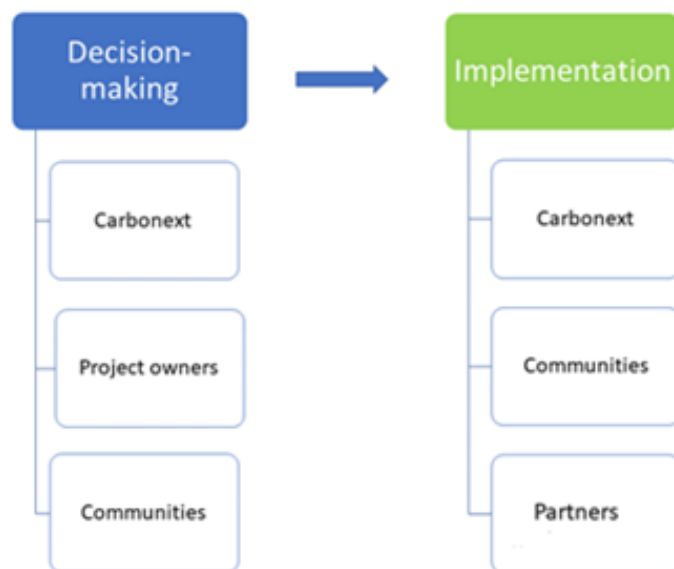


Figure 20 – Governance structure for decision-making and implementation

Given the facilitated communication, any activity to be undertaken by the project will respect the communities' decision as to the details, time and place to be executed.

Cultural and gender information are considered by the proponent. Also encouraging the participation of minority cultural groups and broad female participation.

2.3.11 Anti-Discrimination Assurance (G3.7)

Carbonext has a Code of Ethics and Organizational Conduct in which prohibits any type of abuse or harassment, whether moral, sexual or discriminatory. It does not allow:

- Investments of a sexual or moral nature.
- Exposure of inappropriate material or any other inappropriate conduct.
- Verbal or written offenses.
- Belittling treatment or threats.
- Jokes, nicknames or any offensive reference to sex, race, age, colour and religion.

Owners and others involved the project design are audited within administrative and judicial bodies, through certificates and declarations, in order to verify the involvement with inappropriate conduct that could implicate any risk to the project.

2.3.12 Feedback and Grievance Redress Procedure (G3.8)

Communication Channel

The project proponent understands that stakeholders want and need to be involved in project design, implementation, monitoring and evaluation throughout the project lifetime. Therefore, communication channel was established for stakeholders to continually express their concerns and to solve eventual conflicts and grievances that arise during project planning, implementation, and monitoring. The main communication channel is the project's own email (red.ihoa@carbonext.com.br), which is managed by the Carbonext team.

It is expected that this communication channel will be a mechanism to ensure that the project proponent and all other entities involved in project design and implementation are not involved in or complicit in any form of discrimination or harassment with respect to the project. All denouncements will be available to stakeholders and auditors. In case of conflicts, the project proponent and stakeholders will be free to propose and take any appropriate corrective action. This is the formalized grievance redress procedure that will be used throughout the project lifetime to address disputes with stakeholders, which may arise during project planning and implementation.

The process for receiving, hearing, responding to and attempting to resolve Grievances will be performed within a reasonable time period. It is expected that each grievance is responded in a delay of 7 days, proposing and/or taking corrective actions. This Feedback and Grievance Redress Procedure has three stages:

- 1. The Project Proponent shall attempt to amicably resolve all grievances and provide a written response to the grievances in a manner that is culturally appropriate. (Action must be taken in 7 days).*
- 2. Any grievances that are not resolved by amicable negotiations shall be referred to mediation by a neutral third party. (Action must be taken in 30 days).*
- 3. Any grievances that are not resolved through mediation shall be referred either to a) arbitration, to the extent allowed by the laws of the relevant jurisdiction or b) competent courts in the relevant jurisdiction, without prejudice to a party's ability to submit the grievance to a competent supranational adjudicatory body, if any. (The time to accomplish this stage is dependent on local jurisdiction delays.)*

2.3.13 Accessibility of the Feedback and Grievance Redress Procedure (G3.8)

Stakeholders and community have a direct communication channel with the project team, in which they can give feedbacks or report grievances. Meetings with the communities are held periodically, for the purpose of gathering feedbacks and guarantee the project activities' quality. They are also invited to contact the field workers and monitors whenever they have something to report.

Feedbacks and grievance received during the public comment period will be publicly available in the project description.

2.3.14 Worker Training (G3.9)

During the community consultation, information on the communities' main needs and how the project can assist in the improvement of their life quality and socioeconomic condition was gathered. A relevant set of training activities are being organized and planned to take place periodically to build useful skills and knowledge (see sections 4.5.9 and 0). Also, with the purpose of decrease GHG emission and increase the community participation on the project implementation, workshops and specific trainings on sustainable farming and sustainable forest management will be offered. The trainings will be held by specialized personnel, whenever possible, by local labour, and the participants will receive certificates attesting their participation.

2.3.15 Community Employment Opportunities (G3.10)

The hiring of 2 to 4 community forest monitors is foreseen with the project's climate action scope, to carry out forest surveillance and achieve the project's fundamental goals of avoided deforestation and carbon reduction, as well as providing employment and income for community members.

A community association will be created, which will enable community members to carry out commercial activities such as trading non-timber forest products (NTFPs), artisanal work, or others of their choosing, as well as to make wider community activities official. For example, it is foreseen that partnerships between the public sector and the community association will be key in school refurbishment, basic health facility creation, and other local demands. These activities will be facilitated and intermediated by Carbonext where it is beneficial to do so.

In addition, support for the professional development of 1 or 2 teachers is planned in the project activities' Education scope. All of the activities mentioned above are summarized in Appendix 2.

The positions will be filled with due care to observe Carbonext's procurement standards, as well as CCB concept G3: without any discrimination relating to gender, age, religion, marital status or ethnicity. The selection will be made according to the person that best qualifies for the requirements of the position. Women and elderly will be encouraged to apply for the position. Once hired, the worker(s) will go through a training process and a trial period, that lasts three months. All the workers hired will be officially registered.

2.3.16 Relevant Laws and Regulations Related to Worker's Rights (G3.11)

The IPOÁ Project operates in full compliance with the labor laws of Brazil and strives to set an example of best practice regarding labor terms, conditions, and practices.

Decree-Law No. 5,452, May 1, 1943, is approval of the Consolidation of Labor Laws (CLT), which the then Brazilian president Getúlio Vargas and a group of lawyers and legislators drafted to ensure a series of titles and regulations in the relationship between employers and employees. Since then, the CLT has undergone a series of modifications and improvements.

The main and biggest changes, occurred with the Labor Reform under Law No. 13,467 of July 13, 2017, in order to adapt the legislation to the new labor relations.

The project proponent complies with all laws and regulations covering workers' rights and has no claims against them.

2.3.17 Occupational Safety Assessment (G3.12)

The Occupational Safety Assessment of the IPOÁ REDD+ Project was developed based on the stages of hazard identification and risk analysis, as well as the proposition of prevention, mitigation and contingency measures.

Hazard identification and risk analysis was applied the developed of the IPOÁ REDD+ project. ISO 31000 is the global standard for establishing guidelines and pathways for risk management, and one of the tools that it recommends analyzing them is the "Bow Tie", which, based on identified hazards, defines the risks, causes and consequences of potential events. Once the scenario has been established, prevention and mitigation barriers are outlined for each cause and consequence. That is, the necessary procedures that need to be conducted to prevent the event from happening or to minimize consequences if the event occurs.

Carbonext is structuring a team solely responsible for occupational health and safety, in addition to property and corporate security. Thus, it is possible to guide proponents to follow good practices that improve the health and safety of employees and help establish operational procedures in cases of invasion, deforestation, etc.

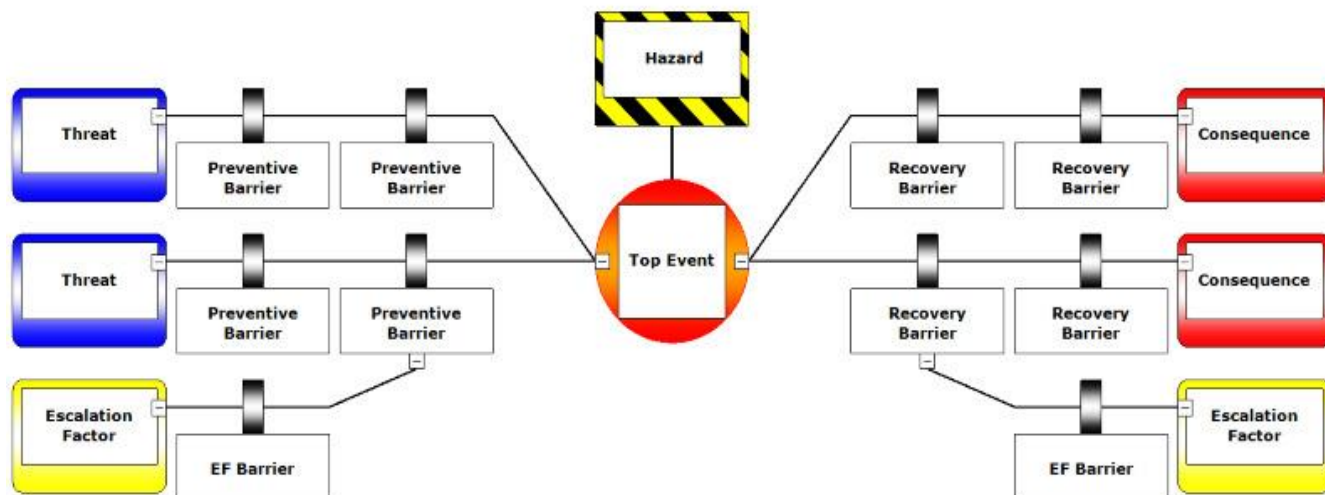


Figure 21. Risk Management: Bow Tie analysis

Hazard Identification

The prevention and mitigation of hazards related to field activities must consider from planning to completion. Thus, the possible scenarios and hazards were identified considering transportation, internal displacement, conditions and physical and biological agents to which the worker may be exposed. For better management, the hazards were categorized as (A) accidental, (B) biological, (P) physical, (C) chemical and (E) ergonomic.

Risk Assessment

To analyze each identified risk, the risk matrix of probability versus severity was used, as shown in *Figure Risk matrix: (probability x severity)*. Five levels of probability and five levels of severity were defined, whose application results in the final risk classification, as shown in *Figure Risk classification*. The risk degree will define the priority for taking prevention and mitigation actions.

Qualitative Risk Matrix		SEVERITY				
		Negligible	Minor	Moderate	Significant	Severe
		1	2	3	4	5
PROBABILITY	Very Likely E					
	Likely D					
	Possible C					
	Unlikely B					
	Very Unlikely A					

Figure 22. Risk matrix: (probability x severity).

	Trivial
	Tolerable
	Moderate
	Substantial
	Intolerable

Figure 23. Risk classification.

Severity values were analyzed for each risk considering the scale and criteria described in *Figure Severity of consequence* below.

Risk Estimation: severity of consequence	
Index	Definition
1	Mild injuries without the need for medical attention, discomfort or discomfort.
2	Reversible serious injury or illness.
3	Irreversible critical injury or illness that may limit functional capacity.
4	Disabling or fatal injury or illness.
5	Multiple deaths or disabilities (>10).

Figure 24. Severity of consequence.

Probability values were analyzed for each risk considering the scale and criteria described in *Figure Categories of exposure* below.

Qualitative Estimation: categories of exposure		
Index	Category	Description
A	Very low exposure	< 5%
B	Low exposure	>5% a <30%
C	Moderate exposure	>30% e <70%
D	Hight exposure	>70%<95%
E	Very hight exposure	>95%

Figure 25. Categories of exposure.

The main prevention and mitigation actions of the identified risks are presented below, considering their causes and priority category.

Prevention, mitigation, and emergency procedures

The proponent developed the occupational health and safety manual, which presents risk prevention and mitigation actions, as well as procedures for emergency situations.

The manual also has contacts for activation in cases of emergency, a training schedule, a checklist for planning activities and work safety equipment. The manual is available for the audit.

2.4. Management Capacity

2.4.1 Project Governance Structures (G4.1)

Carbonext is responsible for all the activities necessary to develop the Project Description according to the standards and methodology compatible with the project; calculation of the generation of carbon credits; assistance for the project validation and verification under VCS/CCB and registration process. The team responsible for the aforementioned activities is divided into four principal interconnected areas:

1. **Legal Department:** Legal team specialized in legal matters of the properties, proponents, land tenure, and others. This analysis is made through certificates from the federal and state court of justice, presence of land conflicts, overlaps with public areas and possible risks of expropriation.
2. **Project Development Team:** Technical team specialized in the development of REDD+ projects, being the main responsible for the project design, selection of activities, determination of specific objectives and to provide technical support to the owner and to monitor those activities are being carried out.
3. **GIS (Geographic Information Systems):** Team responsible for determining and reviewing the project baseline, deforestation dynamics, project leakage and monitoring via satellite images the occurrence of deforestation in the project area.
4. **Social Analysts:** Team responsible for socioeconomic analysis, engagement and articulation with communities and other stakeholders, development of social activities and indicators, in addition to monitoring socio-economic indicators.

It is important to note that the project actions, while designed by Carbonext, involve a participatory process together with communities to ensure that they address the communities' particular context and needs.

Carbonext and the other proponents will collaborate on management manage and execute the activities according to the defined planning/schedule.

The owners of the areas where the project is located are co-responsible, together with Carbonext, for the activities necessary to maintain the forest integrity of the areas according to the norms and methodologies compatible with the project; hiring forest agents and other workers for the project; financing the socio-environmental actions for maintaining and improving the environmental and social quality of the project.

1. **Legal Department:** Legal team specialized in legal issues of properties, bidders, land tenure, and others.
2. **Local Field Team:** Led by the field manager, who is a project employee, and includes people from the community who may be called upon on a case-by-case basis to conduct tasks in the field and are duly remunerated.

The division of responsibilities between Carbonext and the owners is depicted in the figure below.

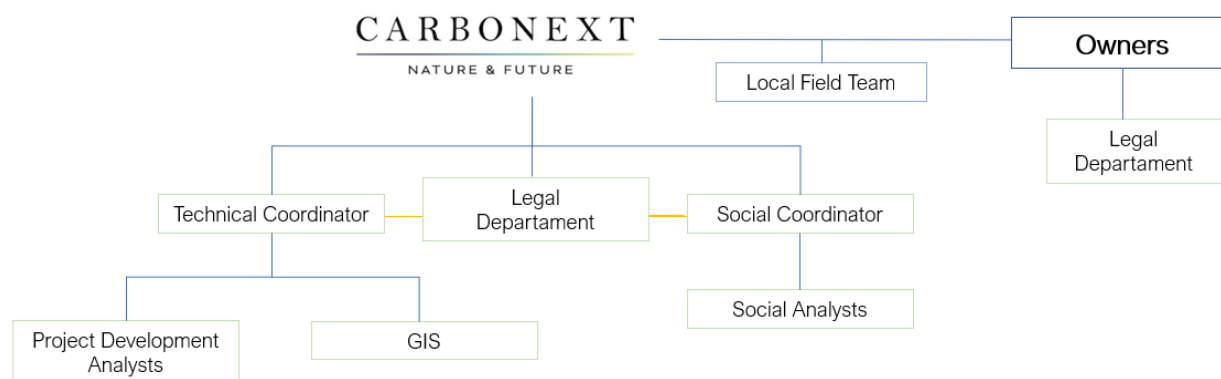


Figure 26. Carbonext Internal Governance Structure in Project Management.

To deal with any invasive deforestation detected mutual contact between the proponents will be maintained continuously during the project period. In case of invasive deforestation events, the project activates the Forest Integrity Plan's Invasive Deforestation Event Protocol⁷⁶ (Annex II), which is as follows:

Carbonext's Technical and SIG teams carries out the monitoring of the Project Areas to detect any deforestation in the Project Area, via the company's CarbonEye system. Upon detection, the Field Manager is dispatched to the site, to ascertain details via application of the Incident Report Form (*formulário de ocorrência*), and the appropriate local environmental, policing and enforcement bureaus may be dispatched, depending on the case analysis by the technical team. The project owners are also kept updated throughout the process above, and their primary areas of co-responsibility are in liaising with police and other departments, activities on the ground, and coordinating the wider network of project neighbours and contacts.

To summarize the essential actions of the Deforestation Event Protocol are as follows:

- i. Intrusion detection by the CarbonEye system, reported in the CarbonEye Event report by the SIG Team.
- ii. Action meeting of the Technical and Social teams.
- iii. Activation of the local Field Manager and field team.
- iv. On-site measurement of the size of the event (width of access roads; description of the level of deforestation, etc).
- v. Placement of signs in the field stating that the area is private property and a REDD+ private forest conservation area.
- vi. Completion of the Incident Report Form by the Field Manager.
- vii. Holding of meetings and conversations with competent local bodies, partners, and community entities each with filling of the Post Deforestation Meeting Report by the Technical Team, Social Team, and Field Team, as appropriate.
- viii. Saving of all above records in specific server folder for permanent record.

⁷⁶ In Portuguese: "Protocolo de ocorrência de invasão"

- ix. Conclusion: appropriate activity implemented to deal with the intrusion event.

2.4.2 Required Technical Skills (G4.2)

In order to carry out REDD+ project planning, development, and monitoring, the required skills include:

- specialized REDD+ standards (VCS + CCB) and implementation knowledge,
- carbon measurement and monitoring,
- forestry/ forest engineering,
- GIS mapping specialized in the Amazon region,
- social development theory, data collection and,
- biodiversity assessment and conservation,
- environmental management,
- oceanography, geoprocessing, and other related fields.

In all the above fields, Carbonext has a specialized team led by the Technical Director and Managers with extensive education and prior experience in key areas. The company will undertake the development and monitoring of all aspects of the REDD+ project. In addition, it has a team of social experts, who have the necessary skills to involve the community and stakeholders, so the team will be able to ensure community participation, designing activities and the monitoring of socio-economic indicators.

For the fauna and flora inventory, a third-party team will be hired to collect data in the field and generate the diagnostic report. Preference will be given to local teams that have knowledge of local conditions, but this activity will only take place after the first verification of the project and emission of carbon credits.

2.4.3 Management Team Experience (G4.2)

Carbonext was founded in 2010 with the aim of preserving the Amazon rainforest by designing REDD+ carbon credit projects. Since then, we have seen three REDD+ projects through validation and verification under VCS standards, representing over 276,000 hectares to date. With five new projects already in advanced phases of development, and approximately fifteen within an ever-expanding pipeline, that figure will grow rapidly over the next five years.

Carbonext has a multidisciplinary technical team consisting of forest engineers, biologists, environmental managers, ecologists, geographers, social scientists, and others who constitute a qualified team of researchers and experts capable of conducting project activities.

Further information about Carbonext can be found on our website: <https://carbonext.com.br/about-us>.

Below are summaries of the expertise of the technical team dedicated to the validation and monitoring of the IPOÁ project.

Name	Janaina Dallan
Position	Carbonext CEO
Education	Degree in Forestry Engineering from Luiz de Queiroz College of Agriculture (ESALQ) and a Master's in business of the environment
Resume summary	Janaína Dallan has been working with the carbon market since 2001 on projects around the world. As CEO of Carbonext, she is directly involved in coordinating the development and implementation of carbon credit projects. Throughout her career, she has managed UNFCCC CDM (Clean Development Mechanisms) projects through several international companies such as Golder Associates, Global Energy Partners, Orbeo / Soci��t�� Generale, One Carbon and Ecofys. Jana��na's involvement Carbonext's projects spans from their very conception, prospecting, and initial groundwork, through technical development, and leading at the CEO level. She is currently a member of the United Nations Framework Convention on Climate Change (UNFCCC), in which she has been a member of the Registration and Issuance Team (RIT) since 2013, she is president of the Brazil Alliance for Nature-Based Solutions and is a member of the experts for the Taskforce for Scaling Voluntary Carbon Markets (TSVCM).
Contact Information	janaina@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/janainadallan/

Name	Luiz Fernando de Moura
Position	Technical Director
Education	Forestry Engineer by Luiz de Queiroz College of Agriculture (ESALQ) the M.Sc. and Ph.D. in Wood technology by the Université Laval (Quebec, Canada).
Resumé summary	Luiz Fernando de Moura coordinates Carbonext's technical team. Luiz's direction has seen all of Carbonext's projects through validation and verification, being heavily involved in their technical development and leading at the Direction level. He participated in the preparation of "Energia Verde Carbonization Project - Mitigation of Methane Emissions in the Charcoal Production of Grupo Queiroz Galvão, Maranhão, Brazil", registered on March 21, 2011. Dr. de Moura also participation as project designer in Florestal Santa Maria REDD+ Project, Fortaleza Ituxi REDD+ project, UNITOR REDD+ project and Evergreen REDD+ project.
Contact Information	luiz.moura@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/luiz-fernando-de-moura-6077089/

Name	Franciele Salvador
Position	Legal Manager e Compliance Officer
Education	Lawyer by the Cenecista College of Joinville

Resumé summary	Franciele Salvador manages Carbonext's legal group, providing strategic support for projects on the Brazilian voluntary carbon market. Previous experience working in large companies, as well as specializations in Business Law, Compliance, Contract: Business Vision and Practice.
Contact Information	franciele.salvador@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/franciele-salvador/

Name	David Swallow
Position	Technical REDD+ Coordinator
Education	Degree in Anthropology from the University of Edinburgh and an MSc in Ecology, Evolution and Conservation from Imperial College London.
Resumé summary	David Swallow has eleven years' experience conducting the technical development of REDD+ project on the Brazilian voluntary carbon market. During his two years at Carbonext, he has managed two of the company's largest projects to date through validation and verification with a technical team of 2 – 4 members. He specializes in REDD project's social aspects, co-benefits, and impact investing.
Contact Information	david@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/david-swallow-6b960b14/

Name	Débora Santos de Oliveira Brum
Position	REDD+ Analyst

Education	Environmental Manager from the University of São Paulo (USP) and Specialization in Business Sustainability from the Federal University of São Carlos (UFSCAR).
Resume summary	Débora Brum has experience with Geographic Information Systems (GIS) in national and international urban registry projects and with Greenhouse Gas (GHG) emissions inventories.
Contact Information	debora.oliveira@carbonext.com.br
LinkedIn	linkedin.com/in/débora-santos-de-oliveira

Name	Fernando Camargo da Silva
Position	REDD+ Analyst
Education	Degree in Forest Engineering by Federal University Of Paraná (UFPR) and MBA in Project Management by Positivo University. Currently Master's degree student in Nature Conservation (UFPR).
Resume summary	Ten years' experience in environmental studies, such as flora inventories, environmental licensing, biodiversity conservation projects, land use, research and development, ecological restoration and environmental disasters, and also working in different biomes of Brazil, such as Amazon Rainforest, Atlantic Rainforest, Pantanal and Pampa. Currently working with REDD+ project development.
Contact Information	fernando.silva@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/fernando-camargo-626164113/

Name	Monique Batista Fascina
Position	REDD+ Analyst

Education	Degree in Biological Sciences by the University Presbyterian Mackenzie (Mackenzie) and post-graduation/MBA in Institute of Administration Foundation (FIA) in Strategic Sustainability Management
Resume summary	Experience in environmental consulting, where Monique had the opportunity to work on several environmental licensing projects, environmental management and supervision, forest inventory aiming at conservation projects and environmental compensation. Acting as Sustainability Analyst, participating in projects for the implementation of socio-environmental certification for companies in the sugar-energy sector (Bonsucro, RSB, Renovabio) and for more sustainable operations. Currently working with REDD+ project development
Contact Information	monique.fascina@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/moniquefascina/

Name	Mariana Farias de Araujo
Position	REDD+ Analyst
Education	Post-graduate degree in Environmental Auditing and Expertise, graduated in Environmental Management from the University of São Paulo (USP) and environmental technician.
Resume summary	Mariana Araujo has experience in management systems, project monitoring, she has worked with supplier monitoring and business analysis for climate governance.
Contact Information	mariana.araujo@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/mariana-farias-araujo/

Name	Vinício Rossi Sugui
Position	REDD+ Analyst
Education	Environmental Engineer (FOC) and M.Sc. in Urban and Industrial Environment (UFPR). MBA in ESG and Impact.
Resume summary	Vinício Sugui has twelve years' experience in environmental projects and programs; environmental management; sustainability; air pollutant emissions; climate changes; environmental construction control; environmental licensing processes.
Contact Information	vinicio.sugui@carbonext.com.br
LinkedIn	www.linkedin.com/in/vinicio-sugui

Name	Rui Almeida
Position	Social Project REDD+ Coordinator
Education	Graduate degree in Education, currently attending a post-graduate course in project management, and a master's course in Conflict Management.
Resume summary	Rui has extensive experience in Amazon-based projects with traditional communities, including quilombolas and indigenous peoples. Experience with the implementation of a financial fund in quilombola communities. Sustainable development, social dialogue, and land and conflict mediation. Training in Nonviolent Communication and advanced knowledge of social dialogue tools, digital technologies, communication methodologies and development of support material for communication projects.
Contact Information	rui.almeida@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/ruicardosoalmeida/

Name	Thiago Manoel Sozinho dos Santos
Position	REDD+ Social Analyst
Education	Degree in Forestry Engineering from Amazonia Federal Rural University and a Master's in forestry economy in Paraná Federal University
Resume summary	Thiago Sozinho has ten years' experience working on the follow themes: protected areas, environment, carbon market, community relations, research, geotechnologies, customer success, project management, BI and Data Science.
Contact Information	thiago.sozinho@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/thiago-sozinho/

Name	Mateus Trez
Position	GIS and Remote Sensing Coordinator
Education	Environmental Engineer from the Piracicaba Engineering School
Resume summary	Mateus Trez is responsible for coordinating team of analysts in the development of REDD+ projects and the prospection and feasibility analysis of new carbon projects in the Amazon. Experience in the development of REDD+ projects with participation in the elaboration of the VCS Santa Maria REDD+ Forest Project, UNITOR REDD+ project and Evergreen REDD+ project.
Contact Information	mateus.trez@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/mateustrez/

Name	Janaína Guidolini
Position	GIS Analyst
Education	Postdoctoral researcher in Environmental Science (Aeronautics Institute of Technology); PhD in Earth System Science (National Institute for Space Research); MSc in Agronomy-Soil Science (São Paulo State University); Graduated in Agronomy and Environmental Management (Federal Institute of Triângulo Mineiro).
Resume summary	Janaína Guidolini is an environmental scientist with over ten years of experience in the environmental area, especially in soil and water management and conservation, geoprocessing for environmental analysis and sustainability.
Contact Information	janaina.guidolini@carbonext.com.br
LinkedIn	https://www.linkedin.com/in/janainaguidolini

Name	Valdenir Lemos Ribeiro
Position	Field technician
Education	Completed high school (2009)
Resume summary	Extensive experience as field service provider in the Amazon Forest environments and their communities; security guard experience for four years; Municipal secretary of environment in Novo Aripuanã, Amazonas (2010 – 2012); Municipal secretary of housing and land regularization in Novo Aripuanã, Amazonas (2017 – 2018).
Contact Information	Valdenir.lr@gmail.com
LinkedIn	https://www.linkedin.com/in/valdenir-ribeiro-2247a3236/

2.4.4 Project Management Partnerships/Team Development (G4.2)

As mentioned in section 2.4.2, for the fauna and flora inventory, a third party will be hired to conduct the data survey in the field. Preference will be given to local teams that understand the local context. These

activities are predicted to begin after the first monitoring period. The proponents already have contact with a biodiversity inventory company with over 20 years of expertise in projects with fauna and flora, with the most advanced resources and a team of specialists. For the inventories, the service provider is expected to apply a scientific methodology, adhere to legislation, and ethical, responsible, and sustainable procedures.

The IPOÁ REDD+ project covers a landscape that involves diverse social, economic, and political actors. For this reason, multi-level collaboration such as PPPs (Public-Private Partnerships) are likely to be implemented in the project activities (see 2.1.11), which will have the community association and the public sector as the parties involved. Such PPPs may take the form of formal signed partnerships, or less legally binding collaboration agreements, MOUs, or similar agreements. Carbonext has already contacted public bodies such as health and education secretariats, and city councilors to establish partnerships and contribute to improvements in the region.

With regard to other capacities needed for project formulation or implementation, no technical or administrative gaps have been identified so far that need to be filled by another entity. However, if during the verification work the need to engage the services of independent professionals or firms to perform the work is identified, the project team will take the necessary steps to find, hire, and then manage any subcontractors. In addition, the project team will determine the time, resources, and level of participation of any outside entities providing services to the project in accordance with internal processes.

2.4.5 Financial Health of Implementing Organization(s) (G4.3)

Carbonext has a financial department that is responsible for taking care of the company's financial health, so that it can run its operation on a regular basis in addition to being able to execute projects without running financial risks. To this end, the company has an annual budgeting process that is periodically reviewed in order to give managers visibility of what the main expenses will be for the period, including expenses related to the execution of projects. In this way, a monetary amount is set aside for each project that will be executed, in order to avoid any cash flow risk.

Thus, the project proponents have the necessary funds to maintain the project activities until the start of the GHG revenue, as presented in the Non-Permanence Risk Report⁷⁷.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (G4.3)

In Brazil, the two main legal instruments that aim to prevent acts of corruption and unethical attitudes are (i) Law 12,846/2013 (Anti-Corruption Law); and (i) Law 8,429/1992 (Administrative Misconduct Law).

Law no. 12,846/2013, known as the Anti-Corruption Law, establishes rules regarding the administrative liability and civil liability of legal entities in the case of practice of acts against the public administration, national and/or foreign. Under Law 8,429/1992, administrative improbity is an immorality qualified by the dishonesty and acts of bad faith by the public agent.

Both practices, in addition to the legal risks, could represent enormous reputational risk to the project and to all that are involved. Due to this risks, Carbonext Legal Due Diligence aims to verify and, if possible, mitigate any involvement of the landowner (partner) with such practices.

The clearance certificates examined by Carbonext's legal team (applicable for all the project proponents/partners and real estate assets) during the Legal Due Diligence process cover a large spectrum of jurisdictional and administrative matters, such as, for example, consultation before State Courts of Justice and Superior Courts, State and Federal Public Prosecutors, Environmental Agencies, Protest Registry Offices, Real Estate Registry Offices, etc., significantly reducing the chances of both debts and/or acts of corruption (regarding the landowner partner or the real estate assets) go unnoticed. Eventual discovery of relevant debts and/or acts of corruption involving the landowner (partner), is one of the many reasons for the termination of the intended partnership, as provided for in the contract.

Carbonext has an Anti-corruption Policy and a code of ethics and organizational conduct in place to ensure non-involvement with corruption⁷⁸. So that employees, customers, partners, and other stakeholders can, with complete security and anonymity, report situations that are not in accordance with our values or with our Code of Ethics and Organizational Conduct and / or Compliance Policy, we created a channel for complaints to record, investigate and promote decisions involving the complaints received, to ensure

⁷⁷ Evidence presented to the auditors.

⁷⁸ Anti-corruption policy and code of ethics and organizational conduct in force: <https://carbonext.com.br/governanca-corporativa/>

compliance and continuous improvement of processes and internal controls. To register complaints, an e-mail can be sent to Compliance@carbonext.com.br.

To the extent that Carbonext's anti-corruption policy also includes partners, it along with the project contracts, are the legal instruments that ensure avoidance of corruption of all the project proponents, listed in section 2.1.3.

2.4.7 Commercially Sensitive Information (Rules 3.5.13 – 3.5.14)

Some information required by the VCS and CCB standards is considered confidential or commercially sensitive and cannot be made public by the project proponent. This information will be supplied to the audit team, but not available for the public. The documents are:

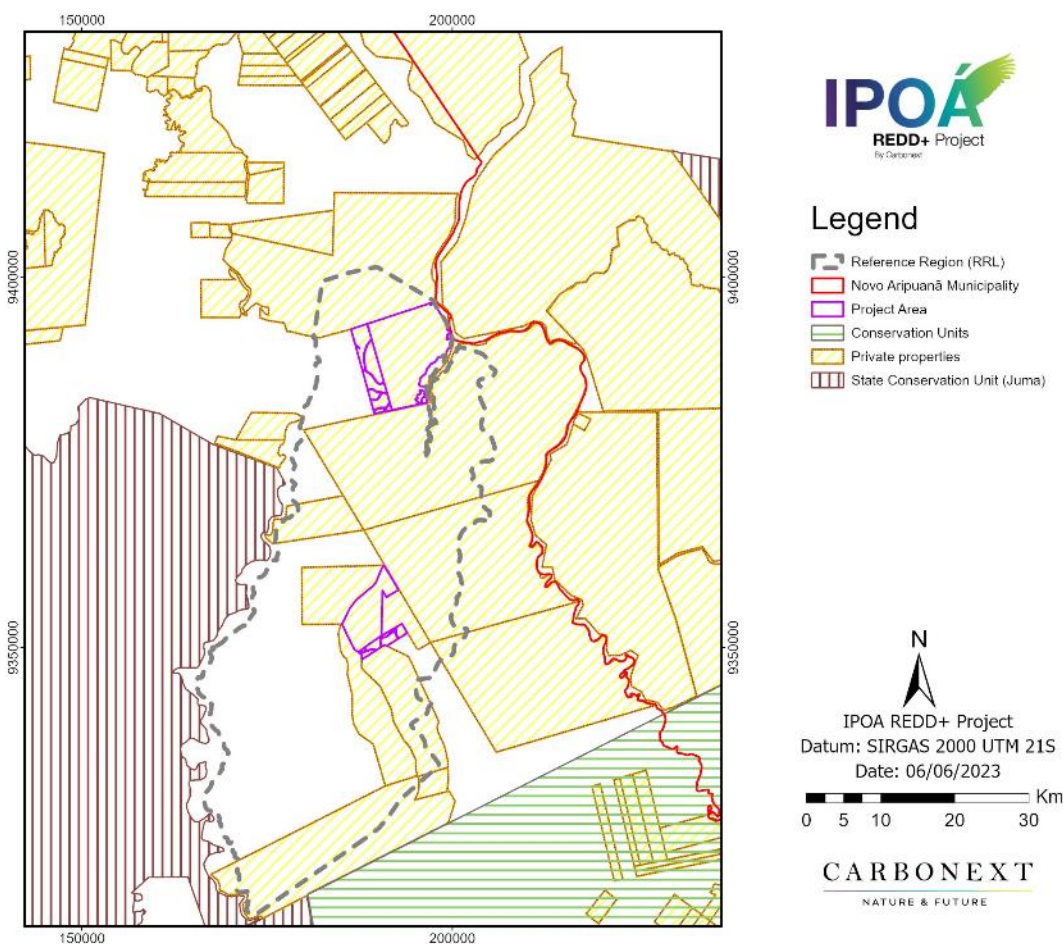
- Agreements and contracts between the parties involved.
- Financial evidence of the proponents' funds for the project.
- Property Rights Documents.

2.5. Legal Status and Property Rights

2.5.1 Statutory and Customary Property Rights (G5.1)

The project area is privately owned by the proponents, who hold all rights of possession, access, and use, as demonstrated in the documentation made available at audit.

The Project Zone encompasses, in addition to the Project Area, the communities of Vila Batista, Boa Esperança and Vale da Benção. Other than the Maanaim project property itself, with which Vila Batista intersects, these communities do not coincide with private areas registered in the Land Management System (SIGEF) maintained by the National Institute for Colonization and Agrarian Reform (INCRA), the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country. The communities are also not located in Federal Conservation Units or Indigenous Territories, characterized as areas of public domain, as shown in Figure below.



It is relevant to cite the various Brazilian squatters' rights laws – *usocapião* in Portuguese – which can be categorized as:

- “Ordinary,” governed by Law Nº10.406⁷⁹;
- “Extraordinary” covered by Art. 1.238 of the Civil Code⁸⁰;
- “Special rural” described under Art 191 of the Federal Constitution⁸¹ and.

⁷⁹ Source: http://www.planalto.gov.br/ccivil_03/leis/2002/10406compilada.htm

⁸⁰ Source: http://www.planalto.gov.br/ccivil_03/leis/2002/10406compilada.htm

⁸¹ Source: http://www.planalto.gov.br/ccivil_03/constituicao/constituicao.htm

- Extrajudicial measures, possible to be carried out in a notary office, and the framework for which is provided by Art. 1.071 of Civil Law nº 13.105/15⁸²;

To cite just some of the laws which affect these three modalities. The clauses of these laws state that if a person possesses an area, in some cases limited to 50 hectares, as if it were their own, for an uninterrupted period of between five and 15 years, depending on the context, they are entitled to acquire the property document.

This is the main legal framework within which legitimate property rights claims may be made within this project's context. However, at the time of writing the present PD, no legal action relating to the ownership or possession of the properties has been identified.

2.5.2 Recognition of Property Rights (G5.1)

All the project lands are private property owned by the project proponents, and their property rights are recognized, respected, and supported by all relevant legal institutions. All the properties have the relevant property documents attesting to their full land-rights, which were made available at audit.

The national public registry of rural properties is carried out with the Rural Environmental Registry ("Cadastro Ambiental Rural" - CAR). An experienced professional was hired by the proponent to carry out the CAR registration for both properties. The CAR of the initial Campos farm was cancelled by the previous owner, the numbering was updated and regularized before the public organizations on behalf of the representative (Silvio Balen). In the same way, the Maanaim property's CAR for was evaluated at the beginning of the documentation verification process. Despite the identification of unconsolidated overlapping areas – a common reality in the Amazon region – the CAR was successfully updated and ratified. It is important to note that the proprietor's ownership is fully backed up by the relevant documentation in the SIGEF system⁸³, which differentiates it from unconsolidated claims such as those identified in the Maanaim area.

The measures designed and implemented by the project to help to secure full community property rights are as follows: Carbonext will repeat the questionnaire and monitor the land-title situation at each monitoring

⁸² Source : http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/lei/l13105.htm

⁸³ Source : Sistema de Gestão Fundiária, SIGEF: <https://sigef.incra.gov.br/>

period. Carbonext commits to guaranteeing the rights of all those with legitimate and legally approved land claims to have those rights upheld and be entitled to corresponding project benefits. This is further illustrated by the specific case discussed in section 2.5.6.

2.5.3 Free, Prior and Informed Consent (G5.2)

The project will not encroach on or affect private property, community property, or government property, given that the project area consists of private land, as discussed in Section 2.5.

The community of Vila Batista is on the proponent's private property, but outside the Project Area (PA) itself. The communities of Boa Esperança and Vale da Benção are not in private areas registered in the Land Management System (SIGEF), which is maintained by the National Institute for Colonization and Agrarian Reform (INCRA) – the Brazilian federal agency whose objective is to maintain the national registry of rural properties in the country.

Property rights are not being affected and project activities will not lead to the removal or relocation of habitation or activities important to their culture and livelihood.

During the stakeholder consultation with the local leaders of the nearby communities, consent for the development of the project was obtained in the field and their signature or fingerprint was collected, with the aim of ensuring that the free, prior, and informed consent of those whose property rights could potentially be affected by the project was obtained through a transparent and consensual process. That is, that given by a fully empowered agent, voluntarily and free of defects in consent, as required by Brazilian law for legal transactions in general.

The project has no interest in or intention of taking possession of any kind of intellectual, cultural, or religious property, also, in the meetings with the technical team of Carbonext it was reinforced that there will be no removal of residents and that the project supports traditional ways of life.

Laws and administrative measures are also not implemented or facilitated by the project, and no change in the way of life of the populations that live in the surroundings of the area is foreseen or desired. Equally, there is no prospects to undertake mining, or incentives for agro-industrial endeavors. All decisions that need to be taken will always consider the rights of the residents of the region without any of their rights being changed or any activity being carried out without consultation of all those directly and indirectly involved.

Decisions about activities are made jointly with the population, through consultations with the surrounding communities, ensuring that the information is passed on transparently, free of coercion, and conditions, and ensuring that all people can contribute regardless of age and gender.

The questionnaires applied by Carbonext for the present CCB project's development – as well as the statements of the consent of the people involved in the consultation process – collect data on the local context, education, health, economy, infrastructure, leisure, culture and beliefs, social organization, threats and opportunities, and the biodiversity of the region.

2.5.4 Property Rights Protection (G5.3)

The implementation and development of the IPOÁ project must not lead to the involuntary removal or relocation of any stakeholder. Indeed, the due rights to property and resulting carbon credit rights are actively facilitated as part of Carbonext's policy, detailed in section 2.5.2.

During the March 2022 field visit, agreements were freely signed by the leaders the three communities, which were made available at audit. This was process was the result of previous education workshops and the commitment of both parties to identify and define the activities that will be developed in the area, and the community representatives' agreement with these objectives as well as the overarching one of forest conservation.

This consultation process ensures that the communities have not been forced to relocate their homes or their activities, which are central to their culture and livelihoods. Equally, the deforestation mitigation measures implemented as part of the project must conducted in constant communication with all partners and stakeholders.

2.5.5 Illegal Activity Identification (G5.4)

The baseline scenario involves a significant threat of illegal logging followed by cattle ranching, which is a predominant activity in the region. In order to reduce it, the project will train between 2 to 4 community monitors to patrol the area according to the Forest Integrity Plan (FIP). This will be led by the Field Manager, with consultation and support from Carbonext. The FIP has the core strategy of implementing preventive, combat and mitigation measures for events that threaten to cause GHG emissions in the Project Area. It includes clear and objective definitions of the stages of the monitoring process, creation of records,

quantifications and reports of corrective actions which aim to mitigate emissions, as well as to comply with the VCS methodologies in the best possible manner.

Additionally, the project continuous, high-frequency remote monitoring using its CarbonEye system, with an expected frequency of 7 days. This allows the project team to identify any deforestation activity and take the necessary measures. For more details see sections: 2.4 Project Governance Structures; 2.4.6 Avoidance of Corruption and Other Unethical Behavior; 3.2 Baseline Emissions; 3.3.3 Monitoring Plan and 4.4 Community Monitoring Plan.

Further Operational Procedures have been developed to address other illegal activities such as: theft, vandalism, embezzlement of resources and/or equipment, etc. These were fully presented at audit.

2.5.6 Ongoing Disputes (G5.5)

There are no legally formalized ongoing disputes in the Project Area, regarding land rights, resource access rights, or other grounds.

Carbonext's legal procedures regarding disputes include due diligence assessment of judicial records in the district where the properties are located. This specific survey covers any ongoing disputes that are discussed in federal and state courts, regardless of the commencement date of the dispute. No ongoing or unresolved disputes that compromise ownership of the property in the last twenty years were identified in this survey.

During two pre-validation visits to the PZ's communities, in March and May 2022, the issue of land rights was addressed in the community questionnaires. A community member, alleged to have a document that supports a potential claim to approximately 800 hectares within the Maanaim property area, however he was unable to produce any evidence to support this. The situation is being monitored in order to grant and facilitate legitimate community claims to land rights, which will be updated at least at each monitoring event. Carbonext's general policy regarding such disputes and resulting changes to carbon credit rights is expounded in section 2.5.2.

The same approach as that given to land disputes will be applied to resource disputes: monitoring by project field personnel at least at each monitoring event.

During the project fieldwork, no disputes were related by communities, regarding in the historical timeframe of ten years prior to the project start date.

In relation to public entities, whether state, municipal or federal, all documents analyzed indicate the legitimacy of the project's land-titles, minimizing the possibility of illegal appropriation of public lands (*grilagem*), which would undermine the development of the project. However, such practices are unlikely to occur, as public lands cannot legally be acquired through squatting, except with the effective participation of the public entity that owns the area, requiring some legislative measure (law, decree, legal order, etc.) expressly legalizing such claims.

Under Brazilian legislation, it can be stated that the certificates from environmental, administrative, jurisdictional, and certifying bodies indicate that both the possession and property rights of all areas involved in the Ipoá REDD+ Project belong to the proponent, being fully legal and extensively documented and proven. In other words, the owner has the full rights to and use the natural resources of the properties that make up the project, which constitute legitimate rights to develop the REDD+ project.

2.5.7 National and Local Laws (G5.6)

There is no specific governing legislation in Brazil for carbon credit projects or even for the carbon credits generated. Therefore, projects must be aligned with the laws and principles in diverse areas of the legal sphere.

There is a hierarchy between norms in Brazilian legislation, starting with the Federal Constitution. All branches of law applicable to carbon projects and the applicable rules have their origin, foundation, and validity in the Federal Constitution.

When we talk about Environmental Law, Article 225 of the Constitution is the basis for all norms, principles, objectives, and policies. The National Land Policy begins with article 184. Furthermore, rights such as possession, property, and free enterprise are also based on the Federal Constitution.

Taking into account the sources in the brief explanation above, there are several directly and indirectly applicable rules, regarding carbon credit projects, each one regulating a specific aspect.

Law 11, 284/2006 - Provides for the management of public forests for sustainable production, under the governance of the Brazilian Forest Service - SFB. Among other topics, this law deals with the management

of public forests for sustainable activities including timber production; Direct timber management; Forest concessions; Management and inspection bodies; and accompanying Principles and concepts⁸⁴.

Law 12,651/2012 (Forest Code) - The Forest Code establishes general rules regarding the protection of vegetation, Permanent Preservation areas and Legal Reserve areas, forest exploitation, the supply of raw materials, control of the origin of forest products and the control and prevention of forest fires. In addition, it stipulates economic and financial instruments to achieve its objectives. It creates the CAR system, which was later regulated by MMA Normative Instruction No. 2 of May 5, 2014⁸⁵.

Regarding the Brazilian Forest Code, the following definitions stand out as being relevant:

“Article 41 - The Federal Executive Power is authorized to institute, without prejudice to the compliance with the environmental legislation, a support and incentive program for environmental conservation, as well as for the adoption of technologies and good practices that reconcile agricultural, livestock, and forest productivity, with a reduction in environmental impacts, as a way of promoting ecologically sustainable development, always observing the criteria of progressiveness, comprising the following categories and lines of action.

*§ 4º - The maintenance activities of the Permanent Preservation Areas, Legal Reserves and Restricted Use Areas are **eligible** for any payments or incentives for environmental services, constituting **additionality** for the purposes of national and international markets for certified reductions in greenhouse gas emissions”.*

Law 9,433/1997 (The National Policy on Water Resources) - The National Policy on Water Resources is based on the premise that water is a limited natural resource, with economic value and pertains to the public domain. Similarly, the management of water resources must always provide for the multiple uses of water, and the management of water resources must be decentralized and count on the participation of the Public sector, users and communities⁸⁶.

⁸⁴ http://www.planalto.gov.br/ccivil_03/_Ato20042006/2006/Lei/L11284.htm

⁸⁵ http://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm

⁸⁶ http://www.planalto.gov.br/ccivil_03/leis/l9433.htm

Law 12,305/2010 (National Solid Waste Policy) - The National Solid Waste Policy brings together the set of principles, objectives, instruments, guidelines, goals and actions adopted by the Federal Government, alone or in cooperation with states, the Federal District, municipalities or individuals, with a view to the integrated management and environmentally appropriate management of solid waste. This is a milestone in Brazilian environmental legislation, as it is the first federal standard created with a focus on the problem of solid waste. Thus, it deals with relevant issues related to social, environmental, and economic interests in practically all activities. It includes as instruments environmental, sanitary, and agricultural monitoring and inspection, technical and financial cooperation between the public and private sectors for the development of research on new products, methods, processes and technologies of management, recycling, reuse, waste treatment and final disposal environmentally sound tailings; scientific and technological research; environmental education, among others⁸⁷.

Law 12,187/09 (National Policy on Climate Change - PNMC) - The PNMC's objectives are to make economic and social development compatible with the protection of the climate system, the reduction of anthropogenic greenhouse gas emissions in relation to their different sources, the strengthening of human removals by sinks of greenhouse gases across the national territory, the implementation of measures to promote adaptation to climate change by the 3 (three) spheres of the Federation, with the participation and collaboration of economic and social agents stakeholders or beneficiaries, in particular those especially vulnerable to its adverse effects; the preservation, conservation and recovery of environmental resources, with particular attention to the great natural biomes considered National Heritage; the consolidation and expansion of legally protected areas and the encouragement of reforestation and recomposition of vegetation cover in degraded areas; and encouraging the development of the Brazilian Emissions Reduction Market – MBRE ⁸⁸.

Decree no. 9,578/2018 - Provides for the National Fund on Climate Change (FNMC), dealt with in Law no. 12,114, of December 9, 2009, and the National Policy on Climate Change, dealt with in Law n. 12,187, of December 29, 2009. Among the topics covered by the decree is the application of FNMC resources to projects to reduce carbon emissions from deforestation and forest degradation, with priority for natural areas threatened with destruction and relevant to conservation strategies biodiversity (art. 7, V)⁸⁹.

⁸⁷ http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2010/lei/l12305.htm

⁸⁸ http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2009/lei/l12187.htm

⁸⁹ http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/decreto/d9578.htm

Resolution no. 001/1986 under CONAMA - Deals with Environmental Licensing. It is an administrative procedure through which the Public Administration establishes conditions and limits certain activities⁹⁰.

Law 9,985/2000 (SNUC Law) - This Law establishes the National System of Nature Conservation Units - SNUC, establishes criteria and norms for the creation, implementation and management of "Conservation Units" (that is, conservation areas) and presents a series of important concepts for the proper understanding of Conservation Units⁹¹.

Law 6,938/1981 (National Environmental Policy) - The National Environmental Policy has the objective of preserving, improving and recovering the environmental quality conducive to life, aiming to ensure conditions for socio-economic development, the interests of national security and the protection of human dignity⁹².

Law 10,406/2002 (Civil Code) - Deals with various rights and obligations, including possession, property and legal business⁹³.

Law 4,504/1964 (Land Statute) - The Land Statute regulates the rights and obligations concerning rural real estate, for the purposes of implementing the Agrarian Reform and promoting the Agricultural Policy. It lists provisions regarding the rights of ownership and property in addition to their use⁹⁴.

Law 6,015/1973 (Public Records Law) - The law deals with public records and, especially, in its chapter V it refers to the registration of rural properties, through which the ownership of rural property is demonstrated⁹⁵.

Land Squatting Laws: Law Nº10.406, Art 1.238 of the Civil Code, Art 191 of the Federal Constitution among other laws which govern Land Squatting are fully discussed under section 2.5.1⁹⁶.

⁹⁰ <http://www.ibama.gov.br/sophia/cnia/legislacao/MMA/RE0001-230186.PDF>

⁹¹ http://www.planalto.gov.br/ccivil_03/leis/l9985.htm

⁹² https://www.planalto.gov.br/ccivil_03/Leis/L6938.htm

⁹³ http://www.planalto.gov.br/ccivil_03/leis/2002/l10406compilada.htm

⁹⁴ http://www.planalto.gov.br/ccivil_03/leis/l4504.htm

⁹⁵ http://www.planalto.gov.br/ccivil_03/leis/l6015compilada.htm

⁹⁶ https://www.planalto.gov.br/ccivil_03/LEIS/2002/L10406compilada.htm

Consolidation of labour laws (CLT): Regulates provisions relating to labour legislation and institutes the Permanent Program for Consolidation, Simplification and bureaucratization of Integral Labour Standards and the National Labour Award, and amends Decree No. 9,580 of November 22, 2018⁹⁷.

Regulatory norms (NR) such as: NR-5 - Internal Commission for Accident Prevention NR-12 - Safety At Work In Machinery And Equipment; NR-24 - Sanitary And Comfort Conditions In The Workplace; NR-26- Safety Signs⁹⁸.

State Laws:

Law 1,532/82: Disciplines the State Policy for the Prevention and Control of Pollution, Improvement and Recovery of the Environment and Protection of Natural Resources⁹⁹.

Law 2,416/96: Provides for the requirements for granting a license for the exploration, processing and industrialization of forest products and by-products for timber purposes¹⁰⁰.

Law 3,135/07: Establishes Amazonas State Policy on Climate Change, Environmental Conservation and Sustainable Development¹⁰¹.

Law 3,785/12: Provides for environmental licensing in the State of Amazonas, revokes Law n. 3,219, of December 28, 2007¹⁰².

Law 4,406/16: Establishes the State Policy for Regularization, provides for the Rural Environmental Registry - CAR, the Rural Environmental Registry System - Sicar-AM, the Environmental Regularization Program - PRA, in the State of Amazonas¹⁰³.

⁹⁷ https://www.planalto.gov.br/ccivil_03/decreto-lei/Del5452.htm; https://www.planalto.gov.br/ccivil_03/_Ato2015-2018/2018/Decreto/D9579.htm#art109

⁹⁸ <https://www.gov.br/trabalho-e-previdencia/pt-br/composicao/orgaos-especificos/secretaria-de-trabalho/inspecao/seguranca-e-saude-no-trabalho/ctpp-nrs/normas-regulamentadoras-nrs>

⁹⁹ <http://www.ipaam.am.gov.br/wp-content/uploads/2021/01/LDE-1.532-82-Pol%C3%ADtica-Estadual-da-Preven%C3%A7%C3%A3o-e-Control-e-da-Polui%C3%A7%C3%A3o.pdf>

¹⁰⁰ https://www.mpam.mp.br/images/stories/lei_2416_96.pdf

¹⁰¹ https://legisla.imprensaoficial.am.gov.br/diario_am/12/2007/6/4939?q=3.135

¹⁰² https://legisla.imprensaoficial.am.gov.br/diario_am/12/2012/7/2494?q=3.785

¹⁰³ https://legisla.imprensaoficial.am.gov.br/diario_am/12/2016/12/1433?q=4.406

Decree no. 10,028/87: State System for Licensing Activities with Potential Impact on the Environment¹⁰⁴.

Decree no. 15,780/94: Amends Art. 57 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532, of July 6, 1982, which provides for the State Licensing System for Activities with Potential of Impact on the Environment¹⁰⁵.

Decree no. 15,842/94: Amends Art. 44 of State Decree No. 10,028, of February 4, 1987, which regulated Law No. 1,532 of July 6, 1982, which provides for the State Licensing System for Activities with Potential of Impact on the Environment¹⁰⁶.

Decree no. 44.716/2021 – Assumes the compromise of the states of Amazonas with the neutrality of carbon, reducing emissions by fifty percent until 2050¹⁰⁷

As such, the Project IPOÁ REDD+ is in compliance with all identified applicable laws, and no cases of non-compliance with laws, statutes and other regulations were identified.

2.5.8 Approvals (G5.7)

Brazil does not currently have a specific authority for the management and approval of projects that generate carbon credits, so there was no need for any such approval. Furthermore, there are no communities officially classed as indigenous or traditional in the Project Zone, eliminating the need for approval at a public hearing or by the community management council.

Although there is no specific need for approval by any administrative body, it is worth mentioning that the project was built and will be conducted strictly in compliance with the law and best practices and,

¹⁰⁴<https://www.mpam.mp.br/attachments/article/4836/Decreto%2010.02887.pdf#:~:text=DECRETO%20N%C2%BA%2010.028%2C%20DE%2004%20DE%20FEVEREIRO%20DE,e%20aplica%C3%A7%C3%A3o%20de%20penalidade%20e%20d%C3%A1%20outras%20provid%C3%AAs>

¹⁰⁵ https://www.mpam.mp.br/images/stories/decreto_15780_94.pdf

¹⁰⁶ https://www.mpam.mp.br/images/stories/decreto_15842_94.pdf

<https://leisestaduais.com.br/am/decreto-n-44716-2021-amazonas-formaliza-a-adesao-do-estado-do-amazonas-as-campanhas-race-to-zero-e-under2-coalition-no-ambito-da-convencao-quadro-das-nacoes-unidas-sobre-mudanca-do-clima-cqnumc-unfccc-e-da-outras-providencias>

¹⁰⁷ <https://leisestaduais.com.br/am/decreto-n-44716-2021-amazonas-formaliza-a-adesao-do-estado-do-amazonas-as-campanhas-race-to-zero-e-under2-coalition-no-ambito-da-convencao-quadro-das-nacoes-unidas-sobre-mudanca-do-clima-cqnumc-unfccc-e-da-outras-providencias>

consequently, within all the regulations and principles of administrative entities in Brazil, such as IBAMA, FUNAI, INCRA, and ICMBIO, thus fully attending to CCB concept G5.7.

Approvals for the project were obtained internally from the proponents themselves, being approved by the representatives of both parties, in addition to legal and technical feasibility analyses.

2.5.9 Project Ownership (G5.8)

Both properties (Maanaim and Campos) are owned by Mrs. Carina Marcia Klein Balen and Mr. Sílvio Antônio Balen. Further details are provided in section 2.1.3, and full documentation regarding ownership was made available at audit.

The project's development and management are a partnership between Carbonext Tecnologia em Soluções Ambientais Ltda. and the aforementioned owners of the Project properties.

2.5.10 Management of Double Counting Risk (G5.9)

The IPOÁ project has not received any form of environmental or social credit, including any form of marketable climate (GHG or renewable energy related certificates) in the past, nor does it intend to do so in the future. For information on strategies to eliminate double counting over the project crediting period, see Section 2.5.15.

2.5.11 Emissions Trading Programs and Other Binding Limits

The present section is not applicable to the project, because the emissions reductions resulting from it will not be used for compliance under any other trading program or mechanism. The current VCS and CCB project is entirely unconnected to any other carbon project or payment for ecosystem service scheme being developed.

2.5.12 Other Forms of Environmental Credit

The project will not claim any other forms of environmental credit under other programs.

2.5.13 Participation under Other GHG Programs

The IPOÁ Project has not been registered under any emissions trading programs outside the VCS, and it will not claim credits for GHG emission reduction under any another program.

2.5.14 Projects Rejected by Other GHG Programs

The IPOÁ Project has not applied for or been rejected by any other GHG programs.

2.5.15 Double Counting (G5.9)

The project neither has generated nor intends to generate any form of GHG-related environmental credits for GHG emission reductions or removals other than those claimed under the VCS Program. The VCS Program has a central project database, which lists each approved project. The VCS Project Database is the central storehouse of information on all projects validated under VCS criteria and their Verified Carbon Units issued under the program. Every VCU can be tracked from issuance to retirement within the database, allowing buyers to ensure online that every credit is real, additional, permanent, independently verified, uniquely numbered and fully traceable. This project has not been registered under any other crediting activity, and no VCUs have been assigned to the project area so far. Thus, any possibility of double counting of credits is eliminated.

3 CLIMATE

3.1. Application of Methodology

3.1.1 Title and Reference of Methodology

Approved VCS Methodology VM0007 “REDD+ Methodology Framework (REDD+ MF)”, Version 1.6, 08 September 2020. At: https://verra.org/wp-content/uploads/2020/09/VM0007-REDDMF_v1.6.pdf (last visited 19 April 2022).

Carbon pool modules:

- VMD0001 “Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools” (CP-AB), v1.1
- VMD0005 “Estimation of carbon stocks in the long-term wood products pool” (CP-W), v1.1

Baseline module:

- VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), v1.3
- VMD0007 “Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation” (BL-UP), v3.3

Leakage modules:

- VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3
- VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), v1.2
- VMD0011 “Estimation of emissions from market-effects” (LK-ME), v1.1

Miscellaneous Modules:

- VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB), v1.2
- VMD0016 Methods for stratification of the project area (X-STR), v1.2
- VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), v2.2

Tools:

- Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD), Version 01, 19 Oct 2007 at:
<https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-02-v1.pdf> (last visited 19 April 2022)

- CDM – Executive Board “Tool for testing significance of GHG emissions in A/R CDM project activities (Version 01)” EB 31. At: <https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-04-v1.pdf> (last visited 19 April 2022).
- AFOLU “Non-Permanence Risk Tool” VCS Version 4, Procedural Document, 19 September 2019, v4.0. At: <http://www.v-c-s.org/program-documents> (last visited 19 April 2022).

3.1.2 Applicability of Methodology

The VM0007 REDD+ Methodology Framework is applicable to both Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD) project types, both of which apply to this project, and which fall under the AFOLU project category “REDD” as defined in the VCS AFOLU Guidance document. This methodology is a compilation of modules and tools that together define the project activity and necessary methodological steps. By choosing the appropriate modules, a project-specific methodology can be constructed.

All Project Activities:

No land areas are registered under the CDM or under any other GHG program (both voluntary and compliance-oriented). Therefore, no area must be excluded from the project area.

All REDD+ Activity Types:

This REDD+ activity is applicable due to the following conditions:

- Land in the project area has qualified as forest for at least the 10 years prior to the project start date.
- Baseline deforestation in the project area fall within the categories of Unplanned deforestation (VCS category AUDD) and Planned deforestation/degradation (VCS category APD).
- Leakage avoidance activities do not include: i) Agricultural lands that are flooded to increase production (e.g., rice paddy); ii) Intensifying livestock production through use of feed-lots and/or manure lagoons.

Avoiding Unplanned Deforestation:

This avoiding unplanned deforestation activities are applicable, as the baseline agents of deforestation: i) clear the land for tree harvesting and ranching; ii) have no documented and uncontested legal right

to deforest the land for these purposes; and (iii) are either residents in the reference region for deforestation or immigrants.

In the baseline scenario of this avoiding unplanned deforestation project activity, post deforestation land use does not constitute reforestation (i.e., pasture is the final land use).

Avoiding Planned Deforestation/Degradation:

This avoiding planned deforestation activity is applicable, considering that conversion of forest lands to a deforested condition is legally permitted until the 20% limit relative to the total property area, as described in *1.4 Compliance with Laws, Statutes and Other Regulatory Frameworks*.

The selected modules, along with their VCS site descriptions and applicability conditions, are detailed below, showing their correspondence to the project-specific conditions:

Carbon pool modules:

VMD0001 “Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools” (CP-AB), v1.1: “This module allows for ex ante estimation of carbon stocks in above- and belowground tree and non-tree woody biomass in the baseline case (for both pre- and post-deforestation stocks) and project case and for ex post estimation of change in carbon stocks in above- and belowground tree biomass in the project case”. This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the framework module REDD-MF.

VMD0005 “Estimation of carbon stocks in the long-term wood products pool” (CP-W), v1.1: “This module allows for ex-ante estimation of carbon stocks in the long-term wood products pool in the baseline case. Carbon stocks treated here are those stocks remaining in wood products after 100 years; the bulk of emissions associated with timber harvest, processing and waste, and eventual product retirement occur within this timeframe, and this module employs the simplifying assumption that the proportion remaining after 100 years is effectively ‘permanent’”. This module is applicable to all cases where wood is harvested for conversion to wood products for commercial markets, for all forest types and age classes. This module is applicable in the baseline, as the wood products pool is included as part of the project boundary as per applicability criteria in the framework module REDD-MF, specifically:

- Timber harvest occurs prior to or in the process of deforestation, and timber is destined for commercial markets.
- The wood products pool is determined to be significant (using T-SIG).

Baseline module:

VMD0006 Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL), v1.3: “This module allows for estimating carbon stock changes and GHG emissions related to planned deforestation, planned forest degradation and planned wetland degradation in the baseline scenario.

The module was revised on 8 September 2020 by Restore America’s Estuaries and Silvestrum Climate Associates to expand its applicability to activities on tidal wetlands.”

The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land. Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries Modules BL-DFW and LK-DFW must be used to determine potential leakage.

VMD0007 “Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation” (BL-UP), v3.3: “This module allows for estimating carbon stock changes and GHG emissions related to unplanned deforestation and unplanned wetland degradation in the baseline scenario. Forest degradation is not considered.

This module was assessed and approved as part of VM0007.

The module was revised on 31 July 2012 by The Field Museum to improve uncertainty estimates in the baseline scenario and allow for new modeling approaches (version 3.0). The revision assessment reports are available on the VMD0017 (X-UNC) page.

The module was revised on 8 September 2011 by The Field Museum to add population as an option for estimating the baseline deforestation rate (version 2.0).

The module was revised on 8 September 2020 by Restore America’s Estuaries and Silvestrum Climate Associates to expand its applicability to activities on tidal wetlands.”

The applicability conditions of the module state that “The module is applicable for estimating baseline emissions from unplanned deforestation (conversion of forest land to non-forest land in the baseline case). The following conditions must be met to apply this module. The forest landscape configuration can be mosaic, transition, or frontier.

The module must be applied to all project activities where the baseline agents of deforestation:

- (i) clear the land for settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agri/aquaculture activities;”
- (ii) “have no documented and uncontested legal right to deforest the land for these purposes; and”
- (iii) “are either resident in the region (reference region—cf. Section 1 below) or immigrants.”

The applicability conditions of this module are met in that:

- i.the landscape configuration is transition, as discussed in section 3.1.3.
- ii.the baseline agents of deforestation clear the land for ranching (see section 3.1.4).
- iii.their practices are predominantly illegal and not within their legal rights, as argued in section 3.1.5, under Scenario 3.

Leakage modules:

VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3:

This module allows for estimating GHG emissions caused by the activity shifting leakage of planned deforestation carbon projects.

The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded because of project implementation. The module is also applicable for estimating the leakage emissions due to activity shifting from non-forested wetlands that are legally authorized and documented to be converted and degraded. Under these situations,

displacement of baseline activities can be controlled and measured directly by monitoring the baseline deforestation or wetland degradation agents or class of agents.

This tool must be used for projects in areas where planned deforestation happens on forested wetlands, regardless of the absence of wetland within the project boundaries.

The module is mandatory if Module BL-PL has been used to define the baseline, and the applicability conditions in Module BL-PL must be complied with in full.

However, for the Ipoá Project the module was not necessary, because where the specific deforestation agent can be identified (land owners), leakage does not need to be considered where it can be demonstrated that the management plans and/or land-use designations of the deforestation agent's other lands (which must be identified by location) have not materially changed as a result of the project (e.g., the deforestation agent has not designated new lands as timber concessions, increased harvest rates in lands already managed for timber, cleared intact forests for agricultural production or increased fertilizer use to enhance agricultural yields).

VMD0010 Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU), v1.2:

This Module provides methods for estimating emissions from displacement of unplanned deforestation (leakage due to activity shifting). This Module is applicable for estimating carbon stock changes and greenhouse gas emissions related to the displacement of activities that cause deforestation of lands outside the Project Area due to the avoided unplanned deforestation in the Project Area. Activities subject to potential displacement are conversion of forest land to grazing lands, crop lands, and other land uses. The module is mandatory if BL-UP has been used to define the baseline and the applicability criteria in BL-UP must be complied with in full.

VMD0011 “Estimation of emissions from market-effects” (LK-ME), v1.1:

“This module allows for estimating GHG emissions caused by the market-effects leakage related to extraction of wood for timber, fuelwood or charcoal in the baseline for carbon projects”.

This module is applicable for calculating market-effects leakage from REDD+ projects that are anticipated to reduce levels of wood harvest substantially and permanently. When REDD +project activities result in reductions in wood harvest, it is likely that production could shift to other areas of the country to compensate for the reduction.

As referenced in the Framework (REDD-MF) the module is mandatory where:

- The process of deforestation involves timber harvesting for commercial markets.
- The baseline is calculated using BL-DFW AND fuel wood or charcoal is harvested for commercial markets.

In all other circumstances the module shall not be used.

When REDD+ project activities result in reductions in wood harvest, it is likely that production could shift to other areas of the country to compensate for the reduction. The module is mandatory where the process of deforestation involves timber harvesting for commercial markets (Commercial markets are here defined as sale of products to end users and public and private companies with sales conducted distant (>50 km) from the project area).

Miscellaneous Modules:

VMD0013 Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB), v1.2:

This module provides a stepwise approach for estimating greenhouse emissions from biomass and peat burning. This module is applicable to REDD+ project activities with emissions from biomass burning.

In the baseline scenario, fire is used to clear the land, and emissions of CO₂, N₂O and CH₄ result. Where used in the baseline, accounting must occur under both the baseline and with project scenarios in both the project area and in the leakage belt. Where fires occur ex-post in areas that coincide with areas deforested or degraded in the baseline case, the module shall be used to account greenhouse gas emissions.

VMD0016 Methods for stratification of the project area (X-STR), v1.2:

This module provides guidance on stratifying the project area into discrete, relatively homogeneous units to improve accuracy and precision of carbon stock and carbon stock change estimates.

Any module referencing strata i shall be used in combination with this module. Strata are only used for pre-deforestation forest classes and are the same in baseline and project cases. Post-deforestation

(conversion) land-uses are not stratified, instead using average post-deforestation stock values (e.g., “Simple Conservative” or “Historical Area-weighted” approaches per BL-UP).

VMD0017 Estimation of uncertainty for REDD+ project activities (X-UNC), v2.2:

This module allows for estimating uncertainty in baseline estimations and in estimations of with-project sequestration, emissions, and leakage. This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of CO₂-e generated from REDD+ and WRC project activities. The module focuses on the following sources of uncertainty:

- Determination of rates of deforestation and degradation.
- Uncertainty associated with estimation of stocks in carbon pools and changes in carbon stocks.
- Uncertainty associated with estimation of peat emissions.
- Uncertainty in assessment of project emissions.

Where an uncertainty value is not known or cannot be simply calculated, a project must justify that it is using an indisputably conservative number and an uncertainty of 0% may be used for this component.

Guidance on uncertainty – a precision target of a 95% confidence interval half-width equal to or less than 15% of the recorded value must be targeted. This is especially important in terms of project planning for measurement of carbon stocks; sufficient measurement plots should be included to achieve this precision level across the measured stocks.

Tools:**A/R Methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD)”**

This tool is applicable when AFOLU activities on the land within the proposed project boundary do not lead to violation of any applicable law even if the law is not enforced. The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario.

Tool for testing significance of GHG emissions in A/R CDM project activities

This tool facilitates the determination of which GHG emissions by sources, possible decreases in carbon pools, and leakage emissions are insignificant for a particular CDM A/R project activity.

The tool shall be used in the application of an A/R CDM approved methodology to an A/R CDM project activity:

- a. To determine which decreases in carbon pools and increases in emissions of the greenhouse gases measured in CO₂ equivalents that result from the implementation of the A/R project activity, are insignificant and can be neglected.
- b. To ensure that it is valid to neglect decreases in carbon pools and increases in GHG emissions by sources stated as being insignificant in the applicability conditions of an A/R CDM methodology.

The above modules and tools are applicable because they meet the applicability conditions of the modules as set out below:

Table 12: Applicability Conditions and Justifications for the REDD+ Methodology Framework Module.

Applicability Condition	Justification
Land in the project area has qualified as forest at least 10 years before the project start date.	The project area complies with this condition, with complete forest cover demonstrated for the year 2010.
The project area can include forested wetlands (such as bottomland forests, floodplain forests, mangrove forests) as long as they do not grow on peat. Peat shall be defined as organic soils with at least 65% organic matter and a minimum thickness of 50 cm. If the project area includes forested wetlands growing on peat (e.g., peat swamp forests), this methodology is not applicable.	As demonstrated in Section 2.1.5, no organic soils (or peatlands) exist within the project area.

Project proponents must be able to show control over the project area and ownership of carbon rights for the project area at the time of verification.	As demonstrated in Sections 2.1.3 and 2.5.9, the project proponents have clear, uncontested title to both property rights and the carbon rights.
Baseline deforestation and baseline forest degradation in the project area fall within one or more of the following categories: Unplanned deforestation (VCS category AUDD); Planned deforestation (VCS category APD); Degradation through extraction of wood for fuel (fuelwood and charcoal production) (VCS category AUDD).	Baseline deforestation in the project area falls within APD and AUD: - APD category, as the agent of deforestation is the project proponent. - AUD category: deforestation agents are cattle ranchers, timber harvesters and infrastructure.
Baselines shall be renewed every 6 years from the project start date.	The baseline will be renewed in October 2028.
All land areas registered under the CDM or under any other carbon trading scheme (both voluntary and compliance-orientated) must be transparently reported and excluded from the project area. The exclusion of land in the project area from any other carbon trading scheme shall be monitored over time and reported in the monitoring reports.	The IPOÁ Project REDD+ is not registered in any carbon trading scheme or program.
If land is not being converted to an alternative use but will be allowed to naturally regrow (i.e., temporarily unstocked), this framework shall not be used.	Forest clearing in the baseline is followed by establishment of pasture, which prevents forest regrowth.

Leakage avoidance activities shall not include: Agricultural lands that are flooded to increase production (e.g., paddy rice); Intensifying livestock production through use of “feed-lots” and/or manure lagoons.	Leakage avoidance activities do not include flooding agricultural land or creating feed-lots or manure lagoons.
--	---

Table 13: VMD0006 Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Planned Deforestation and Planned Degradation”

Applicability Condition	Justification
The module is applicable for estimating the baseline emissions on forest lands (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land.	The proponents requested the IPAAM organ’s permission to clear up to 19.14% of the Maanaim and Camps proprieties, according to the legal limit prescribed in the Brazilian Forest Code. The landowner has a plan to deforest in the next few years, which will not be executed in case of success in obtaining VCUs.
Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries modules BL-DFW and LK-DFW shall be used to determine potential leakage.	No fuelwood collection has been conducted within the project area. This statement can also be confirmed through the public consultation carried out with the community, as none of the interviewees mentioned collecting wood for fuelwood or any other activity.

Table 14: “VMD0007 Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Unplanned Deforestation”

Applicability Condition	Justification
--------------------------------	----------------------

Baseline agents of deforestation shall: (i) clear the land for settlements, crop production (agriculturalist) or ranching, where such clearing for crop production or ranching does not amount to large scale industrial agriculture activities; (ii) have no documented and uncontested legal right to deforest the land for these purposes; and (iii) are either resident in the reference region or immigrants. Under any other condition this framework shall not be used.	The baseline agents of deforestation clear the land for ranching as final land use. These land squatters have no legal right to use or deforest the land. These agents of deforestation are from nearby communities or immigrants looking for land to illegally harvest wood and convert the land for pasture.
Where, pre-project, unsustainable fuelwood collection is occurring within the project boundaries modules BL-DFW and LK-DFW shall be used to determine potential leakage.	No fuelwood collection has been conducted within the project area. This statement can also be confirmed through the public consultation carried out with the community, and none of the interviewees mentioned collecting wood for charcoal or any other activity.

Table 15: VMD0009 Estimation of emissions from activity shifting for avoided planned deforestation/forest degradation and avoided planned wetland degradation (LK-ASP), v1.3

Applicability Condition	Justification
The module is applicable for estimating the leakage emissions due to activity shifting from forest lands that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded as a consequence of project implementation.	The proponents have the official request and the plan to deforest 19.14% of the Maanaim and Campos proprietries.

This tool must be used for projects in areas where planned deforestation happens on forested wetlands, regardless of the absence of wetland within the project boundaries.	There are no forested wetlands within the project area.
The module is mandatory if Module BL-PL has been used to define the baseline, and the applicability conditions in Module BL-PL must be complied with in full.	Module BL-PL was applied, and all the conditions were complied with.

Table 16: A/R Methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”

Applicability Condition	Justification
AFOLU activities the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced	As stated in the section 2.5, the project is in compliance with all relevant laws, statutes, and regulatory frameworks.
The use of this tool to determine additionality requires the baseline methodology to provide for a stepwise approach justifying the determination of the most plausible baseline scenario. Project proponent(s) proposing new baseline methodologies shall ensure consistency between the determination of a baseline scenario and the determination of additionality of a project activity.	The approach to define the baseline scenario using this tool was described in the section 3.1.4.

3.1.3 Project Boundary

Project Area and Leakage Belt

The Project Area (PA) consists of forests in their entirety at the beginning of the project. It has a total area of 17,329.73 hectares, consisting of Avoiding Unplanned Deforestation (AUD) and Avoiding Planned Deforestation (APD) areas, comprising 13,781.78 and 3,547.95 hectares, respectively. The APD design excluded 112.34 hectares of permanent preservation areas.

The delineation of the Leakage Belt (LB) involved maintaining forested areas and excluding all non-forested ones. The Leakage Belt (LB) is of 21,895.26 hectares, which corresponds to more than 90% of the project's Avoiding Unplanned Deforestation (AUD) area (13,781.78 hectares), as required in item 1.1.3 of the VMD0007-BL-UP_v3.3 module. Furthermore, the Leakage Belt complies with all similarity criteria (a-g) required in item 1.1.3 of the aforementioned module. Finally, the design of the Leakage Belt removed overlaps with another REDD+ project (Yellow Ipê) and took into account the probable addition of a new area into the project, in future MR periods.

Table 17. Leakage Belt and Project Area similarity analysis

Vegetation		
Area	type	%
PA	Dse	73.28
	Paas	9.09
	Dbe	16.31
	Dau	1.31
LB	Dse	78.13
	Paas	4.99
	Dbe	16.88
Slope		
Area	type	%
PA	≤ 15%	97.28
	> 15%	2.72
LB	≤ 15%	94.44
	> 15%	5.56

Elevation		
Area	type	%
PA	0 - 500m	100
LB	0 - 500m	100
Soil		
area	type	%
PA	Ferrasols	100
LB	Ferrasols	99.998
	Gleisols	0.002
Road density		
area	type	%
PA	Road	0.00
LB	Road	0.00
Settlement density		
area	type	%
PA	Settlement	0.00
LB	Settlement	0.00
Navigable rivers		
area	type	%
PA	Navigable rivers	0.00
LB	Navigable rivers	0.00

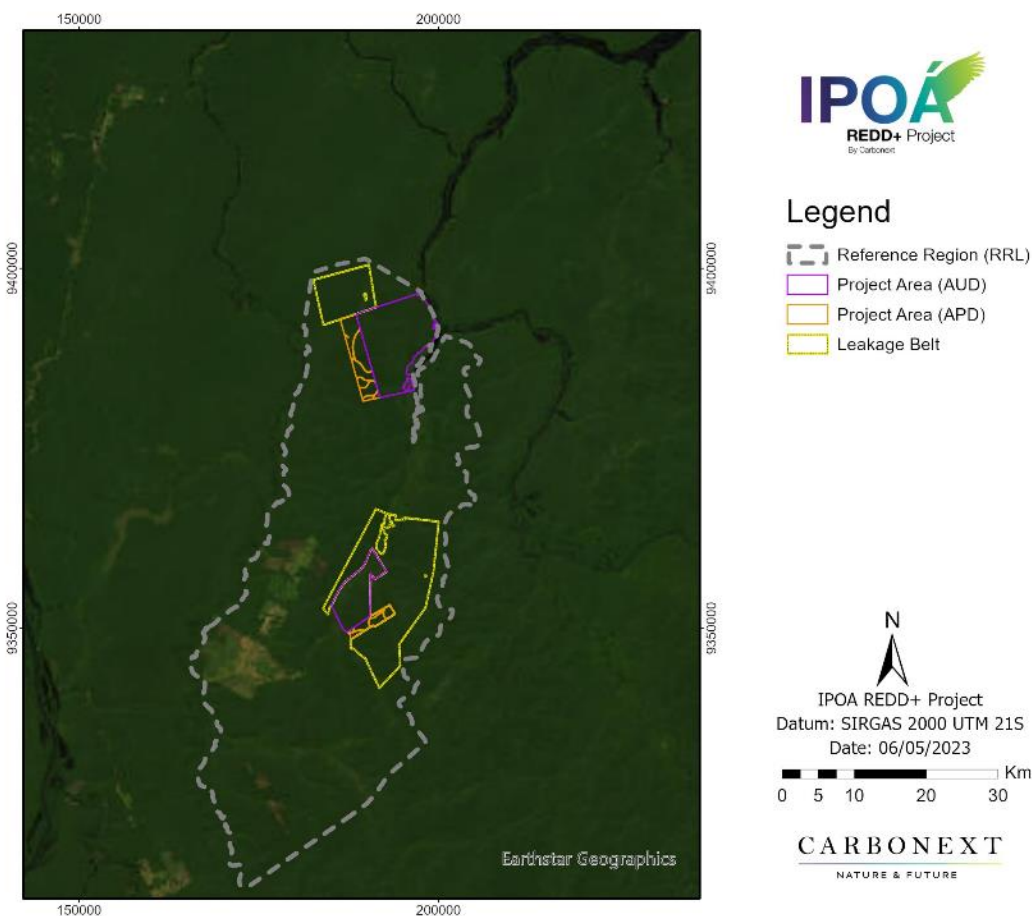


Figure 28 - Ipoá REDD+ project boundaries: Leakage Belt, and Project Area, divided into APD (Avoiding Planned Deforestation), and AUD (Avoiding Unplanned Deforestation)

Reference region for projecting rate of deforestation (RRD)

The Reference Region for projecting rate of deforestation (RRD) was delimited according to the boundaries of watersheds in the region, classified as hierarchical level 6 by the National Water and Sanitation Agency (ANA). In addition, the RRD is entirely located within in the municipality of Novo Aripuanã, Amazonas State, Brazil.

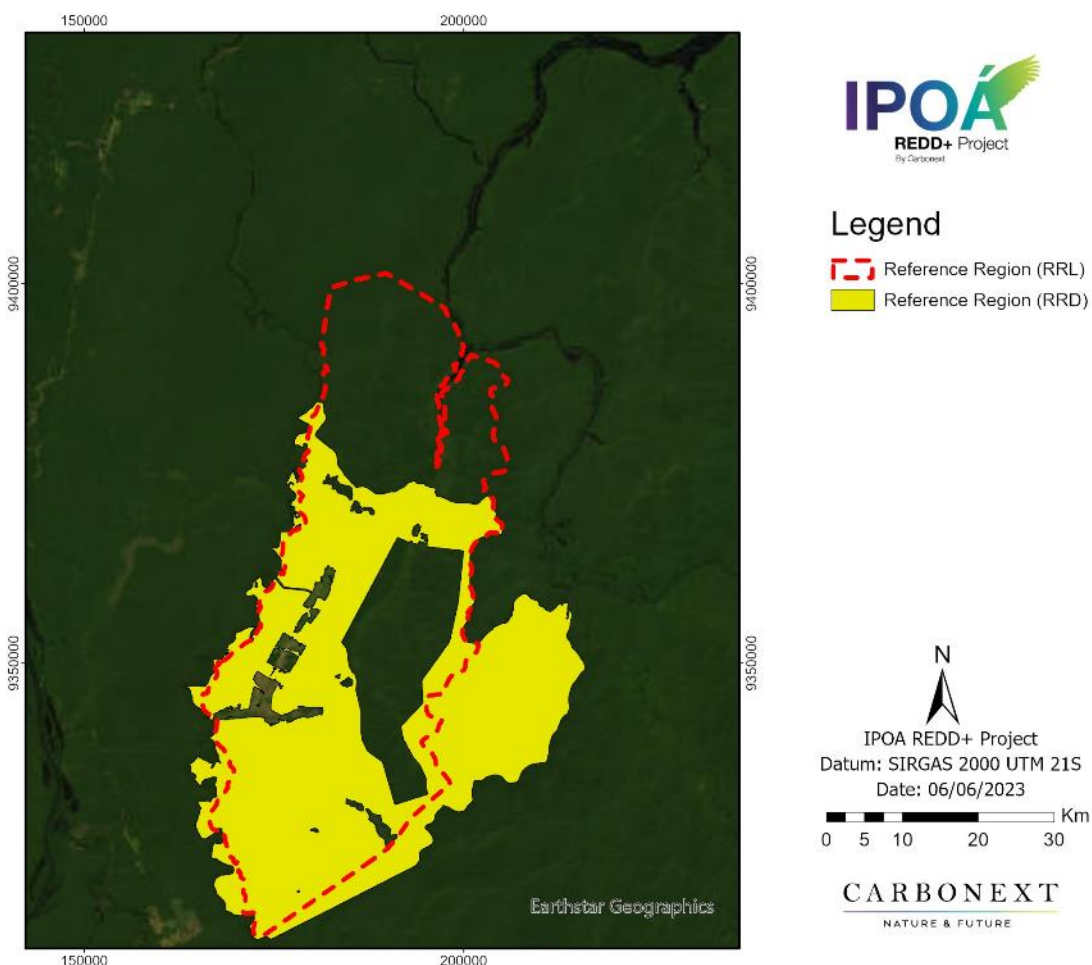


Figure 29. Areas of Reference Region (RRL) and Reference Region (RRD).

The RRD was calculated following the criteria in section 1.1.1.1 of the VMD0007-BL-UP module. The key criteria applied are: the RRD must be 100% forested at the start of the historical reference period and must not encompass the project area and the leakage belt.

Additionally, the section specifies that the area of the RRD must be larger than MREF (Minimum size of reference region for projection rate of deforestation). This is met by the project since the RRD has an area of 157,292.70 hectares and the MREF has an area of 130,873.84 hectares, as described by the following calculations:

$$\text{MREF} = \text{RAF} * \text{PA}$$

$$\text{RAF} = 7500 * \text{PA}^{-0.7}$$

$$\text{RAF} = 7500 * 13,781.78^{-0.7}$$

$$\text{RAF} = 9.4962$$

$$\text{MREF} = 9.4962 * 13,781.78$$

$$\text{MREF} = 130,873.84$$

The boundaries of the RRD were geographically defined based on watersheds, classified as hierarchical level 6 and 7 by the National Water and Sanitation Agency (ANA), and based on the limits of Novo Aripuanã municipality, Amazonas State, Brazil.

Regarding the methodological procedure, the RRD was defined according to the similarity criteria described in section 1.1.1.1, being:

- a) the main agent of deforestation in the RRD is the expansion of livestock, the same as is expected on Project Area;
- b) the landscape factors found on the RRD are similar to the landscape factors found in the PA as shown in the table below);
- c) Transportation networks and human infrastructure found on the RRD are similar to the PA , because, as shown in the table below, both were found to have no presence of road networks;
- d) no further specific social factors regarding the RRD and PA were identified beyond those already mentioned in point “a”;
- e) Policies and regulations having an impact on land-use change patterns within the RRD and the PA are the same, as both contain only private areas. Beyond that, the RDD and PA are under the same jurisdiction, being that of the Novo Aripuanã municipality;
- f) and finally, no planned deforestation for exclusion within the RRD was found.

Table 18 - Project Area and Reference Region similarity analysis

Vegetation	
Area type	%

PA	Dse	73.28
	Paas	9.09
	Dbc	16.31
	Dau	1.31
RRD	Aap	4.23
	Dse	86.26
	Paas	1.17
	Dau	1.43
	Asc	6.58
	Paap	0.34
Slope		
Area	type	%
PA	≤ 15%	97.28
	> 15%	2.72
RRD	≤ 15%	93.62
	> 15%	6.38
Elevation		
Area	type	%
PA	0 - 500m	100
RRD	0 - 500m	100
Soil		
area	type	%
PA	Ferrasols	100
RRD	Ferrasols	80.48
	Gleisols	19.52

Road density		
area	type	%
PA	Road	0.00
RRD	Road	0.00
Settlement density		
area	type	%
PA	Settlement	0.00
RRD	Settlement	0.00
Navigable rivers		
area	type	%
PA	Navigable rivers	0.00
RRD	Navigable rivers	0.00

Reference region for projecting location of deforestation (RRL)

The forest landscape configuration is not frontier, as forests within the Project Area have a physical connection with anthropogenically deforested areas, nor a mosaic, because the forest patches are over 1,000 ha in size, and are not completely surrounded by anthropogenically cleared land. Thus, the forest landscape configuration is transition.

As it is not the case that 25% or more of the project geographic boundary within 50 m of land was anthropogenically deforested within the 10 years prior to the project start date, the location analysis is required.

The reference region for the deforestation location projection (RRL) is a single, continuous polygon, which englobes the Project Area (PA) and the Leakage Belt (LB) and excludes protected areas. At the beginning of the project, the RRL consisted of 12.47% of non-forest and 87.53% forested area. The forest area in the RRD is similar to the RRL (+ 10.80%).

At the beginning of the historical reference period, the RRL consisted of 5.62% non-forest area and 94.38% forested area.

At the beginning of the baseline period, the RRL had a similar proportion of forests suitable for conversion to land-use practices by agents of deforestation as the project area (-7.90%) - the Project Area (PA) is 100% forested area.

The VM0007 Methodology includes the six carbon pools listed in the table below (Carbon sinks), indicating whether they were included or excluded within the proposed AUD and APD project activity, as well as their respective justifications.

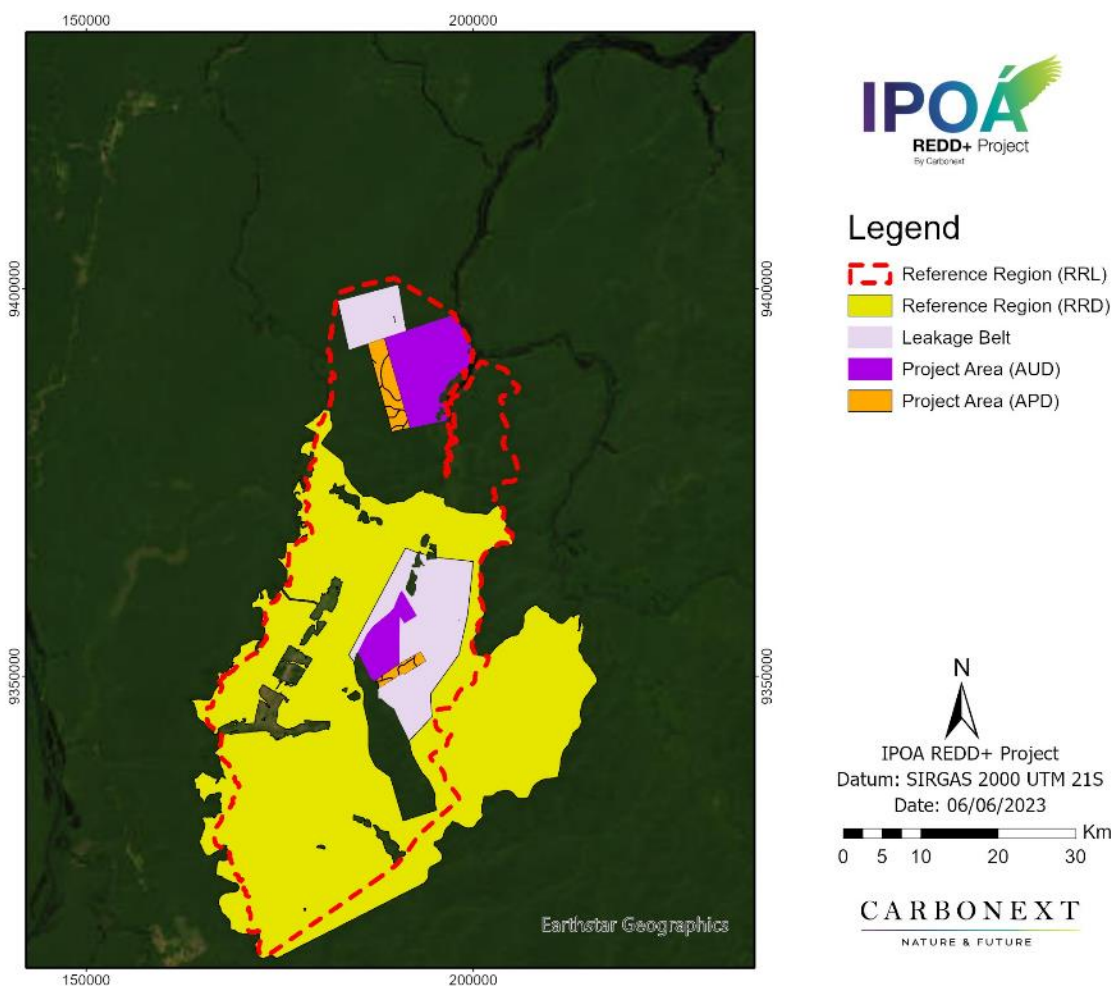


Figure 30. IPOÁ project boundaries

Carbon Pools

The table below shows the relevant carbon reservoirs considered in the scope of the project, according to VM0007 v1.6, table 4 and the applicable modules included therein.

Table 19. Carbon pools

Carbon Pools	Included?	Justification/Explanation
Above-ground tree biomass	Included	Inclusion of aboveground biomass is mandatory in accordance with module VMD0007 – REDD+ MF.
Above-ground non-tree biomass	Excluded	This carbon pool is considered to be optional, and is conservatively excluded. Module CP-AB lists the conditions in which AB non-tree biomass must be included, the first of which is: “Stocks of non-tree aboveground biomass are greater in the baseline than in the project scenario”. In the present project, stocks of non-tree biomass are lower in the post-deforestation land-use than in the project scenario because, the post-deforestation land-use is pasture which does not contain any AB non-tree biomass (see PD section 3.2.1).
Below-ground tree biomass	Included	In the VMD0001-CP-AB module, section 4, inclusion of BB is defined as optional, and it is included as it significant to project activities. In the present project, the BB carbon pool represents between 15.6% and 22.0% of total tree biomass in the various carbon pools, as shown at audit.
Below-ground non-tree biomass	Excluded	Omission is conservative in accordance with module VMD0001-CP-AB, section 4.
Deadwood	Excluded	Omission is conservative. The REDD+ MF module table 4, notes that application of the CP-D module is mandatory “if this carbon pool is greater in baseline (post-deforestation/degradation) than project scenario and significant; otherwise can be conservatively omitted” In the post-deforestation scenario of pastureland, no dead wood is included (see

		section PD 3.2.1, therefore the criterion for conservative omission is met.
Harvested wood products	Included	According to REDD +MF module, table 4, the CP-W module, "Mandatory for the given project activity where the process of deforestation involves timber harvesting for commercial market", and this is the case for the project's baseline scenario (see PD section 3.1.4).
Litter	Excluded	Omitted from both the baseline and project scenarios by choice. According to REDD +MF table 4, the CP-L module is fully optional, and it "can be included or excluded as decided by the project".
Soil organic carbon	Excluded	Omitted from both the baseline and project scenarios by choice. According to REDD +MF table 4, the CP-S module is fully optional, and it "can be included or excluded as decided by the project".

Deforestation emissions were estimated for all forest strata (conservatively excluding non-tree biomass), of which the above- and belowground carbon pools were previously determined by means of a literature survey of data from the Project Area region. It is considered that a certain portion of logged wood is converted into long-term wood products, which serve as carbon pools after deforestation. This was factored into the calculation of net deforestation emissions.

Sources of GHG Emissions

Table 20. GHG sources, sinks and reservoirs for the project and baseline scenarios

Source		Gas	Included or excluded	Justification/Explanation
Baseline	Biomass burning	CO ₂	Excluded	Excluded as recommended by the applied methodology. Counted as carbon stock change.
		CH ₄	Included	Included as non-CO ₂ emissions from biomass burning in the baseline scenario, according to the methodology.

Source		Gas	Included or excluded	Justification/Explanation
		N ₂ O	Included	Included as non-CO ₂ emissions from biomass burning in the baseline scenario, according to the methodology.
		Other	Excluded	No other GHG gases were considered in this project activity.
	Combustion of fossil fuels	CO ₂	Excluded	Conservatively excluded for simplification.
		CH ₄	Excluded	Conservatively excluded for simplification.
		N ₂ O	Excluded	Conservatively excluded for simplification.
		Other	Excluded	Conservatively excluded for simplification.
	Use of Fertilizers	CO ₂	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
		CH ₄	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
		N ₂ O	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
		Other	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
Project	Biomass burning	CO ₂	Excluded	No biomass burning increase is predicted to occur in the project scenario compared to the baseline case. Therefore, considered insignificant.
		CH ₄	Included	Included as non-CO ₂ emissions from biomass burning in the project scenario, according to the methodology.
		N ₂ O	Included	Included as non-CO ₂ emissions from biomass burning in the project scenario, according to the methodology.
		Other	Excluded	No other GHG gases were considered in this project activity.
	Combustion of fossil fuels	CO ₂	Excluded	Excluded for simplification. This is conservative
		CH ₄	Excluded	Excluded for simplification. This is conservative
		N ₂ O	Excluded	Excluded for simplification. This is conservative
		Other	Excluded	Excluded for simplification. This is conservative
	Use of Fertilizers	CO ₂	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.

Source		Gas	Included or excluded	Justification/Explanation
		CH ₄	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
		N ₂ O	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.
		Other	Excluded	There is no occurrence of fertilizer use. Therefore, considered insignificant.

All project installations and management activities, based on the description provided in Section 2.1.11, will take place in the Project Zone, located as shown below. Thus, it is considered that the Project Boundaries are the same as the Project Zone.

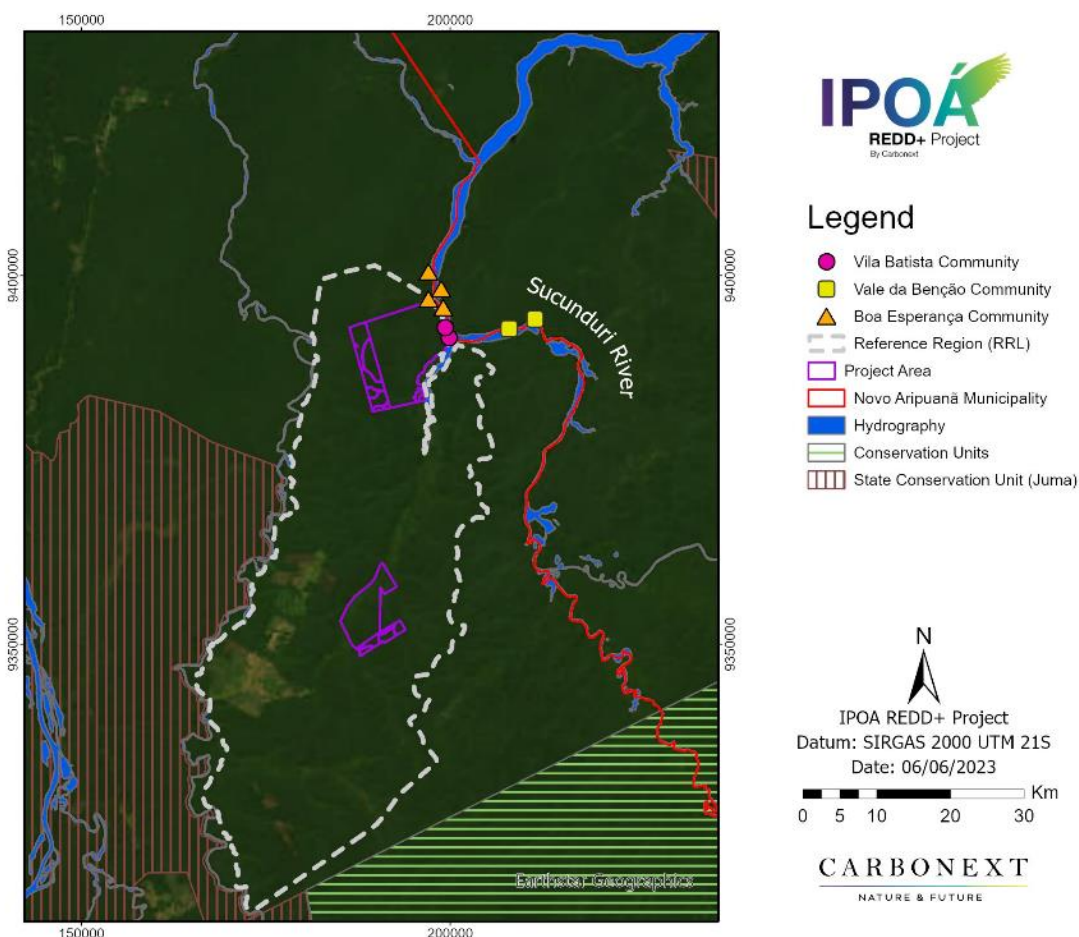


Figure 31 . Project Zone: Project Area and Communities

3.1.4 Baseline Scenario

Forest land is expected to be converted to non-forest land in the baseline scenario, under two mechanisms, or components, of the baseline: Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD). The landowners cannot afford efforts and costs to keep long term vigilance of the project boundaries to avoid unplanned land use change from uncontrolled trespassing and deforestation, without carbon credit income.

The AUD baseline component involves a predicted 652.48 hectares of deforestation over the project lifetime, as further described below.

The APD component is regarding a 3,547.95 ha subsection (19.14%) of the Maanaim and Campos properties, which has documented forest clearing plans. Under the latter, the 19.14% of the property would be clear-cut in the baseline scenario, with wood to be sold on the timber markets, followed by transformation of deforested areas into pasturelands.

Given the context above, the project falls within the AFOLU REDD+ categories of Avoided Unplanned Deforestation (AUD) and Avoided Planned Deforestation (APD).

The stepwise approach of the T-ADD Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities is fully presented in section 3.1.5, below for AUD and APD.

The baseline will be revised every 6 years in accordance with the latest VCS rules.

Avoided Unplanned Deforestation (AUD)

Studies indicate that the municipality of Apuí, the neighbouring municipality of Novo Aripuanã, was the hub of livestock expansion in the southern region of the State of Amazonas. A similar phenomenon has been observed in the project municipality of Novo Aripuanã.

Apuí municipality has the second largest cattle herd size in Amazonas State, second only to Boca do Acre, and cattle ranching represented 51% of the municipal GDP at the time of the most recent government agricultural census (IBGE 2017¹⁰⁸; Carrero et al. 2015¹⁰⁹). This cattle production is strongly linked with illegal deforestation levels and deforestation. Idesam (2011¹¹⁰) note, in connection with the local cattle farmer syndicate representing ~80% of Apuí's cattle farmers, that if their Legal Reserves had just 40% of their required areas, it would be sufficient to render cattle farming inviable. Further studies show the correlation between growth of cattle farming and the curve of deforestation in the municipality is significant (Carrero et al. 2015).

¹⁰⁸ Source: <https://www.ibge.gov.br/estatisticas/economicas/agricultura-e-pecuaria/21814-2017-censo-agropecuaria.html?edicao=21858&t=resultados>

¹⁰⁹ Carrero et al (2015): A Cadeia Produtiva de Carne Bovina no Amazonas.
<http://www.idesam.org.br/publicacao/cadeia-produtiva-corte-amazonas.pdf>

¹¹⁰ Idesam (2011) REDD+ Estudo de Oportunidades para a Região Sul do Amazonas (accessed 13/08/21):
https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas.pdf

A study conducted by Soares (2022) identified that the municipality of Novo Aripuanã is a new hub of expansion of the agricultural frontier, following the dynamics of the search for cheap land, investment in deforestation and agricultural production.¹¹¹

According to MapBiomass¹¹², pasture is the most common alternative land use in the municipality of Novo Aripuanã.

Table 21 - Percentage of land uses in Novo Aripuanã municipality in 2020.

Classes	Area (ha)	% of Total Area
Forest Formation	3,798,780.73	92.25%
Pasture	132,082.99	3.21%
Grassland (Non-Forest Natural Formation)	102,258.54	2.48%
Water	76,906.76	1.87%
Savanna Formation	6,031.63	0.15%
Wetland	1,112.47	0.03%
Urban Area	447.89	0.01%
Temporary Crops	339.47	0.01%

¹¹¹ https://www.teses.usp.br/teses/disponiveis/91/91131/tde-11022022-160839/publico/Pedro_Gandolfo_Soares_versao_revisada.pdf

¹¹² <https://mapbiomas.org/>

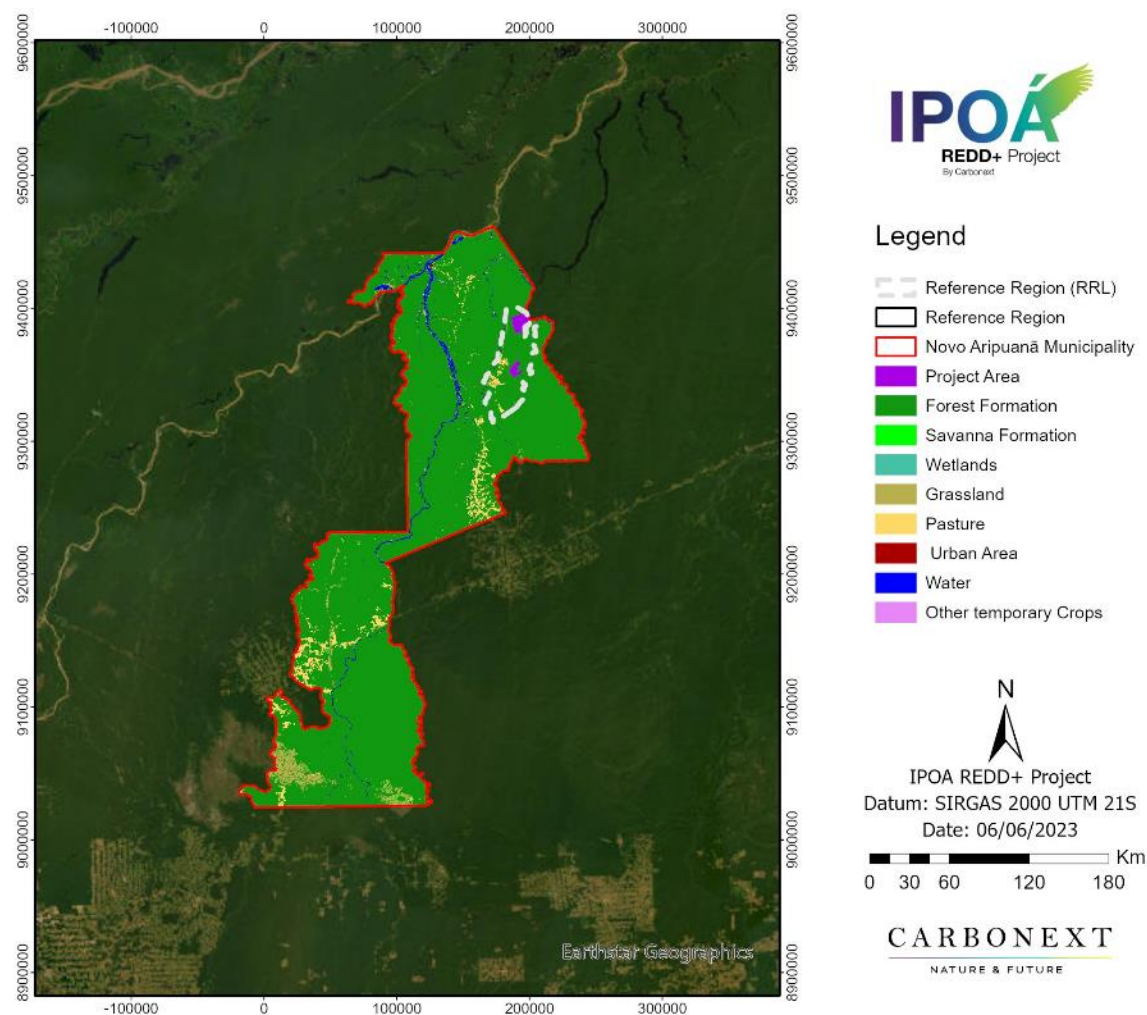


Figure 32. Land-Uses in Novo Aripuanã municipality in 2020.

In 2010, pasturelands covered only 1.25% of the municipal territory. However, in 2020 this percentage rose to 3.21%, totaling 132,082.99 hectares¹¹³. This value represents 99,41% of non-natural land uses in Novo Aripuanã.

¹¹³ <https://mapbiomas.org/>

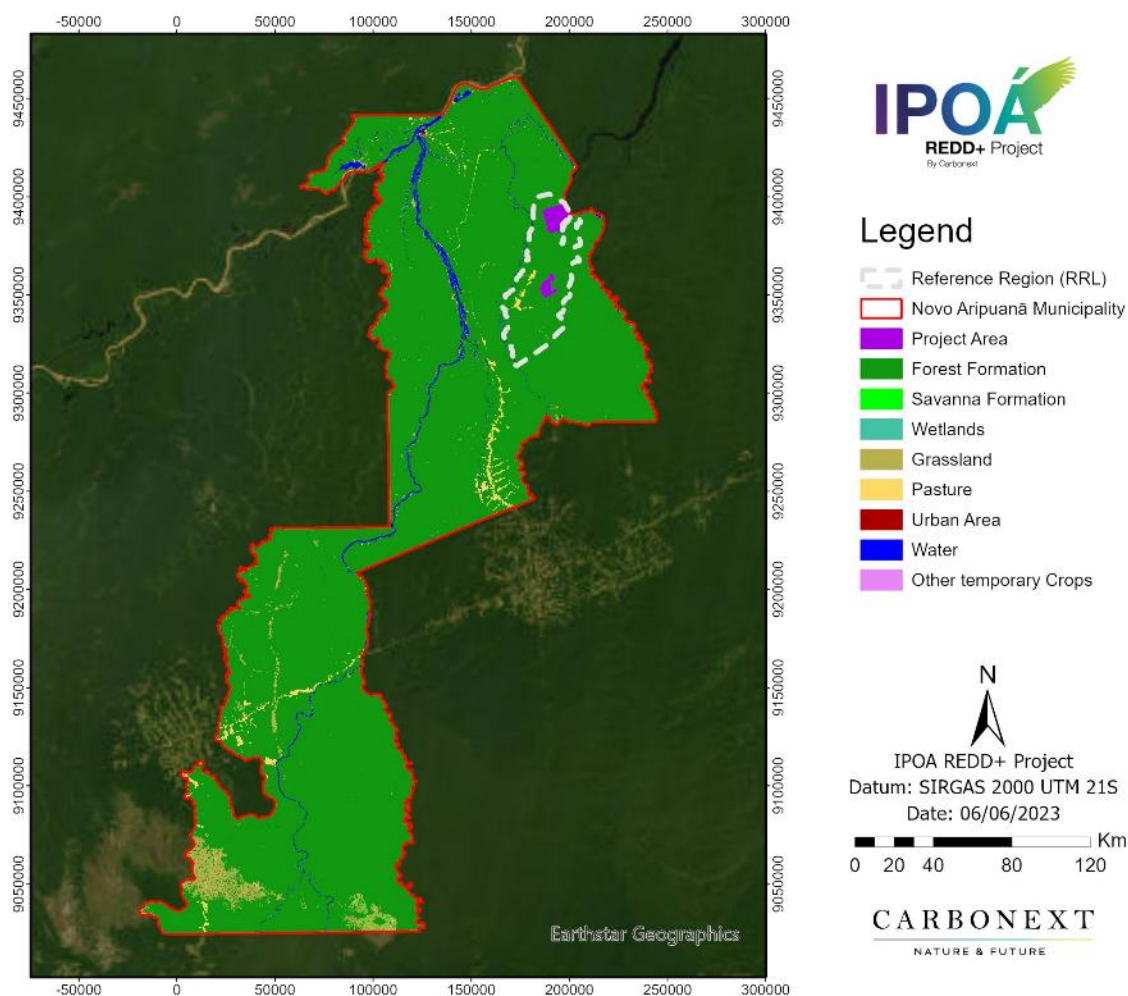


Figure 33. Land-uses in Novo Aripuanã municipality in 2010

This increasing trend was also observed in the number of cattle identified in the municipality of Novo Aripuanã by the agricultural census. In 2010, the number of heads was 13,989, while in 2020 this amount increased to 42,100¹¹⁴, an increase of 300%. As seen in the figure below, the increase in the herd and Novo Aripuanã peaked in 2018.

¹¹⁴ <https://sidra.ibge.gov.br/tabela/3939>

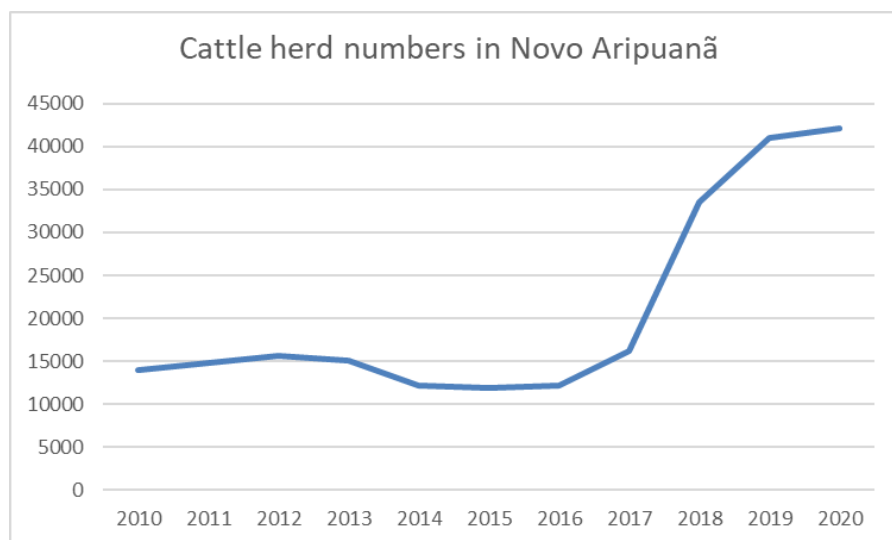


Figure 34. Increase in cattle herd in Novo Aripuanã

As the main deforestation agent in the region, cattle ranching (pasture) is expected to increase in the project region. The percentage of the total area of this land use-class increased from 0.57% in 2010 to 1.67%, in 2020.

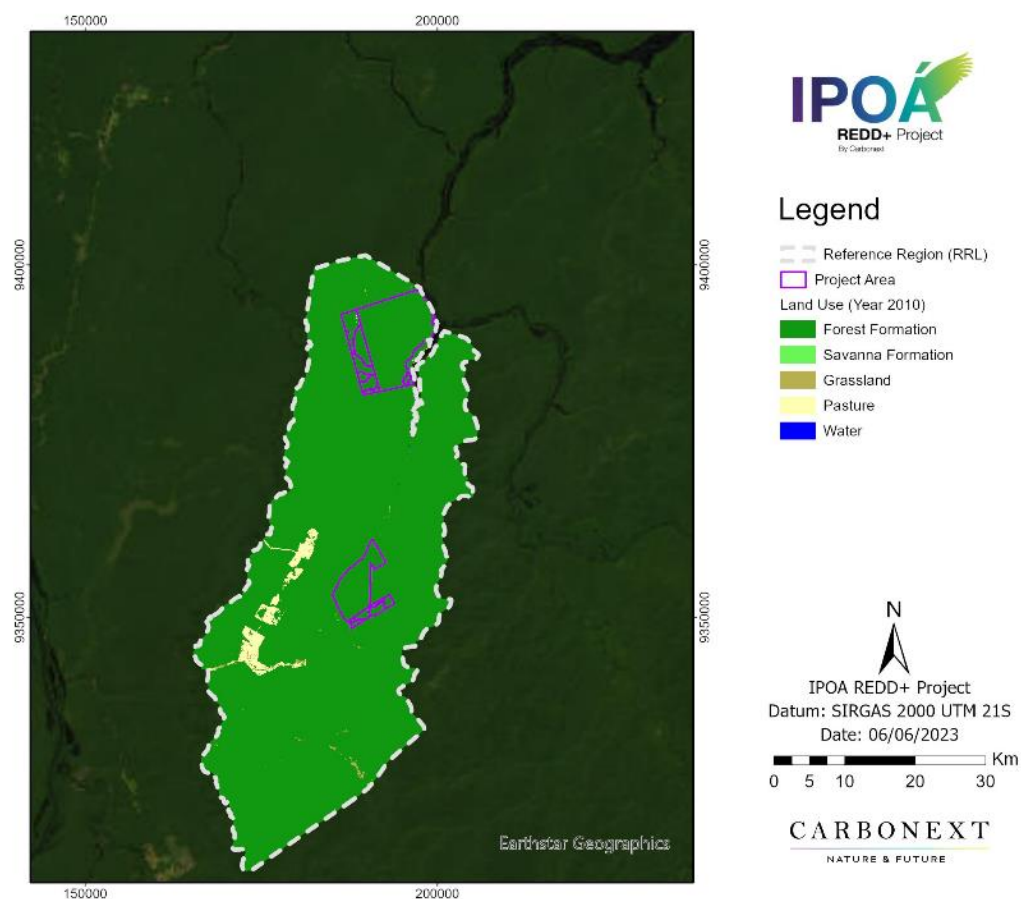


Figure 35. Pasture in Reference region in 2010

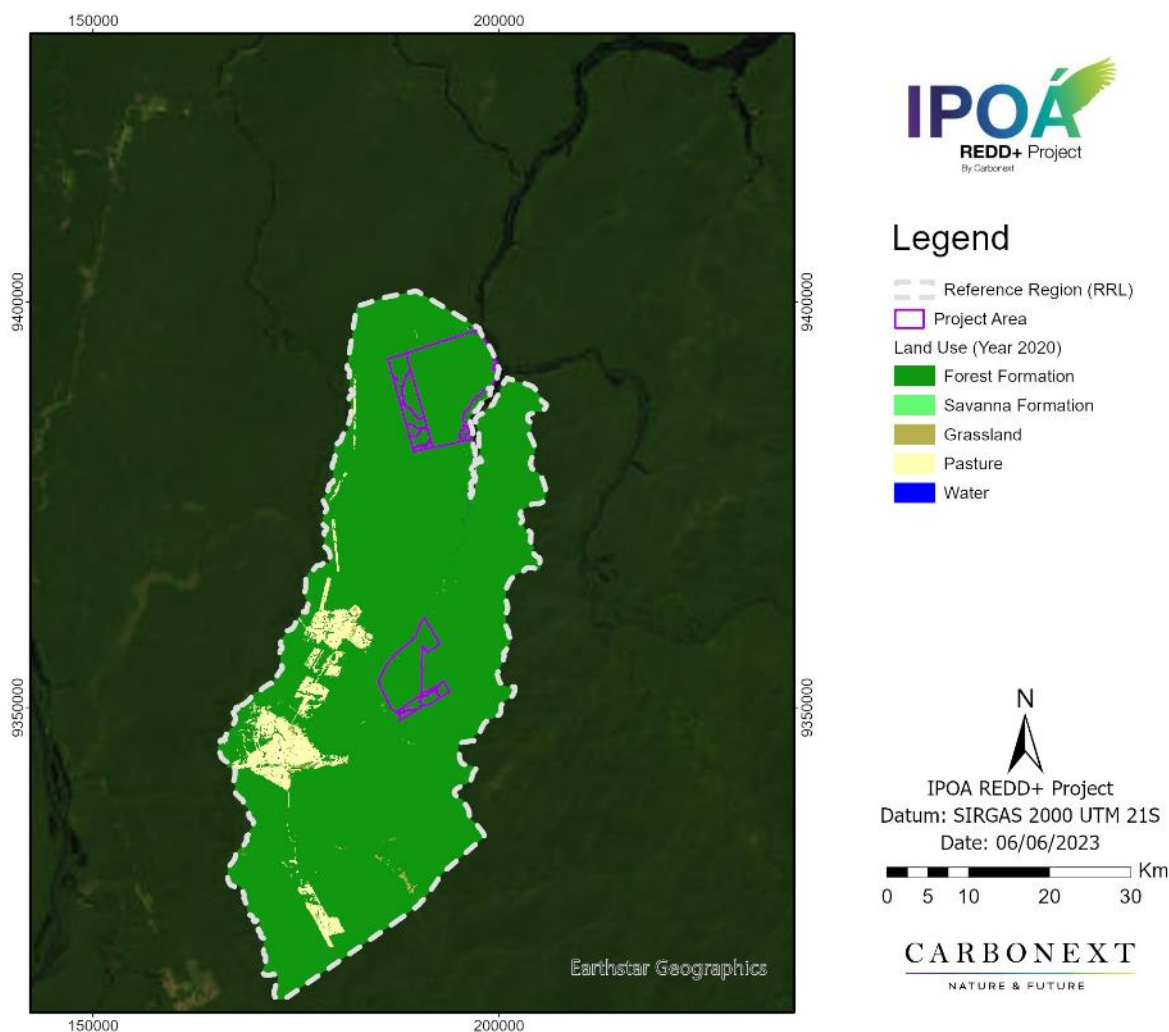


Figure 36. Pasture in Reference region in 2020.

For the reasons outlined above, the proportion of deforestation considered to be transformed into pasturelands ($Cab_{non-treesl}$) in the baseline is 100%.

Therefore low-technology beef cattle ranching is identified as the most probable baseline scenario, being the agent of deforestation and the post-deforestation land-use. The alternative, often overlapping, possible baseline land-uses are presented in Section 3.1.5.

Furthermore, there are several socio-political factors which strongly suggest deforestation rates in the Amazon can be expected to increase even more compared to the current ones. Among the main factors for increased deforestation after 2020, the following can be cited:

Trends in Brazilian legislation: several measures have contributed to slackening of environmental legislation in recent years, mainly under the Bolsonaro administration, with an emphasis on the following aspects:

- **Greater ease of obtaining “forest clearing authorizations”:** An example can be observed in the state of Goiás, which reported a 1,100% increase in the number of permits for deforestation in 2020¹¹⁵. The new environmental licensing in Goiás, plus technologies that facilitate the inspection work of the Secretariat of Environment and Sustainable Development (Semad), in addition to efforts in the analysis of applications, are responsible for increasing the numbers of deforestation permits in the State. According to data from Semad's Environmental Licensing Superintendency, there was a 673% increase of area deforested: 6.5 km² in 2019, to 43.8 km² in 2020. Thus, as occurred in the State of Goiás, the facilitation of the issuance of authorization for the removal of native vegetation can occur at any time in the Amazon Biome. Indeed, attempts at further facilitation have been sought recently (in 2020), as indicated under the next topic.
- **Granting of tacit (or automatic) environmental licensing, in case of delay of the environmental agency:** The controversial automatic release of environmental permits and permits by completion of term, that is, after a period stipulated for the government agency to respond (120 days), was voted for on 29/04/2020, by a virtual plenary. Provisional Measure 915 originally referred to the so-called "Economic Freedom Law" edited by the government, but ended up bringing, within the texts, changes that directly affect the rite of environmental licensing throughout the country. The change could lead to the automatic authorization of forest suppression in the Amazon and Atlantic Forest enforced by delay, and without analysis of the environmental agency. This means that, once the 120-day period is expired, the request would be automatically granted with a “tacit license”¹¹⁶. Fortunately, environmentalists have revised

¹¹⁵ <https://www.meioambiente.gov.br/noticias/2089-emiss%C3%A3o-de-licen%C3%A7as-para-supress%C3%A3o-de-vegeta%C3%A7%C3%A3o-tem-aumento-de-1-100-in-a-year-in-goi%C3%A1s.html>

¹¹⁶ <https://www.correiobraziliense.com.br/app/noticia/brasil/2020/04/29/interna-brasil,849652/camara-pode-aprovar-hoje-licenciamento-ambiental-automatico.shtml>

the Provisional Measure 915, to prevent deforestation licensing from being subject to expiration of term¹¹⁷.

- **Loosening legislation for timber exports:** As reported by Reuters, over 2019 Brazil exported "thousands of cargoes of wood from an Amazonian port without authorization from the federal environmental agency, increasing the risk that they have been extracted from illegally deforested land". The rule change, which scraps IBAMA's authorization for most timber exports, came after five cargoes of wood arrived in US and European ports without these mandatory documents. Foreign authorities contacted Brazil to ask about the missing authorizations, with the head of Ibama in Pará then retroactively granting the authorizations. The problem, however, is much more widespread than just these five shipments. In Pará state, more than half of the roughly 3,000 officially registered shipments in the past year, containing an estimated 54,000 m³ of wood that left one port, was not officially authorized. Companies had requested authorizations from IBAMA for those shipments but exported them before the agency had time to respond. Furthermore, many shipments were exported without seeking IBAMA's approval. Shipments went to the US, the Netherlands, France, Germany, Belgium, and possibly other countries. Before the rules changed, IBAMA was required to authorize all wood exports before they leave port. Even so, most shipments continued to require the proper paperwork to be given the green light, however only a few would be randomly selected for physical inspection ¹¹⁸. Arbitrarily, the president of IBAMA ensured that all future unauthorized exports of wood, previously classified as illegal, became legal: he took advantage of the lack of attention of the press to the topic during Carnival, at the end of February 2020, to quietly revoke a 2011 IBAMA policy that required an authorization from the agency before forest products could receive export licenses. From that date on, such permits would be required only for endangered tree species or in other special circumstances. Given the repeal, the floodgates were opened for large shipments of illegal timber from the Brazilian Amazon to go abroad¹¹⁹. It was also revealed that in February 2020, loggers from Pará asked IBAMA to affect that rule change: the companies wanted to sell wood abroad presenting only the Document of Forest Origin (DFO, "*DOF – Documento de Origem Florestal*" in Portuguese), made by the companies themselves

¹¹⁷ <https://epbr.com.br/ambientalistas-alteram-mp-915-para-prevent-licensing-environmental-by-course-of-time/>

¹¹⁸ <https://www.businesslive.co.za/bd/world/americas/2020-03-04-brazil-may-be-exporting-illegally-deforested-wood/>

¹¹⁹ <https://brasil.mongabay.com/2020/04/ao-afrouxar-leis-de-exportacao-brasil-permite-saida-de-madeira-illegal-da-amazonia/>

and that originally only served to allow the transport of the goods to the port. This change was immediately accepted by the president of Ibama¹²⁰.

- **Legislation favoring landgrabbers:** An analysis conducted by IPAM (Environmental Research Institute of the Amazon) showed that 35% of deforestation in the Amazon between August 2018 and July 2019 was recorded in non-designated areas without information. Regarding land regularization, environmental NGOs warn about two ongoing projects. While Bill 510/21 was being presented in the Senate, the vote on Bill 2633/20¹²¹ was being considered in the House of Representatives. Both derive from the original text of Provisional Measure 910, known as "*MP da Grilagem*" (Landgrabbers' Provisional Measure), which proposes to change the law to favor large occupants of recently invaded public lands. Bill 510/21 once again changes the deadline for public land invasions to be legalized (from 2011 to 2014) and allows large areas (up to 2500 hectares) to be processed without the need for inspection. Indeed, given that land-grabs of undesignated public lands are responsible for more than 1/3 of the deforestation in the country, it is to be expected that amnesty for landgrabbers and illegal deforesters would be an incentive to intensify this practice in the coming years. Bill 2633/20 has a loophole that would allow legalization, via bidding, of public areas invaded after the deadline for occupation provided for by law (i.e., 2014). Of the 49.8 million hectares of forests under state and federal responsibility, but not yet allocated to any category of use, 11.6 million hectares, or 23%, were irregularly declared as rural properties for private use, under the National System of Rural Environmental Registration (CAR). If the entire area registered to date as private property were legalized, 2.2 to 5.5 million hectares could be deforested in the coming years, according to the deforestation limits defined by the Forest Code and taking into account that deforestation is often beyond permitted limits. In recent years, grabbing of uncategorized public forests has increased: in 2019, this was the land category displaying the most forest felled in the Amazon, according to data from the deforestation alert system of INPE (National Institute of Space Research), Deter. The trend continued in 2020. The following are some of the conditions defined by Provisional Measure 910, for appropriation of public lands by individuals: i) the area must be registered in the Rural Environmental Registry (CAR, "*Cadastro Ambiental Rural*"): as it is known, any information can be inputted into the "CAR" system up until the present moment without any veracity checking, and ii) the claimant must be

¹²⁰ <https://g1.globo.com/natureza/noticia/2020/11/17/documentos-mostram-que-ibama-facilitou-exportacao-de-madeira-extraida-ilegalmente.ghtml>

¹²¹ <https://ipam.org.br/35-do-desmatamento-na-amazonia-e-grilagem-indica-analise-do-ipam/>

carrying out agricultural activities in the area (i.e., should preferably have deforested the area)¹²². The Provisional Measure defines that, for areas meeting the requirements above and having up to 15 fiscal modules (i.e., up to 1,650 hectares), the title is to be granted without the need for inspection. Before the Provisional Measure, the exemption from inspection was granted to areas with up to four fiscal modules (i.e., with a maximum of 440 hectares). The exemption from inspection may allow large illegally deforested areas to be taken over by individuals. This is because the Provisional Measure prohibits the regularization only of areas that have been subject to fines or environmental embargoes, and not all environmental violations are known and fined by the government¹²³. Given that Boa Fe farm is surrounded by public lands and that cases of land-grabbing have been proven in the Reference Region, an abnormal increase in deforestation in that region is expected in the coming years. This is primarily because Brazilian legislation increasingly gives every indication that it is highly welcoming to acts of land-grabbing, by granting amnesty to landgrabbers and agents of illegal deforestation.

- **Increased demand for beef in the international market:** China's demand for beef is still a reflection of swine flu, which has decimated between 40% and 60% of the country's pig stock (about one third of the world's pork production). In addition to this conjuncture, China also contributed to the growth of imports, since it was the only major world economy to record economic growth in 2020, even amid the coronavirus pandemic, and a more long-term factor, which is the gradual increase in income of the Chinese population, which results in progressively higher consumption of more expensive protein sources, such as beef. Analysts estimate that the price of beef should remain under pressure for the next few years, due to the livestock cycle: the low supply of oxen is not something that can quickly be solved, because cattle production is a multi-year cycle, given that production beginning today will deliver animals in two, three, or four years¹²⁴. In 2020, Brazil broke its beef export record, with more than 2 million tons sold (8% more than in 2019). For 2021, the projection indicates an increase of 5%

¹²²<https://ipam.org.br/cientistas-mapeiam-grilagem-em-florestas-publicas-na-amazonia/#:~:text=O%20impact%20da%20grilagem%20se,main%20g%C3%A1s%20effect%20estufa>

¹²³ <https://amazonia.org.br/como-a-mp-da-grilagem-pode-mudar-o-mapa-de-regioes-da-amazonia/>

¹²⁴ <https://www.bbc.com/portuguese/brasil-55664305>

on the 2020 values¹²⁵, indicating a strong increasing export trend for the coming years. Beef exports have continued to increase, growing by almost 7% in 2020 and close to 8% in 2021, increasing by more than 15% in the 2020/2021¹²⁶ period. Chinese importers increased their purchases of Brazilian beef by more than 150% in 2020¹²⁷.

It is expected that unplanned deforestation is likely to occur in the Project Area in the absence of the REDD+ Project. In accordance with the predominant characteristics of the region, the baseline scenario is of cattle farming conducting illegal levels of deforestation.

Relating to the illegal deforestation in the scenario, it is expected that the proportion of commercial timber will be sold on timber markets, followed by transformation of deforested areas into pasturelands. The proportion of commercial wood per stratum ($PCOM_i$) was defined in accordance with the volume of merchantable wood in exploitation (m^3/ha) has been based on “Florestal Santa Maria Project”¹²⁸ whose Project Area is located about 120 km South of this Project Area and has the same predominant forest class (Submontane dense rain forest) of this project. Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests).

In the absence of the REDD+ project, it is assumed that the PA would undergo deforestation – to be supplanted by cattle farms – at an average annual rate of 0.158% per year, resulting in a total of 652.48 hectares deforested by 2052.

The rate of deforestation adopted for calculation of the REDD+ Project benefits was obtained from historical measurements of deforestation utilizing Mapbiomas sources, in accordance with the VM0007 methodology requirements.

Above and belowground carbon pools were determined by means of a literature survey regarding the Project region. Considering that the baseline process of deforestation involves timber harvesting for commercial markets, the content of carbon fixed into long-term wood products was also considered in

¹²⁵ <https://revistagloborural.globo.com/Noticias/Criacao/Boi/noticia/2021/01/apos-recorde-brasil-projeta-alta-de-5-nas-exportacoes-de-carne-bovina-em-2021.html#:~:text=Segundo%20Abrasfrigo%2C%20pa%C3%ADs%20alcan%C3%A7ou%20marca,em%20rela%C3%A7%C3%A3o%20ao%20ano%20anterior&text=As%20exports%C3%A7%C3%B5es%20de%20carne%20bovine,by%20fortes%20shipments%20%C3%A0%20China>

¹²⁶ <https://www.avisite.com.br/index.php?page=noticias&id=21284>

¹²⁷ <https://www2.safras.com.br/eng/2020/09/23/meat-exports-in-brazil-will-be-an-important-differential-in-2021/>

¹²⁸ https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdfiku098l71896923

calculation of net deforestation emissions. It is assumed that the Project Activity preserves soil organic carbon and litter pools to a greater extent than BAU activities. However, for conservativeness purposes, the project proponents decided not to include the soil and litter carbon pools in the REDD+ Project benefits.

Fossil fuel emissions were not accounted for in the Reference Area (baseline case) or for the Project Activity. It is assumed that the Project Activity also reduces emissions from fossil fuel burning, in comparison with BAU activities. However, this factor was also not accounted for conservativeness purposes and difficulties in monitoring during the project period.

In summary, regarding IPOÁ's AUD component, SCENARIO 3: cattle farming is the baseline scenario of the present REDD+ project, as further justified by the stepwise application of the T-ADD Tool for Additionality, which is fully presented in section 3.1.5, below.

Avoided Planned Deforestation (APD)

IPOÁ's APD component avoids 3,547,95 ha of planned deforestation from 2022 to 2026.

This component concerns forest lands which have been legally authorized and documented to be converted to non-forest land by the project proponents, plans which and will be abandoned thanks to the present REDD+ Project. The proponents have requested to IPAAM permission to clear up to 19.14% of the Maanaim and Campos propriety, according to the legal limit prescribed in the Brazilian Forest Code. In addition, the landowner has a plan to deforest in the next few years, which will not be executed in case of success in obtaining VCUs.

The total area of deforestation is shown in the maps below, which were presented at audit, equating to 2,666.48 ha in the Manaim property and 881.47 ha in the Campos property, summing to a total of 3,547.95 ha. The planned deforestation occurs between 2022 – 2026 as shown in the tables under the baseline emissions from planned deforestation (APD) heading in section 3.2.1.

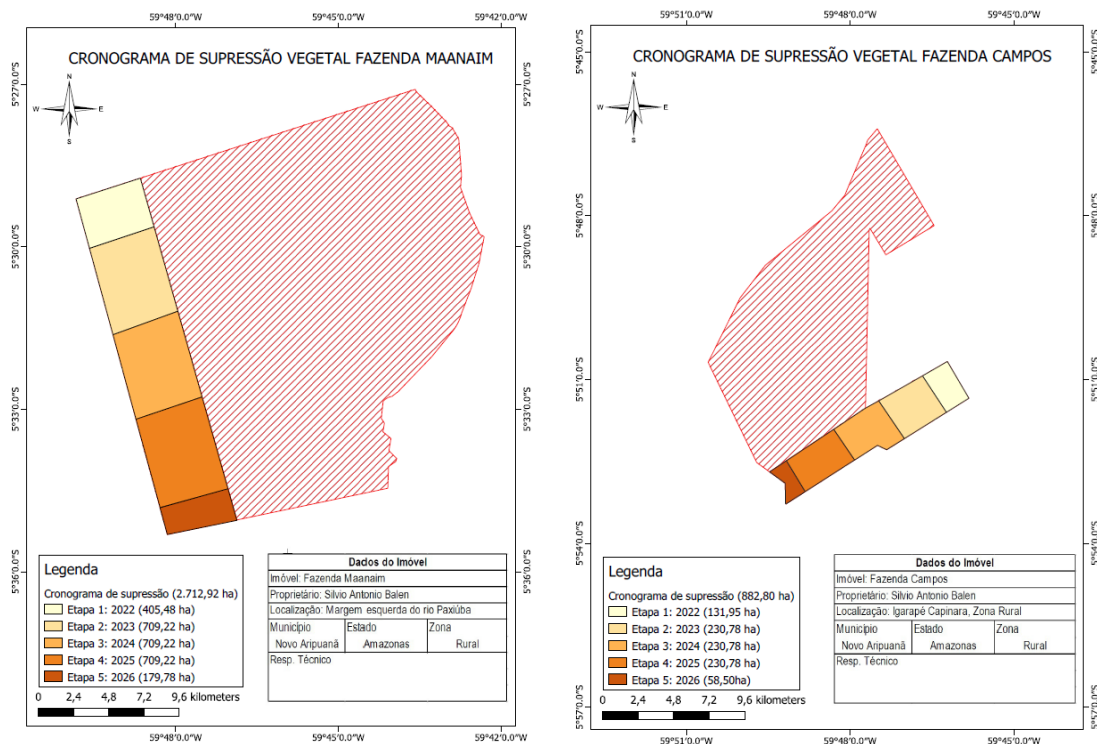


Figure 37. Deforestation schedule in the Maanaim and Campos properties

The following annual activities were planned within the deforestation schedule:

May to October – clear-cut area and harvest timber wood.

November to December – remove any remaining vegetation by burning.

December to January – aerial grass seeding.

July – purchase livestock and begin grazing activities.

For the year 2022, it was expected that the planned deforestation would only occur in the second half of the year, since it would take place in a smaller area considering the necessary steps for licensing.

And regarding the APD Deforestation Agent: The agent of deforestation regarding the Avoided Planned Deforestation component of the project is the proponent landowner.

It is expected that the proportion of commercial timber will be sold on timber markets, followed by transformation of deforested areas into pasturelands. The proportion of commercial wood per stratum (*PCOM*) was defined in accordance with the volume of merchantable wood in exploitation (m^3/ha) has been based on “Florestal Santa Maria Project”¹²⁹ whose Project Area is located about 120 km South of this Project Area and has the same predominant forest class (Submontane dense rain forest) of this project. Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests).

In summary, regarding IPOÁ’s APD component, SCENARIO 3: cattle farming is the baseline scenario of the present REDD+ project, as further justified by the stepwise application of the T-ADD Tool for Additionality, which is fully presented in section 3.1.5, below.

3.1.5 Additionality

According to the procedure defined in the VM0007 methodology, the latest version of the tool referenced in T-ADD was used to identify credible alternative land use scenarios and evaluate both the alternatives and the proposed project scenarios, and to demonstrate the additionality of the project:

“Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities (T-ADD)” (Version 01)”¹³⁰

The tool is applicable to this project, given the following applicability conditions of the tool:

- Conservation of forests within the proposed project boundary performed with or without being registered as the VCS REDD+ project activity¹³¹ shall not lead to violation of any applicable law even if the law is not enforced.
- This tool is not applicable to small - scale afforestation and reforestation project activities.

¹²⁹https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmln902309ksdfiku098l71896923

¹³⁰ Source, CDM (accessed 13/06/22): <https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-02-v1.pdf>

¹³¹ This sentence was adapted from the original version in the AR-AM Tool-02-v1, to fit the context of the present VCS REDD project. The original is: “Forestation of the land within the proposed project boundary performed with or without being registered as the A/R CDM project...”

Regarding the former point, as described in section 2.5 the project is in compliance with all relevant local, regional, and national laws, statutes, and regulatory frameworks, therefore satisfying this applicability condition.

And regarding the second applicability condition, the project does not involve afforestation and reforestation activities, as pointed out in sections 2.1.11 and 2.1.13, therefore fulfilling this applicability condition.

The additionality analysis, applying the T-ADD tool (Version 1), is presented below for AUD and APD activities.

AUD ACTIVITIES ANALYSIS

STEP 0. Preliminary screening based on the starting date of the VCS AFOLU project¹³² activity

According to the step, “If project participants claim that the afforestation or reforestation VCS AFOLU project activity has a starting date after 31 December 1999 but before the date of its registration, then the project participants shall:

- I. Provide evidence that the starting date of the VCS AFOLU project activity was after 31 December 1999, and
- II. Provide evidence that the incentive from the planned sale of CERs was seriously considered in the decision to proceed with the project activity. This evidence shall be based on (preferably official, legal and/or other corporate) documentation that was available to third parties at, or prior to, the start of the project activity.”

Regarding point I. above, the project start date (AUD activities) is defined as 17th March 2022. The project properties already had the explicit contractual intentions of conserving forests and REDD+ projects prior to 17/03/22, however, after 31 December 1999, as evidenced by:

The purchase and sale agreements that shows the acquirence the Campos property on August 21, 2019, and the Maanaim property on October 13, 2019, with the objective of carbon credits generation and forest management.

¹³² The AR-AM Tool-02-v1.PDF is the updated additionality tool recommended by VM0007, however it is intended for use on A/R CDM project activities. Therefore the term “A/R CDM project”, or similar, was altered to “VCS AFOLU project” in all instances (accessed 13/08/21) <https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf>

The contract between Carbonext and the landowner.

After the recognition and analysis stage, Carbonext started the implementation of project activities in the field on March 17, 2022. This event is considered as the Project Start Date.

On this occasion, an experienced social technician from Carbonext, accompanied by a local consultant, visited the Project Area, carrying out public consultations with the communities in the Project Zone, presenting the principles and benefits generated by the REDD+ project. The presentation lecture given to local inhabitants is considered the first activity to generate reductions in the project's GHG emissions, as it made communities aware of preserving the forest and encouraged them to monitor possible illegal deforestation.

Regarding point II, the sale of carbon credits, in this case VERs, was seriously considered in the decision to proceed with the project activity. This is proven by the contract between the landowner and Carbonext being signed (in December 2021) prior to the project start date (17th March 2022). Evidence to this effect was duly provided at audit.

STEP 1. Identification of alternative land use scenarios to the proposed AFOLU project activity

Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity.

a) The identified credible non-excluding alternative land use scenarios to the proposed VCS AFOLU project activity include:

SCENARIO 1: Continuation of the pre-project land use:

Under the pre-project scenario, there are no activities being carried out in the Project Area or Project Zone. Thus, there are no costs or revenues for this scenario.

Scenario 1 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project:

Project activities performed without being registered as the VCS AFOLU project. As there are no economic activities besides REDD+ being implemented, the scenario involves costs but no income. These annual costs are estimated at R\$ 296,620.05 throughout the project. The forest conservation and maintenance activities of Scenario 2 are therefore not financially attractive without carbon credit income. Notwithstanding this fact, Scenario 2 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 3: Cattle farming:

As argued in PD section 3.1.4 pasture represents 99.41% of non-natural land uses in Novo Aripuanã. The trends identified and the proximity to the municipality of Apuí are making Novo Aripuanã the new hub of cattle farming expansion in the south of the State of Amazonas, with a 300% increase in the bovine herd in a decade (2010 - 2020). As the main deforestation agent in the region, cattle ranching (pasture) is expected to increase in the project region. The percentage of the total area of this land-use class increased from 0.57% in 2010 to 1.67% in 2020.

Therefore, this is a credible scenario.

b) All identified land use scenarios above may be deemed realistic and credible, as they currently exist and are technically feasible in the project region. For all land use scenarios, credibility is justified by current BAU practices attested by the literature and local observations.

c) Outcome of Sub-step 1a: The credible alternative land-use scenarios that could have occurred on the land within the project boundary are SCENARIOS 1, 2 and 3, above.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations.

a) The following procedure was applied:

i) Demonstration that all land use scenarios identified in the sub-step 1a are in compliance with all mandatory applicable legal and regulatory requirements:

SCENARIOS 1 and 2: Maintenance of forest on the area is what is specified as legal within the Law: Documentation of legal possession of the property is the only relevant aspect of law regarding No Economic Activity components of the scenario. Thus, this scenario is in compliance with legal and regulatory requirements.

ii) Demonstration that applicable mandatory legal requirements are systematically not enforced and that non-compliance with requirements is widespread:

SCENARIO 3: Cattle farming:

In the AUD case the activity does not meet the mandatory legal requirements. As widely observed, the southern region of the State of Amazonas is the epicenter of illegal deforestation. Novo Aripuanã is among the municipalities with a high incidence of environmental crimes, local political support for predatory activities, and the absence of federal inspection agencies¹³³.

Studies show that in the region, livestock is responsible for 75% of illegal deforestation on public lands in the Amazon, a practice driven by the lack of inspection, land grabbing and the advance of pasture areas¹³⁴.

In Novo Aripuanã, deforestation for the opening of pastures takes place closer to the headquarters of the municipality and to the Acari Settlement Project and also in some isolated points that configure land grabbing and land tenure in remote areas. and difficult to access, especially on the banks of the Acari River¹³⁵.

According to an AMAZON researcher, efforts should be made to combat illegal deforestation in municipalities such as Novo Aripuanã, which are in the region driven by the agricultural expansion known as AMACRO¹³⁶.

¹³³<https://oglobo.globo.com/brasil/nova-fronteira-do-desmatamento-sul-do-amazonas-tem-capitais-das-queimadas-garimpo-ausencia-do-estado-25314282>.

¹³⁴<https://noticias.uol.com.br/meio-ambiente/ultimas-noticias/redacao/2021/10/27/amazonia-87-do-desmate-em-terras-publicas-ocorreu-em-areas-nao-destinadas.htm>

¹³⁵Idesam (2011) REDD+ Estudo de Oportunidades para a Região Sul do Amazonas (accessed 13/08/21): https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas.pdf

¹³⁶<https://amazon.org.br/imprensa/desmatamento-na-amazonia-cresce-quase-70-e-atinge-pior-fevereiro-em-10-anos/>

Finally, Carrero et al. (2015) note that 71% of slaughterhouses in the region are illegal, arguing in favor of the illegality of the beef chain of production being generally.

Therefore, this scenario is not in compliance applicable mandatory legal requirements are systematically not enforced and that this scenario is widespread.

- b) Outcome of Sub-step 1b: It has been demonstrated that SCENARIOS 1, 2 and 3 are plausible alternative land use scenarios to this VCS AFOLU project activity.

Sub-step 1c. Selection of the baseline scenario:

SCENARIO 3: Cattle farming is selected as the baseline scenario for the IPOÁ REDD+ project, as it is the predominant land-use in the project region (see sub-step 1a) and although the cattle farming in region is likely to be illegal (see sub-step 1b), this is not enforced and their illegality is ultimately a factor in favour of this activity.

SCENARIO 3: Cattle farming is the baseline scenario of the present REDD+ project.

Given that the scenarios above have been identified, in accordance with the CDM AR-AM Tool, it is proceeded to Step 2 (Barrier analysis).

STEP 2. Barrier analysis

This step serves to identify barriers and to assess which of the land use scenarios identified in the sub-step 1b are not prevented by these barriers.

Sub-step 2a. Identification of barriers that would prevent the implementation of at least one alternative land use scenarios.

According to T-ADD “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”. The barriers which the previously identified land-use scenarios may potentially face are the following:

Investment barriers:

Debt funding is not available for the land-use scenarios.

No access to international capital markets.

Lack of access to credit.

Institutional barriers:

Risk related to changes in government policies or laws.

Lack of enforcement of forest or land-use-related legislation.

Barriers due to social conditions and land-use practices:

Widespread illegal practices (e.g., illegal grazing, non-timber product extraction and tree felling).

Barriers relating to land tenure, ownership, inheritance, and property rights:

Lack of suitable land tenure legislation and regulation to support the security of tenure.

Possibilities of large price risk due to the fluctuations in the prices of products related to the project activity over the project period in the absence of efficient markets.

Sub-step 2b. Elimination of land use scenarios that are prevented by the identified barriers.

This step aims to determine which land use scenarios identified in sub-step 1b are prevented by at least one of the barriers listed in sub-step 2a.

Investment barriers:

Scenarios 1 and 2: Investment barriers exist in terms of Scenarios 1 and 2, as there are very few options for conservation of forest on private properties, above and beyond legal requirements, without generating any financial returns. The notable exception to this is Private Natural Heritage Reserves (“RPPNs” in Portuguese) which in the State of Amazonas are on average much smaller than the project area (0.2%, on average, of the Project Area as described in Step 4 below) and they generate further costs and no particular incentives or subsidies, as shown in Decree 5746/2006.¹³⁷.

Scenario 3: However, regarding Scenario 3 (Cattle farming), having illegal levels of deforestation generally does not prevent this land-use scenario from obtaining funding. There are numerous funding resources and

¹³⁷ Decree 5746/2006: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm

credit lines for cattle raising activities in Brazil¹³⁸. Moreover, the implementation of cattle raising activities can be funded by initial capital obtained with timber sales after deforestation, and subsequently benefit from speculation of the value of deforested land.

Institutional barriers: This barrier disproportionately hinders legally compliant alternative land-use scenario that do not have revenue streams, as it promotes unfair competition from non-compliant producers (Lack of enforcement of forest or land-use-related legislation), causing uncertainty to investors (Risk related to changes in government policies or laws).

Barriers due to social conditions and land-use practices: This barrier hinders any compliant alternative land-use scenarios (Scenarios 1 – 2), due to widespread illegal practices, which lead to unfair competition from a great portion of regional producers and insecurity related to land tenure (trespassing by squatters). On the other hand, this factor favors cattle raising and pastureland (Scenario 3) which, as previously described, often depends on illegal practices to be financially viable.

Barriers relating to land tenure, ownership, inheritance, and property rights: This barrier hinders the legally compliant land-use scenarios (Scenarios 1 – 2) and represents an incentive to squatters and land-grabbers, which are in many cases associated with cattle farming¹³⁹ Scenario 3.

Sub-step 2c. Determination of baseline scenario (if allowed by the barrier analysis)

As put into context above, Scenario 3 (cattle farming) would not be hindered or prevented by any of the barriers identified in Sub-step 2a. Thus, Scenario 3 is confirmed as the most plausible alternative land-use scenario, in the absence of the project activity.

In this sense, according to the tool:

“Forestation” (or the REDD+ project activity, in this case) “without being registered as a VCS AFOLU project activity” is not included in the list of land use scenarios that are not prevented by any barrier”.

And the list contains only one land use scenario, therefore it is necessary to continue with Step 4: Common practice test.

¹³⁸Sources:<https://www.bcb.gov.br/estabilidadefinanceira/creditorural>; <https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos/agropecuaria>

¹³⁹Sources:https://antigo.mma.gov.br/estruturas/225/_arquivos/9___a_grilagem_de_terras_publicas_na_amaznia_brasileira_225.pdf ;

STEP 4. Common practice test

According to T-ADD “CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”, the previous steps are complemented with an analysis of the extent to which similar activities have already diffused in the geographical area of the proposed VCS AFOLU project activity. This test is a credibility check to demonstrate additionality that complements the barrier analysis (Step 2). Other registered VCS AFOLU project activities, as the Juma Sustainable Development Reserve Project, shall not be included in this analysis.

This analysis took into account similar activities in the project region, being the municipalities of Novo Aripuanã, Amazonas, and the type of land area analyzed were private natural heritage reserves, a land category known as RPPN in Brazil. This is because they are private conservation reserves, and in this sense are comparable to the IPOÁ area.

It was found through the official organ ICMBio, that in the state of Amazonas there are only 14 such areas, with an average of 62.8 ha. The nearest RPPN is located 260 km from the IPOÁ REDD+ project and is of 52.06 ha in area.

Therefore, we can cite at least two essential differences: the private conservation areas are incomparably smaller (0.2%) in size to the IPOÁ REDD+ project, and they are very distant (260 km) from the project area, and not located in the same jurisdictional area (municipality).

In this regard, the proposed VCS REDD+ project activity is not the baseline scenario and, hence, it is additional.

APD ACTIVITIES ANALYSIS

STEP 0. Preliminary screening based on the starting date of the VCS AFOLU project¹⁴⁰ activity

According to the step, “If project participants claim that the afforestation or reforestation VCS AFOLU project activity has a starting date after 31 December 1999 but before the date of its registration, then the project participants shall:

Provide evidence that the starting date of the VCS AFOLU project activity was after 31 December 1999, and

Provide evidence that the incentive from the planned sale of CERs was seriously considered in the decision to proceed with the project activity. This evidence shall be based on (preferably official, legal and/or other corporate) documentation that was available to third parties at, or prior to, the start of the project activity.”

Regarding point I. above, the project start date (APD activities) is defined as 3rd May 2022, as it was the date that the project owner submitted a request to the official organ IPAAM, to obtain permission to deforest a legal proportion of the properties in the Project Area (PA).

The justification is that maintaining the PA forested was no longer judged to be financially viable without carbon credits by the project owner, as of that date.

Regarding point II, the sale of carbon credits, in this case VERs, was seriously considered in the decision to proceed with the project activity. This is proven by the contract between the landowner and Carbonext being signed (in December 2021) prior to the project start date (3th May 2022). Evidence to this effect was duly provided at audit.

STEP 1. Identification of alternative land use scenarios to the proposed AFOLU project activity**Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU project activity.**

- a) The identified credible non-excluding alternative land use scenarios to the proposed VCS AFOLU project activity include:

¹⁴⁰ The AR-AM Tool-02-v1.PDF is the updated additionality tool recommended by VM0007, however it is intended for use on A/R CDM project activities. Therefore the term “A/R CDM project”, or similar, was altered to “VCS AFOLU project” in all instances (accessed 13/08/21) <https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf>

SCENARIO 1: Continuation of the pre-project land use:

Under the pre-project scenario, there are no activities being carried out in the Project Area or Project Zone. Thus, there are no costs or revenues for this scenario.

Scenario 1 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project:

Project activities performed without being registered as the VCS AFOLU project. As there are no economic activities besides REDD+ being implemented, the scenario involves costs but no income. These annual costs are estimated at R\$ 296,620.05 throughout the project. The forest conservation and maintenance activities of Scenario 2 are therefore not financially attractive without carbon credit income. Notwithstanding this fact, Scenario 2 was included in accordance with A/R CDM project activities (T-ADD) requirements.

SCENARIO 3: Cattle farming:

As argued in section 3.1.4 pasture is 99,41% of non-natural land uses in Novo Aripuanã. The trends identified and the proximity to the municipality of Apuí are making Novo Aripuanã the new hub of cattle farming expansion in the south of the State of Amazonas, with a 300% increase in the bovine herd in a decade (2010 - 2020). In the case of the IPOÁ project, the landowners intended to deforest a subsection (19.14%) of the Maanaim and Campos property, which has documented forest clearing plans submitted to the responsible agency for authorization. 3,586.78 ha of the property would be clear-cut in the baseline scenario, with wood to be sold on the timber markets, followed by transformation of deforested areas into pasturelands. Since the activity is a common practice in the region and the project areas are served by navigable rivers, the development of activities and production flow are viable with little need for infrastructure adaptation.

Therefore, this is a credible scenario.

- b) All identified land use scenarios above may be deemed realistic and credible, as they currently exist and are technically feasible in the project region. For all land use scenarios, credibility is justified by current BAU practices attested by the literature and local observations.
- c) Outcome of Sub-step 1a: The credible alternative land-use scenarios that could have occurred on the land within the project boundary are SCENARIOS 1, 2 and 3, above.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations.

- a) The following procedure was applied:

i) Demonstration that all land use scenarios identified in the sub-step 1a are in compliance with all mandatory applicable legal and regulatory requirements:

SCENARIOS 1 and 2: Maintenance of forest on the area is what is specified as legal within the Law: Documentation of legal possession of the property is the only relevant aspect of law regarding No Economic Activity components of the scenario. Thus, this scenario is in compliance with legal and regulatory requirements.

ii) Demonstration that applicable mandatory legal requirements are systematically not enforced and that non-compliance with requirements is widespread:

SCENARIO 3: Cattle farming: Deforestation and cattle farming on the area is specified as legal within the Law in APD case: deforestation of up to 19.14% of Maanain and Campos properties is permitted in accordance with the Brazilian Forest Code, which stipulates that a property should officially allocate 80% of its total area as LR (Legal Reserve) for conservation and maintain Permanent Preservation Areas. The remaining land may be deforested by license. Thus, this scenario is in compliance with legal and regulatory requirements.

- b) Outcome of Sub-step 1b: It has been demonstrated that SCENARIOS 1, 2 and 3 are plausible alternative land use scenarios to this VCS AFOLU project activity.

Sub-step 1c. Selection of the baseline scenario:

SCENARIO 3: Cattle farming is selected as the baseline scenario for the IPOÁ REDD+ project, as it is the predominant land-use in the project region (see sub-step 1a) and deforestation of up to 19.14% of the property for cattle farming is a legal and planned activity (see sub-step 1b).

SCENARIO 3: Small-scale beef cattle farming is the baseline scenario of the present REDD+ project.

Given that the scenarios above have been identified, in accordance with the CDM AR-AM Tool, it is proceeded to Step 2 (Barrier analysis).

STEP 2. Barrier analysis

This step serves to identify barriers and to assess which of the land use scenarios identified in the sub-step 1b are not prevented by these barriers.

Sub-step 2a. Identification of barriers that would prevent the implementation of at least one alternative land use scenarios.

According to T-ADD “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”. The barriers which the previously identified land-use scenarios may potentially face are the following:

Investment barriers:

Debt funding is not available for the for the land-use scenarios.

No access to international capital markets.

Lack of access to credit.

Institutional barriers:

Risk related to changes in government policies or laws.

Lack of enforcement of forest or land-use-related legislation.

Barriers due to social conditions and land-use practices:

Widespread illegal practices (e.g., illegal grazing, non-timber product extraction and tree felling).

Barriers relating to land tenure, ownership, inheritance, and property rights:

Lack of suitable land tenure legislation and regulation to support the security of tenure.

Possibilities of large price risk due to the fluctuations in the prices of products related to the project activity over the project period in the absence of efficient markets.

Sub-step 2b. Elimination of land use scenarios that are prevented by the identified barriers.

This step aims to determine which land use scenarios identified in sub-step 1b are prevented by at least one of the barriers listed in sub-step 2a.

Investment barriers:

Scenarios 1 and 2: Investment barriers exist in terms of Scenarios 1 and 2, as there are very few options for conservation of forest on private properties, above and beyond legal requirements, without generating any financial returns. The notable exception to this is Private Natural Heritage Reserves (“RPPNs” in Portuguese) which in the State of Amazonas are on average much smaller than the project area (0.2%, on average, of the Project Area as described in Step 4 below) and they generate further costs and no particular incentives or subsidies, as shown in Decree 5746/2006¹⁴¹.

Scenario 3: However, regarding Scenario 3 (Cattle farming), access to finance is facilitated, even more so when conducted legally. There are numerous funding resources and credit lines for cattle raising activities in Brazil¹⁴². Moreover, the implementation of cattle raising activities can be funded by initial capital obtained with timber sales after deforestation, and subsequently benefit from speculation of the value of deforested land.

Institutional barriers:

Scenarios 1 and 2: This barrier disproportionately hinders legally compliant alternative land-use scenario that do not have revenue streams, as it promotes unfair competition from non-compliant producers (Lack of

¹⁴¹ Decree 5746/2006: http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm

¹⁴² Sources: <https://www.bcb.gov.br/estabilidadefinanceira/creditorural>;
<https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos/agropecuaria>

enforcement of forest or land-use-related legislation), causing uncertainty to investors (Risk related to changes in government policies or laws).

Scenario 3: This barrier does not hinder scenario 3, since there is no indication that changes in legislation will negatively impact the implementation of legal cattle ranching. On the contrary, in recent years Brazilian environmental policy has become more flexible with the legalization of activities potentially impacting the environment, such as the case of Provisional Measure 1040 of March 29, 2021, which provides automatic environmental licensing for companies and Decree 10,084, of 2019, which revoked the agroecological zoning of sugarcane. The lack of enforcement benefits illegal activities, but does not prevent the execution of legal cattle farming, which are a consolidated market in Brazil.

Barriers due to social conditions and land-use practices: This barrier hinders compliant alternative land-use scenarios without revenues (Scenarios 1 – 2), due to widespread illegal practices, which lead to unfair competition from a great portion of regional producers and insecurity related to land tenure (trespassing by squatters). On the other hand, this factor does not prevent cattle raising and pastureland (Scenario 3) which, as previously described, which are a consolidated market in Brazil, increasingly supported by the country's environmental policy.

Barriers relating to land tenure, ownership, inheritance, and property rights: This barrier hinders the legally compliant land-use scenarios without revenues (Scenarios 1 – 2) and represents an incentive to squatters and land-grabbers. On the other hand, this factor does not prevent cattle raising and pastureland (Scenario 3) which, as previously described, which are a consolidated market in Brazil, increasingly supported by the country's environmental policy.

Sub-step 2c. Determination of baseline scenario (if allowed by the barrier analysis)

As put into context above, Scenario 3 (cattle farming) would not be hindered or prevented by any of the barriers identified in Sub-step 2a. Thus, Scenario 3 is confirmed as the most plausible alternative land-use scenario, in the absence of the project activity.

In this sense, according to the tool:

“Forestation” (or the REDD+ project activity, in this case) “without being registered as a VCS AFOLU project activity” is not included in the list of land-use scenarios that are not prevented by any barrier”

And the list contains only one land use scenario, therefore it is necessary to continue with Step 4: Common practice test.

STEP 4. Common practice test

According to T-ADD “CDM Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities”, the previous steps are complemented with an analysis of the extent to which similar activities have already diffused in the geographical area of the proposed VCS AFOLU project activity. This test is a credibility check to demonstrate additionality that complements the barrier analysis (Step 2). Other registered VCS AFOLU project activities, as the Juma Sustainable Development Reserve Project, shall not be included in this analysis.

This analysis took into account similar activities in the project region, being the municipalities of Novo Aripuanã, Amazonas, and the type of land area analyzed were private natural heritage reserves, a land category known as RPPN in Brazil. This is because they are private conservation reserves, and in this sense are comparable to the IPOÁ area.

It was found through the official organ ICMBio, that in the state of Amazonas there are only 14 such areas, with an average of 62.8 ha. The nearest RPPN is located 260 km from the IPOÁ REDD+ project and is of 52.06 ha in area.

Therefore, we can cite at least two essential differences: the private conservation areas are incomparably smaller (0.2%) in size to the IPOÁ REDD+ project, and they are very distant (260 km) from the project area, and not located in the same jurisdictional area (municipality).

In this regard, the proposed VCS REDD+ project activity is not the baseline scenario and, hence, it is additional.

3.1.6 Methodology Deviations

The methodology's VMD0007-BL-UP_v3.3 module requires that at least one land-tenure variable be included as an input variable, as described in the module's section 3.1.2.

However, in the present project variables related to “actual land tenure and management” are considered not to be applicable, because only private lands are included in the RR (the variable would have only one category: private land).

This is the result of public and protected areas not being included in the RR, in order to respect the following similarity requirement of the aforementioned module, section 1.1.1.1: “Policies and regulations having an impact on land-use change patterns within the RRD and the project area must be of the same type or have an equivalent effect at the start of the historical reference period, taking into account the current level of enforcement”. Duly following this requirement, Conservation Units and Indigenous Territories, which have specific policies and regulations, were excluded from the RR.

To this extent, in order to correctly follow the methodological requirement in section 1.1.1.1, it was considered necessary to disregard the requirement in section 3.1.2.

No further methodology deviations were identified for the Ipoá REDD+ Project.

3.2. Quantification of GHG Emission Reductions and Removals

3.2.1 Baseline Emissions

The net baseline carbon stock changes and greenhouse gas emissions of the project were calculated through two different modules of the VM0007 Methodology. In order to estimate the baseline emissions relating to APD, relating to forested lands with planned deforestation on them (19.14% of the Maanaim and Campos properties according to the legal limit prescribed in the Brazilian Forest Code), the VMD0006 module (BL-PL) “Estimation of Baseline Carbon Stock Changes and Greenhouse Gas Emissions from Planned Deforestation and Planned Degradation” was applied.

Regarding the estimation of baseline emissions from unplanned deforestation, AUD scenario, where agents of deforestation clear the land for ranching as final land use, the VMD0007 module (BL-UP) was applied: “Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation”.

Avoided Unplanned Deforestation (AUD)

Regarding avoided unplanned deforestation (AUD), the baseline net GHG emissions were determined by applying the module VMD0007 according to four parts, such as:

Part 1 DEFINITION OF BOUNDARIES

Part 2 ESTIMATION OF ANNUAL AREAS OF UNPLANNED DEFORESTATION

Part 3 LOCATION AND QUANTIFICATION OF THREAT OF UNPLANNED DEFORESTATION

Part 4 ESTIMATION OF CARBON STOCK CHANGES AND GREENHOUSE GAS EMISSIONS

Location analysis - AUD

The forest landscape configuration is not frontier, as forests within the Project Area have a physical connection with anthropogenically deforested areas, nor mosaic, because the forest patches are over 1,000 ha and are not surrounded by anthropogenically cleared land. Thus, the forest landscape configuration is transition. As there is not 25% or more of the project geographic boundary within 50 m of land that has been anthropogenically deforested within the 10 years prior to the project start date, location analysis is required.

The prerequisites foreseen in the VMD0007 methodology - item 2.1.1 until 2.1.4 - have been fulfilled, to estimate annual unplanned deforestation based on the simple historic method. The historical period is from 2013 to 2022. For the analysis of land use dynamics in the reference region, mapping of 2013 and 2017 by PRODES (Project for Monitoring the Deforestation of the Brazilian Amazon Forest by Satellite) produced by the National Institute for Space Research (INPE) was used. Specifically, rasters from the website (<http://terrabrasilis.dpi.inpe.br>), with 30m x 30m resolution. For 2022, the map was sourced from Sentinel-2 optical images from the March/2022. An accuracy of 92.89% (Kappa) was obtained. The scenes that make up the mosaic are:

0: Image COPENICUS/S2_SR/20220122T141729_20220122T141730_T21MTQ (3 bands)

1: Image COPENICUS/S2_SR/20220512T141739_20220512T141733_T20MRU (3 bands)

2: Image COPENICUS/S2_SR/20220512T141739_20220512T141733_T21MTP (3 bands)

3: Image COPENICUS/S2_SR/20220522T141739_20220522T141735_T20MRU (3 bands)

4: Image COPENICUS/S2_SR/20220522T141739_20220522T141735_T20MRV (3 bands)

5: Image COPENICUS/S2_SR/20220522T141739_20220522T141735_T21MTQ (3 bands)

The historical deforestation rate can be described linearly by the equation $y = -0,0004x + 0,0118$ ($R^2 = 0.02$). Since the linear relationship does not present a significant trend, the historical average of the historical period was adopted, equal to 0.98% (approach 1. / Step 2.2 of VMD0007).

The basic steps to perform the Modeling (*Figure Flowchart for the Quantification of GHG Emission Reductions and Removals*) are as follows:

- 1) Preparation of factor map
- 2) Preparation of risk maps for deforestation
- 3) Selection of the most accurate deforestation risk map and
- 4) Mapping of the locations of future deforestation

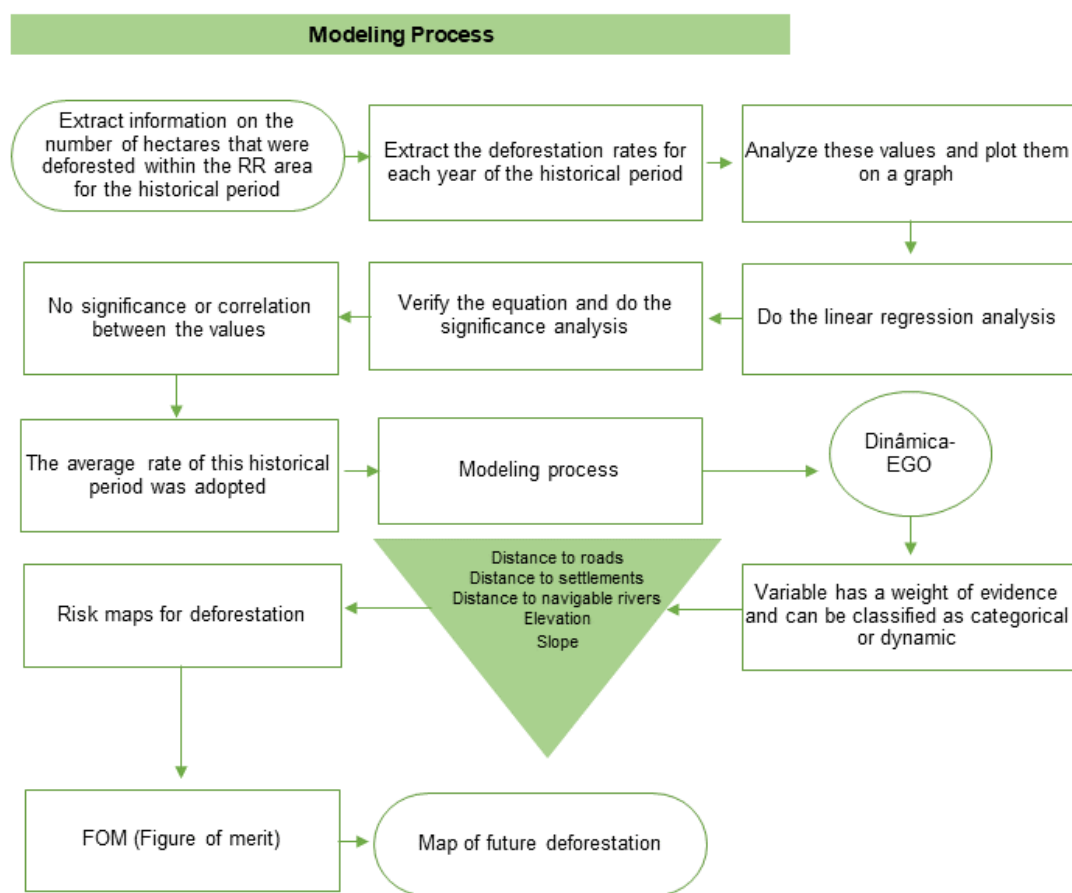


Figure 38. Flowchart for the Quantification of GHG Emission Reductions and Removals

The forest dynamics data and deforestation vectors used to build the project baseline were organized in File Geodatabase format using ArcGIS Pro software. The files are stored in vector and matrix (raster)

format. We reprojected all data to the UTM 21S and Datum SIRGAS 2000 projection standardizing the spatial references.

We found the future deforestation location projection through Dinâmica-EGO software version 6.1.0. The selection for Dinâmica-EGO was made for the following reasons: (1) is widely reported in the scientific literature; (2) it holds a transparent process for input and output of data and parameters processed with the user-friendly graphical interface, and (3) it incorporates the use of appropriate data to explain the location of deforestation.

Our first step was pre-select layers that might be related to the greater chance of deforestation in the reference region and the project area. For example, rivers and roads are usually drivers of deforestation because they are the access roads to forest areas.

The main steps conducted with Dinâmica-EGO at this stage were: organization of maps on land use and land cover and maps with deforestation explanatory factors; model calibration by determining the weight of evidence and analyzing correlations between variables; assessment of model accuracy, and development of deforestation baseline scenarios. We used spatial data with 30 x 30 m pixel size, GeoTIFF format.

This study evaluated the following drivers into account: distance to deforestation, distance from roads, distance from navigable rivers, distance from settlements, elevation, and slope to evaluate the variables and combinations that can better explain the dynamics of historical deforestation in the region, where the main driver is pasture for cattle creation. This step used an empirical approach, defining functions to represent the probability of allocation of deforestation from all key spatial variables. The method used in Dinâmica-EGO is called Weight of Evidence, a Bayesian method. This method requires that the variables used to produce the map of deforestation risk are independent.

Calculating weights of evidence for each variable (or factor) examines their historical influence concerning the deforestation observed. This influence of the variable is obtained from the number of pixels representing forest area destroyed during the analysis period (the chosen calibration period was between 2013 and 2017).

During the analysis of deforestation dynamics in the Reference Region, network model input data (explanatory variables) were all analyzed, despite the possibility of correlation and redundancy of information between them. The only assumption for the Weights of Evidence method is that the input maps have to be spatially independent. Furthermore, to the links to be connected, the transition is the only parameter to be set in the Determine Weights of Evidence Correlation.

The weights of evidence are defined by a Bayesian method, which considers the a posteriori joint probability of deforestation within each class of all explanatory variables. These values represent how the different ranges that make up each predictor variable are related to deforestation. Positive values indicate a correlation with deforestation, and negative values are ranges that have seen little deforestation in the past and, therefore, should have less chance of being deforested in the future. Higher values, whether positive or negative, indicate a greater weight to influence the calculation of the probability of deforestation positively or negatively in an area. From the weights of evidence, the probability of transition, per pixel, from forest to any other anthropic uses is calculated. This probability is calculated based on the sum of all evidence weights that overlap over a given pixel and are dependent on all static and dynamic map combinations.

Different models were tested, all if the allocation pattern of new deforestation considered 31.53 ha for new patches (variance 64.34 ha) and alternating combinations of explanatory variables. We measured the model accuracy by an adaptation of the Fuzzy similarity suggested by the Dinâmica EGO team. This method works on the difference between the simulated and the observed maps, in this case, the simulated 2022 and the observed 2022 (T2), with the initial map (2017). Afterwards, we compare the result of our simulation with the result of a random simulation. The model is considered accurate when the simulated one is superior to the random one. We compare the accuracy of different models and choose the model with the best similarity results.

The model with the highest accuracy included the variables: distance from roads, distance from navigable rivers, distance from settlements, and slope, as well as distance to deforestation, which is a variable automatically included by Dinâmica Ego. Subsequently, we vectorized the raster generated by the model and applied SIG processes to quantify deforestation by forest typology.

Variables related to actual land tenure and management, were considered not to be applicable to the model in the present project. This is because, given that only private lands are included in the RR, the variable would have only one category: private land.

Public and protected areas were not included in the RR in order to respect the following similarity requirement of the methodology's VMD0007BL-UP_v3.3 module, section 1.1.1.1: "Policies and regulations having an impact on land-use change patterns within the RRD and the project area must be of the same type or have an equivalent effect at the start of the historical reference period, taking into account the current level of enforcement". Following this requirement, Conservation Units and Indigenous Territories, which have specific policies and regulations, were excluded from the RR.

The aforementioned module requires that at least one land-tenure variable be included as an input variable and, as this was not conducted in the present project, due to reasons described above, this was included as a methodology deviation in PD section 3.1.6.

Estimation of annual areas of unplanned baseline deforestation in the PA

Table 22 - Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (1)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Submontane Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	66.17	66.17
2029	48.47	114.64
2030	0.00	114.64
2031	0.00	114.64
2032	0.00	114.64
2033	31.75	146.39
2034	29.02	175.41
2035	61.09	236.50
2036	2.88	239.38
2037	0.00	239.38
2038	34.38	273.76
2039	0.00	273.76
2040	30.15	303.91
2041	65.88	369.79
2042	2.74	372.52
2043	0.00	372.52
2044	0.00	372.52
2045	0.00	372.52
2046	0.76	373.29
2047	0.00	373.29
2048	0.36	373.65
2049	0.00	373.65
2050	0.00	373.65
2051	21.44	395.08
2052	0.00	395.08

Table 23. Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (2)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Lowland Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00
2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00
2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 24. Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (3)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area

Dense Tropical Alluvial Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00
2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00
2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 25. Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), per stratum (4)

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Pioneer Formation with fluvial influence without palm trees		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.26	0.26
2023	0.00	0.26
2024	0.00	0.26
2025	0.00	0.26
2026	0.00	0.26
2027	26.35	26.61

2028	27.01	53.62
2029	0.00	53.62
2030	0.00	53.62
2031	0.00	53.62
2032	1.10	54.72
2033	1.49	56.21
2034	38.78	94.99
2035	0.00	94.99
2036	0.00	94.99
2037	0.00	94.99
2038	9.20	104.18
2039	15.51	119.69
2040	50.18	169.87
2041	32.19	202.06
2042	3.49	205.55
2043	6.51	212.05
2044	0.00	212.05
2045	26.77	238.83
2046	0.00	238.83
2047	2.05	240.87
2048	0.00	240.87
2049	0.00	240.87
2050	14.20	255.08
2051	0.00	255.08
2052	2.32	257.40

Table 26. Projection of annual baseline unplanned deforestation for the Project Area (Location Analysis), sum of strata.

Avoided Unplanned Deforestation Annual baseline deforestation for the Project Area Sum of Strata		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.26	0.26
2023	0.00	0.26
2024	0.00	0.26
2025	0.00	0.26
2026	0.00	0.26
2027	92.52	92.78
2028	75.48	168.26
2029	0.00	168.26
2030	0.00	168.26
2031	0.00	168.26
2032	32.85	201.11
2033	30.51	231.62
2034	99.86	331.49
2035	2.88	334.37

2036	0.00	334.37
2037	34.38	368.75
2038	9.20	377.94
2039	45.66	423.60
2040	116.06	539.66
2041	34.93	574.59
2042	3.49	578.07
2043	6.51	584.58
2044	0.00	584.58
2045	27.54	612.11
2046	0.00	612.11
2047	2.41	614.52
2048	0.00	614.52
2049	0.00	614.52
2050	35.64	650.16
2051	0.00	650.16
2052	2.32	652.48

Carbon Stock - AUD

As previously described in section 2.1.5, three forest classes with varying biomass levels are present in AUD component of Project Area: Dense tropical submontane rainforest (Ds) (covering 73.28% of the total area); Dense Tropical Lowland Rainforest (Db) (16.31%); Dense Tropical Alluvial Rainforest (Da) (1.31%); and Pioneer Formation with fluvial influence without palm trees (Pa) (9.09%). Carbons Stock values for all forest types were obtained from literature: Ministério da Ciência, Tecnologia e Inovação, (2015), shown below.

In order to calculate the carbon pools (above and belowground) for each stratum, the Carbons Stock values was multiplied by 44/12 (conversion factor from tC to tCO₂)

The net carbon stock changes in the baseline are equal to the baseline pre-deforestation stock minus the long-term carbon stock after deforestation and minus the baseline stock that enters the wood products pool at the time of deforestation.

Regarding initial and post-deforestation carbon stocks, initial forest biomass data were obtained from peer-reviewed literature applying to the project's geographic region, in accordance biome, and climate regime as in the project area. All values were obtained from the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015). Regarding data per stratum, the total carbon stock numbers were as follows:

- Dense Tropical Submontane Rainforest (code “Ds”): 140.69 tC/ha above ground and 39.7 tC/ha below ground.
- Dense Tropical Lowland Rainforest (code “Db”): 141.10 tC/ha above ground and 39.83 tC/ha below ground.
- Dense Tropical Alluvial Rainforest (code “Da”): 160.18 tC/ha above ground and 45.22 tC/ha below ground.
- Pioneer Formation with fluvial influence without palm trees (code “Pa”): 91.70 tC/ha above ground and 17.01 tC/ha below ground.

Avoided Planned Deforestation (APD)

The baseline net GHG emissions for planned deforestation (APD) were determined as:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

$\Delta C_{BSL,planned}$	Net greenhouse gas emissions in the baseline from planned deforestation up to year t^* ; t CO ₂ -e
$\Delta C_{BSL,i,t}$	Net carbon stock changes in all pools in the baseline stratum i in year t ; t CO ₂ -e
$GHG_{BSL-E,i,t}$	Greenhouse gas emissions as a result of deforestation activities within the project boundary in the baseline stratum i in year t ; t CO ₂ -e yr ⁻¹
i	1, 2, 3, ... M strata
t	1, 2, 3, ... t^* years elapsed since the projected start of the project activity.

The agent of deforestation in APD scenario is a specific individual: The landowner intended to deforest the area for development of agricultural activity, in the absence of the project activity.

As Brazilian law requires government approval for deforestation to occur, the intention to deforest within the project area is demonstrated by evidence of documentation that a request for approval has been filed with the relevant government department for permission to deforest and convert to alternative land use.

The proportion of the total parcel area planned to be deforested does not exceed the legal mandate, since it does not overlap areas protected by the Brazilian forest code, such as Legal Reserves and Permanent Preservation Areas.

The scheduled plan for deforestation, per property, is as presented in section 3.1.4, under the Avoided Planned Deforestation (APD) heading. The deforestation values for each stratum, presented in the tables below, were obtained through analysis of forest types affected per year in the deforestation schedule using IBGE (2021) data, as shown in Figure 8 in the PD's physical parameters section. The resulting baseline deforestation areas per stratum from planned deforestation are presented below.

Table 27 - Annual baseline deforestation for the Project Area from planned deforestation, per stratum (1)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Submontane Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	132.63	132.63
2023	853.15	985.78
2024	940.51	1,926.29
2025	941.29	2,867.58
2026	225.94	3,093.52
2027	0.00	3,093.52
2028	0.00	3,093.52
2029	0.00	3,093.52
2030	0.00	3,093.52
2031	0.00	3,093.52
2032	0.00	3,093.52
2033	0.00	3,093.52
2034	0.00	3,093.52
2035	0.00	3,093.52
2036	0.00	3,093.52
2037	0.00	3,093.52
2038	0.00	3,093.52
2039	0.00	3,093.52

2040	0.00	3,093.52
2041	0.00	3,093.52
2042	0.00	3,093.52
2043	0.00	3,093.52
2044	0.00	3,093.52
2045	0.00	3,093.52
2046	0.00	3,093.52
2047	0.00	3,093.52
2048	0.00	3,093.52
2049	0.00	3,093.52
2050	0.00	3,093.52
2051	0.00	3,093.52
2052	0.00	3,093.52

Table 28. Annual baseline deforestation for the Project Area from planned deforestation, per stratum (2)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Lowland Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	369.07	369.07
2023	76.82	445.88
2024	0.00	445.88
2025	0.00	445.88
2026	0.00	445.88
2027	0.00	445.88
2028	0.00	445.88
2029	0.00	445.88
2030	0.00	445.88
2031	0.00	445.88
2032	0.00	445.88
2033	0.00	445.88
2034	0.00	445.88
2035	0.00	445.88
2036	0.00	445.88
2037	0.00	445.88
2038	0.00	445.88
2039	0.00	445.88
2040	0.00	445.88
2041	0.00	445.88
2042	0.00	445.88
2043	0.00	445.88
2044	0.00	445.88
2045	0.00	445.88
2046	0.00	445.88
2047	0.00	445.88
2048	0.00	445.88
2049	0.00	445.88
2050	0.00	445.88

2051	0.00	445.88
2052	0.00	445.88

Table 29. Annual baseline deforestation for the Project Area from planned deforestation, per stratum (3)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Dense Tropical Alluvial Rainforest		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	0.00	0.00
2027	0.00	0.00
2028	0.00	0.00
2029	0.00	0.00
2030	0.00	0.00
2031	0.00	0.00
2032	0.00	0.00
2033	0.00	0.00
2034	0.00	0.00
2035	0.00	0.00
2036	0.00	0.00
2037	0.00	0.00
2038	0.00	0.00
2039	0.00	0.00
2040	0.00	0.00
2041	0.00	0.00
2042	0.00	0.00
2043	0.00	0.00
2044	0.00	0.00
2045	0.00	0.00
2046	0.00	0.00
2047	0.00	0.00
2048	0.00	0.00
2049	0.00	0.00
2050	0.00	0.00
2051	0.00	0.00
2052	0.00	0.00

Table 30. Annual baseline deforestation for the Project Area from planned deforestation, per stratum (4)

Avoided Planned Deforestation Annual baseline deforestation for the Project Area

Pioneer Formation with fluvial influence without palm trees		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	0.00	0.00
2023	0.00	0.00
2024	0.00	0.00
2025	0.00	0.00
2026	8.54	8.54
2027	0.00	8.54
2028	0.00	8.54
2029	0.00	8.54
2030	0.00	8.54
2031	0.00	8.54
2032	0.00	8.54
2033	0.00	8.54
2034	0.00	8.54
2035	0.00	8.54
2036	0.00	8.54
2037	0.00	8.54
2038	0.00	8.54
2039	0.00	8.54
2040	0.00	8.54
2041	0.00	8.54
2042	0.00	8.54
2043	0.00	8.54
2044	0.00	8.54
2045	0.00	8.54
2046	0.00	8.54
2047	0.00	8.54
2048	0.00	8.54
2049	0.00	8.54
2050	0.00	8.54
2051	0.00	8.54
2052	0.00	8.54

Table 31. Total annual baseline deforestation for the Project Area from planned deforestation, sum of strata

Avoided Planned Deforestation Annual baseline deforestation for the Project Area Sum of Strata		
Year	Total yearly area (ha/year)	Total cumulative area (ha)
2022	501.70	501.70
2023	929.97	1,431.66
2024	940.51	2,372.17
2025	941.29	3,313.46
2026	234.49	3,547.95
2027	0.00	3,547.95
2028	0.00	3,547.95

2029	0.00	3,547.95
2030	0.00	3,547.95
2031	0.00	3,547.95
2032	0.00	3,547.95
2033	0.00	3,547.95
2034	0.00	3,547.95
2035	0.00	3,547.95
2036	0.00	3,547.95
2037	0.00	3,547.95
2038	0.00	3,547.95
2039	0.00	3,547.95
2040	0.00	3,547.95
2041	0.00	3,547.95
2042	0.00	3,547.95
2043	0.00	3,547.95
2044	0.00	3,547.95
2045	0.00	3,547.95
2046	0.00	3,547.95
2047	0.00	3,547.95
2048	0.00	3,547.95
2049	0.00	3,547.95
2050	0.00	3,547.95
2051	0.00	3,547.95
2052	0.00	3,547.95

Carbon Stock - APD

As previously described in section 2.1.5, three forest classes with varying biomass levels are present in APD component of Project Area: Dense tropical submontane rainforest (Ds) (covering 86% of the total area); Dense Tropical Lowland Rainforest (Db) (13%); and Pioneer Formation with fluvial influence without palm trees (Pa) (0.24%). Carbons Stock values for all forest types were obtained from literature: Ministério da Ciência, Tecnologia e Inovação, (2015), shown below.

In order to calculate the carbon pools (above and belowground) for each stratum, the Carbons Stock values was multiplied by 44/12 (conversion factor from tC to tCO₂)

The net carbon stock changes in the baseline is equal to the baseline pre-deforestation stock minus the long-term carbon stock after deforestation and minus the baseline stock that enters the wood products pool at the time of deforestation.

Regarding initial and post-deforestation carbon stocks, initial forest biomass data were obtained from peer-reviewed literature applying to the project's geographic region, in accordance biome, and climate regime as

in the project area. All values were obtained from the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015). Regarding data per stratum, the the total carbon stock numbers were as follows:

- Dense Tropical Submontane Rainforest (code “Ds”): 140.69 tC/ha above ground and 39.7 tC/ha below ground.
- Dense Tropical Lowland Rainforest (code “Db”): 141.10 tC/ha above ground and 39.83 tC/ha below ground.
- Dense Tropical Alluvial Rainforest (code “Da”): 160.18 tC/ha above ground and 45.22 tC/ha below ground.
- Pioneer Formation with fluvial influence without palm trees (code “Pa”): 91.70 tC/ha above ground and 17.01 tC/ha below ground.

Baseline emissions from unplanned deforestation (AUD)

In order to estimate emissions from unplanned deforestation that would occur in the Project Area in the absence of project (i.e. in the baseline case), the annual estimated area of deforestation (see “Projection of annual baseline deforestation for the Project Area”) multiplied by the carbon stocks in forest for each stratum.

Stock changes in aboveground biomass are emitted at the time of deforestation. Following deforestation, emissions from belowground biomass and wood products take place gradually over time. Stock changes in belowground biomass are emitted at an annual of 1/10 of the stock change for 10 years. Carbon stocks entering the wood products pool at the time of deforestation and that are expected to be emitted over 100 years are emitted at an annual rate of 1/20 of the stock for 20 years.

The baseline net GHG emissions from unplanned deforestation (AUD) were determined as follows, in accordance with the VMD0007 BL-UP module, equation 22, which is applicable to terrestrial carbon pools in non-wetland REDD project activities:

$$\begin{aligned}\Delta C_{BSL,i,t} = & A_{unplanned,i,t} * (\Delta C_{AB_{tree},i} - \Delta C_{WP,i} + \Delta C_{AB_{non-tree},i} + \Delta C_{LI,i}) \\ & + \left(\sum_{t-10}^t A_{unplanned,i,t} * (\Delta C_{BB_{tree},i} + \Delta C_{BB_{non-tree},i} + \Delta C_{DW,i}) * \left(\frac{1}{10} \right) \right. \\ & \left. + (\sum_{t-20}^t A_{unplanned,i,t}) * (C_{WP100,i} + \Delta C_{SOC,i}) * \left(\frac{1}{20} \right) \right)\end{aligned}$$

Where:

$\Delta C_{BSL,i,t}$	Sum of the baseline carbon stock change in all terrestrial pools in stratum i in year t ; t CO _{2e} (calculated separately for the project area [PA] and the leakage belt [LB])
$A_{unplanned,i,t}$	Area of unplanned deforestation in forest stratum i in year t ; ha
$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i ; t CO _{2e} ha ⁻¹
$C_{WP100,i}$	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i ; t CO _{2e} ha ⁻¹
$C_{AB,tree,i}$	Baseline carbon stock change in aboveground tree biomass in stratum i ; t CO _{2e} ha ⁻¹
$C_{BB,tree,i}$	Baseline carbon stock change in belowground tree biomass in stratum i ; t CO _{2e} ha ⁻¹
$C_{AB_non-tree,i}$	Baseline carbon stock change in aboveground non-tree biomass in stratum i ; t CO _{2e} ha ⁻¹
$C_{BB_non-tree,i}$	Baseline carbon stock change in belowground non-tree biomass in stratum i ; t CO _{2e} ha ⁻¹
$\Delta C_{DW,i}$	Baseline carbon stock change in dead wood in stratum i ; t CO _{2e} ha ⁻¹
$\Delta C_{LI,i}$	Baseline carbon stock change in litter in stratum i ; t CO _{2e} ha ⁻¹
$\Delta C_{SOC,i}$	Baseline carbon stock change in terrestrial soil organic carbon in stratum i ; t CO _{2e} ha ⁻¹
i	1, 2, 3 ... M strata

t 1, 2, 3, ... t^* years elapsed since the projected start of the project activity.

The results of the baseline AUD calculations is shown in the tables below.

Table 32. Summary of gross baseline emissions from unplanned deforestation within the Project Area, per stratum (1)

Avoided Unplanned Deforestation Gross baseline emissions from deforestation Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	66.17	66.17	35,154.97	35,154.97
2028	48.47	114.64	26,773.39	61,928.36
2029	0.00	114.64	1,768.11	63,696.47
2030	0.00	114.64	1,768.11	65,464.58
2031	0.00	114.64	1,768.11	67,232.69
2032	31.75	146.39	18,635.56	85,868.25
2033	29.02	175.41	17,676.75	103,545.00
2034	61.09	236.50	35,160.06	138,705.06
2035	2.88	239.38	5,177.60	143,882.66
2036	0.00	239.38	3,691.92	147,574.58
2037	34.38	273.76	20,994.09	168,568.66
2038	0.00	273.76	2,552.93	171,121.59
2039	30.15	303.91	18,571.20	189,692.79
2040	65.88	369.79	38,017.64	227,710.43
2041	2.74	372.52	5,487.68	233,198.11
2042	0.00	372.52	3,613.88	236,811.98
2043	0.00	372.52	3,191.31	240,003.29
2044	0.00	372.52	2,301.86	242,305.15
2045	0.76	373.29	2,665.79	244,970.93
2046	0.00	373.29	2,271.71	247,242.64
2047	0.36	373.65	1,905.31	249,147.95
2048	0.00	373.65	1,677.79	250,825.75
2049	0.00	373.65	1,238.80	252,064.54
2050	21.44	395.08	11,668.76	263,733.30
2051	0.00	395.08	570.38	264,303.68
2052	0.00	395.08	543.00	264,846.68

Table 33. Summary of gross baseline emissions from unplanned deforestation within the Project Area, per stratum (2)

Avoided Unplanned Deforestation Gross baseline emissions from deforestation Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 34. Summary of gross baseline emissions from unplanned deforestation within the Project Area, per stratum (3)

Avoided Unplanned Deforestation Gross baseline emissions from deforestation Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂

2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 35. Summary of gross baseline emissions from unplanned deforestation within the Project Area, per stratum (4)

Avoided Unplanned Deforestation				
Gross baseline emissions from deforestation				
Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	89.71	89.71
2023	0.00	0.26	1.86	89.71
2024	0.00	0.26	1.86	91.56
2025	0.00	0.26	1.86	93.42
2026	0.00	0.26	1.86	95.27
2027	26.35	26.61	9,048.47	97.13
2028	27.01	53.62	9,461.60	9,145.60
2029	0.00	53.62	380.67	18,607.20

2030	0.00	53.62	380.67	18,987.86
2031	0.00	53.62	380.67	19,368.53
2032	1.10	54.72	757.28	19,749.19
2033	1.49	56.21	897.73	20,506.47
2034	38.78	94.99	13,711.10	21,404.20
2035	0.00	94.99	672.72	35,115.30
2036	0.00	94.99	672.72	35,788.03
2037	0.00	94.99	508.38	36,460.75
2038	9.20	104.18	3,497.98	36,969.13
2039	15.51	119.69	5,729.72	40,467.12
2040	50.18	169.87	17,743.30	46,196.84
2041	32.19	202.06	11,924.01	63,940.13
2042	3.49	205.55	2,289.55	75,864.14
2043	6.51	212.05	3,342.19	78,153.69
2044	0.00	212.05	912.83	81,495.88
2045	26.77	238.83	10,104.92	82,408.70
2046	0.00	238.83	1,102.90	92,513.62
2047	2.05	240.87	1,782.90	93,616.53
2048	0.00	240.87	1,014.04	95,399.43
2049	0.00	240.87	917.32	96,413.47
2050	14.20	255.08	5,480.73	97,330.79
2051	0.00	255.08	504.41	102,811.52
2052	2.32	257.40	1,279.53	103,315.92

Table 36. Summary of gross baseline emissions from unplanned deforestation within the Project Area, sum of strata

Avoided Unplanned Deforestation Gross baseline emissions from deforestation Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	89.71	89.71
2023	0.00	0.26	1.86	91.56
2024	0.00	0.26	1.86	93.42
2025	0.00	0.26	1.86	95.27
2026	0.00	0.26	1.86	97.13
2027	92.52	92.78	44,203.44	44,300.57
2028	75.48	168.26	36,234.99	80,535.55
2029	0.00	168.26	2,148.78	82,684.33
2030	0.00	168.26	2,148.78	84,833.11
2031	0.00	168.26	2,148.78	86,981.88
2032	32.85	201.11	19,392.84	106,374.72
2033	30.51	231.62	18,574.49	124,949.20
2034	99.86	331.49	48,871.15	173,820.36
2035	2.88	334.37	5,850.33	179,670.69
2036	0.00	334.37	4,364.64	184,035.33

2037	34.38	368.75	21,502.47	205,537.80
2038	9.20	377.94	6,050.91	211,588.71
2039	45.66	423.60	24,300.92	235,889.63
2040	116.06	539.66	55,760.93	291,650.56
2041	34.93	574.59	17,411.69	309,062.25
2042	3.49	578.07	5,903.43	314,965.67
2043	6.51	584.58	6,533.49	321,499.16
2044	0.00	584.58	3,214.68	324,713.85
2045	27.54	612.11	12,770.71	337,484.56
2046	0.00	612.11	3,374.61	340,859.17
2047	2.41	614.52	3,688.22	344,547.38
2048	0.00	614.52	2,691.83	347,239.22
2049	0.00	614.52	2,156.12	349,395.33
2050	35.64	650.16	17,149.49	366,544.82
2051	0.00	650.16	1,074.79	367,619.61
2052	2.32	652.48	1,822.53	369,442.14

Baseline emissions from planned deforestation (APD)

In order to estimate emissions from planned deforestation that would occur in the Project Area in the absence of the project (i.e., in the baseline case), the annual estimated area of deforestation (see “Total annual baseline deforestation for the Project Area from planned deforestation” tables, above), multiplied by the carbon stocks in the forest types within each stratum. As described earlier in section 3.2.1., the strata affected by APD per year were defined in accordance with the scheduled deforestation plan.

Stock changes in aboveground biomass are emitted at the time of deforestation. Following deforestation, emissions from belowground biomass and wood products take place gradually over time. Stock changes in belowground biomass are emitted at an annual of 1/10 of the stock change for 10 years. Carbon stocks entering the wood products pool at the time of deforestation and that are expected to be emitted over 100 years are emitted at an annual rate of 1/20 of the stock for 20 years.

The result of this procedure is shown in the tables below.

Table 37. Summary of gross baseline emissions from planned deforestation within the Project Area, per stratum (1)

Avoided Planned Deforestation Gross baseline emissions from deforestation Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO₂/year	tCO₂
2022	132.63	132.63	70,464.27	70,464.27

2023	853.15	985.78	455,313.10	525,777.37
2024	940.51	1,926.29	514,881.57	1,040,658.94
2025	941.29	2,867.58	529,804.13	1,570,463.07
2026	225.94	3,093.52	164,267.42	1,734,730.49
2027	0.00	3,093.52	47,710.96	1,782,441.45
2028	0.00	3,093.52	47,710.96	1,830,152.42
2029	0.00	3,093.52	47,710.96	1,877,863.38
2030	0.00	3,093.52	47,710.96	1,925,574.34
2031	0.00	3,093.52	47,710.96	1,973,285.31
2032	0.00	3,093.52	45,779.83	2,019,065.14
2033	0.00	3,093.52	33,357.66	2,052,422.80
2034	0.00	3,093.52	19,663.58	2,072,086.38
2035	0.00	3,093.52	5,958.06	2,078,044.44
2036	0.00	3,093.52	2,668.23	2,080,712.67
2037	0.00	3,093.52	2,668.23	2,083,380.90
2038	0.00	3,093.52	2,668.23	2,086,049.13
2039	0.00	3,093.52	2,668.23	2,088,717.36
2040	0.00	3,093.52	2,668.23	2,091,385.59
2041	0.00	3,093.52	2,668.23	2,094,053.82
2042	0.00	3,093.52	2,553.83	2,096,607.65
2043	0.00	3,093.52	1,817.97	2,098,425.62
2044	0.00	3,093.52	1,006.77	2,099,432.39
2045	0.00	3,093.52	194.88	2,099,627.27
2046	0.00	3,093.52	0.00	2,099,627.27
2047	0.00	3,093.52	0.00	2,099,627.27
2048	0.00	3,093.52	0.00	2,099,627.27
2049	0.00	3,093.52	0.00	2,099,627.27
2050	0.00	3,093.52	0.00	2,099,627.27
2051	0.00	3,093.52	0.00	2,099,627.27
2052	0.00	3,093.52	0.00	2,099,627.27

Table 38. Summary of gross baseline emissions from planned deforestation within the Project Area, per stratum (2)

Avoided Planned Deforestation Gross baseline emissions from deforestation Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	369.07	369.07	196,650.72	196,650.72
2023	76.82	445.88	46,639.18	243,289.90
2024	0.00	445.88	6,896.41	250,186.31
2025	0.00	445.88	6,896.41	257,082.73
2026	0.00	445.88	6,896.41	263,979.14
2027	0.00	445.88	6,896.41	270,875.55
2028	0.00	445.88	6,896.41	277,771.97
2029	0.00	445.88	6,896.41	284,668.38
2030	0.00	445.88	6,896.41	291,564.79

2031	0.00	445.88	6.896.41	298,461.21
2032	0.00	445.88	1.506.45	299,967.66
2033	0.00	445.88	384.58	300,352.24
2034	0.00	445.88	384.58	300,736.83
2035	0.00	445.88	384.58	301,121.41
2036	0.00	445.88	384.58	301,505.99
2037	0.00	445.88	384.58	301,890.58
2038	0.00	445.88	384.58	302,275.16
2039	0.00	445.88	384.58	302,659.75
2040	0.00	445.88	384.58	303,044.33
2041	0.00	445.88	384.58	303,428.91
2042	0.00	445.88	66.26	303,495.17
2043	0.00	445.88	0.00	303,495.17
2044	0.00	445.88	0.00	303,495.17
2045	0.00	445.88	0.00	303,495.17
2046	0.00	445.88	0.00	303,495.17
2047	0.00	445.88	0.00	303,495.17
2048	0.00	445.88	0.00	303,495.17
2049	0.00	445.88	0.00	303,495.17
2050	0.00	445.88	0.00	303,495.17
2051	0.00	445.88	0.00	303,495.17
2052	0.00	445.88	0.00	303,495.17

Table 39. Summary of gross baseline emissions from planned deforestation within the Project Area, per stratum (3)

Avoided Planned Deforestation Gross baseline emissions from deforestation Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00

2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 40. Summary of gross baseline emissions from planned deforestation within the Project Area, per stratum (4)

Avoided Planned Deforestation Gross baseline emissions from deforestation Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	8.54	8.54	2,932.62	0.00
2027	0.00	8.54	60.64	2,932.62
2028	0.00	8.54	60.64	2,993.27
2029	0.00	8.54	60.64	3,053.91
2030	0.00	8.54	60.64	3,114.55
2031	0.00	8.54	60.64	3,175.19
2032	0.00	8.54	60.64	3,235.83
2033	0.00	8.54	60.64	3,296.47
2034	0.00	8.54	60.64	3,357.11
2035	0.00	8.54	60.64	3,417.76
2036	0.00	8.54	7.37	3,478.40
2037	0.00	8.54	7.37	3,485.77
2038	0.00	8.54	7.37	3,493.13
2039	0.00	8.54	7.37	3,500.50
2040	0.00	8.54	7.37	3,507.87
2041	0.00	8.54	7.37	3,515.23
2042	0.00	8.54	7.37	3,522.60
2043	0.00	8.54	7.37	3,529.97
2044	0.00	8.54	7.37	3,537.34
2045	0.00	8.54	7.37	3,544.70
2046	0.00	8.54	0.00	3,552.07

2047	0.00	8.54	0.00	3,552.07
2048	0.00	8.54	0.00	3,552.07
2049	0.00	8.54	0.00	3,552.07
2050	0.00	8.54	0.00	3,552.07
2051	0.00	8.54	0.00	3,552.07
2052	0.00	8.54	0.00	3,552.07

Table 41. Summary of gross baseline emissions from planned deforestation within the Project Area, sum of strata

Avoided Planned Deforestation Gross baseline emissions from deforestation Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	501.70	501.70	267,114.99	267,114.99
2023	929.97	1,431.66	501,952.29	769,067.27
2024	940.51	2,372.17	521,777.98	1,290,845.25
2025	941.29	3,313.46	536,700.55	1,827,545.80
2026	234.49	3,547.95	174,096.46	2,001,642.26
2027	0.00	3,547.95	54,668.02	2,056,310.27
2028	0.00	3,547.95	54,668.02	2,110,978.29
2029	0.00	3,547.95	54,668.02	2,165,646.31
2030	0.00	3,547.95	54,668.02	2,220,314.33
2031	0.00	3,547.95	54,668.02	2,274,982.35
2032	0.00	3,547.95	47,346.92	2,322,329.27
2033	0.00	3,547.95	33,802.89	2,356,132.16
2034	0.00	3,547.95	20,108.80	2,376,240.96
2035	0.00	3,547.95	6,403.28	2,382,644.25
2036	0.00	3,547.95	3,060.18	2,385,704.43
2037	0.00	3,547.95	3,060.18	2,388,764.61
2038	0.00	3,547.95	3,060.18	2,391,824.79
2039	0.00	3,547.95	3,060.18	2,394,884.97
2040	0.00	3,547.95	3,060.18	2,397,945.15
2041	0.00	3,547.95	3,060.18	2,401,005.33
2042	0.00	3,547.95	2,627.46	2,403,632.79
2043	0.00	3,547.95	1,825.34	2,405,458.13
2044	0.00	3,547.95	1,014.13	2,406,472.26
2045	0.00	3,547.95	202.25	2,406,674.51
2046	0.00	3,547.95	0.00	2,406,674.51
2047	0.00	3,547.95	0.00	2,406,674.51
2048	0.00	3,547.95	0.00	2,406,674.51
2049	0.00	3,547.95	0.00	2,406,674.51
2050	0.00	3,547.95	0.00	2,406,674.51
2051	0.00	3,547.95	0.00	2,406,674.51
2052	0.00	3,547.95	0.00	2,406,674.51

Emissions from biomass burning in the baseline.

Based on the IPCC 2006 Inventory Guidelines and E-BPB Module (VMD0013), estimating greenhouse gas emissions from biomass burning was determined as follows. The same methodology was applied for both AUD and APD components of the project:

$$E_{BiomassBurn, i, t} = \sum_{g=1}^G ((A_{burn, i, t} * B_{i, t} * COMF_i * G_{g, i}) * 10^{-3}) * GWP_g$$

Where:

$E_{BiomassBurn, i, t}$	Greenhouse gas emissions due to biomass burning as part of deforestation activities in stratum i in year t ; tCO ₂ -e of each GHG (CO ₂ , CH ₄ , N ₂ O)
$A_{burn, i, t}$	Area burnt for stratum i at time t ; ha
$B_{i, t}$	Average aboveground biomass stock before burning stratum i , time t ; tonnes d.m. ha-1
$COMF_i$	Combustion factor for stratum i ; dimensionless (default value derived from Table 2.6 of IPCC, 2006) ¹⁴³
$G_{g, i}$	Emission factor for stratum i for gas g ; kg t-1 dry matter burnt (default values derived from Table 2.5 of IPCC, 2006) ¹⁴⁴

¹⁴³ E-BPB module refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, "Primary tropical moist forest" https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited 20/06/2022).

¹⁴⁴ E-BPB refers to Table 2.5; page 2.47 "Tropical forest" of Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines: For CH₄: 6.8 - 2 = 4.8 g kg-1 dry matter burnt (conservative); For N₂O: 0.20 g kg-1 dry matter burnt (unique value proposed): https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf (last visited in 20/06/2022).

GWP_g	Global warming potential for gas g ; t CO ₂ /t gas g (default values from IPCC, 2022: CH ₄ = 27; N ₂ O = 273) ¹⁴⁵
g	1, 2, 3 ... G greenhouse gases
i	1, 2, 3 ... M strata
t	1, 2, 3 ... t years elapsed since the start of the REDD+ project activity

Table 38 below shows the parameters used to calculate biomass burning for the baseline scenario, as well as results accounted for CH₄ and N₂O emissions generated as a consequence of incomplete biomass burning, over a 30-year period.

Table 42. Parameters used to calculate biomass burning for the baseline scenario.

Parameter	Value	Unit
COMF	0.50	Dimensionless
G_{CH_4}	6.80	g kg ⁻¹ dry matter burnt
GWP_{CH_4}	27.00	Dimensionless
G_{N_2O}	0.20	g kg ⁻¹ dry matter burnt
GWP_{N_2O}	273.00	Dimensionless
B Ds	299.34	t d.m. ha ⁻¹
B Db	300.21	t d.m. ha ⁻¹
B Da	340.81	t d.m. ha ⁻¹
B Pa	195.11	t d.m. ha ⁻¹

Table 43. Summary of baseline biomass burning emissions (CH₄) within the Project Area, AUD component, per stratum (1)

Avoided Unplanned Deforestation

¹⁴⁵ IPCC, 2022: Annex II: Definitions, Units and Conventions [Al Khourdajie et al. (eds)]. In IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [P.R. Shukla et al. (eds.)]. Cambridge University Press, Cambridge, UK and New York, NY, USA. https://www.ipcc.ch/report/ar6/wg3/downloads/report/IPCC_AR6_WGIII_Annex-II.pdf

Biomass Burn Emissions - CH ₄				
Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	66.17	66.17	1,818.30	1,818.30
2028	48.47	114.64	1,332.00	3,150.31
2029	0.00	114.64	0.00	3,150.31
2030	0.00	114.64	0.00	3,150.31
2031	0.00	114.64	0.00	3,150.31
2032	31.75	146.39	872.43	4,022.73
2033	29.02	175.41	797.51	4,820.24
2034	61.09	236.50	1,678.64	6,498.88
2035	2.88	239.38	79.14	6,578.02
2036	0.00	239.38	0.00	6,578.02
2037	34.38	273.76	944.74	7,522.76
2038	0.00	273.76	0.00	7,522.76
2039	30.15	303.91	828.51	8,351.27
2040	65.88	369.79	1,810.27	10,161.54
2041	2.74	372.52	75.19	10,236.73
2042	0.00	372.52	0.00	10,236.73
2043	0.00	372.52	0.00	10,236.73
2044	0.00	372.52	0.00	10,236.73
2045	0.76	373.29	20.99	10,257.72
2046	0.00	373.29	0.00	10,257.72
2047	0.36	373.65	9.89	10,267.62
2048	0.00	373.65	0.00	10,267.62
2049	0.00	373.65	0.00	10,267.62
2050	21.44	395.08	589.08	10,856.69
2051	0.00	395.08	0.00	10,856.69
2052	0.00	395.08	0.00	10,856.69

Table 44. Summary of baseline biomass burning emissions (CH₄) within the Project Area, AUD component, per stratum (2)

Avoided Unplanned Deforestation Biomass Burn Emissions - CH₄ Dense Tropical Lowland Rainforest
--

Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 45. Summary of baseline biomass burning emissions (CH₄) within the Project Area, AUD component, per stratum (3)

Avoided Unplanned Deforestation Biomass Burn Emissions - CH ₄ Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00

2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 46. Summary of baseline biomass burning emissions (CH₄) within the Project Area, AUD component, per stratum (4)

Avoided Unplanned Deforestation Biomass Burn Emissions - CH ₄ Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	4.68	4.68
2023	0.00	0.26	0.00	4.68
2024	0.00	0.26	0.00	4.68
2025	0.00	0.26	0.00	4.68
2026	0.00	0.26	0.00	4.68
2027	26.35	26.61	471.94	476.62
2028	27.01	53.62	483.73	960.35
2029	0.00	53.62	0.00	960.35
2030	0.00	53.62	0.00	960.35
2031	0.00	53.62	0.00	960.35
2032	1.10	54.72	19.73	980.08
2033	1.49	56.21	26.65	1,006.73
2034	38.78	94.99	694.54	1,701.27
2035	0.00	94.99	0.00	1,701.27
2036	0.00	94.99	0.00	1,701.27

2037	0.00	94.99	0.00	1,701.27
2038	9.20	104.18	164.75	1,866.02
2039	15.51	119.69	277.76	2,143.78
2040	50.18	169.87	898.74	3,042.52
2041	32.19	202.06	576.58	3,619.09
2042	3.49	205.55	62.42	3,681.51
2043	6.51	212.05	116.53	3,798.04
2044	0.00	212.05	0.00	3,798.04
2045	26.77	238.83	479.53	4,277.57
2046	0.00	238.83	0.00	4,277.57
2047	2.05	240.87	36.66	4,314.23
2048	0.00	240.87	0.00	4,314.23
2049	0.00	240.87	0.00	4,314.23
2050	14.20	255.08	254.39	4,568.61
2051	0.00	255.08	0.00	4,568.61
2052	2.32	257.40	41.62	4,610.23

Table 47 . Summary of baseline biomass burning emissions (CH₄) within the Project Area, AUD component, sum of strata

Avoided Unplanned Deforestation Biomass Burn Emissions - CH ₄ Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	4.68	4.68
2023	0.00	0.26	0.00	4.68
2024	0.00	0.26	0.00	4.68
2025	0.00	0.26	0.00	4.68
2026	0.00	0.26	0.00	4.68
2027	92.52	92.78	2,290.24	2,294.92
2028	75.48	168.26	1,815.73	4,110.65
2029	0.00	168.26	0.00	4,110.65
2030	0.00	168.26	0.00	4,110.65
2031	0.00	168.26	0.00	4,110.65
2032	32.85	201.11	892.16	5,002.81
2033	30.51	231.62	824.16	5,826.97
2034	99.86	331.49	2,373.18	8,200.15
2035	2.88	334.37	79.14	8,279.29
2036	0.00	334.37	0.00	8,279.29
2037	34.38	368.75	944.74	9,224.03
2038	9.20	377.94	164.75	9,388.78
2039	45.66	423.60	1,106.27	10,495.05
2040	116.06	539.66	2,709.01	13,204.06
2041	34.93	574.59	651.77	13,855.83
2042	3.49	578.07	62.42	13,918.25
2043	6.51	584.58	116.53	14,034.77
2044	0.00	584.58	0.00	14,034.77

2045	27.54	612.11	500.52	14,535.29
2046	0.00	612.11	0.00	14,535.29
2047	2.41	614.52	46.55	14,581.84
2048	0.00	614.52	0.00	14,581.84
2049	0.00	614.52	0.00	14,581.84
2050	35.64	650.16	843.46	15,425.31
2051	0.00	650.16	0.00	15,425.31
2052	2.32	652.48	41.62	15,466.93

Table 48. Summary of baseline biomass burning emissions (N₂O) within the Project Area, AUD component, per stratum (1)

Avoided Unplanned Deforestation Biomass Burn Emissions - N ₂ O Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	66.17	66.17	540.74	540.74
2028	48.47	114.64	396.12	936.86
2029	0.00	114.64	0.00	936.86
2030	0.00	114.64	0.00	936.86
2031	0.00	114.64	0.00	936.86
2032	31.75	146.39	259.45	1,196.30
2033	29.02	175.41	237.17	1,433.47
2034	61.09	236.50	499.20	1,932.67
2035	2.88	239.38	23.54	1,956.21
2036	0.00	239.38	0.00	1,956.21
2037	34.38	273.76	280.95	2,237.16
2038	0.00	273.76	0.00	2,237.16
2039	30.15	303.91	246.39	2,483.55
2040	65.88	369.79	538.35	3,021.90
2041	2.74	372.52	22.36	3,044.26
2042	0.00	372.52	0.00	3,044.26
2043	0.00	372.52	0.00	3,044.26
2044	0.00	372.52	0.00	3,044.26
2045	0.76	373.29	6.24	3,050.50
2046	0.00	373.29	0.00	3,050.50
2047	0.36	373.65	2.94	3,053.44
2048	0.00	373.65	0.00	3,053.44
2049	0.00	373.65	0.00	3,053.44
2050	21.44	395.08	175.18	3,228.62
2051	0.00	395.08	0.00	3,228.62
2052	0.00	395.08	0.00	3,228.62

Table 49. Summary of baseline biomass burning emissions (N₂O) within the Project Area, AUD component, per stratum (2)

Avoided Unplanned Deforestation Biomass Burn Emissions - N ₂ O Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 50. Summary of baseline biomass burning emissions (N₂O) within the Project Area, AUD component, per stratum (3)

Avoided Unplanned Deforestation Biomass Burn Emissions - N ₂ O Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 51. Summary of baseline biomass burning emissions (N₂O) within the Project Area, AUD component, per stratum (4)

Avoided Unplanned Deforestation				
Biomass Burn Emissions - N ₂ O				
Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	1.39	1.39
2023	0.00	0.26	0.00	1.39
2024	0.00	0.26	0.00	1.39
2025	0.00	0.26	0.00	1.39
2026	0.00	0.26	0.00	1.39
2027	26.35	26.61	140.35	141.74
2028	27.01	53.62	143.85	285.59
2029	0.00	53.62	0.00	285.59
2030	0.00	53.62	0.00	285.59
2031	0.00	53.62	0.00	285.59
2032	1.10	54.72	5.87	291.46
2033	1.49	56.21	7.93	299.39
2034	38.78	94.99	206.55	505.93
2035	0.00	94.99	0.00	505.93
2036	0.00	94.99	0.00	505.93
2037	0.00	94.99	0.00	505.93
2038	9.20	104.18	48.99	554.93
2039	15.51	119.69	82.60	637.53
2040	50.18	169.87	267.27	904.80
2041	32.19	202.06	171.47	1,076.27
2042	3.49	205.55	18.56	1,094.83
2043	6.51	212.05	34.65	1,129.48
2044	0.00	212.05	0.00	1,129.48
2045	26.77	238.83	142.60	1,272.09
2046	0.00	238.83	0.00	1,272.09
2047	2.05	240.87	10.90	1,282.99
2048	0.00	240.87	0.00	1,282.99
2049	0.00	240.87	0.00	1,282.99
2050	14.20	255.08	75.65	1,358.64
2051	0.00	255.08	0.00	1,358.64
2052	2.32	257.40	12.38	1,371.02

Table 52. Summary of baseline biomass burning emissions (N₂O) within the Project Area, AUD component, sum of strata

Avoided Unplanned Deforestation				
Biomass Burn Emissions - N ₂ O				
Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	1.39	1.39
2023	0.00	0.26	0.00	1.39
2024	0.00	0.26	0.00	1.39
2025	0.00	0.26	0.00	1.39
2026	0.00	0.26	0.00	1.39
2027	92.52	92.78	681.08	682.48
2028	75.48	168.26	539.97	1,222.45
2029	0.00	168.26	0.00	1,222.45
2030	0.00	168.26	0.00	1,222.45
2031	0.00	168.26	0.00	1,222.45
2032	32.85	201.11	265.32	1,487.76
2033	30.51	231.62	245.09	1,732.86
2034	99.86	331.49	705.75	2,438.61
2035	2.88	334.37	23.54	2,462.14
2036	0.00	334.37	0.00	2,462.14
2037	34.38	368.75	280.95	2,743.10
2038	9.20	377.94	48.99	2,792.09
2039	45.66	423.60	328.99	3,121.08
2040	116.06	539.66	805.62	3,926.70
2041	34.93	574.59	193.83	4,120.52
2042	3.49	578.07	18.56	4,139.09
2043	6.51	584.58	34.65	4,173.74
2044	0.00	584.58	0.00	4,173.74
2045	27.54	612.11	148.85	4,322.59
2046	0.00	612.11	0.00	4,322.59
2047	2.41	614.52	13.84	4,336.43
2048	0.00	614.52	0.00	4,336.43
2049	0.00	614.52	0.00	4,336.43
2050	35.64	650.16	250.83	4,587.26
2051	0.00	650.16	0.00	4,587.26
2052	2.32	652.48	12.38	4,599.64

Table 53. Summary of baseline biomass burning emissions (CH₄ and N₂O) within the Project Area, AUD component, sum of strata

Avoided Unplanned Deforestation

Biomass Burn Emissions Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	6.07	6.07
2023	0.00	0.26	0.00	6.07
2024	0.00	0.26	0.00	6.07
2025	0.00	0.26	0.00	6.07
2026	0.00	0.26	0.00	6.07
2027	92.52	92.78	2,971.33	2,977.40
2028	75.48	168.26	2,355.71	5,333.10
2029	0.00	168.26	0.00	5,333.10
2030	0.00	168.26	0.00	5,333.10
2031	0.00	168.26	0.00	5,333.10
2032	32.85	201.11	1,157.47	6,490.58
2033	30.51	231.62	1,069.25	7,559.83
2034	99.86	331.49	3,078.93	10,638.76
2035	2.88	334.37	102.68	10,741.43
2036	0.00	334.37	0.00	10,741.43
2037	34.38	368.75	1,225.70	11,967.13
2038	9.20	377.94	213.74	12,180.87
2039	45.66	423.60	1,435.26	13,616.13
2040	116.06	539.66	3,514.63	17,130.76
2041	34.93	574.59	845.59	17,976.35
2042	3.49	578.07	80.98	18,057.33
2043	6.51	584.58	151.18	18,208.51
2044	0.00	584.58	0.00	18,208.51
2045	27.54	612.11	649.37	18,857.88
2046	0.00	612.11	0.00	18,857.88
2047	2.41	614.52	60.40	18,918.27
2048	0.00	614.52	0.00	18,918.27
2049	0.00	614.52	0.00	18,918.27
2050	35.64	650.16	1,094.30	20,012.57
2051	0.00	650.16	0.00	20,012.57
2052	2.32	652.48	54.00	20,066.57

Table 54. Summary of baseline biomass burning emissions (CH₄) within the Project Area, APD component, per stratum (1)

Avoided Planned Deforestation Biomass Burn Emissions – CH ₄ Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	132.63	132.63	3,644.59	3,644.59
2023	853.15	985.78	23,444.13	27,088.72

2024	940.51	1,926.29	25,844.60	52,933.32
2025	941.29	2,867.58	25,866.18	78,799.50
2026	225.94	3,093.52	6,208.83	85,008.33
2027	0.00	3,093.52	0.00	85,008.33
2028	0.00	3,093.52	0.00	85,008.33
2029	0.00	3,093.52	0.00	85,008.33
2030	0.00	3,093.52	0.00	85,008.33
2031	0.00	3,093.52	0.00	85,008.33
2032	0.00	3,093.52	0.00	85,008.33
2033	0.00	3,093.52	0.00	85,008.33
2034	0.00	3,093.52	0.00	85,008.33
2035	0.00	3,093.52	0.00	85,008.33
2036	0.00	3,093.52	0.00	85,008.33
2037	0.00	3,093.52	0.00	85,008.33
2038	0.00	3,093.52	0.00	85,008.33
2039	0.00	3,093.52	0.00	85,008.33
2040	0.00	3,093.52	0.00	85,008.33
2041	0.00	3,093.52	0.00	85,008.33
2042	0.00	3,093.52	0.00	85,008.33
2043	0.00	3,093.52	0.00	85,008.33
2044	0.00	3,093.52	0.00	85,008.33
2045	0.00	3,093.52	0.00	85,008.33
2046	0.00	3,093.52	0.00	85,008.33
2047	0.00	3,093.52	0.00	85,008.33
2048	0.00	3,093.52	0.00	85,008.33
2049	0.00	3,093.52	0.00	85,008.33
2050	0.00	3,093.52	0.00	85,008.33
2051	0.00	3,093.52	0.00	85,008.33
2052	0.00	3,093.52	0.00	85,008.33

Table 55 - Summary of baseline biomass burning emissions (CH₄) within the Project Area, APD component, per stratum (2)

Avoided Planned Deforestation Biomass Burn Emissions – CH ₄ Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	369.07	369.07	10,171.29	10,171.29
2023	76.82	445.88	2,117.05	12,288.34
2024	0.00	445.88	0.00	12,288.34
2025	0.00	445.88	0.00	12,288.34
2026	0.00	445.88	0.00	12,288.34

2027	0.00	445.88	0.00	12,288.34
2028	0.00	445.88	0.00	12,288.34
2029	0.00	445.88	0.00	12,288.34
2030	0.00	445.88	0.00	12,288.34
2031	0.00	445.88	0.00	12,288.34
2032	0.00	445.88	0.00	12,288.34
2033	0.00	445.88	0.00	12,288.34
2034	0.00	445.88	0.00	12,288.34
2035	0.00	445.88	0.00	12,288.34
2036	0.00	445.88	0.00	12,288.34
2037	0.00	445.88	0.00	12,288.34
2038	0.00	445.88	0.00	12,288.34
2039	0.00	445.88	0.00	12,288.34
2040	0.00	445.88	0.00	12,288.34
2041	0.00	445.88	0.00	12,288.34
2042	0.00	445.88	0.00	12,288.34
2043	0.00	445.88	0.00	12,288.34
2044	0.00	445.88	0.00	12,288.34
2045	0.00	445.88	0.00	12,288.34
2046	0.00	445.88	0.00	12,288.34
2047	0.00	445.88	0.00	12,288.34
2048	0.00	445.88	0.00	12,288.34
2049	0.00	445.88	0.00	12,288.34
2050	0.00	445.88	0.00	12,288.34
2051	0.00	445.88	0.00	12,288.34
2052	0.00	445.88	0.00	12,288.34

Table 56 - Summary of baseline biomass burning emissions (CH₄) within the Project Area, APD component, per stratum (3)

Avoided Planned Deforestation Biomass Burn Emissions – CH ₄ Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00

2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 57 - S Summary of baseline biomass burning emissions (CH₄) within the Project Area, APD component, per stratum (4)

Avoided Planned Deforestation Biomass Burn Emissions Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	8.54	8.54	152.99	152.99
2027	0.00	8.54	0.00	152.99
2028	0.00	8.54	0.00	152.99
2029	0.00	8.54	0.00	152.99
2030	0.00	8.54	0.00	152.99
2031	0.00	8.54	0.00	152.99
2032	0.00	8.54	0.00	152.99
2033	0.00	8.54	0.00	152.99
2034	0.00	8.54	0.00	152.99
2035	0.00	8.54	0.00	152.99
2036	0.00	8.54	0.00	152.99
2037	0.00	8.54	0.00	152.99
2038	0.00	8.54	0.00	152.99
2039	0.00	8.54	0.00	152.99
2040	0.00	8.54	0.00	152.99
2041	0.00	8.54	0.00	152.99
2042	0.00	8.54	0.00	152.99

2043	0.00	8.54	0.00	152.99
2044	0.00	8.54	0.00	152.99
2045	0.00	8.54	0.00	152.99
2046	0.00	8.54	0.00	152.99
2047	0.00	8.54	0.00	152.99
2048	0.00	8.54	0.00	152.99
2049	0.00	8.54	0.00	152.99
2050	0.00	8.54	0.00	152.99
2051	0.00	8.54	0.00	152.99
2052	0.00	8.54	0.00	152.99

Table 58. Summary of baseline biomass burning emissions (CH₄) within the Project Area, APD component, sum of strata

Avoided Planned Deforestation Biomass Burn Emissions - CH ₄ Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	501.70	501.70	13,815.87	13,815.87
2023	929.97	1,431.66	25,561.19	39,377.06
2024	940.51	2,372.17	25,844.60	65,221.66
2025	941.29	3,313.46	25,866.18	91,087.84
2026	234.49	3,547.95	6,361.82	97,449.65
2027	0.00	3,547.95	0.00	97,449.65
2028	0.00	3,547.95	0.00	97,449.65
2029	0.00	3,547.95	0.00	97,449.65
2030	0.00	3,547.95	0.00	97,449.65
2031	0.00	3,547.95	0.00	97,449.65
2032	0.00	3,547.95	0.00	97,449.65
2033	0.00	3,547.95	0.00	97,449.65
2034	0.00	3,547.95	0.00	97,449.65
2035	0.00	3,547.95	0.00	97,449.65
2036	0.00	3,547.95	0.00	97,449.65
2037	0.00	3,547.95	0.00	97,449.65
2038	0.00	3,547.95	0.00	97,449.65
2039	0.00	3,547.95	0.00	97,449.65
2040	0.00	3,547.95	0.00	97,449.65
2041	0.00	3,547.95	0.00	97,449.65
2042	0.00	3,547.95	0.00	97,449.65
2043	0.00	3,547.95	0.00	97,449.65
2044	0.00	3,547.95	0.00	97,449.65
2045	0.00	3,547.95	0.00	97,449.65
2046	0.00	3,547.95	0.00	97,449.65
2047	0.00	3,547.95	0.00	97,449.65
2048	0.00	3,547.95	0.00	97,449.65
2049	0.00	3,547.95	0.00	97,449.65
2050	0.00	3,547.95	0.00	97,449.65
2051	0.00	3,547.95	0.00	97,449.65
2052	0.00	3,547.95	0.00	97,449.65

Table 59. Summary of baseline biomass burning emissions (N₂O) in the Project Area, APD component, per stratum (1)

Avoided Planned Deforestation Biomass Burn Emissions - N ₂ O Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	132.63	132.63	1,083.85	1,083.85
2023	853.15	985.78	6,971.95	8,055.80
2024	940.51	1,926.29	7,685.81	15,741.61
2025	941.29	2,867.58	7,692.23	23,433.84
2026	225.94	3,093.52	1,846.42	25,280.25
2027	0.00	3,093.52	0.00	25,280.25
2028	0.00	3,093.52	0.00	25,280.25
2029	0.00	3,093.52	0.00	25,280.25
2030	0.00	3,093.52	0.00	25,280.25
2031	0.00	3,093.52	0.00	25,280.25
2032	0.00	3,093.52	0.00	25,280.25
2033	0.00	3,093.52	0.00	25,280.25
2034	0.00	3,093.52	0.00	25,280.25
2035	0.00	3,093.52	0.00	25,280.25
2036	0.00	3,093.52	0.00	25,280.25
2037	0.00	3,093.52	0.00	25,280.25
2038	0.00	3,093.52	0.00	25,280.25
2039	0.00	3,093.52	0.00	25,280.25
2040	0.00	3,093.52	0.00	25,280.25
2041	0.00	3,093.52	0.00	25,280.25
2042	0.00	3,093.52	0.00	25,280.25
2043	0.00	3,093.52	0.00	25,280.25
2044	0.00	3,093.52	0.00	25,280.25
2045	0.00	3,093.52	0.00	25,280.25
2046	0.00	3,093.52	0.00	25,280.25
2047	0.00	3,093.52	0.00	25,280.25
2048	0.00	3,093.52	0.00	25,280.25
2049	0.00	3,093.52	0.00	25,280.25
2050	0.00	3,093.52	0.00	25,280.25
2051	0.00	3,093.52	0.00	25,280.25
2052	0.00	3,093.52	0.00	25,280.25

Table 60 - Summary of baseline biomass burning emissions (N₂O) in the Project Area, APD component, per stratum (2)

Avoided Planned Deforestation Biomass Burn Emissions - N ₂ O
--

Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	369.07	369.07	3,024.79	3,024.79
2023	76.82	445.88	629.58	3,654.37
2024	0.00	445.88	0.00	3,654.37
2025	0.00	445.88	0.00	3,654.37
2026	0.00	445.88	0.00	3,654.37
2027	0.00	445.88	0.00	3,654.37
2028	0.00	445.88	0.00	3,654.37
2029	0.00	445.88	0.00	3,654.37
2030	0.00	445.88	0.00	3,654.37
2031	0.00	445.88	0.00	3,654.37
2032	0.00	445.88	0.00	3,654.37
2033	0.00	445.88	0.00	3,654.37
2034	0.00	445.88	0.00	3,654.37
2035	0.00	445.88	0.00	3,654.37
2036	0.00	445.88	0.00	3,654.37
2037	0.00	445.88	0.00	3,654.37
2038	0.00	445.88	0.00	3,654.37
2039	0.00	445.88	0.00	3,654.37
2040	0.00	445.88	0.00	3,654.37
2041	0.00	445.88	0.00	3,654.37
2042	0.00	445.88	0.00	3,654.37
2043	0.00	445.88	0.00	3,654.37
2044	0.00	445.88	0.00	3,654.37
2045	0.00	445.88	0.00	3,654.37
2046	0.00	445.88	0.00	3,654.37
2047	0.00	445.88	0.00	3,654.37
2048	0.00	445.88	0.00	3,654.37
2049	0.00	445.88	0.00	3,654.37
2050	0.00	445.88	0.00	3,654.37
2051	0.00	445.88	0.00	3,654.37
2052	0.00	445.88	0.00	3,654.37

Table 61. Summary of baseline biomass burning emissions (N₂O) in the Project Area, APD component, per stratum (3)

Avoided Planned Deforestation Biomass Burn Emissions - N ₂ O Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00

2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 62. Summary of baseline biomass burning emissions (N₂O) in the Project Area, APD component, per stratum (4)

Avoided Planned Deforestation Biomass Burn Emissions - N ₂ O Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	8.54	8.54	45.50	45.50
2027	0.00	8.54	0.00	45.50
2028	0.00	8.54	0.00	45.50
2029	0.00	8.54	0.00	45.50
2030	0.00	8.54	0.00	45.50
2031	0.00	8.54	0.00	45.50
2032	0.00	8.54	0.00	45.50
2033	0.00	8.54	0.00	45.50
2034	0.00	8.54	0.00	45.50
2035	0.00	8.54	0.00	45.50

2036	0.00	8.54	0.00	45.50
2037	0.00	8.54	0.00	45.50
2038	0.00	8.54	0.00	45.50
2039	0.00	8.54	0.00	45.50
2040	0.00	8.54	0.00	45.50
2041	0.00	8.54	0.00	45.50
2042	0.00	8.54	0.00	45.50
2043	0.00	8.54	0.00	45.50
2044	0.00	8.54	0.00	45.50
2045	0.00	8.54	0.00	45.50
2046	0.00	8.54	0.00	45.50
2047	0.00	8.54	0.00	45.50
2048	0.00	8.54	0.00	45.50
2049	0.00	8.54	0.00	45.50
2050	0.00	8.54	0.00	45.50
2051	0.00	8.54	0.00	45.50
2052	0.00	8.54	0.00	45.50

Table 63. Summary of baseline biomass burning emissions (N₂O) in the Project Area, APD component, sum of strata

Avoided Planned Deforestation Biomass Burn Emissions – N ₂ O Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	501.70	501.70	4,108.64	4,108.64
2023	929.97	1,431.66	7,601.53	11,710.17
2024	940.51	2,372.17	7,685.81	19,395.98
2025	941.29	3,313.46	7,692.23	27,088.21
2026	234.49	3,547.95	1,891.91	28,980.13
2027	0.00	3,547.95	0.00	28,980.13
2028	0.00	3,547.95	0.00	28,980.13
2029	0.00	3,547.95	0.00	28,980.13
2030	0.00	3,547.95	0.00	28,980.13
2031	0.00	3,547.95	0.00	28,980.13
2032	0.00	3,547.95	0.00	28,980.13
2033	0.00	3,547.95	0.00	28,980.13
2034	0.00	3,547.95	0.00	28,980.13
2035	0.00	3,547.95	0.00	28,980.13
2036	0.00	3,547.95	0.00	28,980.13
2037	0.00	3,547.95	0.00	28,980.13
2038	0.00	3,547.95	0.00	28,980.13
2039	0.00	3,547.95	0.00	28,980.13
2040	0.00	3,547.95	0.00	28,980.13
2041	0.00	3,547.95	0.00	28,980.13
2042	0.00	3,547.95	0.00	28,980.13
2043	0.00	3,547.95	0.00	28,980.13

2044	0.00	3,547.95	0.00	28,980.13
2045	0.00	3,547.95	0.00	28,980.13
2046	0.00	3,547.95	0.00	28,980.13
2047	0.00	3,547.95	0.00	28,980.13
2048	0.00	3,547.95	0.00	28,980.13
2049	0.00	3,547.95	0.00	28,980.13
2050	0.00	3,547.95	0.00	28,980.13
2051	0.00	3,547.95	0.00	28,980.13
2052	0.00	3,547.95	0.00	28,980.13

Table 64 – Summary of baseline biomass burning emissions (CH₄ and N₂O) within the Project Area, APD component, sum of strata

Avoided Planned Deforestation Biomass Burn Emissions Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	501.70	501.70	17,924.52	17,924.52
2023	929.97	1,431.66	33,162.71	51,087.23
2024	940.51	2,372.17	33,530.41	84,617.64
2025	941.29	3,313.46	33,558.41	118,176.05
2026	234.49	3,547.95	8,253.73	126,429.78
2027	0.00	3,547.95	0.00	126,429.78
2028	0.00	3,547.95	0.00	126,429.78
2029	0.00	3,547.95	0.00	126,429.78
2030	0.00	3,547.95	0.00	126,429.78
2031	0.00	3,547.95	0.00	126,429.78
2032	0.00	3,547.95	0.00	126,429.78
2033	0.00	3,547.95	0.00	126,429.78
2034	0.00	3,547.95	0.00	126,429.78
2035	0.00	3,547.95	0.00	126,429.78
2036	0.00	3,547.95	0.00	126,429.78
2037	0.00	3,547.95	0.00	126,429.78
2038	0.00	3,547.95	0.00	126,429.78
2039	0.00	3,547.95	0.00	126,429.78
2040	0.00	3,547.95	0.00	126,429.78
2041	0.00	3,547.95	0.00	126,429.78
2042	0.00	3,547.95	0.00	126,429.78
2043	0.00	3,547.95	0.00	126,429.78
2044	0.00	3,547.95	0.00	126,429.78
2045	0.00	3,547.95	0.00	126,429.78
2046	0.00	3,547.95	0.00	126,429.78
2047	0.00	3,547.95	0.00	126,429.78
2048	0.00	3,547.95	0.00	126,429.78
2049	0.00	3,547.95	0.00	126,429.78
2050	0.00	3,547.95	0.00	126,429.78
2051	0.00	3,547.95	0.00	126,429.78

2052	0.00	3,547.95	0.00	126,429.78
------	------	----------	------	------------

Wood products carbon pool in the baseline

To estimate the biomass carbon of the commercial volume extracted in the process of deforestation, the following equation was applied, which is applicable both for IPOÁ's APD and AUD components, according to "Option 2: Commercial inventory estimation", as recommended in the CP-W:

$$C_{XB,i} = C_{ABtree,i} * \frac{1}{BCEF} * Pcom_i$$

Where:

$C_{XB,i}$	Mean stock of extracted biomass carbon from stratum i ; t CO ₂ -e ha ⁻¹
$C_{ABtree,i}$	Mean aboveground biomass carbon stock in stratum i ; tCO ₂ -e ha ⁻¹
$BCEF$	Biomass conversion and expansion factor (BCEF) for conversion of merchantable volume to total aboveground tree biomass; dimensionless
$Pcom_i$	Commercial volume as a percent of total aboveground volume in stratum i ; dimensionless
i	1,2,3, ... M strata

For calculation of P_{COMi} parameter, the following parameters and sources were used: volume of merchantable wood in exploitation (35.08 m³/ha) from FSM_REDD_MR2_04_03_2021¹⁴⁶; Dmm value (0.59

¹⁴⁶https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashoufifmIn902309ksdfiku098l71896923

t d.m.m⁻³) from Brown et al 1989; above ground carbon stock from MCTI 2015¹⁴⁷; CF value (0.47 t Ct-1 d.m) from IPCC 2006 V4_04_Ch4_Forest_Land¹⁴⁸. For the BCEF parameter, the average of the three proposed factors in CP-AB¹⁴⁹ was applied.

To calculate the biomass carbon entering the wood products pool at the time of deforestation, the following equation was applied.

$$C_{WP,i} = \sum_{ty=s,w,oil,p,o} C_{XB,ty,i} * (1 - WW_{ty})$$

Where:

$C_{WP,i}$	Carbon stock entering the wood products pool from stratum i; t CO ₂ -e ha ⁻¹
$C_{XB,ty,i}$	Mean stock of extracted biomass carbon by class of wood product ty from stratum i; t CO ₂ -e ha ⁻¹
WW_{ty}	Wood waste. The fraction immediately emitted through mill inefficiency by class of wood product ty; dimensionless (0.24 for developing countries; Winjum et al. 1998 cited by CP-W)
Ty	'Wood product class – defined here as sawnwood (s)
i	1,2, 3,... M strata

To calculate the amount of wood products entering the pool at the time of deforestation (CWP,i, calculated in C-WP) that is expected to be emitted over a 100-year timeframe, the following equation was applied.

¹⁴⁷http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

¹⁴⁸https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

¹⁴⁹<https://verra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf>

$$C_{WP100,i} = C_{WP,i} - C_{WP,i} * (1 - SLF_p) * (1 - OF_p)$$

Where:

$C_{WP100,i}$ Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i ; t CO₂-e ha⁻¹

$C_{WP,i}$ Carbon stock entering wood products pool at time of deforestation from stratum i ; t CO₂-e ha⁻¹

SLF_{ty} Fraction of wood products that will be emitted to the atmosphere within 5 years of timber harvest by class of wood product ty ; dimensionless (0.2 for sawnwood; Winjum et al. 1998 cited by CP-W)

OF_{ty} Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty ; dimensionless (0.80 for sawnwood in tropical forests; Winjum et al. 1998 cited by CP-W)

ty 'Wood product class – defined here as sawnwood (s)

i 1,2, 3,... M strata

The parameters used in calculation of wood products carbon pool in the baseline, as well as the results of estimates (sum of strata), are demonstrated in the table below, for the whole project period.

Table 65 - Parameters used to calculate of wood products carbon pool for the baseline scenario.

Parameter	Value	Unit
Cab Ds	515.86	t C m ⁻³
Cab Db	517.37	t C m ⁻³
Cab Da	587.33	t C m ⁻³
Cab Pa	336.23	t C m ⁻³

PCOM Ds	0.07	Dimensionless
PCOM Db	0.07	Dimensionless
PCOMDa	0.06	Dimensionless
PCOM Pa	0.11	Dimensionless
BCEF	1.32	Dimensionless
WWp	0.24	Dimensionless
SLFp	0.20	Dimensionless
Ofp	0.80	Dimensionless

Table 66. Summary of baseline wood products carbon pool from the Project Area, AUD component, per stratum (1)

Avoided Unplanned Deforestation Wood products carbon pool in baseline Dense Tropical Submontane Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00	0.00
2027	66.17	66.17	1,358.87	1,358.87	1,141.45
2028	48.47	114.64	995.45	2,354.32	836.17
2029	0.00	114.64	0.00	2,354.32	0.00
2030	0.00	114.64	0.00	2,354.32	0.00
2031	0.00	114.64	0.00	2,354.32	0.00
2032	31.75	146.39	651.99	3,006.31	547.67
2033	29.02	175.41	596.00	3,602.31	500.64
2034	61.09	236.50	1,254.50	4,856.81	1,053.78
2035	2.88	239.38	59.14	4,915.95	49.68
2036	0.00	239.38	0.00	4,915.95	0.00
2037	34.38	273.76	706.04	5,621.99	593.07
2038	0.00	273.76	0.00	5,621.99	0.00
2039	30.15	303.91	619.17	6,241.15	520.10
2040	65.88	369.79	1,352.87	7,594.03	1,136.41
2041	2.74	372.52	56.19	7,650.22	47.20
2042	0.00	372.52	0.00	7,650.22	0.00
2043	0.00	372.52	0.00	7,650.22	0.00

2044	0.00	372.52	0.00	7,650.22	0.00
2045	0.76	373.29	15.69	7,665.91	13.18
2046	0.00	373.29	0.00	7,665.91	0.00
2047	0.36	373.65	7.39	7,673.30	6.21
2048	0.00	373.65	0.00	7,673.30	0.00
2049	0.00	373.65	0.00	7,673.30	0.00
2050	21.44	395.08	440.23	8,113.53	369.80
2051	0.00	395.08	0.00	8,113.53	0.00
2052	0.00	395.08	0.00	8,113.53	0.00

Table 67. Summary of baseline wood products carbon pool from the Project Area, AUD component, per stratum (2)

Avoided Unplanned Deforestation Wood products carbon pool in baseline Dense Tropical Lowland Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00	0.00

2045	0.00	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00	0.00

Table 68. Summary of baseline wood products carbon pool from the Project Area, AUD component, per stratum (3)

Avoided Unplanned Deforestation Wood products carbon pool in baseline Dense Tropical Alluvial Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00	0.00

2045	0.00	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00	0.00

Table 69. Summary of baseline wood products carbon pool from the Project Area, APD component, per stratum (4)

Avoided Unplanned Deforestation Wood products carbon pool in baseline Pioneer Formation with fluvial influence without palm trees					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.26	0.26	5.37	5.37	4.51
2023	0.00	0.26	0.00	5.37	0.00
2024	0.00	0.26	0.00	5.37	0.00
2025	0.00	0.26	0.00	5.37	0.00
2026	0.00	0.26	0.00	5.37	0.00
2027	26.35	26.61	541.12	546.48	454.54
2028	27.01	53.62	554.64	1,101.12	465.90
2029	0.00	53.62	0.00	1,101.12	0.00
2030	0.00	53.62	0.00	1,101.12	0.00
2031	0.00	53.62	0.00	1,101.12	0.00
2032	1.10	54.72	22.62	1,123.75	19.00
2033	1.49	56.21	30.56	1,154.30	25.67
2034	38.78	94.99	796.35	1,950.65	668.93
2035	0.00	94.99	0.00	1,950.65	0.00
2036	0.00	94.99	0.00	1,950.65	0.00
2037	0.00	94.99	0.00	1,950.65	0.00
2038	9.20	104.18	188.90	2,139.55	158.67
2039	15.51	119.69	318.48	2,458.03	267.52
2040	50.18	169.87	1,030.48	3,488.51	865.60
2041	32.19	202.06	661.09	4,149.60	555.32
2042	3.49	205.55	71.57	4,221.17	60.12
2043	6.51	212.05	133.61	4,354.78	112.23
2044	0.00	212.05	0.00	4,354.78	0.00
2045	26.77	238.83	549.82	4,904.60	461.85

2046	0.00	238.83	0.00	4,904.60	0.00
2047	2.05	240.87	42.03	4,946.63	35.31
2048	0.00	240.87	0.00	4,946.63	0.00
2049	0.00	240.87	0.00	4,946.63	0.00
2050	14.20	255.08	291.68	5,238.31	245.01
2051	0.00	255.08	0.00	5,238.31	0.00
2052	2.32	257.40	47.72	5,286.03	40.09

Table 70. Summary of baseline wood products carbon pool from the Project Area, AUD component, sum of strata

Avoided Unplanned Deforestation Wood products carbon pool in baseline Sum of strata					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.26	0.26	5.37	5.37	4.51
2023	0.00	0.26	0.00	5.37	0.00
2024	0.00	0.26	0.00	5.37	0.00
2025	0.00	0.26	0.00	5.37	0.00
2026	0.00	0.26	0.00	5.37	0.00
2027	92.52	92.78	1,899.99	1,905.36	1,595.99
2028	75.48	168.26	1,550.08	3,455.44	1,302.07
2029	0.00	168.26	0.00	3,455.44	0.00
2030	0.00	168.26	0.00	3,455.44	0.00
2031	0.00	168.26	0.00	3,455.44	0.00
2032	32.85	201.11	674.61	4,130.05	566.68
2033	30.51	231.62	626.56	4,756.62	526.31
2034	99.86	331.49	2,050.85	6,807.46	1,722.71
2035	2.88	334.37	59.14	6,866.60	49.68
2036	0.00	334.37	0.00	6,866.60	0.00
2037	34.38	368.75	706.04	7,572.64	593.07
2038	9.20	377.94	188.90	7,761.54	158.67
2039	45.66	423.60	937.65	8,699.18	787.62
2040	116.06	539.66	2,383.35	11,082.53	2,002.01
2041	34.93	574.59	717.29	11,799.82	602.52
2042	3.49	578.07	71.57	11,871.39	60.12
2043	6.51	584.58	133.61	12,004.99	112.23
2044	0.00	584.58	0.00	12,004.99	0.00
2045	27.54	612.11	565.51	12,570.50	475.03
2046	0.00	612.11	0.00	12,570.50	0.00

2047	2.41	614.52	49.43	12,619.93	41.52
2048	0.00	614.52	0.00	12,619.93	0.00
2049	0.00	614.52	0.00	12,619.93	0.00
2050	35.64	650.16	731.91	13,351.84	614.81
2051	0.00	650.16	0.00	13,351.84	0.00
2052	2.32	652.48	47.72	13,399.56	40.09

Table 71. Summary of baseline wood products carbon pool from the Project Area, APD component, per stratum (1)

Avoided Planned Deforestation Wood products carbon pool in baseline Dense Tropical Submontane Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	132.63	132.63	2,723.71	2,723.71	2,287.92
2023	853.15	985.78	17,520.51	20,244.22	14,717.23
2024	940.51	1,926.29	19,314.45	39,558.66	16,224.13
2025	941.29	2,867.58	19,330.58	58,889.24	16,237.68
2026	225.94	3,093.52	4,640.05	63,529.28	3,897.64
2027	0.00	3,093.52	0.00	63,529.28	0.00
2028	0.00	3,093.52	0.00	63,529.28	0.00
2029	0.00	3,093.52	0.00	63,529.28	0.00
2030	0.00	3,093.52	0.00	63,529.28	0.00
2031	0.00	3,093.52	0.00	63,529.28	0.00
2032	0.00	3,093.52	0.00	63,529.28	0.00
2033	0.00	3,093.52	0.00	63,529.28	0.00
2034	0.00	3,093.52	0.00	63,529.28	0.00
2035	0.00	3,093.52	0.00	63,529.28	0.00
2036	0.00	3,093.52	0.00	63,529.28	0.00
2037	0.00	3,093.52	0.00	63,529.28	0.00
2038	0.00	3,093.52	0.00	63,529.28	0.00
2039	0.00	3,093.52	0.00	63,529.28	0.00
2040	0.00	3,093.52	0.00	63,529.28	0.00
2041	0.00	3,093.52	0.00	63,529.28	0.00
2042	0.00	3,093.52	0.00	63,529.28	0.00
2043	0.00	3,093.52	0.00	63,529.28	0.00
2044	0.00	3,093.52	0.00	63,529.28	0.00
2045	0.00	3,093.52	0.00	63,529.28	0.00
2046	0.00	3,093.52	0.00	63,529.28	0.00
2047	0.00	3,093.52	0.00	63,529.28	0.00

2048	0.00	3,093.52	0.00	63,529.28	0.00
2049	0.00	3,093.52	0.00	63,529.28	0.00
2050	0.00	3,093.52	0.00	63,529.28	0.00
2051	0.00	3,093.52	0.00	63,529.28	0.00
2052	0.00	3,093.52	0.00	63,529.28	0.00

Table 72. Summary of baseline wood products carbon pool from the Project Area, APD component, per stratum (2)

Avoided Planned Deforestation Wood products carbon pool in baseline Dense Tropical Lowland Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	369.07	369.07	7,579.22	7,579.22	6,366.55
2023	76.82	445.88	1,577.54	9,156.76	1,325.13
2024	0.00	445.88	0.00	9,156.76	0.00
2025	0.00	445.88	0.00	9,156.76	0.00
2026	0.00	445.88	0.00	9,156.76	0.00
2027	0.00	445.88	0.00	9,156.76	0.00
2028	0.00	445.88	0.00	9,156.76	0.00
2029	0.00	445.88	0.00	9,156.76	0.00
2030	0.00	445.88	0.00	9,156.76	0.00
2031	0.00	445.88	0.00	9,156.76	0.00
2032	0.00	445.88	0.00	9,156.76	0.00
2033	0.00	445.88	0.00	9,156.76	0.00
2034	0.00	445.88	0.00	9,156.76	0.00
2035	0.00	445.88	0.00	9,156.76	0.00
2036	0.00	445.88	0.00	9,156.76	0.00
2037	0.00	445.88	0.00	9,156.76	0.00
2038	0.00	445.88	0.00	9,156.76	0.00
2039	0.00	445.88	0.00	9,156.76	0.00
2040	0.00	445.88	0.00	9,156.76	0.00
2041	0.00	445.88	0.00	9,156.76	0.00
2042	0.00	445.88	0.00	9,156.76	0.00
2043	0.00	445.88	0.00	9,156.76	0.00
2044	0.00	445.88	0.00	9,156.76	0.00
2045	0.00	445.88	0.00	9,156.76	0.00
2046	0.00	445.88	0.00	9,156.76	0.00
2047	0.00	445.88	0.00	9,156.76	0.00
2048	0.00	445.88	0.00	9,156.76	0.00
2049	0.00	445.88	0.00	9,156.76	0.00
2050	0.00	445.88	0.00	9,156.76	0.00
2051	0.00	445.88	0.00	9,156.76	0.00
2052	0.00	445.88	0.00	9,156.76	0.00

Table 73. Summary of baseline wood products carbon pool from the Project Area, APD component, per stratum (3)

Avoided Planned Deforestation Wood products carbon pool in baseline Dense Tropical Alluvial Rainforest					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00	0.00

Table 74. Summary of baseline wood products carbon pool from the Project Area, APD component, per stratum (4)

Avoided Planned Deforestation Wood products carbon pool in baseline					
--	--	--	--	--	--

Pioneer Formation with fluvial influence without palm trees					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	0.00	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00	0.00
2026	8.54	8.54	175.41	175.41	147.35
2027	0.00	8.54	0.00	175.41	0.00
2028	0.00	8.54	0.00	175.41	0.00
2029	0.00	8.54	0.00	175.41	0.00
2030	0.00	8.54	0.00	175.41	0.00
2031	0.00	8.54	0.00	175.41	0.00
2032	0.00	8.54	0.00	175.41	0.00
2033	0.00	8.54	0.00	175.41	0.00
2034	0.00	8.54	0.00	175.41	0.00
2035	0.00	8.54	0.00	175.41	0.00
2036	0.00	8.54	0.00	175.41	0.00
2037	0.00	8.54	0.00	175.41	0.00
2038	0.00	8.54	0.00	175.41	0.00
2039	0.00	8.54	0.00	175.41	0.00
2040	0.00	8.54	0.00	175.41	0.00
2041	0.00	8.54	0.00	175.41	0.00
2042	0.00	8.54	0.00	175.41	0.00
2043	0.00	8.54	0.00	175.41	0.00
2044	0.00	8.54	0.00	175.41	0.00
2045	0.00	8.54	0.00	175.41	0.00
2046	0.00	8.54	0.00	175.41	0.00
2047	0.00	8.54	0.00	175.41	0.00
2048	0.00	8.54	0.00	175.41	0.00
2049	0.00	8.54	0.00	175.41	0.00
2050	0.00	8.54	0.00	175.41	0.00
2051	0.00	8.54	0.00	175.41	0.00
2052	0.00	8.54	0.00	175.41	0.00

Table 75. Summary of baseline wood products from the Project Area, APD component, sum of strata

Avoided Planned Deforestation Wood products carbon pool in baseline Sum of strata					
Year	ha/year	ha (cumulative)	CWP tCO ₂ /year	CWP tCO ₂ (cumulative)	CWP ₁₀₀ tCO ₂
2022	501.70	501.70	10,302.93	10,302.93	8,654.46
2023	929.97	1,431.66	19,098.05	29,400.98	16,042.36
2024	940.51	2,372.17	19,314.45	48,715.42	16,224.13
2025	941.29	3,313.46	19,330.58	68,046.00	16,237.68
2026	234.49	3,547.95	4,815.46	72,861.46	4,044.98
2027	0.00	3,547.95	0.00	72,861.46	0.00
2028	0.00	3,547.95	0.00	72,861.46	0.00

2029	0.00	3,547.95	0.00	72,861.46	0.00
2030	0.00	3,547.95	0.00	72,861.46	0.00
2031	0.00	3,547.95	0.00	72,861.46	0.00
2032	0.00	3,547.95	0.00	72,861.46	0.00
2033	0.00	3,547.95	0.00	72,861.46	0.00
2034	0.00	3,547.95	0.00	72,861.46	0.00
2035	0.00	3,547.95	0.00	72,861.46	0.00
2036	0.00	3,547.95	0.00	72,861.46	0.00
2037	0.00	3,547.95	0.00	72,861.46	0.00
2038	0.00	3,547.95	0.00	72,861.46	0.00
2039	0.00	3,547.95	0.00	72,861.46	0.00
2040	0.00	3,547.95	0.00	72,861.46	0.00
2041	0.00	3,547.95	0.00	72,861.46	0.00
2042	0.00	3,547.95	0.00	72,861.46	0.00
2043	0.00	3,547.95	0.00	72,861.46	0.00
2044	0.00	3,547.95	0.00	72,861.46	0.00
2045	0.00	3,547.95	0.00	72,861.46	0.00
2046	0.00	3,547.95	0.00	72,861.46	0.00
2047	0.00	3,547.95	0.00	72,861.46	0.00
2048	0.00	3,547.95	0.00	72,861.46	0.00
2049	0.00	3,547.95	0.00	72,861.46	0.00
2050	0.00	3,547.95	0.00	72,861.46	0.00
2051	0.00	3,547.95	0.00	72,861.46	0.00
2052	0.00	3,547.95	0.00	72,861.46	0.00

Pasture carbon pools in the baseline

Regarding calculation of the carbon pool remaining in pasturelands after deforestation, a conservative value from the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015) of 27.8 tCO₂/ha ($CAB_{non-tree,post,i} + CBB_{non-tree,post,i}$) was applied¹⁵⁰ for both AUD and APD components. The proportion of baseline deforestation converted to pasture was defined as 100%. The table below summarizes the results obtained for pasture carbon pools in the baseline scenario, over the 30 years of the project lifetime:

Table 76. Summary of baseline pasture carbon pool in the Project Area, AUD component, per stratum (1)

Avoided Unplanned Deforestation Annual baseline pasture carbon pool for the Project Area Dense Tropical Submontane Rainforest

¹⁵⁰ Source: http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf (Table 5, is the value of 7.57 tC/ha, for the Amazon biome, multiplied by the 44/12 conversion factor)

Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	66.17	66.17	1,836.65	1,836.65
2028	48.47	114.64	1,345.44	3,182.09
2029	0.00	114.64	0.00	3,182.09
2030	0.00	114.64	0.00	3,182.09
2031	0.00	114.64	0.00	3,182.09
2032	31.75	146.39	881.23	4,063.31
2033	29.02	175.41	805.55	4,868.87
2034	61.09	236.50	1,695.57	6,564.44
2035	2.88	239.38	79.94	6,644.38
2036	0.00	239.38	0.00	6,644.38
2037	34.38	273.76	954.27	7,598.65
2038	0.00	273.76	0.00	7,598.65
2039	30.15	303.91	836.86	8,435.52
2040	65.88	369.79	1,828.53	10,264.05
2041	2.74	372.52	75.95	10,340.00
2042	0.00	372.52	0.00	10,340.00
2043	0.00	372.52	0.00	10,340.00
2044	0.00	372.52	0.00	10,340.00
2045	0.76	373.29	21.20	10,361.21
2046	0.00	373.29	0.00	10,361.21
2047	0.36	373.65	9.99	10,371.20
2048	0.00	373.65	0.00	10,371.20
2049	0.00	373.65	0.00	10,371.20
2050	21.44	395.08	595.02	10,966.22
2051	0.00	395.08	0.00	10,966.22
2052	0.00	395.08	0.00	10,966.22

Table 77. Summary of baseline pasture carbon pool in the Project Area, AUD component, per stratum (2)

Avoided Unplanned Deforestation Annual baseline pasture carbonpool for the Project Area Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00

2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 78. Summary of baseline pasture carbon pool in the Project Area, AUD component, per stratum (3)

Avoided Unplanned Deforestation Annual baseline pasture carbonpool for the Project Area Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00

2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 79. Summary of baseline pasture carbon pool in the Project Area, AUD component, per stratum (4)

Avoided Unplanned Deforestation Annual baseline pasture carbonpool for the Project Area Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.26	0.26	7.25	7.25
2023	0.00	0.26	0.00	7.25
2024	0.00	0.26	0.00	7.25
2025	0.00	0.26	0.00	7.25
2026	0.00	0.26	0.00	7.25
2027	26.35	26.61	731.37	738.62
2028	27.01	53.62	749.65	1,488.27
2029	0.00	53.62	0.00	1,488.27
2030	0.00	53.62	0.00	1,488.27
2031	0.00	53.62	0.00	1,488.27
2032	1.10	54.72	30.58	1,518.85
2033	1.49	56.21	41.30	1,560.15
2034	38.78	94.99	1,076.34	2,636.49
2035	0.00	94.99	0.00	2,636.49
2036	0.00	94.99	0.00	2,636.49
2037	0.00	94.99	0.00	2,636.49
2038	9.20	104.18	255.31	2,891.80
2039	15.51	119.69	430.46	3,322.26
2040	50.18	169.87	1,392.79	4,715.05
2041	32.19	202.06	893.53	5,608.58
2042	3.49	205.55	96.73	5,705.31
2043	6.51	212.05	180.58	5,885.90
2044	0.00	212.05	0.00	5,885.90
2045	26.77	238.83	743.13	6,629.03

2046	0.00	238.83	0.00	6,629.03
2047	2.05	240.87	56.81	6,685.84
2048	0.00	240.87	0.00	6,685.84
2049	0.00	240.87	0.00	6,685.84
2050	14.20	255.08	394.23	7,080.07
2051	0.00	255.08	0.00	7,080.07
2052	2.32	257.40	64.50	7,144.57

Table 80. Summary of baseline pasture carbon pool in the Project Area, AUD component, sum of strata

Avoided Unplanned Deforestation Annual baseline pasture carbonpool for the Project Area Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.26	0.26	7.25	7.25
2023	0.00	0.26	0.00	7.25
2024	0.00	0.26	0.00	7.25
2025	0.00	0.26	0.00	7.25
2026	0.00	0.26	0.00	7.25
2027	92.52	92.78	2,568.02	2,575.27
2028	75.48	168.26	2,095.09	4,670.36
2029	0.00	168.26	0.00	4,670.36
2030	0.00	168.26	0.00	4,670.36
2031	0.00	168.26	0.00	4,670.36
2032	32.85	201.11	911.81	5,582.16
2033	30.51	231.62	846.86	6,429.02
2034	99.86	331.49	2,771.91	9,200.93
2035	2.88	334.37	79.94	9,280.87
2036	0.00	334.37	0.00	9,280.87
2037	34.38	368.75	954.27	10,235.15
2038	9.20	377.94	255.31	10,490.46
2039	45.66	423.60	1,267.32	11,757.78
2040	116.06	539.66	3,221.32	14,979.10
2041	34.93	574.59	969.48	15,948.58
2042	3.49	578.07	96.73	16,045.32
2043	6.51	584.58	180.58	16,225.90
2044	0.00	584.58	0.00	16,225.90
2045	27.54	612.11	764.34	16,990.24
2046	0.00	612.11	0.00	16,990.24
2047	2.41	614.52	66.80	17,057.04
2048	0.00	614.52	0.00	17,057.04
2049	0.00	614.52	0.00	17,057.04
2050	35.64	650.16	989.25	18,046.29
2051	0.00	650.16	0.00	18,046.29
2052	2.32	652.48	64.50	18,110.79

Table 81. Summary of baseline pasture carbon pool in the Project Area, APD component, per stratum (1)

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	132.63	132.63	3,681.36	3,681.36
2023	853.15	985.78	23,680.64	27,362.00
2024	940.51	1,926.29	26,105.32	53,467.32
2025	941.29	2,867.58	26,127.12	79,594.44
2026	225.94	3,093.52	6,271.46	85,865.90
2027	0.00	3,093.52	0.00	85,865.90
2028	0.00	3,093.52	0.00	85,865.90
2029	0.00	3,093.52	0.00	85,865.90
2030	0.00	3,093.52	0.00	85,865.90
2031	0.00	3,093.52	0.00	85,865.90
2032	0.00	3,093.52	0.00	85,865.90
2033	0.00	3,093.52	0.00	85,865.90
2034	0.00	3,093.52	0.00	85,865.90
2035	0.00	3,093.52	0.00	85,865.90
2036	0.00	3,093.52	0.00	85,865.90
2037	0.00	3,093.52	0.00	85,865.90
2038	0.00	3,093.52	0.00	85,865.90
2039	0.00	3,093.52	0.00	85,865.90
2040	0.00	3,093.52	0.00	85,865.90
2041	0.00	3,093.52	0.00	85,865.90
2042	0.00	3,093.52	0.00	85,865.90
2043	0.00	3,093.52	0.00	85,865.90
2044	0.00	3,093.52	0.00	85,865.90
2045	0.00	3,093.52	0.00	85,865.90
2046	0.00	3,093.52	0.00	85,865.90
2047	0.00	3,093.52	0.00	85,865.90
2048	0.00	3,093.52	0.00	85,865.90
2049	0.00	3,093.52	0.00	85,865.90
2050	0.00	3,093.52	0.00	85,865.90
2051	0.00	3,093.52	0.00	85,865.90
2052	0.00	3,093.52	0.00	85,865.90

Table 82. Summary of baseline pasture carbon pool in the Project Area, APD component, per stratum (2)

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	369.07	369.07	10,244.04	10,244.04

2023	76.82	445.88	2,132.20	12,376.24
2024	0.00	445.88	0.00	12,376.24
2025	0.00	445.88	0.00	12,376.24
2026	0.00	445.88	0.00	12,376.24
2027	0.00	445.88	0.00	12,376.24
2028	0.00	445.88	0.00	12,376.24
2029	0.00	445.88	0.00	12,376.24
2030	0.00	445.88	0.00	12,376.24
2031	0.00	445.88	0.00	12,376.24
2032	0.00	445.88	0.00	12,376.24
2033	0.00	445.88	0.00	12,376.24
2034	0.00	445.88	0.00	12,376.24
2035	0.00	445.88	0.00	12,376.24
2036	0.00	445.88	0.00	12,376.24
2037	0.00	445.88	0.00	12,376.24
2038	0.00	445.88	0.00	12,376.24
2039	0.00	445.88	0.00	12,376.24
2040	0.00	445.88	0.00	12,376.24
2041	0.00	445.88	0.00	12,376.24
2042	0.00	445.88	0.00	12,376.24
2043	0.00	445.88	0.00	12,376.24
2044	0.00	445.88	0.00	12,376.24
2045	0.00	445.88	0.00	12,376.24
2046	0.00	445.88	0.00	12,376.24
2047	0.00	445.88	0.00	12,376.24
2048	0.00	445.88	0.00	12,376.24
2049	0.00	445.88	0.00	12,376.24
2050	0.00	445.88	0.00	12,376.24
2051	0.00	445.88	0.00	12,376.24
2052	0.00	445.88	0.00	12,376.24

Table 83. Summary of baseline pasture carbon pool in the Project Area APD component, per stratum (3)

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00

2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 84. Summary of baseline pasture carbon pool in the Project Area, APD component, per stratum (4)

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	8.54	8.54	237.09	237.09
2027	0.00	8.54	0.00	237.09
2028	0.00	8.54	0.00	237.09
2029	0.00	8.54	0.00	237.09
2030	0.00	8.54	0.00	237.09
2031	0.00	8.54	0.00	237.09
2032	0.00	8.54	0.00	237.09
2033	0.00	8.54	0.00	237.09
2034	0.00	8.54	0.00	237.09
2035	0.00	8.54	0.00	237.09
2036	0.00	8.54	0.00	237.09
2037	0.00	8.54	0.00	237.09
2038	0.00	8.54	0.00	237.09
2039	0.00	8.54	0.00	237.09
2040	0.00	8.54	0.00	237.09

2041	0.00	8.54	0.00	237.09
2042	0.00	8.54	0.00	237.09
2043	0.00	8.54	0.00	237.09
2044	0.00	8.54	0.00	237.09
2045	0.00	8.54	0.00	237.09
2046	0.00	8.54	0.00	237.09
2047	0.00	8.54	0.00	237.09
2048	0.00	8.54	0.00	237.09
2049	0.00	8.54	0.00	237.09
2050	0.00	8.54	0.00	237.09
2051	0.00	8.54	0.00	237.09
2052	0.00	8.54	0.00	237.09

Table 85. Summary of baseline pasture carbon pool in the Project Area, APD component, sum of strata

Avoided Planned Deforestation Annual baseline pasture carbonpool for the Project Area Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂ (cumulative)
2022	501.70	501.70	13,925.40	13,925.40
2023	929.97	1,431.66	25,812.84	39,738.23
2024	940.51	2,372.17	26,105.32	65,843.55
2025	941.29	3,313.46	26,127.12	91,970.67
2026	234.49	3,547.95	6,508.55	98,479.22
2027	0.00	3,547.95	0.00	98,479.22
2028	0.00	3,547.95	0.00	98,479.22
2029	0.00	3,547.95	0.00	98,479.22
2030	0.00	3,547.95	0.00	98,479.22
2031	0.00	3,547.95	0.00	98,479.22
2032	0.00	3,547.95	0.00	98,479.22
2033	0.00	3,547.95	0.00	98,479.22
2034	0.00	3,547.95	0.00	98,479.22
2035	0.00	3,547.95	0.00	98,479.22
2036	0.00	3,547.95	0.00	98,479.22
2037	0.00	3,547.95	0.00	98,479.22
2038	0.00	3,547.95	0.00	98,479.22
2039	0.00	3,547.95	0.00	98,479.22
2040	0.00	3,547.95	0.00	98,479.22
2041	0.00	3,547.95	0.00	98,479.22
2042	0.00	3,547.95	0.00	98,479.22
2043	0.00	3,547.95	0.00	98,479.22
2044	0.00	3,547.95	0.00	98,479.22
2045	0.00	3,547.95	0.00	98,479.22
2046	0.00	3,547.95	0.00	98,479.22
2047	0.00	3,547.95	0.00	98,479.22
2048	0.00	3,547.95	0.00	98,479.22
2049	0.00	3,547.95	0.00	98,479.22
2050	0.00	3,547.95	0.00	98,479.22
2051	0.00	3,547.95	0.00	98,479.22

2052	0.00	3,547.95	0.00	98,479.22
------	------	----------	------	-----------

Uncertainty

The X-UNC Estimation of Uncertainty for REDD+ Project Activities module (VMD017) assesses the following sources of uncertainty applicable to this project: i) Uncertainty in Projection of Baseline Rate of Deforestation or Degradation; ii): Uncertainty of Emissions and Removals in Project Area in Baseline Scenario.

Uncertainty in Projection of Baseline Rate of Deforestation

AUD component: It is assumed that there is zero uncertainty in baseline rate of deforestation, as numbers are equal to a long-term average (BL-UP; which is the case in this project, where deforestation rate was taken as the average of the reference period). The regression analysis was not applied because it was statistically insignificant, once $R^2 = 0.17$. Thus, average deforested area was used, in hectares per year, in the historical reference period.

APD component: It is assumed that there is zero uncertainty in baseline rate of deforestation, as number are based on actual deforestation plan (BL-PL; which is the case in this project, as a planned deforestation plan is available).

Uncertainty of Emissions and Removals in Project Area in Baseline Scenario

Initial forest biomass data were obtained from the National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015). To calculate the uncertainty associated with total biomass with average values by phytophysiology of the Amazon, the methodology from the IPCC's Good Practice Guidance and Uncertainty Management in National Greenhouse Gas Inventories (2000) was applied in the National GHG Inventory. According to this document, uncertainties must have a confidence level of 95%.

Uncertainty across combined strata was assessed as follows:

$$Uncertainty_{REDD-BSL,SS} = \frac{\sqrt{\sum_{i=1}^M (U_{REDD-BSL,SS,i} \times E_{REDD-BSL,SS,i})^2}}{\sum_{i=1}^M E_{REDD-BSL,SS,i}}$$

Where:

$Uncertainty_{REDD-BSL,SS}$ Total uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario (%)

$U_{REDD-BSL,SS,i}$ Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in stratum i in the REDD baseline scenario (%)

$E_{REDD-BSL,SS,t,i}$ Sum of combined carbon stocks and GHG sources in the REDD baseline scenario (t CO₂-e)

i 1, 2, 3, ... M strata

Three different uncertainty levels ($U_{REDD-BSL,SS,i}$) were identified by the National Inventory is verified within the Project Area: 3.38%, 3.97% and 5.02%. For the $E_{REDD-BSL,SS,t,i}$ parameter, the weighted average of carbon stock from the different strata was used.

The result is 2.41% of uncertainty. Given that uncertainty was estimated to be under 15% for all forest types, no deduction resulted for uncertainty, as the allowable uncertainty in this methodology REDD+ MF is +/- 15% at a 95% confidence level.

3.2.2 Project Emissions

Given that some deforestation may continue to occur within the project boundary in the project case, despite the project's conservation activities, an estimate of the total net GHG emissions reductions due to unplanned deforestation continuing in the with-project scenario was made.

Regarding project emissions, the VM0007 Methodology Framework provides the following guidelines: "The project proponent must present conservative ex-ante estimations of the total net GHG emissions reductions of the project activity. For ex-ante estimations of specific parameters, refer to the parameter tables in the appropriate modules." In accordance with the latter indication, module VMD0015 MREDD, step 2, was applied to calculate ex-ante GHG emission reduction and unplanned deforestation, and a 10% ex-ante estimate for deforestation and GHG emissions reductions in the project scenario.

The assumptions adopted were that the surveillance activities will be able to avoid unplanned deforestation in the Project Area at a 90% effectiveness rate. This is considered to be conservative, given the project's comprehensive deforestation control measures described in the PD's Theory of Change climate scope (see section 6.2) and within the Forest Integrity Plan (FIP). In addition, data regarding monitored project emissions of Carbonext's active REDD projects in the south of Amazonas were collected (Ituxi, Evergreen and Unitor REDD+ Projects), which validate that on average project emissions range between 0 and 5% of AUD baselines. Therefore, Carbonext conservatively adopted a 10% ex-ante estimate of deforestation in the with-project case.

The calculation of net change in carbon stock as a result of with-project deforestation in the project area was made according to the formula below, from the VMD0015 MREDD module. The following tables summarize the results obtained for the total net emissions in the carbon stock in the project area.

$$\Delta C_{P, DefPA, i, t} = \sum_{u=1}^U \left(A_{DefPA, u, i, t} * \Delta C_{pools, F, DefPA, u, i, t} \right)$$

Table 86. Summary of annual ex ante project emissions in Project Area, AUD component, per stratum (1)

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	6.62	6.62	3,515.50	3,515.50
2028	4.85	11.46	2,677.34	6,192.84
2029	0.00	11.46	176.81	6,369.65
2030	0.00	11.46	176.81	6,546.46
2031	0.00	11.46	176.81	6,723.27
2032	3.17	14.64	1,863.56	8,586.82
2033	2.90	17.54	1,767.68	10,354.50
2034	6.11	23.65	3,516.01	13,870.51
2035	0.29	23.94	517.76	14,388.27
2036	0.00	23.94	369.19	14,757.46
2037	3.44	27.38	2,099.41	16,856.87
2038	0.00	27.38	255.29	17,112.16
2039	3.02	30.39	1,857.12	18,969.28

2040	6.59	36.98	3,801.76	22,771.04
2041	0.27	37.25	548.77	23,319.81
2042	0.00	37.25	361.39	23,681.20
2043	0.00	37.25	319.13	24,000.33
2044	0.00	37.25	230.19	24,230.51
2045	0.08	37.33	266.58	24,497.09
2046	0.00	37.33	227.17	24,724.26
2047	0.04	37.36	190.53	24,914.80
2048	0.00	37.36	167.78	25,082.57
2049	0.00	37.36	123.88	25,206.45
2050	2.14	39.51	1,166.88	26,373.33
2051	0.00	39.51	57.04	26,430.37
2052	0.00	39.51	54.30	26,484.67

Table 87. Summary of annual ex ante project emissions in Project Area, AUD component, per stratum (2)

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00

2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 88. Summary of annual ex ante project emissions in Project Area, AUD component, per stratum (3)

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 89. Summary of annual ex ante project emissions in Project Area, AUD component, per stratum (4)

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	8.97
2024	0.00	0.03	0.19	9.16
2025	0.00	0.03	0.19	9.34
2026	0.00	0.03	0.19	9.53
2027	2.63	2.66	904.85	9.71
2028	2.70	5.36	946.16	914.56
2029	0.00	5.36	38.07	1,860.72
2030	0.00	5.36	38.07	1,898.79
2031	0.00	5.36	38.07	1,936.85
2032	0.11	5.47	75.73	1,974.92
2033	0.15	5.62	89.77	2,050.65
2034	3.88	9.50	1,371.11	2,140.42
2035	0.00	9.50	67.27	3,511.53
2036	0.00	9.50	67.27	3,578.80
2037	0.00	9.50	50.84	3,646.08
2038	0.92	10.42	349.80	3,696.91
2039	1.55	11.97	572.97	4,046.71
2040	5.02	16.99	1,774.33	4,619.68
2041	3.22	20.21	1,192.40	6,394.01
2042	0.35	20.55	228.96	7,586.41
2043	0.65	21.21	334.22	7,815.37
2044	0.00	21.21	91.28	8,149.59
2045	2.68	23.88	1,010.49	8,240.87
2046	0.00	23.88	110.29	9,251.36
2047	0.20	24.09	178.29	9,361.65
2048	0.00	24.09	101.40	9,539.94
2049	0.00	24.09	91.73	9,641.35
2050	1.42	25.51	548.07	9,733.08
2051	0.00	25.51	50.44	10,281.15
2052	0.23	25.74	127.95	10,331.59

Table 90. Summary of annual ex ante project emissions in Project Area, AUD component – sum of strata

Avoided Unplanned Deforestation Annual ex ante emission of deforestation in PA Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	9.16
2024	0.00	0.03	0.19	9.34

2025	0.00	0.03	0.19	9.53
2026	0.00	0.03	0.19	9.71
2027	9.25	9.28	4,420.34	4,430.06
2028	7.55	16.83	3,623.50	8,053.56
2029	0.00	16.83	214.88	8,268.43
2030	0.00	16.83	214.88	8,483.31
2031	0.00	16.83	214.88	8,698.19
2032	3.29	20.11	1,939.28	10,637.47
2033	3.05	23.16	1,857.45	12,494.92
2034	9.99	33.15	4,887.12	17,382.04
2035	0.29	33.44	585.03	17,967.07
2036	0.00	33.44	436.46	18,403.53
2037	3.44	36.87	2,150.25	20,553.78
2038	0.92	37.79	605.09	21,158.87
2039	4.57	42.36	2,430.09	23,588.96
2040	11.61	53.97	5,576.09	29,165.06
2041	3.49	57.46	1,741.17	30,906.22
2042	0.35	57.81	590.34	31,496.57
2043	0.65	58.46	653.35	32,149.92
2044	0.00	58.46	321.47	32,471.38
2045	2.75	61.21	1,277.07	33,748.46
2046	0.00	61.21	337.46	34,085.92
2047	0.24	61.45	368.82	34,454.74
2048	0.00	61.45	269.18	34,723.92
2049	0.00	61.45	215.61	34,939.53
2050	3.56	65.02	1,714.95	36,654.48
2051	0.00	65.02	107.48	36,761.96
2052	0.23	65.25	182.25	36,944.21

The Project Area does not carry out any wood harvesting activities at the time of audit, therefore the tables below have no further ex-ante project emissions to report.

Table 91. Logging Gap in the Project Area, sum of strata

Year	Total area exploited per year (ha/year)	Extracted wood (m ³ /ha)	Total extracted wood (t/year)	Damage (tCO ₂ /Year)	Gross emission (tCO ₂ /Year)
2022	0.0	0.0	0.0	0.0	0.0
2023	0.0	0.0	0.0	0.0	0.0
2024	0.0	0.0	0.0	0.0	0.0
2025	0.0	0.0	0.0	0.0	0.0
2026	0.0	0.0	0.0	0.0	0.0
2027	0.0	0.0	0.0	0.0	0.0
2028	0.0	0.0	0.0	0.0	0.0
2029	0.0	0.0	0.0	0.0	0.0
2030	0.0	0.0	0.0	0.0	0.0
2031	0.0	0.0	0.0	0.0	0.0
2032	0.0	0.0	0.0	0.0	0.0
2033	0.0	0.0	0.0	0.0	0.0
2034	0.0	0.0	0.0	0.0	0.0
2035	0.0	0.0	0.0	0.0	0.0
2036	0.0	0.0	0.0	0.0	0.0
2037	0.0	0.0	0.0	0.0	0.0
2038	0.0	0.0	0.0	0.0	0.0
2039	0.0	0.0	0.0	0.0	0.0
2040	0.0	0.0	0.0	0.0	0.0
2041	0.0	0.0	0.0	0.0	0.0
2042	0.0	0.0	0.0	0.0	0.0
2043	0.0	0.0	0.0	0.0	0.0
2044	0.0	0.0	0.0	0.0	0.0
2045	0.0	0.0	0.0	0.0	0.0
2046	0.0	0.0	0.0	0.0	0.0
2047	0.0	0.0	0.0	0.0	0.0
2048	0.0	0.0	0.0	0.0	0.0
2049	0.0	0.0	0.0	0.0	0.0
2050	0.0	0.0	0.0	0.0	0.0
2051	0.0	0.0	0.0	0.0	0.0
2052	0.0	0.0	0.0	0.0	0.0

If sustainable forest management activities are eventually carried out in the project area, emissions will be accounted using the M-REDD module.

Regarding other types of greenhouse gasses in the project scenario, in accordance with the methodology's VMD0013 module, "5 Procedures" section: "Where used in the baseline, accounting must occur under both the baseline and project scenarios in both the project area and in the leakage belt". Therefore, to match the baseline scenario, emissions of CH₄ and N₂O were accounted for ex-ante in the project scenario, as detailed in the tables below. In the present project, such GHG emissions from biomass burning are considered to originate from the following category, provided by the methodology: "2. Periodical burning of grassland or agricultural land after deforestation".

The applicable equation is as follows.

$$E_{biomassburn,i,t} = \sum_{g=1}^G \left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \quad (1)$$

Where:

$E_{biomassburn,i,t}$	Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO ₂ , CH ₄ , N ₂ O) (t CO _{2e})
$A_{burn,i,t}$	Area burnt for stratum i in year t (ha)
$B_{i,t}$	Average aboveground biomass stock before burning stratum i , year (t d.m. ha ⁻¹)
$COMF_i$	Combustion factor for stratum i (unitless)
$G_{g,i}$	Emission factor for stratum i for gas g (kg t ⁻¹ d.m. burnt)
GWP_g	Global warming potential for gas g (t CO ₂ /t gas g)
g (unitless)	1, 2, 3 ... G greenhouse gases including carbon dioxide ¹ , methane and nitrous oxide
i	1, 2, 3 ... M strata (unitless)
t	1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

Finally, it is noted that, according to the module, “Where fires occur ex-post, the module must be used to account GHG emissions”, and this will be applicable at each monitoring event.

The resulting estimate of ex-ante emissions from burning in the project scenario are presented in the table below.

Table 88. Projection of ex-ante estimated project GHG emissions from biomass and peat burning (E-BPB), per stratum (1)

Ex-ante Project Emissions Biomass Burning Emissions Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	6.62	6.62	235.90	235.90
2028	4.85	11.46	172.81	408.72
2029	0.00	11.46	0.00	408.72
2030	0.00	11.46	0.00	408.72
2031	0.00	11.46	0.00	408.72
2032	3.17	14.64	113.19	521.90
2033	2.90	17.54	103.47	625.37
2034	6.11	23.65	217.78	843.16
2035	0.29	23.94	10.27	853.42
2036	0.00	23.94	0.00	853.42
2037	3.44	27.38	122.57	975.99
2038	0.00	27.38	0.00	975.99
2039	3.02	30.39	107.49	1,083.48
2040	6.59	36.98	234.86	1,318.34
2041	0.27	37.25	9.76	1,328.10
2042	0.00	37.25	0.00	1,328.10
2043	0.00	37.25	0.00	1,328.10
2044	0.00	37.25	0.00	1,328.10
2045	0.08	37.33	2.72	1,330.82
2046	0.00	37.33	0.00	1,330.82
2047	0.04	37.36	1.28	1,332.11
2048	0.00	37.36	0.00	1,332.11
2049	0.00	37.36	0.00	1,332.11

2050	2.14	39.51	76.43	1,408.53
2051	0.00	39.51	0.00	1,408.53
2052	0.00	39.51	0.00	1,408.53

Table 89. Projection of ex-ante estimated project GHG emissions form biomass and peat burning (E-BPB), per stratum (2)

Ex-ante Project Emissions Biomass Burning Emissions Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00

2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 92. Projection of ex-ante estimated project GHG emissions form biomass and peat burning (E-BPB), per stratum (3)

Ex-ante Project Emissions Biomass Burning Emissions Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00

2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 91. Projection of ex-ante estimated project GHG emissions from biomass and peat burning (E-BPB), per stratum (4)

Ex-ante Project Emissions Biomass Burning Emissions Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	0.61	0.61
2023	0.00	0.03	0.00	0.61
2024	0.00	0.03	0.00	0.61
2025	0.00	0.03	0.00	0.61
2026	0.00	0.03	0.00	0.61
2027	2.63	2.66	61.23	61.84
2028	2.70	5.36	62.76	124.59
2029	0.00	5.36	0.00	124.59
2030	0.00	5.36	0.00	124.59
2031	0.00	5.36	0.00	124.59
2032	0.11	5.47	2.56	127.15
2033	0.15	5.62	3.46	130.61
2034	3.88	9.50	90.11	220.72
2035	0.00	9.50	0.00	220.72
2036	0.00	9.50	0.00	220.72
2037	0.00	9.50	0.00	220.72
2038	0.92	10.42	21.37	242.09
2039	1.55	11.97	36.04	278.13
2040	5.02	16.99	116.60	394.73
2041	3.22	20.21	74.80	469.54
2042	0.35	20.55	8.10	477.63
2043	0.65	21.21	15.12	492.75
2044	0.00	21.21	0.00	492.75
2045	2.68	23.88	62.21	554.97
2046	0.00	23.88	0.00	554.97

2047	0.20	24.09	4.76	559.72
2048	0.00	24.09	0.00	559.72
2049	0.00	24.09	0.00	559.72
2050	1.42	25.51	33.00	592.73
2051	0.00	25.51	0.00	592.73
2052	0.23	25.74	5.40	598.13

Table 92. Projection of ex-ante estimated project GHG emissions from biomass and peat burning (E-BPB)
– sum of strata.

Ex-ante Project Emissions Biomass Burning Emissions Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	0.61	0.61
2023	0.00	0.03	0.00	0.61
2024	0.00	0.03	0.00	0.61
2025	0.00	0.03	0.00	0.61
2026	0.00	0.03	0.00	0.61
2027	9.25	9.28	297.13	297.74
2028	7.55	16.83	235.57	533.31
2029	0.00	16.83	0.00	533.31
2030	0.00	16.83	0.00	533.31
2031	0.00	16.83	0.00	533.31
2032	3.29	20.11	115.75	649.06
2033	3.05	23.16	106.93	755.98
2034	9.99	33.15	307.89	1,063.88
2035	0.29	33.44	10.27	1,074.14
2036	0.00	33.44	0.00	1,074.14
2037	3.44	36.87	122.57	1,196.71
2038	0.92	37.79	21.37	1,218.09
2039	4.57	42.36	143.53	1,361.61
2040	11.61	53.97	351.46	1,713.08
2041	3.49	57.46	84.56	1,797.64
2042	0.35	57.81	8.10	1,805.73
2043	0.65	58.46	15.12	1,820.85
2044	0.00	58.46	0.00	1,820.85
2045	2.75	61.21	64.94	1,885.79

2046	0.00	61.21	0.00	1,885.79
2047	0.24	61.45	6.04	1,891.83
2048	0.00	61.45	0.00	1,891.83
2049	0.00	61.45	0.00	1,891.83
2050	3.56	65.02	109.43	2,001.26
2051	0.00	65.02	0.00	2,001.26
2052	0.23	65.25	5.40	2,006.66

3.2.3 Leakage

Projections of deforestation and baseline emissions were calculated in the Leakage Belt using the same criteria as in the Project Area, following the VMD0007 BL-UP and VMD0010 LK-ASU modules.

This type of leakage is understood to pertain only to AUD, because the VMD0009 LK-ASP module does not include discussion baseline deforestation in the leakage belt.

Therefore, the projections of baseline emissions from deforestation in the Leakage Belt relate to unplanned deforestation, presented per stratum.

Table 93. Summary of gross baseline emissions from deforestation in the Leakage Belt, per stratum (1)

Gross baseline emissions from deforestation in Leakage Belt Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	34.58	34.58	18,370.52	18,370.52
2026	0.00	34.58	533.28	18,903.80
2027	20.22	54.80	11,275.36	30,179.16
2028	26.92	81.71	15,144.81	45,323.97
2029	29.28	110.99	16,813.81	62,137.78
2030	23.83	134.81	14,370.38	76,508.16
2031	49.31	184.12	28,277.39	104,785.56
2032	31.22	215.35	19,429.11	124,214.67
2033	60.57	275.92	35,501.30	159,715.98
2034	71.59	347.51	42,289.56	202,005.54
2035	63.29	410.80	38,480.63	240,486.17
2036	58.65	469.44	36,989.59	277,475.76
2037	4.10	473.55	8,623.04	286,098.81
2038	33.38	506.93	23,850.67	309,949.48

2039	66.13	573.06	41,335.47	351,284.94
2040	60.00	633.06	38,754.69	390,039.63
2041	19.54	652.61	17,465.49	407,505.12
2042	21.24	673.85	18,213.03	425,718.15
2043	37.39	711.23	26,237.53	451,955.68
2044	57.36	768.59	36,383.28	488,338.95
2045	11.24	779.83	11,813.38	500,152.34
2046	0.00	779.83	5,162.11	505,314.45
2047	25.61	805.44	18,693.35	524,007.80
2048	10.50	815.94	10,548.24	534,556.04
2049	4.98	820.92	6,792.39	541,348.42
2050	2.42	823.34	4,610.47	545,958.89
2051	26.07	849.41	16,886.21	562,845.09
2052	8.93	858.33	7,845.02	570,690.11

Table 94. Summary of gross baseline emissions from deforestation in the Leakage Belt, per stratum (2)

Gross baseline emissions from deforestation in Leakage Belt Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00

2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 95. Summary of gross baseline emissions from deforestation in the Leakage Belt, per stratum (3)

Gross baseline emissions from deforestation in Leakage Belt Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 96. Summary of gross baseline emissions from deforestation in the Leakage Belt, per stratum (4)

Gross baseline emissions from deforestation in Leakage Belt Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	22.72	22.72	7,801.02	0.00
2029	0.47	23.19	322.22	7,801.02
2030	7.04	30.23	2,582.93	8,123.24
2031	10.92	41.15	3,963.24	10,706.17
2032	0.00	41.15	292.16	14,669.41
2033	0.00	41.15	292.16	14,961.57
2034	16.18	57.34	5,848.30	15,253.73
2035	0.00	57.34	407.05	21,102.04
2036	1.62	58.96	963.41	21,509.09
2037	0.00	58.96	418.56	22,472.50
2038	14.86	73.81	5,378.00	22,891.06
2039	36.10	109.92	12,774.38	28,269.06
2040	6.02	115.94	2,659.75	41,043.44
2041	12.56	128.50	4,879.47	43,703.20
2042	13.50	142.00	5,290.81	48,582.67
2043	21.71	163.71	8,206.37	53,873.47
2044	0.33	164.05	918.43	62,079.85
2045	4.87	168.92	2,480.75	62,998.28
2046	0.00	168.92	831.55	65,479.03
2047	4.02	172.94	2,210.93	66,310.58
2048	0.12	173.06	789.59	68,521.52
2049	11.07	184.13	4,323.61	69,311.11
2050	0.03	184.16	569.09	73,634.72
2051	0.00	184.16	470.51	74,203.81
2052	0.02	184.18	392.74	74,674.31

Table 97. Summary of gross baseline emissions from deforestation in the Leakage Belt in all strata

Gross baseline emissions from deforestation in Leakage Belt Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	34.58	34.58	18,370.52	18,370.52
2026	0.00	34.58	533.28	18,903.80
2027	20.22	54.80	11,275.36	30,179.16
2028	49.64	104.43	22,945.83	53,124.99
2029	29.74	134.18	17,136.02	70,261.02
2030	30.87	165.05	16,953.32	87,214.33
2031	60.23	225.28	32,240.64	119,454.97
2032	31.22	256.50	19,721.27	139,176.25
2033	60.57	317.07	35,793.46	174,969.71
2034	87.77	404.84	48,137.87	223,107.58
2035	63.29	468.13	38,887.68	261,995.25
2036	60.27	528.40	37,953.01	299,948.26
2037	4.10	532.50	9,041.60	308,989.86
2038	48.24	580.75	29,228.67	338,218.54
2039	102.23	682.98	54,109.85	392,328.38
2040	66.03	749.00	41,414.44	433,742.83
2041	32.10	781.11	22,344.96	456,087.78
2042	34.74	815.85	23,503.84	479,591.62
2043	59.10	874.95	34,443.90	514,035.52
2044	57.69	932.64	37,301.71	551,337.23
2045	16.11	948.75	14,294.13	565,631.37
2046	0.00	948.75	5,993.66	571,625.03
2047	29.63	978.38	20,904.29	592,529.32
2048	10.62	989.00	11,337.83	603,867.15
2049	16.05	1,005.05	11,115.99	614,983.14
2050	2.45	1,007.50	5,179.55	620,162.69
2051	26.07	1,033.57	17,356.71	637,519.41
2052	8.94	1,042.51	8,237.76	645,757.16

Market Leakage – LK-ME

The Market Leakage module was understood to be relevant to both AUD and APD components of the baseline, therefore hectare values in the calculations below sum the two.

Similarly to the Reference Area and Project Area, the Leakage Belt is also subject to risks of land grabbing by illegal organizations (i.e. family-scale land-grabber associations, property documentation forgers), mostly supported by unscrupulous sawmills and political interests. Total GHG emissions due to market-effects

leakage through decreased timber harvest is equal to the summed emissions from timber harvest in the baseline case potentially displaced plus a leakage factor:

$$LK_{MarketEffects,timber} = \sum_{i=1}^M (LF_{ME} \times LK_{MAF} \times AL_{T,i})$$

Where:

$LK_{MarketEffects,Timber}$ Total GHG emissions due to market- effects leakage through decreased timber harvest; t CO₂-e

LF_{ME} Leakage factor for market-effects calculations (dimensionless)

LF_{MAF} Leakage management adjustment factor (dimensionless)

$AL_{T,i}$ Summed emissions from timber harvest in stratum i in the baseline case potentially displaced through implementation of carbon project; t CO₂-e

i 1,2, 3... M strata

The leakage factor (LFME) was adopted as 0.2, given that the mean merchantable biomass as a proportion of total aboveground tree biomass (PML) for dense forest in Amazon Biome is more than 15% greater than merchantable biomass as a proportion of total aboveground tree biomass for the project boundaries (PMP).

For calculation PMP parameter, the volume of merchantable wood in exploitation (in m³/ha) was based on “Florestal Santa Maria Project” (35.08 m³/ha), whose Project Area is located about 120 km south of this Project Area and has the same predominant forest class represented within it (According to Figure 22 of FSM VCS-PD): Submontane dense tropical forest (“Dse”)”, in the IBGE classification. Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests). This value was multiplied by the mean wood density of commercially harvested species (0.59 t d.m.m⁻³) and divided by the aboveground biomass for each forest type (Dse:299.34 t/ha; Dbe:300.21 t/ha; Dau: 340.80 t/ha). The values obtained for PMP are:

Table 98. PMP parameters values.

Parameter	Value	Unit
PMP Dse	0.069	Dimensionless
PMP Dbe	0.069	Dimensionless
PMP Dau	0.061	Dimensionless
PMP Pa	0.106	Dimensionless

PML parameter was obtained from the literature by deduction from the Volume Expansion Factor (VEF) parameter for dense forest in the Legal Amazon. As described in the table on page 20 of VMD0011 LK-ME v1.1, merchantable trees have a DBH greater than 30 cm. The VEF value presented by Brown, S., Lugo, A.E., 1992 for dense forest in the Legal Amazon is 1.25. From the $PML=1/VEF$ deduction, the value obtained for the PML parameter is 0.8.

When evaluating the Leakage Belt deforestation factors and general forest characteristics (stratification, forest type, carbon stocks), the same parameters used for the Project Area are assumed to be equal in the Leakage Belt. This assumption is valid based on previous similarity analysis, which demonstrates that the Leakage Belt area and the Project Area present similarities in the following criteria: Politics and Legislation, Soil and Landscape, Climate and Vegetation, and Accessibility.

The total volume that would have been logged in the baseline in the Project Area, across strata and time periods, is estimated as follows. The baseline is interpreted to include both AUD and APD activity types:

$$AL_{T,i} = \sum_{t=1}^t (C_{BSL,XBT,i,t})$$

Where:

$AL_{T,i}$ Summed emissions from timber harvest in stratum i in the baseline case potentially displaced through implementation of carbon project; t CO₂-e

$C_{BSL,XBT,i,t}$ Carbon emission due to displaced timber harvests in the baseline scenario in stratum i in time t ; t CO₂-e

i	1,2, 3... M strata
t	1, 2, 3, ... t years elapsed since the projected start of the REDD+ project activity

The carbon emission due to the displaced logging has two components: the biomass carbon of the extracted timber and the biomass carbon in the forest damaged in the process of timber extraction:

$$C_{BSL, XBT, i, t} = ([V_{BSL, XE, i, t} * D_{mn} * CF] + [V_{BSL, XE, i, t} * LDF] + [V_{BSL, XE, i, t} * LIF]) * \frac{44}{12}$$

Where:

$C_{BSL, XBT, i, t}$	Carbon emission due to displaced timber harvests in the baseline scenario in stratum i in time t ; t CO ₂ -e
$V_{BSL, XE, i, t}$	Volume of timber projected to be extracted from within the project boundary during the baseline in stratum i at time t ; m ³
D_{mn}	Mean wood density of commercially harvested species; t d.m.m ⁻³
CF	Carbon fraction of biomass for commercially harvested species j ; t C td.m ⁻¹
LDF	Logging damage factor; t C m ⁻³ (default 0.53 t C m ⁻³ for broadleaf and mixed forests ¹⁵¹)
LIF	Logging infrastructure factor; t C m ⁻³ (default 0.29 t C m ⁻³)
i	1,2, 3... M strata
t	1, 2, 3, ... t years elapsed since the projected start of the REDD+ project activity

Table 99. Parameters used to calculate market leakage for the baseline scenario.

¹⁵¹ Source: <https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.0.pdf> (pg 4)

Parameter	Value	Unit
Merchantable wood	35.080	m ³ /ha
Dmm	0.590	t d.m.m-3
CF	0.470	t Ct-1 d.m
LDF	0.530	t C m-3
LIF	0.290	t C m-3

The tables below summarize the results of market leakage estimates for the IPOÁ REDD+ Project, including both the AUD and APD baseline components in the calculations.

Table 100. Total leakage due to market effects per stratum (1)

Annual Leakage Market Effects - Timber Harvest Dense Tropical Submontane Rainforest					
Year	ha/year	ha (cumulative)	V _{BSL} m ³	C _{BSL,XBT} tCO ₂ /year	AL _T *LF _{ME} tCO ₂ /year
2022	0.00	0.00	0.00	0.00	0.0
2023	0.00	0.00	0.00	0.00	0.0
2024	0.00	0.00	0.00	0.00	0.0
2025	0.00	0.00	0.00	0.00	0.0
2026	0.00	0.00	0.00	0.00	0.0
2027	66.17	66.17	2,321.23	9,339.30	1,867.9
2028	48.47	114.64	1,700.42	6,841.53	1,368.3
2029	0.00	114.64	0.00	0.00	0.0
2030	0.00	114.64	0.00	0.00	0.0
2031	0.00	114.64	0.00	0.00	0.0
2032	31.75	146.39	1,113.73	4,481.02	896.2
2033	29.02	175.41	1,018.09	4,096.23	819.2
2034	61.09	236.50	2,142.93	8,621.95	1,724.4
2035	2.88	239.38	101.03	406.49	81.3
2036	0.00	239.38	0.00	0.00	0.0
2037	34.38	273.76	1,206.05	4,852.46	970.5
2038	0.00	273.76	0.00	0.00	0.0
2039	30.15	303.91	1,057.66	4,255.43	851.1
2040	65.88	369.79	2,310.98	9,298.06	1,859.6
2041	2.74	372.52	95.99	386.20	77.2
2042	0.00	372.52	0.00	0.00	0.0
2043	0.00	372.52	0.00	0.00	0.0
2044	0.00	372.52	0.00	0.00	0.0
2045	0.76	373.29	26.80	107.82	21.6
2046	0.00	373.29	0.00	0.00	0.0
2047	0.36	373.65	12.63	50.81	10.2
2048	0.00	373.65	0.00	0.00	0.0

2049	0.00	373.65	0.00	0.00	0.0
2050	21.44	395.08	752.01	3,025.66	605.1
2051	0.00	395.08	0.00	0.00	0.0
2052	0.00	395.08	0.00	0.00	0.0

Table 101. Total leakage due to market effects per stratum (2)

Annual Leakage Market Effects - Timber Haverst Dense Tropical Lowland Rainforest					
Year	ha/year	ha (cumulative)	V _{BSL} m ³	C _{BSL,XBT} tCO ₂ /year	AL _T *LF _{ME} tCO ₂ /year
2022	0.00	0.00	0.00	0.00	0.0
2023	0.00	0.00	0.00	0.00	0.0
2024	0.00	0.00	0.00	0.00	0.0
2025	0.00	0.00	0.00	0.00	0.0
2026	0.00	0.00	0.00	0.00	0.0
2027	0.00	0.00	0.00	0.00	0.0
2028	0.00	0.00	0.00	0.00	0.0
2029	0.00	0.00	0.00	0.00	0.0
2030	0.00	0.00	0.00	0.00	0.0
2031	0.00	0.00	0.00	0.00	0.0
2032	0.00	0.00	0.00	0.00	0.0
2033	0.00	0.00	0.00	0.00	0.0
2034	0.00	0.00	0.00	0.00	0.0
2035	0.00	0.00	0.00	0.00	0.0
2036	0.00	0.00	0.00	0.00	0.0
2037	0.00	0.00	0.00	0.00	0.0
2038	0.00	0.00	0.00	0.00	0.0
2039	0.00	0.00	0.00	0.00	0.0
2040	0.00	0.00	0.00	0.00	0.0
2041	0.00	0.00	0.00	0.00	0.0
2042	0.00	0.00	0.00	0.00	0.0
2043	0.00	0.00	0.00	0.00	0.0
2044	0.00	0.00	0.00	0.00	0.0
2045	0.00	0.00	0.00	0.00	0.0
2046	0.00	0.00	0.00	0.00	0.0
2047	0.00	0.00	0.00	0.00	0.0
2048	0.00	0.00	0.00	0.00	0.0
2049	0.00	0.00	0.00	0.00	0.0
2050	0.00	0.00	0.00	0.00	0.0
2051	0.00	0.00	0.00	0.00	0.0
2052	0.00	0.00	0.00	0.00	0.0

Table 102. Total leakage due to market effects per stratum (3)

Annual Leakage Market Effects - Timber Haverst Dense Tropical Alluvial Rainforest					
--	--	--	--	--	--

Year	ha/year	ha (cumulative)	V _{BSL} m ³	C _{BSL,XBT}	AL _T *LF _{ME}
				tCO ₂ /year	tCO ₂ /year
2022	0.00	0.00	0.00	0.00	0.0
2023	0.00	0.00	0.00	0.00	0.0
2024	0.00	0.00	0.00	0.00	0.0
2025	0.00	0.00	0.00	0.00	0.0
2026	0.00	0.00	0.00	0.00	0.0
2027	0.00	0.00	0.00	0.00	0.0
2028	0.00	0.00	0.00	0.00	0.0
2029	0.00	0.00	0.00	0.00	0.0
2030	0.00	0.00	0.00	0.00	0.0
2031	0.00	0.00	0.00	0.00	0.0
2032	0.00	0.00	0.00	0.00	0.0
2033	0.00	0.00	0.00	0.00	0.0
2034	0.00	0.00	0.00	0.00	0.0
2035	0.00	0.00	0.00	0.00	0.0
2036	0.00	0.00	0.00	0.00	0.0
2037	0.00	0.00	0.00	0.00	0.0
2038	0.00	0.00	0.00	0.00	0.0
2039	0.00	0.00	0.00	0.00	0.0
2040	0.00	0.00	0.00	0.00	0.0
2041	0.00	0.00	0.00	0.00	0.0
2042	0.00	0.00	0.00	0.00	0.0
2043	0.00	0.00	0.00	0.00	0.0
2044	0.00	0.00	0.00	0.00	0.0
2045	0.00	0.00	0.00	0.00	0.0
2046	0.00	0.00	0.00	0.00	0.0
2047	0.00	0.00	0.00	0.00	0.0
2048	0.00	0.00	0.00	0.00	0.0
2049	0.00	0.00	0.00	0.00	0.0
2050	0.00	0.00	0.00	0.00	0.0
2051	0.00	0.00	0.00	0.00	0.0
2052	0.00	0.00	0.00	0.00	0.0

Table 103. Total leakage due to market effects per stratum (4)

Annual Leakage Market Effects - Timber Haverst Pioneer Formation with fluvial influence without palm trees					
Year	ha/year	ha (cumulative)	V _{BSL} m ³	C _{BSL,XBT}	AL _T *LF _{ME}
				tCO ₂ /year	tCO ₂ /year
2022	0.26	0.26	9.17	36.88	7.4
2023	0.00	0.26	0.00	0.00	0.0
2024	0.00	0.26	0.00	0.00	0.0
2025	0.00	0.26	0.00	0.00	0.0
2026	0.00	0.26	0.00	0.00	0.0
2027	26.35	26.61	924.34	3,719.01	743.8
2028	27.01	53.62	947.43	3,811.94	762.4

2029	0.00	53.62	0.00	0.00	0.0
2030	0.00	53.62	0.00	0.00	0.0
2031	0.00	53.62	0.00	0.00	0.0
2032	1.10	54.72	38.65	155.49	31.1
2033	1.49	56.21	52.20	210.02	42.0
2034	38.78	94.99	1,360.32	5,473.17	1,094.6
2035	0.00	94.99	0.00	0.00	0.0
2036	0.00	94.99	0.00	0.00	0.0
2037	0.00	94.99	0.00	0.00	0.0
2038	9.20	104.18	322.67	1,298.25	259.7
2039	15.51	119.69	544.03	2,188.86	437.8
2040	50.18	169.87	1,760.26	7,082.31	1,416.5
2041	32.19	202.06	1,129.28	4,543.58	908.7
2042	3.49	205.55	122.25	491.88	98.4
2043	6.51	212.05	228.23	918.26	183.7
2044	0.00	212.05	0.00	0.00	0.0
2045	26.77	238.83	939.20	3,778.82	755.8
2046	0.00	238.83	0.00	0.00	0.0
2047	2.05	240.87	71.80	288.89	57.8
2048	0.00	240.87	0.00	0.00	0.0
2049	0.00	240.87	0.00	0.00	0.0
2050	14.20	255.08	498.24	2,004.65	400.9
2051	0.00	255.08	0.00	0.00	0.0
2052	2.32	257.40	81.52	327.98	65.6

Table 104. Total leakage due to market effects: sum of strata

Annual Leakage Market Effects - Timber Haverst Sum of strata					
Year	ha/year	ha (cumulative)	V _{BSL} m ³	C _{BSL,XBT} tCO ₂ /year	AL _T *LF _{ME} tCO ₂ /year
2022	0.26	0.26	9.17	36.88	7.4
2023	0.00	0.26	0.00	0.00	0.0
2024	0.00	0.26	0.00	0.00	0.0
2025	0.00	0.26	0.00	0.00	0.0
2026	0.00	0.26	0.00	0.00	0.0
2027	92.52	92.78	3,245.57	13,058.31	2,611.7
2028	75.48	168.26	2,647.86	10,653.47	2,130.7
2029	0.00	168.26	0.00	0.00	0.0
2030	0.00	168.26	0.00	0.00	0.0
2031	0.00	168.26	0.00	0.00	0.0
2032	32.85	201.11	1,152.38	4,636.52	927.3
2033	30.51	231.62	1,070.29	4,306.24	861.2
2034	99.86	331.49	3,503.26	14,095.12	2,819.0
2035	2.88	334.37	101.03	406.49	81.3
2036	0.00	334.37	0.00	0.00	0.0
2037	34.38	368.75	1,206.05	4,852.46	970.5

2038	9.20	377.94	322.67	1,298.25	259.7
2039	45.66	423.60	1,601.69	6,444.29	1,288.9
2040	116.06	539.66	4,071.24	16,380.37	3,276.1
2041	34.93	574.59	1,225.27	4,929.79	986.0
2042	3.49	578.07	122.25	491.88	98.4
2043	6.51	584.58	228.23	918.26	183.7
2044	0.00	584.58	0.00	0.00	0.0
2045	27.54	612.11	966.00	3,886.64	777.3
2046	0.00	612.11	0.00	0.00	0.0
2047	2.41	614.52	84.43	339.70	67.9
2048	0.00	614.52	0.00	0.00	0.0
2049	0.00	614.52	0.00	0.00	0.0
2050	35.64	650.16	1,250.25	5,030.30	1,006.1
2051	0.00	650.16	0.00	0.00	0.0
2052	2.32	652.48	81.52	327.98	65.6

Unplanned Deforestation Displaced from the Project Area to the Leakage Belt (Step 3 - LK-ASU)

Leakage displaced from the Project Area to the Leakage Belt is understood to apply only to the AUD component of the baseline, as it is discussed in the VMD0010 LK-ASU module, while it is absent from the VMD0009 module LK-ASP. Therefore, the explanation below relates only to the AUD baseline.

The following procedure from section 5.1.4.1 of the methodology's VMD0010 LK-ASU module was applied:

“Estimate the carbon stock changes and greenhouse gas emissions in the Leakage Belt that are expected to occur due to the implementation of the REDD project activity and that would not occur in the baseline case. This must be done by multiplying the estimated baseline carbon stock changes and greenhouse gas emissions for the project area by a factor < 1.0 representing the % of deforestation expected to be displaced into the Leakage Belt”.

Based on the expected effectiveness of the proposed REDD+ project activities, the carbon stock changes and greenhouse gas emissions in the Leakage Belt that are expected to occur due to the implementation of the REDD+ project activity (that would not occur in the baseline case) were conservatively estimated as being 10% higher in the project case. It is assumed that this factor is valid for the IPOÁ REDD+ Project, given that the project proponent will adopt a series of activities for leakage mitigation, as previously mentioned in this VCS/CCB-PD.

The results pertaining to leakage displaced from the Project Area to the Leakage Belt relating to unplanned deforestation, are presented below per stratum.

Table 105. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt per stratum (1)

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	6.62	6.62	3,515.50	3,515.50
2028	4.85	11.46	2,677.34	6,192.84
2029	0.00	11.46	176.81	6,369.65
2030	0.00	11.46	176.81	6,546.46
2031	0.00	11.46	176.81	6,723.27
2032	3.17	14.64	1,863.56	8,586.82
2033	2.90	17.54	1,767.68	10,354.50
2034	6.11	23.65	3,516.01	13,870.51
2035	0.29	23.94	517.76	14,388.27
2036	0.00	23.94	369.19	14,757.46
2037	3.44	27.38	2,099.41	16,856.87
2038	0.00	27.38	255.29	17,112.16
2039	3.02	30.39	1,857.12	18,969.28
2040	6.59	36.98	3,801.76	22,771.04
2041	0.27	37.25	548.77	23,319.81
2042	0.00	37.25	361.39	23,681.20
2043	0.00	37.25	319.13	24,000.33
2044	0.00	37.25	230.19	24,230.51
2045	0.08	37.33	266.58	24,497.09
2046	0.00	37.33	227.17	24,724.26
2047	0.04	37.36	190.53	24,914.80
2048	0.00	37.36	167.78	25,082.57
2049	0.00	37.36	123.88	25,206.45
2050	2.14	39.51	1,166.88	26,373.33
2051	0.00	39.51	57.04	26,430.37
2052	0.00	39.51	54.30	26,484.67

Table 106. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt per stratum (2)

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂

2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 107. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt per stratum (3)

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00

2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 108. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt per stratum (4)

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	8.97
2024	0.00	0.03	0.19	9.16
2025	0.00	0.03	0.19	9.34
2026	0.00	0.03	0.19	9.53
2027	2.63	2.66	904.85	9.71
2028	2.70	5.36	946.16	914.56
2029	0.00	5.36	38.07	1,860.72
2030	0.00	5.36	38.07	1,898.79
2031	0.00	5.36	38.07	1,936.85
2032	0.11	5.47	75.73	1,974.92
2033	0.15	5.62	89.77	2,050.65
2034	3.88	9.50	1,371.11	2,140.42
2035	0.00	9.50	67.27	3,511.53
2036	0.00	9.50	67.27	3,578.80

2037	0.00	9.50	50.84	3,646.08
2038	0.92	10.42	349.80	3,696.91
2039	1.55	11.97	572.97	4,046.71
2040	5.02	16.99	1,774.33	4,619.68
2041	3.22	20.21	1,192.40	6,394.01
2042	0.35	20.55	228.96	7,586.41
2043	0.65	21.21	334.22	7,815.37
2044	0.00	21.21	91.28	8,149.59
2045	2.68	23.88	1,010.49	8,240.87
2046	0.00	23.88	110.29	9,251.36
2047	0.20	24.09	178.29	9,361.65
2048	0.00	24.09	101.40	9,539.94
2049	0.00	24.09	91.73	9,641.35
2050	1.42	25.51	548.07	9,733.08
2051	0.00	25.51	50.44	10,281.15
2052	0.23	25.74	127.95	10,331.59

Table 109. Estimation of Unplanned Deforestation Displaced from the Project Area to the Leakage Belt, sum of strata

Avoided Unplanned Deforestation Annual ex ante deforestation displaced from PA to LB Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	8.97	8.97
2023	0.00	0.03	0.19	9.16
2024	0.00	0.03	0.19	9.34
2025	0.00	0.03	0.19	9.53
2026	0.00	0.03	0.19	9.71
2027	9.25	9.28	4,420.34	4,430.06
2028	7.55	16.83	3,623.50	8,053.56
2029	0.00	16.83	214.88	8,268.43
2030	0.00	16.83	214.88	8,483.31
2031	0.00	16.83	214.88	8,698.19
2032	3.29	20.11	1,939.28	10,637.47
2033	3.05	23.16	1,857.45	12,494.92
2034	9.99	33.15	4,887.12	17,382.04
2035	0.29	33.44	585.03	17,967.07
2036	0.00	33.44	436.46	18,403.53
2037	3.44	36.87	2,150.25	20,553.78
2038	0.92	37.79	605.09	21,158.87
2039	4.57	42.36	2,430.09	23,588.96
2040	11.61	53.97	5,576.09	29,165.06
2041	3.49	57.46	1,741.17	30,906.22
2042	0.35	57.81	590.34	31,496.57
2043	0.65	58.46	653.35	32,149.92
2044	0.00	58.46	321.47	32,471.38

2045	2.75	61.21	1,277.07	33,748.46
2046	0.00	61.21	337.46	34,085.92
2047	0.24	61.45	368.82	34,454.74
2048	0.00	61.45	269.18	34,723.92
2049	0.00	61.45	215.61	34,939.53
2050	3.56	65.02	1,714.95	36,654.48
2051	0.00	65.02	107.48	36,761.96
2052	0.23	65.25	182.25	36,944.21

Activity Shifting Leakage (LK-APD)

In certain cases, activity shifting leakage may be applicable to the APD component of REDD project baselines. VMD0009 LK-ASP is the mandatory leakage module applicable in cases where module VMD0006 BL-PL has been used to define the baseline.

However, the Ipoá project's APD baseline is understood to be exempt from Activity Shifting leakage, because the BL-PL module states: "where the specific deforestation agent can be identified, leakage does not need to be considered where it can be demonstrated that the management plans and/or land-use designations of the deforestation agent's other lands (which must be identified by location) have not materially changed as a result of the project (e.g., the deforestation agent has not designated new lands as timber concessions, increased harvest rates in lands already managed for timber, cleared intact forests for agricultural production or increased fertilizer use to enhance agricultural yields)."

This is argued to be the case for the Ipoá Project, as no new management plans, deforestation requests, or any other changes in the land-uses of the owner's other properties has occurred since the Ipoá project's start, as fully proven at audit.

Leakage outside the Leakage Belt (Step 4 - LK-ASU)

Regarding the activity shifting leakage relating to the unplanned component of the baseline, the following procedure was applied according to step 4 of module LK-ASU. Immigrants prevented from migrating into and deforesting the project area are conservatively assumed to migrate to an alternative forest area and to cause deforestation in the alternative area. The alternative forest area could be within the Leakage Belt, or it could be elsewhere in the country. The proportion migrating to the Leakage Belt is calculated as the area

of the Leakage Belt as a proportion of the total available forest area nationally (AVFOR). AVFOR was estimated as follows:

$$AVFOR = TOTFOR - PROTFOR - MANFOR$$

Where:

AVFOR Total available national forest area for unplanned deforestation; ha

TOTFOR Total available national forest area; ha

PROTFOR Total area of fully protected forests nationally; ha

MANFOR Total area of forests under active management nationally; ha

The total available national forest is estimated by National Forest Information System SNIF¹⁵², representing 497,962,509 ha.

According to ISA¹⁵³, PROTFOR is estimated at 63,755,583 ha. MANFOR is estimated at 1,400,000, according to IBAMA¹⁵⁴. In this context, AVFOR is estimated at 432,806,926.00 ha.

The proportion of Leakage Belt area related to the total available national forest area ($PROP_{LB}$) is calculated by dividing LBFOR (21,895.26 ha) by AVFOR (432,806,926 ha). This procedure results in $PROP_{LB}$.

¹⁵² <https://snif.florestal.gov.br/pt-br/os-biomas-e-suas-florestas>

¹⁵³ <https://uc.socioambiental.org/pt-br/paineldedados>

¹⁵⁴ Source [http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em-2019-totalizou-39-mil-hectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20ibama%20em%202019%20totalizou%2039%20mil%20hectares,-Publicado%3A%20Sexta%2C%2021&text=Bras%C3%ADlia%20\(21%2F02%2F2020,metros%20c%C3%BAbicos%20de%20madeira%20nativa](http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em-2019-totalizou-39-mil-hectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20ibama%20em%202019%20totalizou%2039%20mil%20hectares,-Publicado%3A%20Sexta%2C%2021&text=Bras%C3%ADlia%20(21%2F02%2F2020,metros%20c%C3%BAbicos%20de%20madeira%20nativa)

The proportion of Leakage Belt area related to the total available national forest area ($PROP_{LB}$) is calculated by dividing Leakage Belt area (LBFOR: 21,895.26 ha) by AVFOR. This procedure results in $PROP_{LB}$ equal to 0.000051 (dimensionless).

The average carbon stock across the Leakage Belt (C_{LB} ; 523.73 tCO₂/ha; based on similarity analysis, data from the Project Area was applied to Leakage Belt area) and the average carbon stock for all available forest area outside the Leakage Belt (C_{OLB} ; 578,1 tCO₂/ha) were taken for calculation of the proportional difference in carbon stocks between areas of forest available for unplanned deforestation both inside and outside the Leakage Belt ($PROP_{CS}$). $PROP_{CS}$ is calculated by dividing the stock outside the Leakage Belt (C_{OLB}) by the stock inside the Leakage Belt (C_{LB}), which results in a value of 1.10.

The proportion of baseline deforestation caused by immigrating population ($PROP_{IMM}$) was estimated for a period from 2014 to 2022. For calculating $PROP_{IMM}$, local data¹⁵⁵ were used for births, deaths, and population. It is then assumed that the total annual population growth in a given municipality is attributed to i) births and ii) immigration. Thus, by subtracting the number of annual births from the total annual population growth, it is possible to infer the number of immigrants. According to the number of immigrants, we have inferred the proportion of deforestation attributed to immigrant agents ($PROP_{IMM}$) as 0.20%.

The proportional leakage for areas with immigrating populations (LK_{PROP}) was then equal to the immigrating proportion multiplied by the proportion of available national forest area outside the Leakage Belt multiplied by the proportional difference in stocks between forests inside and outside the Leakage Belt.

$$LK_{PROP} = PROP_{IMM} * (1 - PROP_{LB}) * PROP_{CS}$$

Where:

LK_{PROP} Proportional leakage for areas with immigrating populations; proportion

$PROP_{IMM}$ Estimated proportion of baseline deforestation caused by immigrating population, proportion.

¹⁵⁵ Sources: <http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinasc/cnv/nvam.def> ;
<https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html>

$PROP_{LB}$	area of forest available for unplanned deforestation as a proportion of the total national forest area available for unplanned deforestation; proportion
$PROP_{CS}$	Proportional difference in stocks between areas of forest available for unplanned deforestation both inside and outside the Leakage Belt; proportion
LK^{PROP}	was estimated in 0.0024

Leakage due to the proportion of the baseline deforestation actors who are displaced to areas outside the Leakage Belt was therefore equal to the change in stocks in the baseline scenario minus the change in stocks in the project scenario multiplied by the proportional leakage factor for areas with immigrating populations:

$$\Delta C_{LK-ASU,OLB} = (\Delta C_{BSL,LK,unplanned} - \Delta C_{P,LB}) * LK_{PROP}$$

Where:

$\Delta C_{LK-ASU,OLB}$	Net CO ₂ emissions due to unplanned deforestation displaced outside the Leakage Belt; t CO ₂ -e
$\Delta C_{BSL,LK,unplanned}$	Net CO ₂ equivalent emissions in the baseline from unplanned deforestation in the leakage belt; t CO ₂ -e
$\Delta C_{P,LB}$	Net CO ₂ equivalent emissions within the leakage belt in the project case; t CO ₂ -e
LK_{PROP}	Proportional leakage for areas with immigrating populations; proportion

For the difference in emissions from unplanned deforestation within the Leakage Belt in the baseline and in the project case, a 10% factor was considered (i.e. $\Delta C_{P,LB}$ was considered to be 10% higher than $\Delta C_{BSL,LK,unplanned}$). It is assumed that this factor is valid for the IPOÁ REDD+ Project, given that the project proponent will adopt a series of activities for leakage mitigation, as previously mentioned in this VCS/CCB-PD.

The table below summarizes the parameters used for calculation of Leakage outside the Leakage Belt.

Table 110. Parameters used to calculate leakage market for the baseline scenario.

Parameter	Value	Unit
AVFOR	432,806,926.00	ha
TOTFOR	497,962,509.00	ha
PROTFOR	63,755,583.00	ha
MANFOR	1,400,000.00	ha
PROP _{LB}	0.000051	Dimensionless
LBFOR	21,895.26	ha
PROP _{CS}	1.23	Dimensionless
CO _{LB}	578.10	t C m ⁻³
C _{LB}	470.19	t C m ⁻³
LK _{PROP}	0.002488	Dimensionless
PROP _{IMM}	0.0020	Dimensionless

The tables below summarize the results obtained for calculation of Leakage outside the Leakage Belt, relating only to the unplanned deforestation component of the baseline.

Table 111. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (1)

Annual Leakage Outside of Leakage Belt Dense Tropical Submontane Rainforest			
Year	$\Delta C_{BSL,LK,unplanned}$	$\Delta C_{P,LB}$	$\Delta C_{LK-ASU,OLB}$
	tCO ₂ /year	tCO ₂ /year	tCO ₂ /year
2022	0.00	0.00	0.00
2023	0.00	0.00	0.00
2024	0.00	0.00	0.00
2025	18,370.52	0.00	37.18
2026	533.28	0.00	1.08
2027	11,275.36	3,515.50	15.70
2028	15,144.81	2,677.34	25.23
2029	16,813.81	176.81	33.67
2030	14,370.38	176.81	28.73
2031	28,277.39	176.81	56.87
2032	19,429.11	1,863.56	35.55

2033	35,501.30	1,767.68	68.27
2034	42,289.56	3,516.01	78.47
2035	38,480.63	517.76	76.83
2036	36,989.59	369.19	74.11
2037	8,623.04	2,099.41	13.20
2038	23,850.67	255.29	47.75
2039	41,335.47	1,857.12	79.90
2040	38,754.69	3,801.76	70.74
2041	17,465.49	548.77	34.24
2042	18,213.03	361.39	36.13
2043	26,237.53	319.13	52.45
2044	36,383.28	230.19	73.17
2045	11,813.38	266.58	23.37
2046	5,162.11	227.17	9.99
2047	18,693.35	190.53	37.45
2048	10,548.24	167.78	21.01
2049	6,792.39	123.88	13.50
2050	4,610.47	1,166.88	6.97
2051	16,886.21	57.04	34.06
2052	7,845.02	54.30	15.77

Table 112. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (2)

Annual Leakage Outside of Leakage Belt Dense Tropical Lowland Rainforest			
Year	$\Delta C_{BSL,LK,unplanned}$ tCO ₂ /year	$\Delta C_{P,LB}$ tCO ₂ /year	$\Delta C_{LK-ASU,OLB}$ tCO ₂ /year
2022	0.00	0.00	0.00
2023	0.00	0.00	0.00
2024	0.00	0.00	0.00
2025	0.00	0.00	0.00
2026	0.00	0.00	0.00
2027	0.00	0.00	0.00
2028	0.00	0.00	0.00
2029	0.00	0.00	0.00
2030	0.00	0.00	0.00
2031	0.00	0.00	0.00
2032	0.00	0.00	0.00
2033	0.00	0.00	0.00
2034	0.00	0.00	0.00
2035	0.00	0.00	0.00
2036	0.00	0.00	0.00
2037	0.00	0.00	0.00
2038	0.00	0.00	0.00
2039	0.00	0.00	0.00

2040	0.00	0.00	0.00
2041	0.00	0.00	0.00
2042	0.00	0.00	0.00
2043	0.00	0.00	0.00
2044	0.00	0.00	0.00
2045	0.00	0.00	0.00
2046	0.00	0.00	0.00
2047	0.00	0.00	0.00
2048	0.00	0.00	0.00
2049	0.00	0.00	0.00
2050	0.00	0.00	0.00
2051	0.00	0.00	0.00
2052	0.00	0.00	0.00

Table 113. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (3)

Annual Leakage Outside of Leakage Belt Dense Tropical Alluvial Rainforest			
Year	$\Delta C_{BSL,LK,unplanned}$ tCO ₂ /year	$\Delta C_{P,LB}$ tCO ₂ /year	$\Delta C_{LK-ASU,OLB}$ tCO ₂ /year
2022	0.00	0.00	0.00
2023	0.00	0.00	0.00
2024	0.00	0.00	0.00
2025	0.00	0.00	0.00
2026	0.00	0.00	0.00
2027	0.00	0.00	0.00
2028	0.00	0.00	0.00
2029	0.00	0.00	0.00
2030	0.00	0.00	0.00
2031	0.00	0.00	0.00
2032	0.00	0.00	0.00
2033	0.00	0.00	0.00
2034	0.00	0.00	0.00
2035	0.00	0.00	0.00
2036	0.00	0.00	0.00
2037	0.00	0.00	0.00
2038	0.00	0.00	0.00
2039	0.00	0.00	0.00
2040	0.00	0.00	0.00
2041	0.00	0.00	0.00
2042	0.00	0.00	0.00
2043	0.00	0.00	0.00
2044	0.00	0.00	0.00
2045	0.00	0.00	0.00
2046	0.00	0.00	0.00
2047	0.00	0.00	0.00

2048	0.00	0.00	0.00
2049	0.00	0.00	0.00
2050	0.00	0.00	0.00
2051	0.00	0.00	0.00
2052	0.00	0.00	0.00

Table 114. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, per stratum (4)

Annual Leakage Outside of Leakage Belt Pioneer Formation with fluvial influence without palm trees			
Year	$\Delta C_{BSL,LK,unplanned}$	$\Delta C_{P,LB}$	$\Delta C_{LK-ASU,OLB}$
	tCO ₂ /year	tCO ₂ /year	tCO ₂ /year
2022	0.00	8.97	-0.02
2023	0.00	0.19	0.00
2024	0.00	0.19	0.00
2025	0.00	0.19	0.00
2026	0.00	0.19	0.00
2027	0.00	904.85	-1.83
2028	7,801.02	946.16	13.87
2029	322.22	38.07	0.58
2030	2,582.93	38.07	5.15
2031	3,963.24	38.07	7.94
2032	292.16	75.73	0.44
2033	292.16	89.77	0.41
2034	5,848.30	1,371.11	9.06
2035	407.05	67.27	0.69
2036	963.41	67.27	1.81
2037	418.56	50.84	0.74
2038	5,378.00	349.80	10.18
2039	12,774.38	572.97	24.69
2040	2,659.75	1,774.33	1.79
2041	4,879.47	1,192.40	7.46
2042	5,290.81	228.96	10.24
2043	8,206.37	334.22	15.93
2044	918.43	91.28	1.67
2045	2,480.75	1,010.49	2.98
2046	831.55	110.29	1.46
2047	2,210.93	178.29	4.11
2048	789.59	101.40	1.39
2049	4,323.61	91.73	8.56
2050	569.09	548.07	0.04
2051	470.51	50.44	0.85
2052	392.74	127.95	0.54

Tabel 115. Estimation of Unplanned Deforestation Displaced from the Project Area to Outside the Leakage Belt, Sum of Strata

Annual Leakage Outside of Leakage Belt Sum of Strata			
Year	$\Delta C_{BSL,LK,unplanned}$ tCO ₂ /year	$\Delta C_{P,LB}$ tCO ₂ /year	$\Delta C_{LK-ASU,OLB}$ tCO ₂ /year
2022	0.00	8.97	-0.02
2023	0.00	0.19	0.00
2024	0.00	0.19	0.00
2025	18,370.52	0.19	37.18
2026	533.28	0.19	1.08
2027	11,275.36	4,420.34	13.87
2028	22,945.83	3,623.50	39.10
2029	17,136.02	214.88	34.25
2030	16,953.32	214.88	33.88
2031	32,240.64	214.88	64.81
2032	19,721.27	1,939.28	35.99
2033	35,793.46	1,857.45	68.68
2034	48,137.87	4,887.12	87.53
2035	38,887.68	585.03	77.52
2036	37,953.01	436.46	75.93
2037	9,041.60	2,150.25	13.95
2038	29,228.67	605.09	57.93
2039	54,109.85	2,430.09	104.59
2040	41,414.44	5,576.09	72.53
2041	22,344.96	1,741.17	41.70
2042	23,503.84	590.34	46.37
2043	34,443.90	653.35	68.39
2044	37,301.71	321.47	74.84
2045	14,294.13	1,277.07	26.34
2046	5,993.66	337.46	11.45
2047	20,904.29	368.82	41.56
2048	11,337.83	269.18	22.40
2049	11,115.99	215.61	22.06
2050	5,179.55	1,714.95	7.01
2051	17,356.71	107.48	34.91
2052	8,237.76	182.25	16.30

Regarding other types of greenhouse gasses considered in leakage, in accordance with the methodology's VMD0013 module, "5 Procedures" section: "Where used in the baseline, accounting must occur under both the baseline and project scenarios in both the project area and in the leakage belt". Therefore, emissions of CH₄ and N₂O relating to unplanned deforestation displaced from the project area to the leakage belt were included, in order to match the baseline scenario, as detailed in the tables below.

In the present project, such GHG emissions from biomass burning are considered to originate from the following category, provided by the methodology: “2. Periodical burning of grassland or agricultural land after deforestation”. The applicable equation for the calculations are as follows.

$$E_{biomassburn,i,t} = \sum_{g=1}^G \left(\left(A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i} \right) \times 10^{-3} \right) \times GWP_g \quad (1)$$

Where:

$E_{biomassburn,i,t}$	Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO ₂ , CH ₄ , N ₂ O) (t CO _{2e})
$A_{burn,i,t}$	Area burnt for stratum i in year t (ha)
$B_{i,t}$	Average aboveground biomass stock before burning stratum i , year (t d.m. ha ⁻¹)
$COMF_i$	Combustion factor for stratum i (unitless)
$G_{g,i}$	Emission factor for stratum i for gas g (kg t ⁻¹ d.m. burnt)
GWP_g	Global warming potential for gas g (t CO ₂ /t gas g)
g (unitless)	1, 2, 3 ... G greenhouse gases including carbon dioxide ¹ , methane and nitrous oxide
i	1, 2, 3 ... M strata (unitless)
t	1, 2, 3, ... t^* time elapsed since the start of the project activity (years)

Furthermore, it is noted that, according to the module, “Where fires occur ex-post, the module must be used to account GHG emissions”, and this will be applicable to the leakage deforestation displaced from the Project Area to the Leakage Belt at each monitoring event.

The results of the calculations of the ex-ante estimated emissions from burning (summing both gasses CH₄ and N₂O) related to the leakage deforestation displaced from the Project Area to the Leakage Belt, are presented in the tables below.

Table 116. Ex-ante estimated emissions from biomass burning related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt, per stratum (1).

Ex-ante Leakage Emissions Biomass Burning Emissions Dense Tropical Submontane Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	6.62	6.62	235.90	235.90
2028	4.85	11.46	172.81	408.72
2029	0.00	11.46	0.00	408.72
2030	0.00	11.46	0.00	408.72
2031	0.00	11.46	0.00	408.72
2032	3.17	14.64	113.19	521.90
2033	2.90	17.54	103.47	625.37
2034	6.11	23.65	217.78	843.16
2035	0.29	23.94	10.27	853.42
2036	0.00	23.94	0.00	853.42
2037	3.44	27.38	122.57	975.99
2038	0.00	27.38	0.00	975.99
2039	3.02	30.39	107.49	1,083.48
2040	6.59	36.98	234.86	1,318.34
2041	0.27	37.25	9.76	1,328.10
2042	0.00	37.25	0.00	1,328.10
2043	0.00	37.25	0.00	1,328.10
2044	0.00	37.25	0.00	1,328.10
2045	0.08	37.33	2.72	1,330.82
2046	0.00	37.33	0.00	1,330.82
2047	0.04	37.36	1.28	1,332.11
2048	0.00	37.36	0.00	1,332.11
2049	0.00	37.36	0.00	1,332.11
2050	2.14	39.51	76.43	1,408.53
2051	0.00	39.51	0.00	1,408.53
2052	0.00	39.51	0.00	1,408.53

Table 117. Ex-ante estimated emissions from biomass burning related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt, per stratum (2).

Ex-ante Leakage Emissions Biomass Burning Emissions Dense Tropical Lowland Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 118. Ex-ante estimated emissions from biomass burning related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt, per stratum (3).

Ex-ante Leakage Emissions Biomass Burning Emissions Dense Tropical Alluvial Rainforest				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.00	0.00	0.00	0.00
2023	0.00	0.00	0.00	0.00
2024	0.00	0.00	0.00	0.00
2025	0.00	0.00	0.00	0.00
2026	0.00	0.00	0.00	0.00
2027	0.00	0.00	0.00	0.00
2028	0.00	0.00	0.00	0.00
2029	0.00	0.00	0.00	0.00
2030	0.00	0.00	0.00	0.00
2031	0.00	0.00	0.00	0.00
2032	0.00	0.00	0.00	0.00
2033	0.00	0.00	0.00	0.00
2034	0.00	0.00	0.00	0.00
2035	0.00	0.00	0.00	0.00
2036	0.00	0.00	0.00	0.00
2037	0.00	0.00	0.00	0.00
2038	0.00	0.00	0.00	0.00
2039	0.00	0.00	0.00	0.00
2040	0.00	0.00	0.00	0.00
2041	0.00	0.00	0.00	0.00
2042	0.00	0.00	0.00	0.00
2043	0.00	0.00	0.00	0.00
2044	0.00	0.00	0.00	0.00
2045	0.00	0.00	0.00	0.00
2046	0.00	0.00	0.00	0.00
2047	0.00	0.00	0.00	0.00
2048	0.00	0.00	0.00	0.00
2049	0.00	0.00	0.00	0.00
2050	0.00	0.00	0.00	0.00
2051	0.00	0.00	0.00	0.00
2052	0.00	0.00	0.00	0.00

Table 119. Ex-ante estimated emissions from biomass burning related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt, per stratum (4).

Ex-ante Leakage Emissions Biomass Burning Emissions				
Pioneer Formation with fluvial influence without palm trees				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	0.61	0.61
2023	0.00	0.03	0.00	0.61
2024	0.00	0.03	0.00	0.61
2025	0.00	0.03	0.00	0.61
2026	0.00	0.03	0.00	0.61
2027	2.63	2.66	61.23	61.84
2028	2.70	5.36	62.76	124.59
2029	0.00	5.36	0.00	124.59
2030	0.00	5.36	0.00	124.59
2031	0.00	5.36	0.00	124.59
2032	0.11	5.47	2.56	127.15
2033	0.15	5.62	3.46	130.61
2034	3.88	9.50	90.11	220.72
2035	0.00	9.50	0.00	220.72
2036	0.00	9.50	0.00	220.72
2037	0.00	9.50	0.00	220.72
2038	0.92	10.42	21.37	242.09
2039	1.55	11.97	36.04	278.13
2040	5.02	16.99	116.60	394.73
2041	3.22	20.21	74.80	469.54
2042	0.35	20.55	8.10	477.63
2043	0.65	21.21	15.12	492.75
2044	0.00	21.21	0.00	492.75
2045	2.68	23.88	62.21	554.97
2046	0.00	23.88	0.00	554.97
2047	0.20	24.09	4.76	559.72
2048	0.00	24.09	0.00	559.72
2049	0.00	24.09	0.00	559.72
2050	1.42	25.51	33.00	592.73
2051	0.00	25.51	0.00	592.73
2052	0.23	25.74	5.40	598.13

Table 120. Ex-ante estimated emissions from biomass burning related to Unplanned Deforestation Displaced from the Project Area to the Leakage Belt (sum of strata).

Ex-ante Leakage Emissions Biomass Burning Emissions Sum of Strata				
Year	ha/year	ha (cumulative)	tCO ₂ /year	tCO ₂
2022	0.03	0.03	0.61	0.61
2023	0.00	0.03	0.00	0.61
2024	0.00	0.03	0.00	0.61
2025	0.00	0.03	0.00	0.61
2026	0.00	0.03	0.00	0.61
2027	9.25	9.28	297.13	297.74
2028	7.55	16.83	235.57	533.31
2029	0.00	16.83	0.00	533.31
2030	0.00	16.83	0.00	533.31
2031	0.00	16.83	0.00	533.31
2032	3.29	20.11	115.75	649.06
2033	3.05	23.16	106.93	755.98
2034	9.99	33.15	307.89	1,063.88
2035	0.29	33.44	10.27	1,074.14
2036	0.00	33.44	0.00	1,074.14
2037	3.44	36.87	122.57	1,196.71
2038	0.92	37.79	21.37	1,218.09
2039	4.57	42.36	143.53	1,361.61
2040	11.61	53.97	351.46	1,713.08
2041	3.49	57.46	84.56	1,797.64
2042	0.35	57.81	8.10	1,805.73
2043	0.65	58.46	15.12	1,820.85
2044	0.00	58.46	0.00	1,820.85
2045	2.75	61.21	64.94	1,885.79
2046	0.00	61.21	0.00	1,885.79
2047	0.24	61.45	6.04	1,891.83
2048	0.00	61.45	0.00	1,891.83
2049	0.00	61.45	0.00	1,891.83
2050	3.56	65.02	109.43	2,001.26
2051	0.00	65.02	0.00	2,001.26
2052	0.23	65.25	5.40	2,006.66

3.2.4 Net GHG Emission Reductions and Removals

Net GHG Emission Reductions and Removals can be summarized as the “Estimated baseline emissions” minus the “Estimated project emissions” minus the “Estimated leakage emissions”, whose components are presented below.

Estimate baseline emissions

- = Baseline emissions from unplanned deforestation
- + Emissions from biomass burning in the baseline
- Sum of wood products and pasture carbon pools in the baseline case

Estimate project emissions

- = Emissions arising from logging gap in Project Area
- + Emissions from constructing logging infrastructure for removal of timber in Project Area
- Wood Products Carbon Pool in Project Area

Estimate leakage emissions

- = (Sum of Market Leakage and Leakage outside the Leakage Belt)
- * Leakage Factor

Table 121. Net GHG emission reductions from the IPOÁ REDD project, detailing: baseline; project emissions; and leakage.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
2022	260,894	10	14,179	246,706
2023	490,206	0	26,252	463,954
2024	509,890	0	26,549	483,341
2025	524,803	0	26,609	498,194
2026	171,028	0	6,620	164,407
2027	97,375	4,717	7,343	85,314
2028	89,614	3,859	6,029	79,726
2029	56,817	215	249	56,353
2030	56,817	215	249	56,353

2031	56,817	215	280	56,322
2032	66,311	2,055	3,018	61,237
2033	51,973	1,964	2,894	47,115
2034	67,236	5,195	8,102	53,940
2035	12,217	595	754	10,868
2036	7,425	436	512	6,476
2037	24,128	2,273	3,257	18,598
2038	8,881	626	944	7,310
2039	26,591	2,574	3,967	20,051
2040	56,731	5,928	9,276	41,527
2041	19,631	1,826	2,853	14,952
2042	8,444	598	743	7,102
2043	8,196	668	921	6,607
2044	4,229	321	396	3,511
2045	12,292	1,342	2,146	8,805
2046	3,375	337	349	2,688
2047	3,632	375	484	2,773
2048	2,692	269	292	2,131
2049	2,156	216	238	1,703
2050	16,523	1,824	2,837	11,861
2051	1,075	107	142	825
2052	1,764	188	270	1,307
Total	2,719,762	38,951	158,755	2,522,057

The total net greenhouse gas emissions reductions corresponding to the AUD and APD components of the baseline are presented in the table below. Emissions from both APD and AUD presented in Table below, which takes into account the balance of: Carbon stock changes from initial land-use (forest) to post-deforestation land-use (pasture), wood products, biomass burning, and ex-ante estimated leakage.

Table 122. Net GHG emission reductions from the IPOÁ REDD+ project, detailing: AUD and APD components.

Year	Net AUD emissions reductions (tCO ₂)	Net APD emissions reductions (tCO ₂)
2022	-14,105	260,811

2023	-26,250	490,204
2024	-26,548	509,889
2025	-26,607	524,801
2026	-6,619	171,026
2027	30,646	54,668
2028	25,058	54,668
2029	1,685	54,668
2030	1,685	54,668
2031	1,654	54,668
2032	13,891	47,347
2033	13,312	33,803
2034	33,831	20,109
2035	4,465	6,403
2036	3,416	3,060
2037	15,538	3,060
2038	4,250	3,060
2039	16,991	3,060
2040	38,467	3,060
2041	11,891	3,060
2042	4,474	2,627
2043	4,782	1,825
2044	2,497	1,014
2045	8,603	202
2046	2,688	0
2047	2,773	0
2048	2,131	0
2049	1,703	0
2050	11,861	0
2051	825	0
2052	1,307	0
Total	160,293	2,361,764
Total net AUD + APD	2,522,057	

In order to calculate credits eligible for trading, known as VCUs, the “Estimated net GHG emission reductions or removals” presented above are subject to the subtraction of the project’s buffer credits, as denoted by module VM0007 REDDMF’s equation 19, below.

$$VCU_t = (Adjusted_NER_{REDD+,t_2} - Adjusted_NER_{REDD+,t_1}) - Buffer_{Total} \quad (19)$$

Where:

VCU_t	Number of Verified Carbon Units at year $t = t_2 - t_1$ (VCU)
$Adjusted_NER_{REDD+,t_2}$	Total net GHG emission reductions of the REDD+ project activity up to year t_2 and adjusted to account for uncertainty (t CO ₂ e)
$Adjusted_NER_{REDD+,t_1}$	Total net GHG emission reductions of the REDD+ project activity up to year t_1 and adjusted to account for uncertainty (t CO ₂ e)

It is clarified that the net GHG emissions already presented have been undergone a 2.41% for uncertainty, as presented in section 3.2.1.

In the present project, the buffer proportion, calculated in accordance with the AFOLU_Non-Permanence_Risk-Tool is set at 10%, subdivided into 5 for internal, 0 for external, and 2 for natural risk types, which is duly rounded up to 10%. The full breakdown of all aspects of non-permanence risk is presented in the separate VCS-Non-Permanence-Risk-Report.

The resulting tradable credits, totaling 2,268,424 VCUs, are presented in Appendix 3 of the present PD.

3.3. Monitoring

3.3.1 Data and Parameters Available at Validation

Data / Parameter	CF
Data unit	tCt/td.m ⁻¹
Description	Carbon fraction of dry matter in t Ct ⁻¹ d.m
Source of data	Values from the literature (e.g. IPCC 2006 INV GLs AFOLU Chapter 4 Table 4.3 ¹⁵⁶) shall be used if available, otherwise default value of 0.47 t C t ⁻¹ d.m. can be used
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	The default value was used for conservativeness purposes.
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage
Comments	If new and more accurate carbon fraction data become available, these can be used to estimate the net anthropogenic GHG emission reduction of the subsequent fixed baseline period.

Data / Parameter	44/12
Data unit	Dimensionless
Description	Carbon mass to CO ₂ e mass conversion factor
Source of data	From scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU ¹⁵⁷
Value applied	44/12

¹⁵⁶https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

¹⁵⁷https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

Justification of choice of data or description of measurement methods and procedures applied	Conversion from C to CO ₂ based on molecular weights
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage
Comments	IPCC standard value

Data / Parameter	WWty
Data unit	Dimensionless
Description	Fraction of extracted biomass effectively emitted to the atmosphere during production by class of wood product ty
Source of data	Default value for developing countries: CP-W - page 14 ¹⁵⁸ .
Value applied	0.24
Justification of choice of data or description of measurement methods and procedures applied	As per CP-W.
Purpose of data	Calculation of baseline emissions
Comments	Parameter values to be updated if new empirically based peer-reviewed findings become available.

Data / Parameter	SLFty
Data unit	Dimensionless
Description	Fraction of wood products that will be emitted to the atmosphere within 5 years of production by class of wood product ty

¹⁵⁸<https://verra.org/wp-content/uploads/2018/03/VMD0005-CP-W-v1.1.pdf>

Source of data	As per CP-W: Sawnwood, page 13 ¹⁵⁹ .
Value applied	0.20
Justification of choice of data or description of measurement methods and procedures applied	Default conservative value prescribed by CP-W.
Purpose of data	Calculation of baseline emissions
Comments	Parameter values to be updated if new empirically based peer-reviewed findings become available

Data / Parameter	OFty
Data unit	Dimensionless
Description	Fraction of wood products that will be emitted to the atmosphere between 5 and 100 years of timber harvest by class of wood product ty
Source of data	The OFty is the complementary number of SLFty: the sum of both parameters must equal 1 (i.e., 100%).
Value applied	0.80
Justification of choice of data or description of measurement methods and procedures applied	As per CP-W.
Purpose of data	Calculation of baseline emissions
Comments	Parameter values to be updated if new empirically based peer-reviewed findings become available.

Data / Parameter	PCOMi
Data unit	Dimensionless

¹⁵⁹<https://verra.org/wp-content/uploads/2018/03/VMD0005-CP-W-v1.1.pdf>

Description	Commercial volume as a percent of total aboveground volume in stratum i.
Source of data	For calculation of this parameter, the following parameters and sources were used: - Volume of merchantable wood in exploitation (35.08 m³/ha) from FSM_REDD_MR2_04_03_2021¹⁶⁰. - Dmm value (0.59 t d.m.m⁻³) from Brown et al 1989. - Above ground carbon stock from MCTI 2015¹⁶¹. - CF value (0.47 t Ct-1 d.m) from IPCC 2006 V4_04_Ch4_Forest_Land¹⁶².
Value applied	Submontane Dense Tropical Rainforest: 0,07814 Lowland Dense Tropical Rainforest: 0,07791 Alluvial Dense Tropical Rainforest: 0,06863 Pioneer Formation with fluvial influence without palm trees: 0,1161
Justification of choice of data or description of measurement methods and procedures applied	Volume of merchantable wood in exploitation (m³/ha) has been based on "Florestal Santa Maria Project" (35.08 m³/ha), whose Project Area is located about 120 km South of this Project Area and has the same predominant forest class represented within it (According to Figure 22 of FSM VCS-PD¹⁶³): Submontane dense tropical forest ("Ds"), in the IBGE classification, which are the same as those predominating in this project. Age classes are assumed to be similar, as both sites had not been deforested or managed until the date of their forest inventories (i.e., both were primary forests).
Purpose of data	Calculation of baseline emissions
Comments	Updated at the time of baseline revision (at least every 10 years). Application of the commercial percentage of total volume introduces the simplifying assumption (and conservative, as it is only used in

¹⁶⁰https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashouifmIn902309ksdfiku098I71896923

¹⁶¹http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

¹⁶²https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_04_Ch4_Forest_Land.pdf

¹⁶³https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqowrnoiuomnckjashouifmIn902309ksdfiku098I71896923 (last visited in 10/09/2021)

	the ex-ante baseline calculations) that all commercial stocks are extracted (i.e., perfect efficiency).
--	---

Data / Parameter	Cab_tree
Data unit	t CO ₂ -e ha ⁻¹
Description	Mean aboveground biomass carbon stock in stratum i
Source of data	National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015) ¹⁶⁴ , Table 14. Values multiplied by 44/12 conversion parameter. Regarding data per stratum, the carbon stock numbers were as follows: - Dense Tropical Submontane Rainforest (code “Ds”): 140.69 tC/ha. - Dense Tropical Lowland Rainforest (code “Db”): 141.10 tC/ha. - Dense Tropical Alluvial Rainforest (code “Da”): 160.18 tC/ha. - Pioneer Formation with fluvial influence without palm trees (code “Pa”): 91.70 tC/ha.
Value applied	- Submontane Dense Tropical Rainforest: 515.86. - Lowland Dense Tropical Rainforest: 517.37. - Alluvial Dense Tropical Rainforest: 587.33. - Pioneer Formation with fluvial influence without palm trees: 336.23.
Justification of choice of data or description of measurement methods and procedures applied	Official data from Brazilian government.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Cbb_tree
Data unit	t CO ₂ -e ha ⁻¹

¹⁶⁴http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

Description	Mean belowground biomass carbon stock in stratum i
Source of data	National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015)¹⁶⁵, Table 14. Values multiplied by 44/12 conversion parameter. Regarding data per stratum, the carbon stock numbers were as follows: - Dense Tropical Submontane Rainforest (code “Ds”): 39.7 tC/ha. - Dense Tropical Lowland Rainforest (code “Db”): 39.83 tC/ha. - Dense Tropical Alluvial Rainforest (code “Da”): 45.22 tC/ha. - Pioneer Formation with fluvial influence without palm trees (code “Pa”): 17.01 tC/ha.
Value applied	- Submontane Dense Tropical Rainforest: 145.60. - Lowland Dense Tropical Rainforest: 146.04. - Alluvial Dense Tropical Rainforest: 165.81. - Pioneer Formation with fluvial influence without palm trees: 62.37.
Justification of choice of data or description of measurement methods and procedures applied	Official data from Brazilian government.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Pasture carbon pool
Data unit	t CO ₂ e
Description	Pasture carbon pool in the baseline scenario
Source of data	National GHG Inventory (Ministério da Ciência, Tecnologia e Inovação, 2015) ¹⁶⁶ , Table 5. Value (7.57 tC/ha) multiplied by 44/12 conversion parameter.
Value applied	27.76

¹⁶⁵http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

¹⁶⁶http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf

Justification of choice of data or description of measurement methods and procedures applied	Official data from Brazilian government.
Purpose of data	Calculation of baseline emissions
Comments	Calculation based on country-specific values.

Data / Parameter	BCEF
Data unit	Dimensionless
Description	Biomass conversion and expansion factor for conversion of commercial wood volume per unit area to total aboveground tree biomass per unit area
Source of data	According to CP-AB ¹⁶⁷ - page 14, being the average of the three proposed factors.
Value applied	1.32
Justification of choice of data or description of measurement methods and procedures applied	BCEF was applied for conversion of merchantable volume to total aboveground tree biomass
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage
Comments	N/A.

Data / Parameter	COMF
Data unit	Dimensionless
Description	Combustion factor for stratum i (vegetation type)

¹⁶⁷<https://verra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf>

Source of data	E-BPB refers to Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, "Primary tropical moist forest" ¹⁶⁸
Value applied	0.50
Justification of choice of data or description of measurement methods and procedures applied	Value was applied according to E-BPB: Table 2.6 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4 Agriculture, Forestry and Other Land Use, Chapter 2, "Primary tropical moist forest"
Purpose of data	Calculation of baseline emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	Ggi
Data unit	g kg-1 dry matter burnt
Description	Emission factor for stratum i for gas g
Source of data	Defaults can be found in Volume 4, Chapter 2, of the IPCC 2006 Inventory Guidelines in table 2.5 (see Annex 2: emission factors for various types of burning for CH ₄ and N ₂ O). ¹⁶⁹
Value applied	GCH ₄ = 6.8 GN ₂ O = 0.2
Justification of choice of data or description of measurement methods and procedures applied	Default values from IPCC 2006.
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A.

¹⁶⁸https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf

¹⁶⁹https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf

Data / Parameter	Deforestation
Data unit	Ha
Description	Maps of forest cover areas converted into non-forest areas
Source of data	Measured through data from Mapbiomas project
Value applied	Yearly variable: deforestation values are presented for the Reference Region, Leakage Belt and Project Area (projections) in this VCS/CCB-PD
Justification of choice of data or description of measurement methods and procedures applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for the monitoring period, other sources will be consulted such as PRODES, or a classification of imagery (Landsat8) will be carried out to measure deforested area: Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very high.

Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Dmm
Data unit	t d.m.m ⁻³
Description	Mean wood density of commercially harvested species
Source of data	Source: Brown, S., A. J. R. Gillespie, and A. E. Lugo, 1989. Biomass estimation methods for tropical forests with applications to forest inventory data. Forest Science, 35:881- 902. See pg. 890, Table 4, Moist
Value applied	0.59
Justification of choice of data or description of measurement methods and procedures applied	Country-specific data obtained from the same biome.
Purpose of data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A.

Data / Parameter	LDF
Data unit	t C m ⁻³
Description	Factor for calculating the biomass of dead wood created during logging operations per cubic meter extracted (i.e. Logging Damage Factor)
Source of data	VMD0011 LK-ME ¹⁷⁰
Value applied	0.53

¹⁷⁰<https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.1.pdf>

Justification of choice of data or description of measurement methods and procedures applied	It is the default value for broadleaf and mixed forests, according to module VMD0011 LK-ME. Default value for broadleaf and mixed forests of 0.53 t C m ⁻³ from 774 logging gaps measured by Winrock International in Bolivia, Belize, the Republic of Congo, Brazil, and Indonesia may be used for tropical broadleaf forests
Purpose of data	Calculation of leakage emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	LIF
Data unit	t C m ⁻³
Description	Factor for calculating the emissions arising from the creation of logging infrastructure (roads, skid trails and decks) during logging operations per cubic meter extracted
Source of data	LK-ME, pg 17 ¹⁷¹
Value applied	0.29
Justification of choice of data or description of measurement methods and procedures applied	Conservative default value of 0.29 t CO ₂ -e m ⁻³ calculated from 1,839 hectares of logging concessions analysed by Winrock International in the Republic of Congo and Brazil, may be used for tropical broadleaf forests.
Purpose of data	Calculation of leakage emissions Calculation of project emissions
Comments	N/A.

Data / Parameter	LF_{me}
Data unit	Dimensionless
Description	Leakage factor for market effects calculations
Source of data	VMD0011 (LK-ME)
Value applied	0.2

¹⁷¹<https://verra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.1.pdf>

Justification of choice of data or description of measurement methods and procedures applied	The mean merchantable biomass as a proportion of total aboveground tree biomass for each forest type (PML) is more than 15% greater than the merchantable biomass as a proportion of total aboveground tree biomass within the project boundaries (PMP).
Purpose of data	Calculation of leakage
Comments	N/A.

Data / Parameter	C_{LB}
Data unit	tCO ₂ e ha ⁻¹
Description	Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation inside the leakage belt
Source of data	Project Area data
Value applied	470.19
Justification of choice of data or description of measurement methods and procedures applied	Based on similarity analysis, data from the Project Area was applied to Leakage Belt area. It was taken a weighted average of aboveground biomass inside the Project Area.
Purpose of data	Calculation of leakage emissions
Comments	N/A.

Data / Parameter	C_{OLB}
Data unit	t CO ₂ e ha ⁻¹
Description	Area-weighted average aboveground tree carbon stock for forests available for unplanned deforestation outside the leakage belt
Source of data	Saatchi, R.A. Houghton, R.C. dos Santos Alvalá, J.V. Soares, and Yifan Yu. Distribution of Aboveground Live Biomass in the Amazon Basin. 2007. As per LK-ASU, the number is derived from peer-reviewed literature that is appropriate, as it considers the same biome.
Value applied	578.1

Justification of choice of data or description of measurement methods and procedures applied	The value of 578,1 tCO ₂ /ha is the total mean biomass for floodplain forest in the Amazon basin of 157.66 tC/ha (according to Table 7 the study mentioned before), multiplied by 44/12, the conversion factor from tC to tCO ₂ .
Purpose of data	Calculation of leakage emissions
Comments	N/A.

Data / Parameter	DLF
Data unit	%
Description	Displacement Leakage Factor
Source of data	Local assessment
Value applied	10
Justification of choice of data or description of measurement methods and procedures applied	If deforestation agents do not participate in leakage prevention activities and project activities, the Displacement Factor shall be 100%. Where leakage prevention activities are implemented, the factor shall be equal to the proportion of the baseline agents estimated to be given the opportunity to participate in leakage prevention activities and project activities. The project design team estimates that all communities in Project Zone will participate on deforestation prevention activities, as Sustainability and Conservation Courses, that will disseminate sustainable practices in the use of forest and agriculture resources, preventing leakage emissions. Thus, the "Displacement Leakage Factor" (DLF) was conservatively defined as 10%.
Purpose of data	Calculation of leakage
Comments	This value is an ex-ante estimate. Accurate and actual values will be monitored and reported in verification periods

Data / Parameter	GWP
Data unit	Dimensionless
Description	GWP of CH ₄ and N ₂ O
Source of data	Sixth Report IPCC (AR6) IPCC, 2022: Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844.
Value applied	27 for CH₄ 273 for N₂O
Justification of choice of data or description of measurement methods and procedures applied	The IPCC default value was used for conservativeness purposes.
Purpose of Data	Calculation of baseline emissions Calculation of project emissions Calculation of leakage emissions
Comments	N/A

3.3.2 Data and Parameters Monitored

Data / Parameter	<i>Aburn,i,t</i>
Data unit	<i>Ha</i>
Description	Area burnt in stratum <i>i</i> at time <i>t</i>
Source of data	Remote sensing data
Description of measurement methods and procedures to be applied	It is considered that burning is the common practice in the region, and that all deforested area undergoes burning in a given moment ¹⁷² . The Amazon Biome comprises a great amount of biomass, and post- deforestation residuals are very abundant. There is no technology known and economically feasible to allow mechanical skidding or disposal of deforestation residual biomass. Selling of these residuals is not feasible, as transportation is more expensive than the residual material itself. In this context, the only feasible alternative for cattle ranchers for land clearing is setting fire in those residues, getting rid of them thoroughly, quickly, and cheaply. According to an article in the "Globo" newspaper: "Basically, whenever you deforest, you burn it. This is a viable way to get rid of all the weeds. Burning also helps prepare the soil for planting. It serves to reduce soil acidity, a problem in the Amazon, and deposits nutrients" - Philip Fearnside, American biologist and scientist ¹⁷³ .
Frequency of monitoring/recording	Areas burnt will be monitored yearly, examination will occur prior to any verification event.
Value applied	The value will be calculated ex-post for every verification period.
Monitoring equipment	Remote sensing

¹⁷²<https://cnr.ncsu.edu/news/2019/09/amazon-rainforest-fires-everything-you-need-to-know/> (last visited in 20/09/2022)

¹⁷³<https://g1.globo.com/natureza/noticia/2019/08/23/queimadas-e-desmatamento-estao-relacionados-na-amazonia-entenda.gh.html> (last visited in 20/09/2022)

QA/QC procedures to be applied	Best practices in remote sensing Land use / land change map for monitoring period; Overlay land use / land change map with fire alert's location data from INPE-BDQUEIMADAS (http://www.inpe.br/queimadas/abasFogo.php) into the period; Quantify pixels of deforested areas over fire alerts. Monitor burned forest areas
Purpose of data	• Calculation of project emissions • Calculation of baseline emissions • Calculation of leakage emissions
Calculation method	As burning of biomass is the common practice in the region, it was considered that all the deforested areas were burnt – deforestation cycle includes burning.
Comments	Remote sensing

Data / Parameter	ADistPA
Data unit	Ha
Description	Area impacted by natural disturbance in the project stratum i converted to natural disturbance stratum q in year t; ha
Source of data	Remote Sensing imagery combined with ground verification or GPS coordinates
Description of measurement methods and procedures to be applied	Minimum monitoring unit shall be equal to a minimum of 11 Landsat pixels or one hectare.

Frequency monitoring/recording of	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value applied	The value will be calculated ex-post for every verification period.
Monitoring equipment	Remote sensing and GIS tools
QA/QC procedures to be applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other sources will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area: Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the mapping classification should be over 90%, considered very high.
Purpose of data	Calculation of project emissions
Calculation method	Ex ante, estimations of emissions from natural disturbances shall be based on historic incidence of such event in the Project region
Comments	N/A

Data / Parameter	ABSLPAt
Data unit	Ha
Description	Annual area of baseline deforestation in the project area at year t
Source of data	Remote sensing data and GIS
Description of measurement methods and procedures to be applied	Forest cover change due to deforestation is monitored through periodic assessment of classified satellite imagery covering the project area provided by Mapbiomas project.
Frequency of monitoring/recording	Annually
Value applied	The value will be calculated ex-post for every verification period.
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other sources will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area: Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix.

	The minimum accuracy of the classification mapping should be over than 90%, considered very high). Land-use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September.
Purpose of data	• Calculation of project emissions • Calculation of baseline emissions
Calculation method	Analysis of satellite images and maps
Comments	N/A

Data / Parameter	<i>VEXT z i t</i>
Data unit	m ³
Description	Volume extracted from logging stratum z in stratum i in year t
Source of data	Records and reports documenting amount of wood extracted within the project boundary. Official data of timber harvest reports from logging in sustainable forest management. Local field inventory.
Description of measurement methods and procedures to be applied	All logged trees are previously inventoried.
Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post.
Monitoring equipment	The same equipment applied in forest inventory.
QA/QC procedures to be applied	Forest inventory data can be crosschecked with official data of timber harvest reports from logging in sustainable forest management.
Purpose of data	Calculation of project emissions
Calculation method	Official data of timber harvest reports from logging in sustainable forest management and timber inventory.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions

	and knowing that the management need to have the FSC certification. It is being listed here for precaution.
--	---

Data / Parameter	CEXT z i t
Data unit	tCO ₂ -e
Description	Biomass carbon stock of timber extracted within the project boundary for logging stratum z, in stratum i in year t;
Source of data	Official data of timber volume reports from logging in sustainable forest management
Description of measurement methods and procedures to be applied	Official data of timber volume reports from logging in sustainable forest management will be converted to tCO ₂ -e, considering wood basic density and carbon fraction, multiplied by 44/12. All logged trees are previously inventoried.
Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post.
Monitoring equipment	The logged trees are previously inventoried.
QA/QC procedures to be applied	Forest inventory data can be crosschecked with official data of timber volume reports from logging in sustainable forest management.
Purpose of data	Calculation of project emissions
Calculation method	Official data of timber volume reports from logging in sustainable forest management (VEXT z i t) will be converted to tCO ₂ -e, considering wood basic density and carbon fraction, multiplied by 44/12.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	CLG i t
Data unit	tCO ₂ -e

Description	Actual net project emissions arising in the logging gap, in stratum i in year t
Source of data	Official data of timber volume reports from logging in sustainable forest management
Description of measurement methods and procedures to be applied	Biomass carbon stock of timber extracted within the project boundary ($C_{EXT, i, t}$) is summed to volume of logs in sustainable forest management ($V_{EXT, i, t}$) multiplied by LDF (Logging damage factor), to estimate both amounts of emissions of timber extracted and emissions caused for logging damage.
Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post.
Monitoring equipment	The logged trees are previously inventoried.
QA/QC procedures to be applied	Forest inventory data can be crosschecked with official data of timber volume reports from logging in sustainable forest management.
Purpose of data	Calculation of project emissions
Calculation method	Biomass carbon stock of timber extracted within the project boundary ($C_{EXT, i, t}$) is summed to volume of logs in sustainable forest management ($V_{EXT, i, t}$) multiplied by LDF (Logging damage factor), to estimate both amounts of emissions of timber extracted and emissions caused for logging damage.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	ADECKS_{i,t}
Data unit	Ha
Description	Area of logging decks in stratum i in year t
Source of data	Field measurements or reported measurements such as postharvest assessment reports and post-harvest maps that are based on field measurements. Remote sensing data and GIS.
Description of measurement methods	Systematic sampling must take place to ensure all decks within the area logged are identified and a conservative estimate of area produced.

and procedures to be applied	
Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post
Monitoring equipment	According to the Annual Operational Plans of the Sustainable Management Plan.
QA/QC procedures to be applied	The measured area of logging decks in current logging gaps will be compared with those of previous logging gaps.
Purpose of data	Calculation of project emissions
Calculation method	Data obtained from annual forest management.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	<i>AROAD_{i,t}</i>
Data unit	Ha
Description	Area of roads in stratum i in year t
Source of data	Reported measurements such as post-harvest assessment reports, post-harvest maps that are based on field measurements, or Annual Operational Plans of the Sustainable Management Plan. Remote sensing data and GIS.
Description of measurement methods and procedures to be applied	The area of roads created may be based on the length of roads multiplied by the average width of roads. The length of all roads created during selective logging within a stratum i in year t must be measured through systematically sampling the entire area logged to produce a conservative estimate of the length of roads created. The average width of the roads may be estimated through systematically sampling road width. Sufficient number of measurements of road width shall be measured to achieve a precision of equal or less than 15% of the mean at the 95%

	confidence interval. Where different categories of roads exist, different average road widths must be used.
Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post
Monitoring equipment	According to the Annual Operational Plans of the Sustainable Management Plan.
QA/QC procedures to be applied	The measured area of roads will be crosschecked with GIS analysis.
Purpose of data	Calculation of project emissions
Calculation method	Estimations of emissions from road creation shall be based on logging management plans or average width of roads and length of roads produced for logging in the farm. GIS analysis will support final conclusions.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	Lsk
Data unit	M
Description	Length of skid trail sk.
Source of data	Reported measurements such as post-harvest assessment reports, post-harvest maps that are based on field measurements, or Annual Operational Plans of the Sustainable Management Plan.
Description of measurement methods and procedures to be applied	The length of all skid trails within a stratum i in year t may be estimated through using systematic sampling with a random start of the entire area logged or within a sampled known logged area within the project boundary to produce a conservative estimate of the length of skid trails created. Where sampling is used, the total length of all skid trails can then be equal to the mean length of skid trails per unit area multiplied by the total area logged

Frequency of monitoring/recording	To be determined, according to the decision listed in the 'Comments' section.
Value applied	The value will be calculated ex-post
Monitoring equipment	According to the Annual Operational Plans of the Sustainable Management Plan.
QA/QC procedures to be applied	The measured length of skid trails in current logging gaps will be compared with those of previous logging gaps.
Purpose of data	Calculation of project emissions
Calculation method	Estimations of emissions from skid trail creation shall be based on logging management plans or average length and number of skid trails produced due to logging in the farm.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	CAB_tree, dest
Data unit	tCO ₂ -e ha-1
Description	Carbon stock in aboveground tree biomass assumed to be killed per unit area resulting from the creation of the skid trail in stratum i.
Source of data	Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data.
Description of measurement methods and procedures to be applied	As indicated in module CP-AB.
Frequency of monitoring/recording	The maximum size tree assumed to be killed during skid trail creation must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event.

Value applied	Currently N/A, to be calculated ex-post, after SFMP implementation.
Monitoring equipment	On-the-ground measurement, data according to the Annual Operational Plans of the Sustainable Management Plan, and scientific data review.
QA/QC procedures to be applied	In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. Furthermore, the measured area of skid trails will be compared with those of previous skid trails.
Purpose of data	Calculation of project emissions
Calculation method	Multiplication of skid trail dimensions by carbon stock in accordance with parameter CAB_tree, described in 4
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	CBB_tree,dest
Data unit	tCO ₂ -e ha-1
Description	Carbon stock in belowground tree biomass assumed to be killed per unit area resulting from the creation of the skid trail in stratum i
Source of data	Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data.
Description of measurement methods and procedures to be applied	As indicated in module CP-AB.
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value applied	Currently N/A, to be calculated ex-post, after SFMP implementation.

Monitoring equipment	On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study.
QA/QC procedures to be applied	In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs. Furthermore, the measured area of skid trails will be compared with those of previous skid trails.
Purpose of data	Calculation of project emissions
Calculation method	Multiplication of skid trail dimensions by carbon stock in accordance with parameter CBB_tree, described in section 4.1
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	CS/U,ig
Data unit	t C
Description	Carbon stock of trees snapped, leaning, or uprooted in logging gap lg
Source of data	Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data.
Description of measurement methods and procedures to be applied	If the forest management team data is used: Measurements (e.g. DBH, H) shall be taken on all trees (>10 cm DBH) that have been significantly damaged and are assumed to be killed by the felling of the timber tree. This shall include all trees where the bark of the stem has been damaged, where the stem has been snapped off, where the stem has been pushed over, and those trees that have been uprooted. The above and belowground carbon stock of each tree must be estimated using the same allometric regression equation and root to shoot ratio used in the module for estimating the carbon pool in trees (CP-AB) in the baseline scenario.
Frequency of monitoring/recording	Measurement shall take place one time prior to any verification event.
Value applied	Currently N/A, to be calculated ex-post, after SFMP implementation.
Monitoring equipment	On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study.

	If peer-reviewed study data is used, the value of carbon stock of trees snapped, leaning, or uprooted due to logging will be applied in the logging area, that will be measured ex-post, when applicable, using remote sensing and GIS.
QA/QC procedures to be applied	In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs.
Purpose of data	Calculation of project emission
Calculation method	Literature data from reputed sources will be used and cross- checked with field data from forest management team, if available.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	CB,ig
Data unit	t C
Description	Carbon stock of branches knocked to the ground as a direct consequence of tree felling in logging gap Ig
Source of data	Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data.
Description of measurement methods and procedures to be applied	As indicated in VM0007's M-REDD module.
Frequency of monitoring/recording	Measurement shall take place prior to every verification event.
Value applied	Currently N/A, to be calculated ex-post, after SFMP implementation.
Monitoring equipment	On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study.
QA/QC procedures to be applied	In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs.
Purpose of data	Calculation of project emissions

Calculation method	Literature data from reputed sources will be used and cross- checked with field data from forest management team, if available.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	Wskid
Data unit	M
Description	Mean width of skid trails
Source of data	Publicly available scientific data, Annual Operational Plans of SFMPs, and forest management team data
Description of measurement methods and procedures to be applied	The average width of skid trails created within a stratum i can be based on reported widths; a conservative estimate based on machinery used; or additional field measurements. Reported width: the average width of field measured skid trails within FSC Post Harvest Assessment reports from logging that has taken place within the project area; Conservative estimate: Width edge of tires on largest skidder type * 140% can only be used where skidder type(s) is known and used to create all skid trails. (Skidder defined as heavy vehicle used in logging operation to create path and pull cut trees to logging road); Field measurements: average of field sampling based on systematic sampling with a random start. The width of the skid trail shall be equal to the distance between trees undamaged by skid trail creation on one side to trees undamaged by skid trail creation on the other side. Sufficient number of measurements of road width shall be measured to achieve a precision of equal or less than 15% of the mean at the 95% confidence interval.
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value applied	Currently N/A, to be calculated ex-post, after SFMP implementation.

Monitoring equipment	On-the-ground measurement, Annual Operational Plans of the Sustainable Management Plan, and peer-reviewed data study.
QA/QC procedures to be applied	In the case of forest management team data, QA/QC procedures will be in accordance with those stipulated by the SFMPs.
Purpose of data	Calculation of project emissions
Calculation method	Literature data from reputed sources will be used and cross-checked with field data from forest management team, if available.
Comments	The parameter will only be measured if the project proponents decide to have forest management, depending on the market conditions and knowing that the management need to have the FSC certification. It is being listed here for precaution.

Data / Parameter	Leakage outside
Data unit	tCO ₂ /year
Description	Net CO ₂ emissions due to unplanned deforestation displaced outside the leakage belt up to year t*
Source of data	Remote sensing data and GIS. Secondary data.
Description of measurement methods and procedures to be applied	As described in section 3.2.3 and 3.3.3.
Frequency of monitoring/recording	Annually
Value applied	EX ante calculation as per LK-ASU.
Monitoring equipment	Remote sensing and GIS. Secondary data.
QA/QC procedures to be applied	N/A.
Purpose of data	Calculation of leakage
Calculation method	As per LK-ASU, Step 4.
Comments	N/A.

Data / Parameter	MANFOR
Data unit	ha
Description	Total area of forests under active management nationally
Source of data	Official country-specific data from IBAMA ¹⁷⁴ .
Description of measurement methods and procedures to be applied	As per LK-ASU, a demonstration is required that areas will be protected against deforestation. Such a demonstration must include the existence of forest guards in sufficient numbers to prevent illegal colonization and an active management plan detailing harvest plans and return intervals, and/or evidence that the concession owner has previously evicted illegal colonists/squatters from the forest areas. Ex-ante it can be assumed that MANFOR must remain constant.
Frequency of monitoring/recording	Annually
Value applied	1.400,000
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	PROTFOR
Data unit	ha
Description	Total area of fully protected forests nationally
Source of data	Official country-specific data from ISA ¹⁷⁵ .

¹⁷⁴ Source [http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em-2019-totalizou-39-mil-hectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20ibama%20em%202019%20totalizou%2039%20mil%20hectares,-Publicado%3A%20Sexta%2C%2021&text=Bras%C3%ADlia%20\(21%2F02%2F2020,metros%20c%C3%BAbicos%20de%20madeira%20nativa](http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em-2019-totalizou-39-mil-hectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20ibama%20em%202019%20totalizou%2039%20mil%20hectares,-Publicado%3A%20Sexta%2C%2021&text=Bras%C3%ADlia%20(21%2F02%2F2020,metros%20c%C3%BAbicos%20de%20madeira%20nativa)

¹⁷⁵ <https://uc.socioambiental.org/pt-br/paineldedados>

Description of measurement methods and procedures to be applied	As per LK-ASU, a demonstration is required that areas will be protected against deforestation. Such a demonstration is made by government mechanisms and national polices. Ex-ante it can be assumed that PROTFOR must remain constant.
Frequency of monitoring/recording	Annually
Value applied	63,755,583
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	TOTFOR
Data unit	ha
Description	Total available national forest area
Source of data	The total available national forest is estimated by National Forest Information System SNIF ¹⁷⁶
Description of measurement methods and procedures to be applied	As per LK-ASU, forest areas suitable for conversion to livestock. Ex ante it can be conservatively assumed that TOTFOR must remain constant for the baseline period.
Frequency of monitoring/recording	Annually
Value applied	486.461,572
Monitoring equipment	N/A.

¹⁷⁶ <https://snif.florestal.gov.br/pt-br/os-biomas-e-suas-florestas>

QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	ΔCP, LB
Data unit	tCO ₂ e
Description	Net greenhouse gas emissions within the leakage belt in the project case
Source of data	As per Module M-REDD
Description of measurement methods and procedures to be applied	As per Module M-REDD.
Frequency of monitoring/recording	Annually
Value applied	To be measured ex-post.
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	PROPI_{MM}
Data unit	Proportion
Description	Estimated proportion of baseline deforestation caused by immigrating population
Source of data	Official country-specific (government) data. The proportion of baseline deforestation caused by immigrating population (PROPI _{MM}) was estimated for a period from 2014 to 2022. For

	calculating $PROP_{IMM}$, local data ¹⁷⁷ were used for births, deaths, and population. It is then assumed that the total annual population growth in a given municipality is attributed to: i) births and ii) immigration. Thus, by subtracting the number of annual births from the total annual population growth, it is possible to infer the number of immigrants.
Description of measurement methods and procedures to be applied	Estimated as proportion of the area deforested in the past 5 years by population that migrated into the leakage belt and project area in the past 5 years (all areas within 2 km of the boundaries of the project area and the leakage belt must be considered here).
Frequency of monitoring/recording	Annually
Value applied	0.20
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	ADefLB,i,t
Data unit	ha
Description	Area of recorded deforestation in the leakage belt in the project case in stratum i in year t
Source of data	As per Module M-REDD. Satellite imagery.

¹⁷⁷ Sources : <http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinasc/cnv/nvam.def> ;
<https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html>

Description of measurement methods and procedures to be applied	As per Module M-REDD. Satellite imagery analysis.
Frequency of monitoring/recording	Annually
Value applied	To be measured ex-post.
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	ADefPA,I,u,t
Data unit	ha
Description	Area of recorded deforestation in the project area in the project case in stratum i converted to land use u in year t
Source of data	As per Module M-REDD. Satellite imagery.
Description of measurement methods and procedures to be applied	As per Module M-REDD. Satellite imagery analysis.
Frequency of monitoring/recording	Annually
Value applied	To be measured ex-post.
Monitoring equipment	N/A.
QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

Data / Parameter	RFt
Data unit	%
Description	Risk factor used to calculate VCS buffer credits
Source of data	• VCS Non-Permanence Risk Report (v3.1), • Remote sensing data and GIS, • Supervisor report. • Literature data.
Description of measurement methods and procedures to be applied	All sources of data from the VCS Non-Permanence Risk Report will be used to measure the various risk factors.
Frequency of monitoring/recording	Annually
Value applied	10
Monitoring equipment	VCS-approved AFOLU Non-Permanence Risk Tool
QA/QC procedures to be applied	Literature data from reputed sources will be used and critically checked. When possible, the average of two or more sources will be used.
Purpose of data	Calculation of project emissions
Calculation method	All the risk factors described in the VCS Risk Report were assessed.
Comments	N/A

Data / Parameter	Project Forest Cover Monitoring Map
Data unit	ha
Description	Map showing the location of forest land within the project area at the beginning of each monitoring period. If within the Project Area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event
Source of data	Remote sensing in combination with GPS data collected during ground truthing

Description of measurement methods and procedures to be applied	The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value applied	To be calculated ex-post
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other sources will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area: Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very good.
Purpose of data	Project Emission

Calculation method	Remote sensing and GIS
Comments	N/A

Data / Parameter	Leakage Belt Forest Cover Monitoring Map
Data unit	ha
Description	Map showing the location of forest land within the leakage belt area at the beginning of each monitoring period. Only applicable where leakage is to be monitored in a leakage belt
Source of data	Remote sensing in combination with GPS data collected during ground truthing
Description of measurement methods and procedures to be applied	The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy.
Frequency of monitoring/recording	Must be monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years examination must occur prior to any verification event
Value applied	To be calculated ex-post
Monitoring equipment	Remote sensing and GIS
QA/QC procedures to be applied	The Mapbiomas project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. Mapbiomas data are applicable for use in this project, according to the criteria listed below (Methodology VM0007): i) Mapbiomas data covers the entire project area, leakage belt and reference region. ii) Mapbiomas data cover the entire reference period (beginning, middle and end) of the fixed baseline period. iii) Mapbiomas monitors conversion of forest land to non-forest land. iv) Monitoring occurred during the entire fixed baseline period. In case of unavailability of Mapbiomas data for monitoring period, other sources will be consulted such as PRODES or a classification of imagery (Landsat8) will be carried out to measure deforested area:

	Land use and land cover mapping is assessed using images with spatial resolution superior to 30 meters. Image acquisition is performed during the period of low incidence of clouds and rainfall in the region, within July and September. Images undergo geometric correction by means of geo-referencing, using topographic maps as reference or USG-NASA orthorectified images. For analysis of areas with cloud cover, visual interpretation of radar image would be performed. Evaluation of classification accuracy is performed by analyzing the overall accuracy and kappa index obtained from a confusion matrix. The minimum accuracy of the classification mapping should be over than 90%, considered very good
Purpose of data	<i>Leakage emissions</i>
Calculation method	Remote sensing and GIS
Comments	N/A

Data / Parameter	ADefLB,i,u,t
Data unit	ha
Description	Area of recorded deforestation in the leakage belt in stratum i converted to land use u in year t
Source of data	As per Module M-REDD v2.2. Remote sensing imagery.
Description of measurement methods and procedures to be applied	As per Module M-REDD. Satellite imagery analysis.
Frequency of monitoring/recording	Annually
Value applied	To be measured ex-post.
Monitoring equipment	N/A

QA/QC procedures to be applied	As per Section 9.3 of REDD+ MF or other VCS methodology that uses this module.
Purpose of data	Calculation of leakage emissions
Calculation method	N/A.
Comments	N/A.

3.3.3 Monitoring Plan

This Monitoring Plan was developed according to Methodology VM0007 Version 1.6. The methodology encompasses four main monitoring tasks:

- i) Monitoring of project implementation;
- ii) Monitoring of actual carbon stock changes and GHG emissions within the project area;
- iii) Monitoring of leakage; and
- iv) Ex post calculation of net anthropogenic GHG emissions reductions.

This Monitoring Plan describes how these tasks will be implemented. For each task, the monitoring plan includes the following aspects:

- a) Technical description of the monitoring tasks;
- b) Data to be collected;
- c) Overview of data collection procedures;
- d) Quality control and quality assurance procedures;
- e) Data archiving; and
- f) Organization and responsibilities of the parties involved in all the above.

General information

Carbonext will perform the monitoring of carbon stocks from deforestation rates and greenhouse gas emissions produced or avoided during the REDD+ project, so it has a specialized and experienced REDD+ team. This includes social experts, who have the necessary skills to involve the community and stakeholders, ensuring community participation, activity design and socio-economic indicator monitoring, as described in 2.4.3.

The main community stakeholders in this project are the communities Vila Batista, Vale da Benção and Boa Esperança. The communication channel between the stakeholders and the proponent is between the secretary and spokesperson Valdenir Lemos Ribeiro, hired by the Project Proponent.

Given the above, the organizational structure below is very simple, where the Project Proponents (Silvio Balen and Carina Balen) and the Project Designer (Carbonext) work together and exist a communication channel direct between site spokesperson Valdenir Lemos Ribeiro, and the technical developer team. The technical team is responsible for planning the activities and providing the guidance, while the site spokesperson is responsible for executing the project's actions and collective their respective items of evidence.

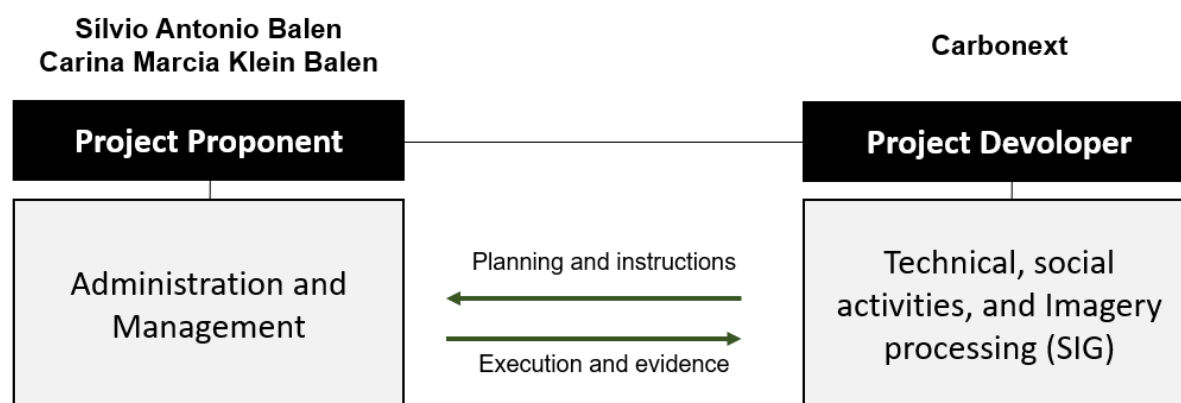


Figure 39 - General overview of parties involved in monitoring activities.

For the IPOÁ REDD+ Project, the PD activities (historical analysis of land use change) were carried out using GIS tools by Carbonext's team of experts using data from the PRODES project¹⁷⁸. Thus, we foresee that the other MRs will follow this strategy in order to guarantee methodological consistency. In case of unavailability of data from the PRODES system, we will consult the Mapbiomas base¹⁷⁹. In case of unavailability of data from Mapbiomas, we will perform a supervised classification of optical images from Landsat or Sentinel 2 orbital sensors. If there is a high incidence of clouds in the mentioned images, the

¹⁷⁸ <http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>

¹⁷⁹ <https://mapbiomas.org/>

strategy will be to classify radar (SAR) images available from the Sentinel 1 sensor and thus measure all changes in land use within the period defined for the MR according to the requirements demanded by the methodology.

As a complementary strategy for monitoring deforestation, Carbonext has implemented a deforestation detection and warning system called CarbonEye. Since the area has difficult access, the adoption of this system increases the chances of detecting possible changes in land use (deforestation), because it is programmed to perform continuous analysis as soon as an optical or radar image of the project area is available. Because it is an alert system, its process is designed to be sensitive enough to detect any change in the forest, undergo a rapid internal validation process, and issue alerts for monitoring agents to confirm and take appropriate action to contain the problem as efficiently as possible. The system is available for the IPOÁ project since 06/2022.

Monitoring of Actual Carbon Stock Changes and GHG Emissions within the Project Area

a) Technical description of the monitoring tasks.

Project activities implemented within the project area will be monitored to be consistent with the management plans of the project area and the PD.

b) Data to be collected.

Monitoring of deforestation-avoidance activities will be performed by means of evaluations and using satellite imagery to continuously inspect the forest condition within the Project Area, as described in 3.3.2. All images, maps and records generated during project implementation should be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

c) Overview of data collection procedures.

All invoices, receipts and contracts related to activities implemented in this REDD+ project shall be conserved in printed version. Whenever possible, documentation should be kept in electronic format.

d) Quality control and quality assurance procedures.

The project proponent will train personnel to collect and keep all documentation in a sure place. All electronic documentation should also be sent to CARBONEXT to further security and checking.

e) Data archiving

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification. Backup copies of files should be available in the project proponent facilities, as well as in CARBONEXT facilities.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

The project proponent is responsible to implement this monitoring item. CARBONEXT is coresponsible in archiving, and quality control and quality assurance procedures, as described in 3.3.2.

Monitoring of land-use and land-cover change within the project area

a) Technical description of the monitoring tasks.

Monitoring of land-use and land-cover change within the Project Area will be performed annually by analyzing satellite imagery of the Project Area.

Imagery analysis of land use will be based on secondary information from PRODES project, referring mainly to the classes "Forest Land" and "Non-Forest Land"

b) Data to be collected.

The categories of change that will be subject to MRV-A (monitoring, reporting, verification, and accounting) are "Area of forest land converted to non-forest land", "Area of forest land undergoing carbon stock decrease" and "Area of forest land undergoing carbon stock increase". In this context, data to be collected consists of annual land use cover processed by PRODES project for the entire land coverage of Project Area and other such as Leakage belt and reference region.

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

c) Overview of data collection procedures.

The PRODES project contributes to understand the deforestation dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

PRODES data cover the entire project area, leakage belt and reference region.

PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.

PRODES monitors conversion of forest land to non-forest land.

Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures.

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

With the help of the "Create Random Points" tool in ArcGIS Pro, at least 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy). Land use classes must have higher values than 90% for the accuracy, as required in VMD0007 3.3 *methodology*.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT GIS expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

Monitoring of carbon stock changes and non-CO₂ emissions from forest fires

a) Technical description of the monitoring tasks.

Monitoring of carbon stocks is mandatory in the following cases:

Within the project area:

- a) Areas subject to significant carbon stock decrease in the project scenario according to the ex-ante assessment. These will be areas subject to controlled deforestation and planned harvest activities, such as logging, fuel wood collection and charcoal production. In these areas, carbon stock changes must be estimated at least once after each harvest event.
- b) Areas subject to unplanned and significant carbon stock decrease, e.g., due to uncontrolled forest fires and other catastrophic events. In these areas, carbon stock losses must be estimated as soon as possible after the catastrophic event.

Within leakage management areas:

- a) Areas subject to planned and significant carbon stock decrease in the project scenario according to the ex-ante assessment. In these areas, carbon stocks must be estimated at least once after the planned event that caused the carbon stock decrease.
- b) Data to be collected.

The results of monitoring activity data and carbon stocks must be reported using the same formats and tables used for the ex-ante assessment, according to Methodology VM0007 (the applicability of each table must be evaluated ex post, in the Monitoring Report).

Non-CO₂ emissions from forest fires are subject to monitoring and accounting, when significant.

Decreases in carbon stocks and increases in GHG emissions (e.g., in case of forest fires) due to natural disturbances (such as hurricanes, earthquakes, volcanic eruptions, tsunamis, flooding, drought, fires, tornados or winter storms) or man-made events, including those over which the project proponent has no control (such as acts of terrorism or war), are subject to monitoring and must be accounted under the project scenario, when significant.

If the area (or a sub-set of it) is affected by natural disturbances or man-made events generated VCUs in past verifications, the total net change in carbon stocks and GHG emissions in the area(s) that generated VCUs must be estimated, and an equivalent amount of VCUs must be cancelled from the VCS buffer.

c) Overview of data collection procedures

The PRODES project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

PRODES data cover the entire project area, leakage belt and reference region.

PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.

PRODES monitors conversion of forest land to non-forest land.

Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

With the help of the "Create Random Points" tool in ArcGIS PRO, 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to

generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy).

Land use classes must have higher values than 90% accuracy for the accuracy and Kappa index, as required in VM0007 1.6 methodology.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

Monitoring of Leakage

a) Technical description of the monitoring tasks

The sources of leakage identified as significant in the ex-ante assessment are subject to monitoring. Two sources of leakage are potentially subject to monitoring:

- I. Decrease in carbon stocks and increase in GHG emissions associated with leakage prevention activities;
- II. Decrease in carbon stocks and increase in GHG emissions in due to activity displacement leakage.

This Project Activity does not involve decrease in carbon stocks and increase in GHG emissions associated with leakage prevention activities. In this project, leakage prevention activities do not involve any carbon stock reduction due to deforestation or additional emissions caused by increased grazing activities. In this

case, only the decrease in carbon stocks and increase in GHG emissions in due to activity displacement leakage will be monitored.

b) Data to be collected.

Deforestation above the baseline in the leakage belt area will be considered activity displacement leakage.

The result of the ex-post estimations of carbon stock changes must be reported using the same table formats used in the ex-ante assessment of baseline carbon stock changes in the leakage belt.

Where strong evidence can be collected that deforestation in the leakage belt is attributable to deforestation agents that are not linked to the Project Area, the detected deforestation will not be attributed to the Project Activity and considered leakage. The operational entity verifying the monitoring data shall determine whether the documentation provided by the project proponent represents sufficient evidence to consider the detected deforestation as not attributable to the Project Activity and therefore not leakage.

To estimate the increased GHG emissions due to forest fires in the leakage belt area the assumption is made that forest clearing is done by burning the forest. The parameter values used to estimate emissions shall be the same used for estimating forest fires in the baseline, except for the initial carbon stocks (Cab, Cdw) which shall be those of the initial forest classes burned in the leakage belt area.

Report the result of the estimations using the same table formats used in the ex-ante assessment of baseline GHG emissions from forest fires in the project:

- Parameters used to calculate emissions from forest fires in the leakage belt area
- Ex post estimated non-CO₂ emissions from forest fires in the leakage belt area.

c) Overview of data collection procedures.

The PRODES project contributes to understand the land use dynamics in Brazil. The data generated by this program is used in this project. PRODES data are applicable for use in this project, according to the criteria listed below (Methodology VM0007):

- i) PRODES data cover the entire project area, leakage belt and reference region.
- ii) PRODES data cover the entire reference period (beginning, middle and end) of the fixed baseline period.

- iii) PRODES monitors conversion of forest land to non-forest land.
- iv) Monitoring occurred during the entire fixed baseline period.

d) Quality control and quality assurance procedures.

The validation of land-use data used for modeling of land use will be performed by using the confusion matrix, in order to calculate the overall index of success by period and by class. Three specific classes will be used: forest, accumulated deforestation and other (hydrography, not forest, clouds, roads, residues, unclassified objects, and others).

The validation will be performed by using satellite images from Landsat 8 TM (orbit 230/64) to cover the total geographic scope of the Project Area and Reference Region. Other optical sensors could be use for accuracy assessment in case of cloud cover such as Sentinel 2 and CBERS.

With the help of the "Create Random Points" tool in ArcGIS 10.0, at least 100 random points will be generated for each class / year as samples for evaluation, using satellite images as reference, making it possible to generate a confusion matrix for calculation of the accuracy indexes, and the Kappa index (indicators for validation of mapping accuracy). Land use classes must have higher values than 90% for the accuracy and Kappa index, as required in VM0007 1.6 methodology.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

Ex Post Calculation of Net Anthropogenic GHG Emission Reduction

a) Technical description of the monitoring tasks.

The calculation of ex post net anthropogenic GHG emission reductions is similar to the ex-ante calculation with the only difference that ex post estimated carbon stock changes and GHG emissions must be used in the case of the project scenario and leakage.

b) Data to be collected.

Report the ex post estimated net anthropogenic GHG emissions and calculation of Verified Carbon Units (VCUt, and VBCT) using the same table format used for the ex-ante assessment:

- Ex post estimated net anthropogenic GHG emission reductions and VCUs

c) Overview of data collection procedures.

Data collection procedures are the same as described in previous steps. This step involves compilation of data from previous procedures to calculate ex post net anthropogenic GHG emission reduction.

d) Quality control and quality assurance procedures.

A map showing Cumulative Areas Credited within the project area shall be updated and presented to VCS verifiers at each verification event. The cumulative area cannot generate additional VCUs in future periods.

e) Data archiving.

All maps and records generated during project implementation will be conserved and made available to VCS verifiers at verification for inspection to demonstrate that the AUD project activity has actually been implemented.

All documents and records will be kept in a secure retrievable manner for at least two years after the end of the project crediting period.

f) Organization and responsibilities of the parties involved in all the above.

All satellite imagery assessments will be performed either in-house by CARBONEXT SIG expert time. CARBONEXT is responsible for reporting and data archiving, according to VM0007, and for providing assistance during verification audits. The Project Proponent is co-responsible for data archiving.

Revisiting the Baseline Projections for Future Fixed Baseline Period

According to VCS standard, the baseline will be revisited every 6 years. Thus, the first revision of the baseline is scheduled for 2028. For this purpose, the following tasks will be carried out:

- Updating information on agents, drivers, and underlying causes of deforestation, which involves the following sub-tasks:

- Collecting information that is relevant to understand deforestation agents, drivers and underlying causes;

- Redoing step 3 of the ex-ante methodology, as specified in the methodology;

- Recalibrating the model for projection of future deforestation, using new “Factor Maps” for the subsequent fixed baseline period.

- Adjusting the land-use and land-cover change component of the baseline, which involves reassessing the following components of the baseline projections:

- The annual areas of baseline deforestation.

- The location of baseline deforestation.

- Adjustment of the annual areas of baseline deforestation.

- Adjustment of the location of the projected baseline deforestation.

Adjusting, as needed, the carbon component of the baseline (this task will only be carried out if more accurate methods for carbon stocks estimates are available in the occasion of baseline revision).

3.3.4 Dissemination of Monitoring Plan and Results (CL4.2)

The dissemination of the results of the monitoring and verification will be published on the internet through the Verra platform page as well as on Carbonext’s website. Given that some local communities don’t yet have access to the internet, during the onsite visits print versions of the summary results will be made available to community members and provided to any individual or institution, upon request. The established frequency is annually or before any verification event, or project visits. To enhance the importance and effectiveness of the project, an Operating Procedure on Monitoring Report is being implemented. Other means of dissemination may also be adopted to give total transparency to the project.

3.4. Optional Criterion: Climate Change Adaptation Benefits

This item does not apply to the present project at this time.

3.4.1 Regional Climate Change Scenarios (GL1.1)

This item does not currently apply to the present project.

3.4.2 Climate Change Impacts (GL1.2)

This item does not currently apply to the present project.

3.4.3 Measures Needed and Designed for Adaptation (GL1.3)

This item does not currently apply to the present project.

4 COMMUNITY

4.1. Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CM1.1)

Social Aspects:

Three river-dwelling (*ribeirinho*) communities¹⁸⁰ are located within the Project Zone (PZ). Throughout the project's development, socio-economic questionnaires¹⁸¹ were applied to the families, to gather data and information from all the communities located within the Project Zone.

¹⁸⁰ *Ribeirinho* populations: residents of the banks of the rivers in the Amazon region.

¹⁸¹ The questionnaires and other social evidence will be made available to the project auditors.

Vila Batista is the community that is closest to the Project Area, having an estimated population of 35 people, belonging to four families, who have lived there for many years. The community leader is known as “*Carlinhos*” (full name Moacir Batista da Silva). Vila Batista, unlike other communities, is not centralized, that is, the houses in this community are very far apart, as can be seen in Figure below. Basically, Vila Batista is composed of two points: the first is where Carlinhos’ family live, the second point are the other two houses. There is also a fourth family that lives at the junction of the Sucunduri river with the *Igarapé Capinarana*, which belongs to the Vila Batista community. This family’s location falls within a private area identified in the SIGEF with unknown owner, so the house is not included in the Project Zone. But his family was interviewed, consulted, and counted as part of the community of Vila Batista, and therefore will participate in the project’s actions that involve that community.

The PZ’s second community is Boa Esperança, of which the leader is Magno Batista da Silva, Carlinhos’ brother. The community has existed for three years and is located about 5 km from Fazenda do Maanaim with an estimated population of 29 people around eight families. The houses are concentrated in a single geographical area, and the community is growing because it is relatively new.

More than ten years ago, the communities of Vila Batista and Boa Esperança were a single community, located together. However, disagreements among family members caused the rupture of the community. Thus, a part of the family, led by Carlinhos, stayed where the houses in Vila Batista are today, and another part, led by Magno, went to the District of Canumã, in the municipality of Borba. Years later they founded the Boa Esperança community.

The third riverside community Vale da Benção is led by Bráulio Pontes Machado and is located approximately 9 km from Fazenda do Maanaim, it used to be a larger community, but over time people moved to nearby towns, and now there are only five families, about 30 people.

None of the three communities have a formalized community association or cooperative, but all the residents recognize their respective leaders.

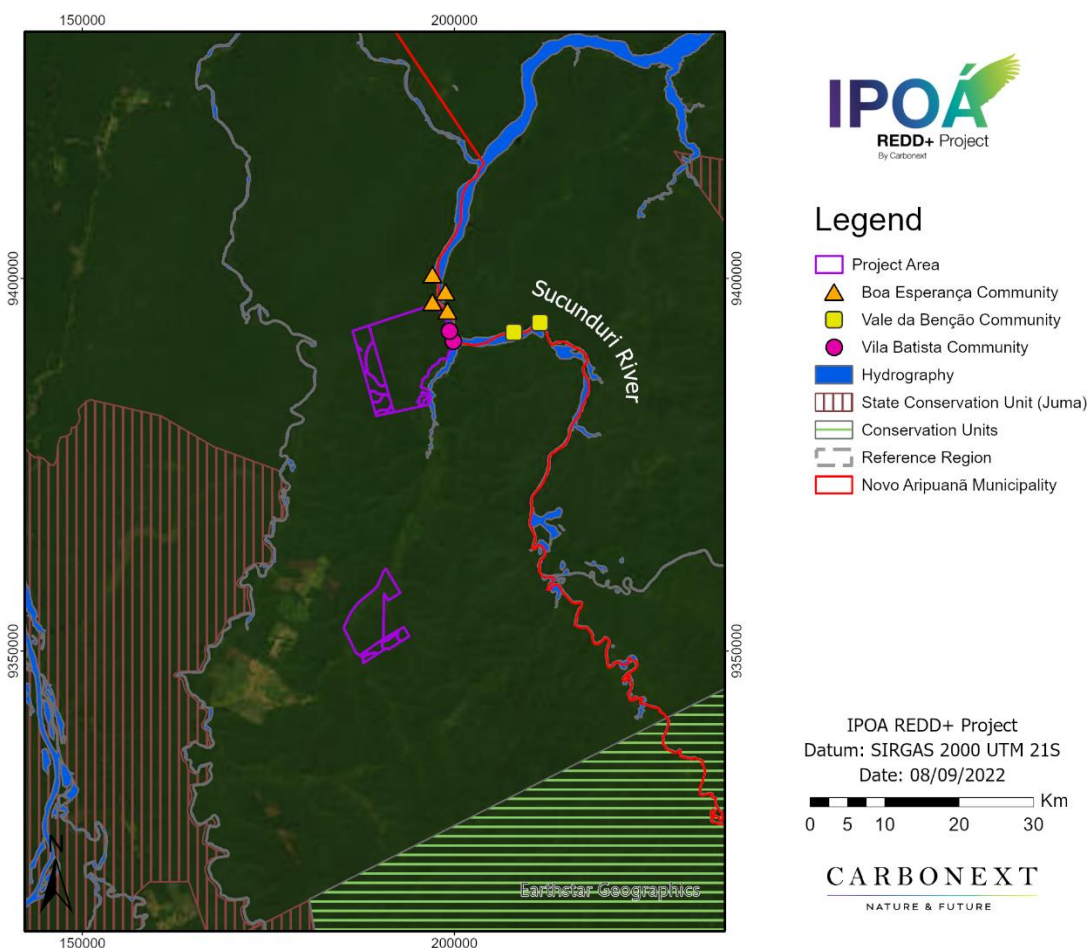


Figure 40. Residents' homes in each community in the Project Zone.

The communities have little or no infrastructure, being the set of fundamental services for the socio-economic development of a region, including sanitation, transportation, electricity, or telecommunication. The communities have no electricity from the municipal grid, however, in the homes of community leaders there are solar panels with low power only for basic use such as night lighting, or charging a cell phone, etc. In addition, one of the houses has a generator that used to be used during the night. Some residents of the communities have cell phones, but as there is no telephone coverage, few have carrier chips, and those who have them usually only have telephone signal or internet access when they go to larger cities like the municipalities of Borba and Novo Aripuanã. In addition, in the community of Arara where Internet access is available for sale (R\$ 3.00 per hour of use). In March 2022, the municipality of Novo Aripuanã

registered 10,045 mobile phone accesses and has a service density of 38.5 accesses per 100 habitants, a value 57% lower than the density in the state of Amazonas¹⁸².

There are no police stations, bank branches, communication antennae, or health centers operating in any of the communities. All community members are native speakers of Portuguese.

As far as education is concerned, according to the socioeconomic questionnaires applied to the residents, most people in the communities are illiterate. According to the socioeconomic questionnaire¹⁸³, of the 13 families interviewed, about 70 people covered by these questionnaires, obtained the following numbers: 2% did higher education, 5% finished high school, 1% Elementary 2¹⁸⁴, 28% finished Elementary 1¹⁸⁵ and 38% are illiterate. In Vila Batista there is a school, which is deteriorated and, according to reports, the students do not have classes during the whole school year. The Vale da Benção community used to have a school, but it no longer exists. Boa Esperança during the writing of this Project Description was articulating with the municipal government of Borba to obtain support and the necessary authorization for a resident of the community to be hired by the municipality and teach the classes. Authorization and support were granted in September 2022, and since then the community has started to have classes for its children in Elementary 1.

¹⁸² Anatel - Mobile Telephony Access

¹⁸³ The questionnaires and other social evidence will be made available to the project auditors.

¹⁸⁴ 6th to 8th grade.

¹⁸⁵ 1st to 5th grade.



Figure 41. Boa Esperança School.

The main leisure activity in the communities are the soccer games played among the men of the communities. In terms of religion, all are evangelical Christians from the Assembly of God, more details are provided in the item below about religion. To access other places, the communities use canoes, small wooden boats with engines known as *rabetas*¹⁸⁶, or other small boats.

It is worth mentioning that there are also no communities or community groups, or quilombos or indigenous groups, within the Fazenda Campos property.

During the visit to the communities conducted in March 2022, the main economic activities of the communities were studied, which are based on the extraction of copaiba oil and nuts, cassava planting for flour production, and vegetable production. In addition, there were reports from residents of wood harvesting for the community's own use for building houses and other objects, but they do not have authorization or a cutting permit from the responsible bodies. Through third parties, we discovered that there is the sale of

¹⁸⁶ A *rabeta* is a small boat, a basic canoe that uses a simple outboard engine capable of carrying up to 3 or 4 adults.

timber, which is taken to the District of Canumã, Borba, and Novo Aripuanã to be sold. Another source of income is the payment generated by the Brazil Assistance Program¹⁸⁷.

The wild-harvesting activities described above are mainly carried out by men, while work within the household is the main responsibility of women. This unequal participation in work that generates financial income is the basis for a gender pay gap which, as found in field studies (as further discussed below), which is also reflected in literature on certain traditional amazon communities¹⁸⁸. While both genders carry out the work for production of manioc flour, a job carried out collectively by the whole family, this is not remunerated work.

Diversity within the community:

To diagnose the communities, we applied socioeconomic questionnaires in the three communities mentioned above: Vila Batista, Boa Esperança and Vale da Benção. Unfortunately, at the time, part of the communities is in the municipalities of Borba and Novo Aripuanã, and some men from the community were engaged in timber extraction near the Camaiu River. Because of this, we did not apply the questionnaires to all community member, but we will consider the data obtained from these questionnaires to quantitatively describe some aspects of the communities.

Taking into account the total number of residents in the three communities, according to data obtained during the community visits in April, March and October 2022¹⁸⁹, a total of 94 habitants were counted in the three communities: 35 residents of Vila Batista, 29 in Boa Esperança, and 30 in Vale da Benção. 70 residents were covered by the questionnaires applied during the site visits, corresponding to 74% of the total number of people residing in the communities, and twelve families, or 71% of the total number of families.

Regarding gender diversity in the communities, through the questionnaires applied, it was observed that 24% of the people evaluated are women, 34% are men, and 37% are children and teenagers, the age group

¹⁸⁷ Auxílio Brasil — Português (Brasil) (www.gov.br)

¹⁸⁸ https://go.gale.com/ps/i.do?id=GALE%7CA565057147&sid=googleScholar&v=2.1&it=r&linkaccess=abs&issn=15168182&p=AONE&sw=w&userGroupName=nysl_me_jfkens

¹⁸⁹ The questionnaires and other social evidence will be made available to the project auditors.

defined by the Law of the Statute of the Child and Adolescent¹⁹⁰. About 4% correspond to families for which there is no information.

Regarding the education of the residents, of the total number of people evaluated by the socioeconomic questionnaires¹⁹¹, there was the evaluation of the level of schooling in the three communities in the IPOÁ Project Zone and concluded that: 38% of people are illiterate in the communities, 28% have completed Elementary 1, 1% Elementary 2, 5% High School, and only 2% have a Higher Education. Based on this data, there was an analysis of the level of schooling by gender, as presented in the table below.

It was considered extremely important to understand the level of education of these women in order to plan and develop activities that generate impacts and broaden the perspectives of these women.

Table 124. Percentage of men and women by education level.

Illiterate		Elementary 1		Elementary 2		High School		Higher Education	
Women	Men	Women	Men	Women	Men	Women	Men	Women	Men
7%	9%	5%	15%	1%	0%	2%	3%	1%	1%

Regarding the income of the families of the evaluated communities, it is noteworthy that in our evaluation women do not work in the forest, a fact consistent with studies by Rodrigues et al. 2015 who cite that in certain communities men contribute more to the financial income, and that women hold mainly knowledge linked to home, and the production of cassava flour, which does not fit a logic "exclusionary of capitalism", but of common processing.

That is, they do not have paid work and do not contribute to the family financial income, but rather to their subsistence. In addition, as described in the economic aspects below, the families' income is not very diversified and highly dependent on the extraction of the Copaiba oil and the Brazil nuts.

¹⁹⁰ L8069 (planalto.gov.br).

¹⁹¹ The questionnaires and other social evidence will be made available to the project auditors.

Economic aspects:

During the 2022 visits to the communities that are in the project's zone of influence, socioeconomic surveys were conducted to diagnose these communities. In the questionnaires applied, we understood the economic dynamics, and the main sources of income of the communities.

The main economic activities of the communities are based on the extraction of copaiba oil and Brazil nuts from the forest; the planting of cassava for flour production and the extraction of wood on demand, and the communities do not have licenses or authorization to harvest wood. Besides the sale of the products, the residents of the communities also use these products for the community's own consumption.

The extraction of copaiba oil and Brazil nuts is done only by men, who enter the forests to search for areas to do the extraction. The communities of Vila Batista, Boa Esperança and Vale da Benção sell almost 70% of the copaiba oil in the community of Arara, which is closer to the city of Novo Aripuanã, and the rest is sold in the municipality of Borba or sells to boat dealers. The price of the oil is given by the quantity in kilograms of product, and the price of a kilo of copaiba oil, according to reports, varies between R\$38.00 to R\$50.00 per kg. The nuts, on the other hand, are sold in 20-liter cans and the price per can in Novo Aripuanã is as high as R\$140.00.

According to Santana et al.¹⁹², the Brazil nut tree (*Bertholletia excelsa*) has represented, since 1950, the main non-timber forest product of the Amazon region and its extraction generates work and income for more than 25 thousand families, besides contributing to the dynamics and diversity of the flora and fauna, as well as to climate regulation.

The price of nuts becomes strongly determined by demand, since supply varies very little due to the productive capacity of Brazil nut groves that are at their limit. Thus, market price adjustments respond to the variation in consumer income (SANTANA et al., 2017).

In the production of cassava flour, women help men in production and in the fields. Besides the flour they sell in the cities or also in the merchant boats that navigate the Sucunduri River. The residents produce

¹⁹² Santana, A. C. de; Santana, Á. L.; Santana, Á. L.; Martins, C. M. Valoração e sustentabilidade da castanha-do-brasil na Amazônia. Revista de Ciências Agrárias, 60, 77-89, 2017. Available: <http://periodicos.ufra.edu.br/index.php/ajaes/article/view/2690>.

other by-products derived from cassava, such as tapioca starch for personal consumption, as can be seen in the Figure below.



Figure 42. Vila Batista resident making Tapioca Gum.

The harvesting of wood usually happens after the men of the community receive an invitation to do this work or some kind of order. During one of the field visits to the area in April 2022, some of the men from Boa Esperança and Vila Batista had moved from their communities to carry out this illegal timber extraction work near the Camaiú River. As in some cases they are commissioned to extract the wood, the value and destination of the wood are agreed upon before the cut, when it is not commissioned, the sale of wood normally occurs in the cities, such as in the District of Canumã, Borba, and Novo Aripuanã.

Some wood is also extracted for the manufacture of canoes, for which the use of Itauba wood is common. The woods that, according to reports, are the ones they normally extract for sale are: *Ipê amarelo*, *Angelim*

pedra, Envira pente de macaco, Louro rosa, among others. Many of these are registered in the Juma RDS Management Plan¹⁹³.

It is worth mentioning that wood was harvested from the Manaaim property many years in the past, but the last harvests for commercialization were outside the project area.



Figure 43. Wood Cutting for Canoe Production.

Another source of income for some families in the communities is the payment generated by the Brazil Aid Program¹⁹⁴ of the Federal Government of Brazil. Generally, the benefited families wait two months for the

¹⁹³ https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf

¹⁹⁴ Auxílio Brasil — Português (Brasil) (www.gov.br).

payment to travel to the municipalities of Borba and Novo Aripuanã to receive the benefit, since the travel expenses are significant.

As described, the without – project scenario presents socioeconomic indicators, which are characteristic of the region with low income and welfare conditions due to the limited agricultural activities, lack of improved production techniques, difficulties in offtake, as well as in access to the consumer market. According to data from IBGE¹⁹⁵ the municipality of Novo Aripuanã has 52% of the population with monthly income of up to half the minimum wage per person. In the socioeconomic questionnaire applied to families in the communities during the community visit in 2022, it was found that none of the families assessed have a monthly income per capita of half the minimum wage¹⁹⁶.

Medical Aspects:

There are no health care centers or health agents in the communities. For medical care, the residents have to go to the towns of Borba or Novo Aripuanã. Borba is easier to access as it is possible to get to the town by *rabeta* boat, but according to reports from the residents themselves, this trip takes practically the whole day. The communities do not have a faster means of transportation, such as the water ambulance (*ambulancha*) to get to the nearest healthcare facility.

A municipal medical center used to be located in the Bom Jardim community, at approximately 20km distance from the IPOÁ communities. It was run by Municipal Health Secretariat with participation by a specialized Civil Society Organization, however in recent years it has been inactive, with no Community Health Agent available, and a lack of maintenance of the structure and medical supplies. To this extent, the IPOÁ Project Zone's communities have no access to permanent Basic Health Facilities at project start.

Once a year the communities have access to medical care by the CSO at a distance of approximately 100km. This project began in 2011, and aims to bring medical care, dental care, and hygiene practices to the riverside populations of the Amazon Basin. In addition to these services, residents reported in interviews that clothes, and hygiene kits were also donated.

¹⁹⁵ IBGE | Cidades@ | Amazonas | Novo Aripuanã | Panorama.

¹⁹⁶ Considering the 2022 Minimum Wage of R\$ 1,212.00. Source: <https://www.gov.br/esocial/pt-br/noticias/novo-salario-minimo-2022-veja-como-registrar-o-reajuste-no-esocial-domestico>.

The boat belonging to CSO used to sail along some rivers, including the Sucunduri river, providing care to the communities there (see figure below). However, in recent years, their nearest stopping point has been the Arara community, located 107 km away from the IPOÁ REDD+ Project. At the moment, the IPOÁ community members need to travel to receive medical care, implying significant costs and difficulty of access, as well as a lack of specialized dental care, which only the communities targeted by the CSO can receive.



Figure 44. Rivers navigated by the specialized health CSO

The diseases that most affect the region are malaria and influenza. In the present pandemic period (2020 – 2022), some residents have shown symptoms, but none have taken tests to identify whether or not there was an infection caused by the SARS-CoV-2 virus (coronavirus - COVID-19). All the residents have had the COVID-19 vaccine, but some adults have not had the 3rd or 4th dose.

Psychological Aspects:

In the interviews conducted in 2022¹⁹⁷, 100% of the residents of the three communities evaluated reported liking living in the communities because of the security, tranquillity, abundance of food, and because of the forest and the river. From the workshop with the women, it was possible to observe that the younger women

¹⁹⁷ The questionnaires and other social evidence will be made available to the project auditors.

between 14 and 20 years old, would prefer to live in the city, mainly because of better health and education conditions.

In the interviews 91% of the interviewees reported having a good relationship with the other communities and only 9% reported that the relationship between the communities is regular.

Environmental aspects:

The communities do not have an association and do not have community cooperatives, the community members recognize their leaders even in an informal way. At the time of the visit, there was no document indicating the leadership of these people. All members of the communities are native Portuguese speakers. As described under “Economic Aspects” above, the community is not supplied with electricity from the grid.

None of the above-mentioned communities have potable water supply systems; sewage collection and treatment systems, nor the collection of solid waste generated. The main source of water supply for these communities is the Sucunduri river, through which the health agents of the Unified Health System - SUS perform the distribution of hypochlorite for domestic water treatment. The solid residues generated are burned and the human waste is thrown into an individual black well.

In the Boa Esperança and Vila Batista communities, there were reports of dangerous situations while carrying out their economic activities, such as:

- Encounters with snakes while working in the forest
- Falling branches while extracting Copaíba oil
- Injuries due to the use of the chainsaw or machete

The community has not yet reported any risk of fires or landslides. However, the community of Boa Esperança has part of the community flooded by the rising level of the Sucunduri riverbed.

The main forest species¹⁹⁸ observed in the field were *Ipê*, *Itaúba*, *Louro Rosa*, *Castanheira* and *Copaíba*. The most sighted terrestrial animals are the white-lipped peccary, monkeys, capuchin monkeys, paca,

¹⁹⁸ Popular names of forest and animal species.

armadillo, and agouti. The birds mentioned were parrot, Macaw, Curassow, Cujubim and Hoatzin¹⁹⁹. The main small watercourses are the *Capinarana* and *Das Pedras* streams.

Spiritual Aspects:

All communities are Protestant Christian, the community is located in the Vila Batista community. Their custom is to attend this church on at least a weekly basis to its annual calendar of events.

The residents of all three communities declare themselves to be Protestant Christians, members of the Assembly of God.

The Vale da Benção community has a church where services are held on Tuesdays and Thursdays. On Sundays the services are held in the church of Vila Batista, which is outside the project area. However, the habitants of Vila Batista are building a new church inside the project area. The residents of Boa Esperança attend services in other communities, but they are interested in having a church within their own community.

¹⁹⁹ *Cigana* or *jacu-cigano* in Portuguese



Figure 45. Church in the Vale da Benção community.



Figure 46. Building the new church in Vila Batista.

The minister responsible for these churches lives in the Arara community, and the residents themselves conduct the services in his absence. Every year they hold the church's anniversary party.

4.1.2 Interactions between Communities and Community Groups (CM1.1)

In contact with the communities, we noticed in the reports of the residents of Vila Batista and Boa Esperança, that some years ago the leaders of both communities had a conflict that resulted in the separation of the community of Vila Batista, and years later from this separation there was the creation of the community of Boa Esperança. As the leaders of the current Vila Batista and Boa Esperança are brothers, and their wives are also sisters, the communities have contact with each other and a good relationship, but they prefer to treat each other as separate communities.

In questionnaires²⁰⁰ applied in the three riverside communities of this project, we asked them how they classified their relationship among the communities, and we obtained the following results: 91% of the interviewees reported having a good relationship with the other communities, and only 9% reported that the relationship among the communities is regular.

It is common in these contexts for neighbors or relatives to form strong bonds of mutual respect, mutual aid, reciprocal provision of services, and solidarity. However, this solid base of local social relations has not been sufficient to induce robust collective action capable of expanding the supply of collective public goods. In addition, knowledge related to the place where they live, including fauna, flora, important places to the community, climate, and dynamics of change, have been passed on from generation to generation, orally.

4.1.3 High Conservation Values (CM1.2)

Based on the study of the local socio-economic scenario and the reality of the communities observed on the ground so far, we identified in section 2.3.4 the main risks to communities, and some of them also put at risk the areas of high conservation value for the communities, which are a) deforestation pressure for

²⁰⁰ The questionnaires and other social evidence will be made available to the project auditors.

grazing/land use change; b) changes in policies and laws governing external actors; and c) disputes and changes in land rights or access/use rights that could cause displacement of the community by potential invaders.

The actions to prevent and mitigate these risks are set out in section 2.3.4 and in the *Adaptation Management Plan*, but in summary the actions taken will focus on surveillance and monitoring. If deforestation alerts are observed via satellite or in the field, action will be taken such as recording a report and specific plans for landscape restoration if necessary.

The following table highlights the areas of high conservation value for communities according to the Resource Network's High Conservation Value Identification Guide, and the table of activities to be implemented to conserve the identified HCVs and avoid negative social impacts.:

High Conservation Value	HCV 4: Ecosystem Services
Qualifying Attribute	According to the HCV Resource Network's High Conservation Value Identification Guide, HCV 4 refers to basic ecosystem services in critical situations, including protection of water sources and erosion control of vulnerable soils and slopes. Ecosystem services are the benefits that people obtain from ecosystems, such as provisioning services, supporting services, regulating services, and cultural services. An ecosystem service is critical where a disruption of the service causes a threat of severe, catastrophic, or cumulative negative impact on the well-being, health, or survival of local communities, the functioning of important infrastructure, or other HCVs. For the present project, HCV 4 will be defined as support and regulation services. Climate regulation²⁰¹: The Sucunduri River and forest are ecosystems that provide numerous important services to local communities.

²⁰¹ CARACENA, Susana Elizabeth López. A Oferta do serviço ecossistêmico de regulação climática em florestas tropicais: o estado da arte e as implicações da conversão da floresta em campo antrópico. 2019. Dissertação (Mestrado em Ecologia: Ecossistemas Terrestres e Aquáticos) - Instituto de Biociências, Universidade de São Paulo, São Paulo, 2019. doi:10.11606/D.41.2019.tde-25062019-090304. Acesso em: 2022-11-22.

	Local climate regulation is one of these services provided and has a direct impact on natural environmental conditions, influencing for example the local temperature, rainfall regime, water availability, water and air quality, etc. The REDD+ project will be an important project for the conservation of these services, since with the scenario of deforestation and land use for cattle ranching, there would be obstacles for the environment, such as erosion, soil impoverishment, silting of rivers, etc. Soil formation²⁰²: Soil is a complex system which is in a constant process of formation – a slow process essential for its maintenance and renewal. Ecosystem services are dependent on soil in aspects related to the soil's key functions, summarized below: providing the basis for biodiversity, the habitat of micro-organisms; providing the key medium for plant growth, which are the basis of a healthy diet for humans and animals; the storage, regulation and release of nutrients and essential elements; action in the water cycle and in the compartmentalization of water, directing it to underground reservoirs; provision of raw materials; and creation of carbon pools. Forest conservation is the key means provided by the project to avoid degradation and loss of soil. Protection of water sources²⁰³: Water is critical for human use, including for drinking water, food preparation, hygiene, irrigation, and fishing. Degradation of terrestrial landscapes can affect water bodies. There are no viable or readily available alternatives to water. Forest conservation acts as a barrier to prevent degradation or pollution of water bodies, so forest maintenance is crucial to the well-being of the communities that use this resource.
Focus Area	Project Zone

High Conservation Value	HCV 5: Needs of Communities
-------------------------	-----------------------------

²⁰² VEZZANI, Fabiane Machado. Solos e os serviços ecossistêmicos (Soils and the Ecosystem Services). Revista Brasileira de Geografia Física, [S.l.], v. 8, p. 673-684, dez. 2015. ISSN 1984-2295. Disponível em: <<https://periodicos.ufpe.br/revistas/rbgfe/article/view/233637/27226>>. Acesso em: 22 nov. 2022. doi:<https://doi.org/10.26848/rbgf.v8.0.p673-684>.

²⁰³ <http://institutoaua.org.br/agua-como-esse-servico-ecossistemico-beneficia-o-ser-humano/>.

Qualifying Attribute	According to the HCV Resource Network's High Conservation Value Identification Guide, HCV 5 refers to sites and resources that are fundamental to meeting the basic needs of local people, identified through dialogue with these communities or indigenous peoples. From the socioeconomic questionnaires and public consultation carried out, the ecosystem services of vital importance to the communities, were identified: Food resources: forests provide the communities with many of their basic food needs, including animal protein, fruit, and nuts, etc. Fresh water. Medicines: oils, roots, and other NTFPs. Construction materials: timber and fiber. These resources and ecosystem services, vital to communities' survival and well-being, are protected by the REDD+ project.
Focus Area	Project Zone

High Conservation Value	HCV 6: Cultural values
Qualifying Attribute	According to the HCV Resource Network's High Conservation Value Identification Guide, HCV 6 refers to areas, resources, habitats, and landscapes of special cultural, archaeological, or historical significance at global or national levels, and/or of critical cultural, ecological, economic, or religious/sacred importance for the traditional culture of local communities, indigenous peoples, or traditional populations, identified in cooperation with these communities or populations. From the socio-economic questionnaires applied and the public consultation carried out, it was identified that:

	Areas of extraction of copaiba oil: HCV of cultural value because the extraction process, use of the product, and the small-scale commercialization have importance for the communities' traditional culture and knowledge. For the community, copaiba oil has innumerable uses, for example, it is widely used as an anti-flu oil, anti-inflammatory, and scar-healing medicinal recipes. The oil is sold in the nearby larger communities and to boats that pass through the region, constituting one of the main sources of income for community residents. Brazil nut collection: The process of collecting Brazil nuts is a traditional and typical process in the project region. The collection, consumption, and small-scale commercialization has cultural importance for communities, as there is much associated traditional knowledge. For example, as well as consuming the nuts, the residents produce flour from the nuts to use in typical recipes in the region such as "beiju"²⁰⁴. The sale of the nuts is carried out in the larger communities and is part of the residents' income sources.
Focus Area	Project Zone

4.1.4 Without-Project Scenario: Community (CM1.3)

In the scenario of the communities without the presence of the REDD+ IPOÁ project, the communities' lack of access to several basic welfare conditions including health, education, communication, and income generation, would be expected to continue. These aspects are summarized below.

²⁰⁴ Beiju is a typically Brazilian delicacy, of indigenous origin, made with tapioca (starch extracted from cassava, usually granulated), which when spread on a hot plate or frying pan coagulates and turns into a type of pancake or dry pancake.

Health: Due to the communities' distance from the nearest town, being about 80 km from Novo Aripuanã²⁰⁵, and their dependence on small boats for river transport, project residents do not have easy access to medical care, medicines and treatments, as described in Section 4.1.1.

It is possible that in the absence of the project, a basic health facility may have been created within the PZ, and that it would have a water ambulance capability. However, the duration and continuity of these activities have been shown historically to be short-lived, including: limited ability for the public sector to maintain a BHF, keep it constantly supplied with medicine and an operational water ambulance service .

Also, critically, in the without-project scenario the BHF would not be equipped with communication and electrification technology, enabling tele-medicine.

Education: There is little access to quality education, although two communities have school buildings (see Section 4.1.1), they are in a precarious state of disrepair, in need of renovation, and robust assistance and training for students and teachers.

Communication: The communities do not have any access to communication on-site: There is no telephone or internet signal in the region, making it impossible for the communities to communicate.

Employment and income: Socio-economic indicators in the communities would remain low, due to the lack of employment opportunities, especially fixed contract positions, leading to low current levels of income, insufficient agricultural production techniques, difficulties in production offtake, and access to markets. Precarious wellbeing conditions result from these limited circumstances.

4.2. Net Positive Community Impacts

4.2.1 Expected Community Impacts (CM2.1)

The tables below describe the impacts expected to result from the project activities in the with-project scenario. The design of these activities involved a participatory process, in the form of a workshop, in which the residents themselves identified activities which could address their main needs, as described in section 2.1.11. This is important for the definition of the final category of the tables below, changes in well-being, a

²⁰⁵ Distance from the Project's Maanaim property.

concept defined by the CCB SBIA Manual²⁰⁶ as being related to wealth and poverty, but which is best framed in terms of the communities' own understanding.

Community Group	All Project Zone community members will benefit from communication and electrification project activities.
Impact(s)	The expected wide-ranging impacts are: - an increase in community members with access to the internet, telecommunications and electricity; - telecommunication and internet impacts for educational and health premises; - positive impacts on work, environmental preservation, health and safety, leisure, and entertainment; - solar power will in increase in community members with access to electric light; electricity for refrigeration; electronic and IT equipment; and ventilation, among others.
Type of Benefit/Cost/Risk	Planned benefits: Planned activities: installation of rural communication antenna equipment, as well as solar power in all PZ communities. The installation of communication equipment is designed to enable direct communication between the project communities and the outside world, for the first time, with wide-ranging benefits related the areas of education, work, environmental preservation, health and safety, leisure, and entertainment, as described in impacts above. Furthermore, the installation of solar power will bring wide ranging benefits beyond those of communication already mentioned: provision of electric light; access to electricity for refrigeration; electronic and IT equipment; and ventilation, among others. Potential risks:

²⁰⁶ https://verra.org/wp-content/uploads/2016/05/SBIA_Part_2.pdf

	- Internal community conflicts of interest relating to access to communication and electrification equipment - Damage, loss, or misuse of communication or electrical equipment - Potential accidents involving electrical or solar equipment
Change in Well-being	Internet services was the most-voted activity in the community participatory workshop, and solar energy was third, evidence was provided at audit. So this project activity area improves well-being according to the communities' own understanding of the term. Furthermore, the communication and electrification project enables and facilitates other activity areas including health, education, work, and environmental preservation and is therefore a central pillar of the projects' activities and their impacts on wellbeing. An increase in overall quality of life and wellbeing is expected from all the above.

Community Group	All Project Zone community members will benefit from healthcare project activities.
Impact(s)	Projected impacts include: - Improved community access to healthcare services. - A healthier community population. - Contribution to good health and well-being (SDG 3).
Type of Benefit/Cost/Risk	Planned benefits: The planned activities in healthcare will deliver benefits relating to: - Access to communication and electrification within the Basic Health Facility (BHF). - Tele-medicine services provided. - Improved supply of medicine for the BHF. - Continuously operational emergency transport. Potential risks:

	- Internal community conflicts of interest relating to access to healthcare initiatives - Risks inherent to various healthcare procedures, including tele-medicine or emergency transport. - Potential conflicts between stakeholders, service providers and the law.
Change in Well-being	The impacts mentioned in the section above aim to contribute to good health and well-being (SDG 3). Increased healthcare was the second most-voted activity in the community participatory workshop, so this project activity area improves wellbeing in according to the communities' own understanding.

Community Group	Project Zone community members of school age will benefit from education project activities.
Impact(s)	Projected impacts in the education spectrum include: - Access to communication and electrification within educational facilities. - Improved training for teachers: both basic and additional. - Improved quality of lessons for students. - Support for supply of educational material for the school. - Educational service in the PZ brought above regional level.
Type of Benefit/Cost/Risk	Planned benefits: The planned activities in education will deliver benefits relating to: - Improved teacher training. - Improved teaching equipment and resources. Potential risks: - Internal community conflicts of interest relating to access to education. - Damage, loss, or misuse of educational equipment. - Any situation resulting from the project's education initiatives creating discomfort or distress for students and resulting in complaints to the project's contact line
Change in Well-being	Given the benefits and impacts above, the projected changes in wellbeing from these education activities are identified as: - Communities with an education level above the regional average. - Communities with improved choices for their futures.

	- Contribution to quality education within the PZ (SDG 4).
--	--

Community Group	Project Zone community members of professional age will benefit from Occupational Health and Safety project activities.
Impact(s)	The projected impacts of these activities include: - An increase in people following occupational health and safety guidelines - Reduction in work-related accidents
Type of Benefit/Cost/Risk	Planned benefits: The planned activities will deliver benefits relating to: - Provision of training and workshops on Occupational health and safety Potential risks: - Damage, loss, or misuse of health and safety equipment - Unavoided risks relating to occupational hazards
Change in Well-being	The change in wellbeing expected to be achieved are of a population more aware of occupational risks related to health and safety in wellbeing, and to make a Contribution to decent work and economic growth (SDG 8).

Community Group	Project Zone community members of professional age will benefit from Training project activities.
Impact(s)	The projected impacts of these activities include: - an increase in community knowledge and adoption of sustainability and conservation practices - an increase in range of sustainability and bioeconomy employment opportunities available to community members
Type of Benefit/Cost/Risk	Planned benefits:

	The planned activities will deliver benefits relating to: - Provision of professional courses relating to Sustainability and Conservation. Potential risks: - Any accidents occurring during training courses - Any situation resulting from the project's training initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line - Accidents occurring during training courses
Change in Well-being	The change in wellbeing expected to be achieved are of contribution to the creation of new job opportunities in a sector that was previously inaccessible to PZ communities. In a wider sense, this contributes to a new bioeconomy in the region, a population empowered with autonomy, leadership and innovation in sustainability and conservation activities, and a contribution to sustainable communities (SDG 11).

Community Group	Project Zone community members of professional age will benefit from the project's Employment actions.
Impact(s)	The projected impacts of these activities include: - An increase in range of employment opportunities available to community members; - Community association creation; - Enabling community entrepreneurship; - Facilitation of services provided by the public sector including health and education.
Type of Benefit/Cost/Risk	Planned benefits: The planned activities will deliver benefits relating to: - Provision of work and income for hired community members; - Improved ability for communities to demand and obtain public services. Potential risks: - Any accidents occurring during training - Community conflict due to benefitting one community before another

	- Any situation resulting from the project initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line
Change in Well-being	The expected changes in wellbeing are: enabling community entrepreneurship; facilitation and guaranteeing of public services which are due in the region; and strengthening of community governance and autonomy. Concrete manifestations of these impacts in the PZ's communities will be increased job opportunities and income. These will reduce the baseline negative impacts of low income, and social services. This is expected to contribute to development in line with SDG number 8.3 and to creation of sustainable communities as captured by SDG 11.

Community Group	Project Zone community members of professional age will benefit from Fire brigade training and Deforestation prevention and mitigation actions.
Impact(s)	The projected impacts of these activities include: - An increase in range of employment opportunities available to community members; - Reduction in deforestation and fire events within the PZ.
Type of Benefit/Cost/Risk	Planned benefits: The planned activities will deliver benefits relating to: - Provision of work and income for hired community members - Reduction of negative effects on communities – including ecological, environmental, and social – from deforestation and fire. Potential risks: - Risks to forest monitors from potentially aggressive deforestation agents invading the property - Risks from fire while carrying out firefighting activities - Any accidents occurring during training

	- Any situation resulting from the project initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line
Change in Well-being	The change in wellbeing expected to be achieved are of contribution to the creation of new job opportunities in a sector for PZ communities, and reduction in negative impacts on communities from deforestation and fire, including ecological, environmental, and social. In a wider sense, this contributes to a new bioeconomy in the region, a population empowered with autonomy, leadership and innovation in sustainability and conservation activities, and a contribution to sustainable communities (SDG 11).

4.2.2 Negative Community Impact Mitigation (CM2.2)

During the public consultation processes, field visits, and interviews, the community groups consulted returned primarily positive feedback. They didn't identify any potential risks that project could pose to them; rather, they perceived that the project activities would only help develop the region and its communities and protect the local ecosystems.

Nonetheless, measures have been designed to mitigate any negative impacts on the wellbeing of community groups (as identified in 4.2 and 4.4), in accordance with the precautionary principle, and to maintain or enhance HCV attributes related to community wellbeing.

At Carbonext these procedures are primarily constituted by the overarching Forest Integrity Plan, various Operational Procedures (OPs)²⁰⁷, and a Responsibility Agreement, further described below:

- i. OP for community conflict resolution
- ii. OP for stakeholder communication
- iii. OP for irregular or illegal community activity (Annex III of FIP)
- iv. OP for community health and safety at work
- v. OP for definition of alternative land-uses (Annex IV of FIP)
- vi. FIP – Forest Integrity Plan

²⁰⁷ *Procedimentos Operacionais* in Portuguese

- vii. Responsibility Agreement and Workshop
- viii. OP for deforestation events (Annex II of FIP)

The procedures above constitute the core of the management process, which is in place to identify any risks or negative impacts to the community and mitigate them. These OPs have been designed for the following respective purposes: (1, above) to mitigate internal community conflict ; (2) to facilitate communication between stakeholders and the proponent; (3) to avoid conflict between stakeholders and the law; (4) to avoid injury or illness and promote wellbeing within the work activities conducted by the communities, such as wild-harvesting; (5) to promote environmental management and provision of established productive areas; (6) to promote environmental protection and community compliance with laws and best practices, and to avoid illegal trespassing and deforestation; (7) to establish responsibility for care and maintenance of equipment and infrastructure provided as part of the project, as well as safety procedures relating to their use and installation; (8) define procedures to be implemented in case of deforestation events, including procedures on the ground, and with police and other organs.

Constant communication between the project team, monitoring, and feedback from community members to the project team will ensure that any complaints about negative impact on communities will be identified and mitigated. In these ways, it will be ensured that the project and activities will bring positive impacts to the community.

4.2.3 Net Positive Community Well-Being (CM2.3, GL1.4)

Project activities involve predicted improvements in various areas of community wellbeing, as identified in 4.2, which, despite all due good intentions, may have negative impacts as described in 4.2.2, and correspond to the without project scenario as described in 4.1.1. The net positive wellbeing regarding the areas of Communication and Electrification; Healthcare; Education; Occupational Health and Safety; and Training project activities are presented in the table below:

Table 125. Net Positive Community Well-Being

Project strategic area	Without-Project Scenario	Community	Expected Community Impacts – summary	Possible Negative Community Impacts	Net positive community well-being
Communication and electrification	The communities have no electricity from the municipal grid, however, in the homes of community leaders there are solar panels with low power only for basic use such as night lighting, or charging a cell phone, etc.		Increase in community access to internet and electricity.	Internal community conflicts of interest. Damage, loss, or misuse of communication or electrification equipment Potential accidents involving electrical or solar equipment	An Operational Procedure is in place to address internal conflicts, to be implemented by Carbonext's internal social team for mediation between the community, proponents, and stakeholders. However, the expected benefits are of considerable impact for all project zone residents. Therefore, the benefits to communities in the short-, medium-, and long-term, are predicted to be more significant, and outweigh the possible negative impacts.
Healthcare	Most likely no medical center or health agent in the region, services only being provided in the neighboring municipalities of Borba and Novo Aripuanã. If a medical center were active, it would most likely be temporary, poorly supplied, and water ambulance capability poorly maintained.		Improve quality of health services through communication and electrification, tele-medicine, medicine procurement services, and support for water ambulance public services.	Internal community conflicts of interest. Potential inherent risks regarding healthcare procedures themselves. Potential conflicts between stakeholders, service providers and the law.	The area of healthcare is complex and involves considerable social and inherent risks, and the actions are carried out in partnership with public and private partners. Nonetheless, Carbonext aims to mitigate the potential impacts to communities, through the OPs described in 4.2.2 in order to guarantee that the net positive impacts of the healthcare actions on communities are positive. The transformative impacts planned by these actions are predicted to overcome

				the potential negative ones, by facilitating essential public services in the communities, as well as complementing them with private actors.
Education	Lack of infrastructure for school-age children to complete basic education, inequality between men and women, a lack of additional training and education for teachers and students, as well as basic materials.	School infrastructure and equipment provision, communication and electrification facilities, teacher and student education and training.	Internal community conflicts of interest. Damage, loss, or misuse of educational equipment. Any situation resulting from the project's initiatives creating discomfort or distress for students and resulting in complaints to the project's contact line	Potential internal conflicts between community stakeholders are mediated by Operational Procedures, as well as the creation of partnerships that establish the responsibilities of each of the project's participants. It is considered that the wide-ranging planned impacts these actions regarding the community's complete education outweigh the potential negative impacts.
Occupational Health and Safety	No health and safety training or equipment are available in the communities.	Provide safety equipment and improve knowledge of occupational health and safety.	Damage, loss, or misuse of health and safety equipment Unavoided risks relating to occupational hazards	The risks associated with the trainings are mitigated by the term of shared responsibility between the community and the project, since damage, loss and misuse will be a shared responsibility. The materials will be provided, and the trainings held can be repeated if the need to update the community is identified.

Training	No training or best-practices relevant to community sustainable work practices are available in the Project Zone.	Increase in community knowledge and adoption of sustainability and conservation practices. Increase in range of employment opportunities available to community members, especially in sustainability and bioeconomy.	Internal conflicts of interest. Any accidents occurring during training courses. Any situation resulting from the project's training initiatives creating discomfort or distress for participants and resulting in complaints to the project's contact line	The project aims to improve existing community means of subsistence and income generation as well as create new ones. The mitigation of potential community conflicts is addressed by the relevant OP, and the FIP aims to promote forest integrity and minimize deforestation and invasion-related conflict. Taking these aspects into account, it is considered that the positive planned impacts of the project's training initiatives outweigh the potential negative impacts.
Employment	Lack of employment opportunities, especially fixed contract positions, leading to low levels of income. Lack of community organizations to promote services including health, education and employment, as well as low representation of women in community decisions.	Increased community job opportunities and income. Increased capacity for the community to govern and obtain vital public-sector services and infrastructure (i.e. health and education) for themselves. Increase in community members (with extra support for women) becoming members and occupying official positions in association.	Internal conflicts of interest. Low participation of women in community organizations. Decision-making carried out by few community members.	The benefits associated with increased community job opportunities, governance and organization promote well-being and quality of life. Despite the potential risks identified, the OPs identified in section 4.2.2 provide means of mitigating them, and the net benefits of the employment activities is predicted to be positive.

		Increase in people hired and carrying out surveillance rounds. Increase in number of community members employed in sustainable economy.		
--	--	---	--	--

All the activities above are in accordance with the descriptions in Appendix 1: Project Activities, and have indicators to monitor their performance described in Appendix 2: Theory of Change.

4.2.4 High Conservation Values Protected (CM2.4)

The following aspects of social high conservation values were identified, as noted in section 4.1.3 :

- HCV 4 - With the conservation/maintenance of the forest in the Project Zone, key Ecosystem services of support and regulation fundamental to the communities in the Project Zone will be maintained and/or improved, such as: Climate regulation, soil formation and the protection of water sources. This HCV is described in the CCB Standard under point CM1.2a.
- HCV 5 - With the conservation/maintenance of the forest in the Project Area in addition to maintaining and/or enhancing ecosystem services, the supply of primary resources to meet basic needs and prevent deterioration of community well-being will be maintained, including food and protein; timber and fiber; and fresh water. All are resources so important to communities that they would be unlikely to have a substitute. This HCV is described in the CCB Standard under point CM1.2b
- HCV 6 - With forest conservation/maintenance in the Project Area areas, resources, habitats, and landscapes of special cultural, archaeological, or regional historical significance, and/or of critical cultural, ecological, economic, or religious/sacred importance to the traditional culture of local communities will be conserved, such as copaiba oil extraction areas and Brazil nut gathering areas. This HCV is described in the CCB Standard under point CM1.2c.

The key parameters which will permit the monitoring of the three HCVs in each MR are presented below. They are predicted to be collected through remote and field activities, such as mapping community land-uses;

Table 93. HCVs benefitted by the project and key parameters for their monitoring

Relevant HCV	Key parameters	Monitoring Means
--------------	----------------	------------------

HCV 4	Hectares of remaining forest in the PA at the end of each MR; Deforested areas cultivated by communities in the PZ (hectares) ²⁰⁸ ; Areas disturbed by human activity in the PZ (hectares) N° of initiatives promoting community forest preservation, or other aspects of ecosystem services, in the PZ.	Remote monitoring Project data GIS
HCV 5 HCV 6	N° or size of areas that are fundamental for the livelihoods of communities mapped. Such as areas for: Fishing; Copaiba oil collection; Brazil nut collection; Timber harvesting; Churches	Participatory community mapping workshops Remote monitoring GIS

The areas covered under HCV 5 are of common use and are fundamental for the continuation of the community way of life for substantial components of cultural life as described by concepts CM1.2b²⁰⁹ CM1.2c²¹⁰, given that the areas in question are where people conduct their work, family, and leisure activities.

The participatory mapping approach aims to consult the community about the areas of high conservation value, identifying and monitoring them. Furthermore, certain activities of the project intend to contribute to the protection and enhancement of HCV5 and HCV6, such as the formation of the community association.

Meanwhile the Forest Integrity Plan and accompanying Operational Procedures, which comprise the core of the present REDD project, intend to guarantee the conservation of forest and ecosystem services contemplated by HCV 4.

To this extent, the following positive effects on HCVs are aimed for:

²⁰⁸ It will be key to measure this indicator for HCV 4 within and outside areas foreseen for expansion of community land-uses.

²⁰⁹ Areas that are fundamental for the livelihoods of communities - Such as for essential food, fuel, fodder, medicines, or building materials without readily available alternatives.

²¹⁰ Areas that are critical for the traditional cultural identity of communities - Such as areas of cultural, ecological, economic or religious significance identified in collaboration with the communities.

- Strengthened community geographical knowledge of their HCV areas, and their consequent monitoring conservation;
- Community members able to visualize and plan their land-uses;
- Communities enabled to document local knowledge;
- Community protection and enhancement of HCVs 5 and 6 through project actions such as the formation of the community association;
- Continuous remote monitoring enabling forest conservation (HCV 4).

Furthermore, any potential negative impacts can be reported by community members through established project communication channels, and subsequent mitigation actions will be taken to reduce and mitigate any negative impacts.

4.3. Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (CM3.1)

During the initial research stages of the project, a list of institutional stakeholders and their potential relationships with the IPOÁ project was drawn up (See section *Stakeholder Descriptions (G1.6, G1.13)*). Most of the institutions identified are related to municipal governments, municipal and state secretariats, and institutes.

In summary, it is found that this project tends to generate positive impacts on these stakeholders with the following being considered to be the main impacts:

- Increased awareness of environmental and biodiversity conservation;
- The maintenance of environmental quality;
- The conservation of ecosystem services of regulation, support, provisioning, etc;
- Promotion of increased community well-being;
- Partnership with public and/or private institutions;
- Strengthening ties between communities and other stakeholders.

Regarding these institutional stakeholders, it is considered that the actions planned within the IPOÁ REDD+ project have the potential to generate positive impacts and synergize with or directly facilitate their objectives. The municipal education, health and environment secretariats are examples of such institutions.

Such positive impacts are expected to be achieved through the establishment of cooperative relationships between Carbonext and the institutional stakeholders, supported by laws and ethics.

Regarding community stakeholders, During the initial diagnoses of the Project Zone and surrounding communities surveys, no stakeholders other than those described in the Project Zone were identified that had a direct relationship to the Project Zone and that used or depended on the resources within it.

However, later on in the project development, it was observed, through satellite monitoring and reports from residents, that a neighboring property within the Project Reference Region was conducting deforestation around and within the Project Area without authorization from the owner. With the intensification of command and control in the PZ, negative impacts for stakeholders who carry out illegal deforestation and/or logging for commercialization without authorization, within the project areas will be to some extent inevitable.

On the other hand, such continuous monitoring of the PZ will continue to ensure forest integrity, conserve biodiversity, and maintain the wellbeing of the communities that directly use the Project Zone.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

Stakeholders carrying out illegal deforestation within the PZ and its surroundings, who may suffer to impacts that are to some extent negative (please see 4.3.3. below for net effect of these impacts), will be attempted to be engaged in proactive political forums. Carbonext is planning to organize events with local and state governments, in which potentially hostile actors will be invited, in order to converse in a neutral environment, in a non-hostile fashion, to truly effect behavioural and cultural change, while mitigating any potentially negative impacts on the other stakeholders.

4.3.3 Net Impacts on Other Stakeholders (CM3.3)

The potential negative impacts on stakeholders who carry out illegal deforestation, mentioned in 4.2.2, are only negative in a short-term sense. Meanwhile, in a broader sense, the actions of the IPOÁ REDD+ Project will encourage these stakeholders and their businesses to come into line with the law, through the protection of the PZ, and the forums and events mentioned in section 4.2.2.

4.4. Community Impact Monitoring

4.4.1 Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The monitoring plan below duly identifies the communities, community groups and other stakeholders to be monitored, variables to be monitored, types of measurements and sampling methods, and the frequency of monitoring and reporting for each type and method.

The monitoring plan addresses the activities to be implemented in the communities, developed with their participation, or those that may pose some risk to them, as is the case of the forest conservation activities (see section 2.1.11). From this list, performance indicators were defined for each activity within respective scope groupings, identifying community groups, and taking into account performance indicators required for Gold Level for exceptional community benefits (GL2), specifically:

- Indicators of well-being impacts (identified below with acronym **WB**);
- Risks for smallholder/community members;
- Indicators of impacts on women (identified with the term “**women**” or “**by gender/age**” next to each indicator).

The risks identified in 4.2.3, are monitored through a variety of indicators such as “Number of complaints received via the communication channel”; “number of occurrences of accidents at work”; or “Number of occurrences of accidents during the boat trip”.

It is important to note that these risks are mitigated through the Operational Procedures and other documents listed in 4.2.2. For example, installed equipment is subject to a responsibility agreement with the community, which defines procedures relating to any loss or damage to the equipment, as well as defining best-practices regarding their safe use and installation.

The indicators were furthermore defined in alignment with the Brazilian targets of the Sustainable Development Goals - SDGs (see section 2.1.12) and on the Social Progress Index that aim to monitor the benefits to communities.

The community monitoring plan aims is to monitor the project's positive impacts, as well as risks along the project lifetime.

Table 126. Social parameters to be monitored.

Scope	Activity	Performance Indicators	Risk Indicators	Source of data and method	Target Community Groups	Frequency
Healthcare	Basic Health Facility (BHF).	Electricity and telecommunication infrastructure installed and functioning in the BHF. N° people (by gender/age) receiving ongoing tele-medicine medical care (WB). N° of medicines and medical supplies procured for the BHF (multiple indicators) (WB).	N° of complaints via the communication channel. N° of accidents during healthcare procedures. N° of records of legal complaints involving partners or stakeholders. N° of reports regarding community conflict.	Project's own field data Third-party project partners' data.	All residents (men, women, youth, and children) of the PZ communities.	Annual

	Support for health transport	Nº equipment items and operational supplies purchased/ transported for water ambulance. Nº of community training events relating to water transport. Nº people transported for medical using project supported water ambulance. (WB) Nº and effectiveness of meetings carried out to facilitate community-government PPP	Nº of complaints via the communication channel. Nº of accidents during activities. Nº of records of legal complaints involving partners or stakeholders. Nº of reports regarding community conflict.	Project's own field data. Public sector project partners' data.	All residents (men, women, youth, and children) of the PZ communities.	Permanent
Education	School infrastructure and equipment	Nº of electricity and telecommunication equipment installed and maintained within education facilities. Nº of educational equipment and material purchased or transported. (WB) Nº and effectiveness of meetings carried out to facilitate community-government PPP.	Nº of complaints via the communication channel.	Project's own field data. Public sector project partners' data.	Teachers and students from the schools involved.	Annual.

	Teacher and student education and training	Nº enrolments in training and educational courses. (WB) Sum spent (R\$) on teacher and student training and courses Progress in key subject areas (by gender/age) of students and teachers (WB) Presence/absence in class roll call (by gender/ age (WB)) Student satisfaction survey (WB)	Nº of complaints via the communication channel.	Project's own field data Third-party project partners' data	Teachers (by gender/age) and students (by gender/year) from the schools involved.	Annual
Occupational Health and Safety	Training and equipment: Occupational health and safety.	Nº people (by gender/age) trained in best practices for occupational health and safety (WB) .	Nº of equipment items damaged, lost, or misused. Nº of occurrences of unavaoided accidents relating to occupational hazards.	Project's own field data Public sector project partner's data Third-party project partners' data.	Workers involved in wild-harvesting and crop production.	Annual.
Communication and Electrification	Installation and maintenance of infrastructure	Nº equipment items purchased and transported.	Nº of equipment items damaged, lost, or misused.	Project's own field data	All PZ residents able to use	Annual.

	and equipment for electricity, telecommunication, and internet access	Nº equipment items installed and maintained in community buildings (WB) . Nº of people (by gender/age) with access to electricity and communication technology (WB) . Nº of community members (by gender/age) trained in: i) communication technology use and ii) maintenance (WB) .	Nº of accidents involving or related to communication and electrification technology. Nº of complaints via the communication channel. Nº of reports regarding community conflict.	Third-party project partner's data	digital technology.	
Employment	Hiring and training forest monitors.	Nº community members identified (by gender/age) for forest monitoring work (WB) Nº people (by gender/age) trained in forest monitoring practices (WB) Nº of monthly surveillance rounds conducted. Nº of reports, protocols, and alerts sent from monitors to PP. Sum spent (R\$) on monitor salaries (WB)	Nº of complaints via the communication channel. Nº of deployments of community forest monitors to areas of potential risk Quality of risk mitigation actions and meetings taken Nº of accidents at work.	Project's own field data Third-party project partner's data	Community workers involved in forest monitoring work. Other community members benefitted by forest monitoring work	Annual
	Community association creation	Consultant identified for community association creation. Consultant hired for community association creation. Association constitution initiated, finalized, or maintained.	Nº of complaints via the communication channel (by gender/age) . Nº of participants engaged in the association (by gender/age) .	Project's own field data Third-party project partner's data	Community members involved in association activity.	Annual

		Nº of association members and official positions in association. (WB) Nº of women promoted to become members and official positions in association. (WB)	Nº of deployments of community firefighters to areas of potential risk Quality of risk mitigation actions and meetings taken Percentage of women in relation to men in the association and related meetings.			
Training	Sustainable and Conservation Courses.	Nº people (by gender/age) educated in Sustainability and Conservation practices (WB) .	Nº of complaints via the communication channel. Nº of accidents during courses.	Project's own field data Third-party project partner's data	All PZ residents of at least school age interested in sustainability and conservation practices.	Annual
Fire brigade training (according to PIF)	Community firefighting work	Nº of fire-fighting training events held. Nº of community firefighters identified and trained.	Nº of complaints via the communication channel. Nº of complaints via the communication channel. Nº of deployments of community firefighters to areas of potential risk Quality of risk mitigation actions and meetings taken	Project's own field data Third-party project partners' data	Community workers involved in firefighting work Other community members benefitted by firefighting work	Annual

			Nº of accidents at work.			
Forest conservation	Community deforestation events	Nº of community deforestation events visited in the field. Nº and effectiveness of meetings carried out to combat deforestation (WB). Nº of infrastructure items constructed for deforestation combat. Nº of equipment items and vehicles purchased for deforestation combat. Nº of protocol applications regarding community deforestation (OP – illegal community activity; OP – Zoning of alternative land-uses, etc.). (WB).	Nº of community deforestation events detected/visited in the field Nº of actions to minimize negative community impacts Nº of protocol applications regarding community deforestation (OP – illegal community activity; OP – Zoning of alternative land-uses, etc.).. Nº of police or other official reports filed against community deforesters	Project's own field data Third-party project partners' data	Community members carrying out deforestation Other community members affected by community deforestation events	Annual

The following forms will be applied after training event or meetings, for data collection purposes:

- **Attendance List:** Contains the participant's name, type of affiliation with the project, and signature.
- **Reaction Assessment:** participants will be able to register their perceptions regarding the content presented, permitting suggestions regarding training and assessment of future training content. This assessment can be done anonymously and is not mandatory.

All communication with workers and families will be registered in meeting reports, which may be signed by all parties involved. The reports should be produced according to the monitoring frequency adopted. Reports must clearly inform the results of the variables involved in the monitoring, by means of photos, interviews, and other documents.

All records generated during project implementation will be retained and made available to the VCS verifiers for verification, demonstrating that the IPOÁ project activity was duly implemented. All documents and records will be kept securely for at least two years after the end of the monitoring cycle.

The Project Proponents are responsible for the financing, implementation and monitoring of the social aspects and indicators within the project credit period.

4.4.2 Monitoring Plan Dissemination (CM4.3)

The monitoring plan will be made available online through annual VCS Monitoring Reports, on the IPOÁ REDD+ Project website, for the voluntary carbon market audience. Additionally, a copy of the Monitoring Report will be emailed to all identified stakeholders.

For the institutional stakeholders, a summary in English and Portuguese of the results of each monitoring report will be produced and disseminated through e-mails to the institutional bodies already identified in the present PD, or those to be identified throughout the project's execution. In addition, all the monitoring reports verified with the results of the period analyzed, will be available on the Verra website and can be consulted at any time.

For the residents of the communities, the method and accessibility of communication methods is extremely important. With this in mind, presentation workshops will be conducted in the field by a Carbonext Social Analyst at the end of each monitoring period. They will be delivered in Portuguese, in accessible language, and with visual prompts. The first monitoring plan dissemination workshop is foreseen after MR1 is completed.

4.5. Optional Criterion: Exceptional Community Benefits

4.5.1 Exceptional Community Criteria (GL2.1)

Attending to Gold Level criterion GL2.1.b, in Brazil there is no official monetary poverty line, according to the World Bank²¹¹. For comparison purposes, the World Bank uses three poverty lines depending on the income level of the countries. These are:

- US\$ 1.90/day for poorest countries - International Extreme Poverty Line
- US\$ 3.20/day for lower-middle income countries
- US\$ 5.50/day for upper-middle income countries (group to which Brazil belongs) - National Poverty Line

Applying current exchange rates, according to the poverty line established by the World Bank, an individual (per capita) living on less than R\$ 882.75/month is considered poor in Brazil. Extreme poverty, on the other hand, is therefore below the value of R\$304.95/month. Regarding the interviews carried out in the communities, a sample of 74% of the total population living in the Project Zone, representing 71% of the total families, was covered the socio-economic questionnaires. This survey of the average family income permitted the development of the following economic scenario for the region:

- Average monthly salary 17 families evaluated is R\$ 531.25 (Five hundred thirty-one reais and twenty-five cents), representing under ½ of minimum wage. The same is observed when analyzing the per capita income at family level, where each would be below ½ minimum wage.
- 100% of the families evaluated have an income below the poverty line established by the World Bank.
- 94% of the families evaluated are with an income below the extreme poverty line established by the World Bank.

According to data from IBGE, 2010²¹² on the municipality of Novo Aripuanã, 52.1% of the total population of the municipality have a monthly per capita income of up to 1/2 minimum²¹³ wage. To this extent, more than half the municipal population is eligible for registration in the Federal Government's Single Registry of

²¹¹ <https://biblioteca.ibge.gov.br/visualizacao/livros/liv101760.pdf>

²¹² <https://cidades.ibge.gov.br/brasil/am/novo-aripuanã/panorama>

²¹³ <https://brasilindicadores.com.br/salario-minimo/>

Social Programs database, CadÚnico, which is the gateway to access a series of welfare benefits such as Auxílio Brasil²¹⁴.

Practically all the interviewed families of the Project Zone have access to this welfare benefit, and even with such assistance, as shown in the data presented above, the communities are in a situation of extreme poverty, thus fitting the criteria for CCB Gold Level – Community classification.

Over the lifetime of the project, this data will be continually assessed in order to capture its evolution over time.

4.5.2 Short-term and Long-term Community Benefits (GL2.2)

The IPOÁ REDD+ Project aims to provide short, medium, and long-term net positive benefits for communities located in the Project Zone, in such a way as to address all community-identified needs, share benefits equitably among all demographics, and leverage public and private partners to maximize impacts.

Please refer to Section 2.1.11 and Appendix 2, for the definitions of the project's outputs, outcomes and impacts as an approach to assessing the short, medium, and long-term impacts of project activities. Below is a description of each term:

Outputs are the short-term tangible results of project activities and usually take the form of products or services provided during the life of the project and as a direct result of project funding. Outputs are fairly easy to observe, measure and verify, and are therefore commonly used as indicators.

Outcomes are the direct intended results arising from the outputs. They are short- and medium-term. They are the changes experienced by the project participants and/or the physical environment and are less tangible and easier to measure than outputs.

Impacts are the end results sought by the project, especially with regard to net social change. They can occur as a direct or indirect result of project outputs.

²¹⁴ Auxílio Brasil — Português (Brasil) (www.gov.br)

The welfare impact indicators of these short, medium, and long-term activities have been incorporated into community impact monitoring plans, based on already consolidated tools and methodologies, such as the Sustainable Development Goals (SDGs) indicators.

4.5.3 Community Participation Risks (GL2.3)

The measurement of community wellbeing through impact and performance indicators will enable the reporting and quantitative monitoring of the project's benefits and positive impacts on communities, as well tracking of potential negative impacts.

For example, risk indicators of such negative impacts include: low meeting attendance, degradation of inter and intra-community relationships, increase in community dissent with the project and its actions, an increase in complaints about the project, decrease in local income, and increase in work accidents. Identification of changes in the indicators compared to the initial project scenario, will permit the identification and assessment of negative impacts, and the implementation of appropriate action, and according to the project's established Operational Procedures.

Any community participation risks will be identified and communicated through a participatory process as described in section 2.3.4 as per the guidance in the manual for REDD+ Projects: Social and Biodiversity Impact Assessment (SBIA) - Part 1 ²¹⁵. This assessment will be carried out through regular project management workshops by Carbonext's social team with community representatives, and community members for joint construction of risk mitigation actions, and if necessary, reconfiguration of the Theory of Change, activity design, etc. Strategies and actions will be implemented to avoid negative impacts, giving priority to communities with greater vulnerability to them.

4.5.4 Marginalized and/or Vulnerable Community Groups (GL2.4)

According to the Socio-Economic Diagnosis, all the communities visited in the Reference Region are considered marginalized and vulnerable due to the difficulty of access to public policies that guarantee basic rights such as infrastructure in health, education, transportation, communication, as shown in item

²¹⁵ https://verra.org/wp-content/uploads/2016/05/SBIA_Part_1.pdf

4.1.1. The communities involved in the IPOÁ REDD+ Project have a low level of social organization, which weakens the local conditions to seek for these fundamental rights.

In this sense, an important measure for the access to public policies and the guarantee of exceptional rights in the communities is the empowerment of the community, based on the strengthening and consolidation of social organizations, aiming at the effective participation of community members in decision making, contributing to the improvement of socio-economic aspects by the community members themselves.

Monitoring of the number of people benefiting from the REDD+ Project will be used to monitor whether these families are receiving net positive impacts. Ultimately, the Project is designed and will be implemented to generate net positive impacts for all families living in the selected communities. And subsequently, a communication channel will be established, to record, discuss, and resolve situations related to the project.

Community Group 1	<i>Communities in the Project Zone: Vila Batista, Boa Esperança e Vale da Benção.</i>
Net positive impacts	<i>Project activities are expected to generate net positive impacts on the well-being of all marginalized and/or vulnerable groups identified in the communities involved in the project, as shown in the items below. Communities have been part of the design of the project activities and will be considered in decision-making processes where appropriate, thereby strengthening local governance.</i>
Benefit access	<i>No barriers or risks have been identified that may prevent benefits going to marginalized and/or vulnerable small producers/community members, as the decision on implementation of activities and generation of benefits will be a responsibility of the project proponents. Community action is at the core of this project and the proponents have demonstrated commitment to the implementation of the project activities, and to the generation of well-being for the residents of the communities involved.</i>
Negative impacts	<i>During the life of the project, community members, including marginalized and/or vulnerable smallholders/community members, will be consulted about the project. This consultation will aim to identify any suggestions and complaints about the project activities, as well as an ongoing diagnosis of impacts on communities and project activities. In the event of the identification of any negative impacts on any community members, including marginalized and/or vulnerable members, Carbonext has a team of social experts trained to mitigate the</i>

	<i>impacts. However, no project activities are expected to generate negative impacts.</i>
--	---

4.5.5 Net Impacts on Women (GL2.5)

During the social diagnosis in the communities in the Project Zone, 23 women were counted. This number corresponds to 24% of the total number of people included in the questionnaires applied. For the public consultation meeting held in April 2022 we invited all those present in the communities regardless of gender and as a result we had the presence of 10 women, corresponding to about 42% of those present.

The women present in the communities participated in the same period in a workshop exclusively for them, so that they could talk about the habits they have, the reasons why they live in that region, their dreams, and the main points that they consider could be improved. This activity was developed to strengthen the relationship with these women, encourage their participation in decision-making, and understand the main points for developing actions with them. This participation was important so that they could participate and influence the design of the project activities that were carried out with a prioritization matrix (see section 2.1.11).

In the social survey, it was identified that the main occupation of women in the communities is domestic work and help men in agricultural activities and in the production of manioc flour, usually these jobs are not paid. Thus, it is demonstrated that women do not have a large share in the family income, as well as financial autonomy.

Certain The project activities were designed to help improve the wellbeing of women, as seen in the table below, with the main core being to encourage women's participation in all project activities.

Table 127. Activities Designed for Positive Impact on Women.

Activity	Net Impact on Women
Employment	Increase the participation of women in professional activities by promoting their work as forest monitors and being participants or occupying leadership roles

	in the community association, increasing, job opportunities, income, and professional capacities.
Healthcare	Improvement of womens' healthcare, by increasing the access to healthcare and health information by tele medicine access.
Training	Promotion of women's participation in training events, including: forest conservation, and sustainable and improved agricultural techniques.
Communication and electrification	Increase women's access to information and communication technologies, expanding knowledge and increasing the use of enabling technologies.

4.5.6 Benefit Sharing Mechanisms (GL2.6)

The means through which project benefits are distributed to communities is through its socio-economic, biodiversity, climate and other initiatives. Members of the involved communities participated fully and effectively in defining the decision making process, designing actions and benefits through a participatory process and meetings with their representatives, as already demonstrated from section 2.3 and section 2.1.11.

As already described, the communities participated in an activity prioritization workshop, where in small groups they were encouraged to discuss among themselves the activities and acquisitions that the communities would need or consider important for local development. The result of these discussions were about 25 posters with the demands of the community members. The next step was to make the same number of stickers available to all adults and young people present so that each one could place the stickers on the demands they considered to be most urgent.

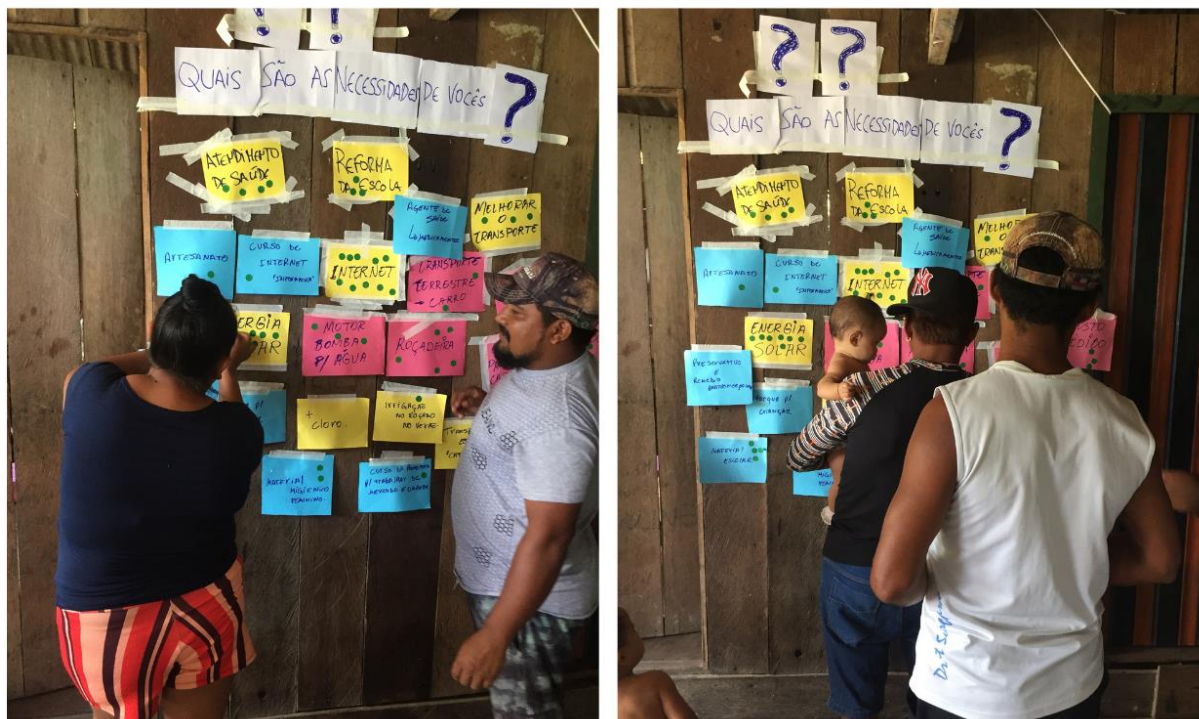


Figure 47. Community members participating in defining activities.

This prioritization activity was the basis for the definition of the activities presented, and thus the expected benefits that will be shared by all the communities. The expected benefits are presented in section 4.2 and will be available according to the frequency described in the monitoring plan (i.e., distributed during site visits) to the members of the communities involved as described in section 4.4.

A proportion of project profits will be reinvested in the region, through the activity's areas listed below. This amount equates to 5.5% of the total credits sold²¹⁶. And the proportion of the total budget that each action represents is as follows:

²¹⁶ Calculations made with generation estimates.

Table 128. Proportion of the cost of activities to the total budget.²¹⁷

	Scope	Activity	Proportion
Social	Healthcare	Basic Health Facility (BHF)	0,1%
		Health transport programme	1,1%
	Education	School infrastructure and equipment	1,1%
		Teacher and student education and training	2,2%
	Occupational Health and Safety	Training and workshop: Occupational health and safety	1,1%
		Personal protective equipment for worker health and safety	2,2%
	Communication and Electrification	Installation, maintenance and/or facilitation of electricity and internet access	3,5%
	Employment	Hiring monitors	6,2%
		Community association creation	1,7%
	Training	Sustainable Practices and Conservation Courses	3,2%
		Fire brigade training (according to PIF)	1,7%
Biodiversity	Fauna and flora study initiatives	Fauna Inventory	5,5%
		Flora Inventory	4,9%
	Biodiversity Research, Monitoring, conservation activities or	Fauna and Flora research, monitoring conservation program	4,3%
	High Conservation Value areas	Survey on High Conservation Value areas	0,2%
Climate	Forest Conservation	Remote Monitoring	0,0%

²¹⁷ Costs and budget will be made available in validation and checks.

		Deforestation prevention and mitigation actions.	4,3%
Other	Specific social voluntary actions	Annual actions to strengthen community relations, such as: local festivities, cultural activities, sports activities, delivery of food baskets, donations of various kits, etc.	1,7%

4.5.7 Benefits, Costs, and Risks Communication (GL2.7)

The predicted benefits, costs and risks of the project were communicated to the community during the field visits. During the two field visits in 2022, a member of the social team personally visited households in the PZ to communicate the proposed benefits of the Project and the potential costs and risks of participation.

Meetings were held in the communities applying active listening to clarify the questions and concerns of the community members about the Project. The project's vision is to maintain an open channel of communication, so that the mapped communities have the opportunity to discuss the benefits, costs, and risks of the project, according to their own perspectives.

In addition, the project proponents a member of the project social team will be in continuous contact with the communities through visits scheduled and consented to by community leaders and representatives. Other forms of communication will also be used for this purpose, such as written reports and social media where applicable. In addition, a communication channel is being implemented to maintain an open channel of communication with the community.

4.5.8 Governance and Implementation Structures (GL2.8)

The project is governed and implemented by Carbonext and the other proponents, who are co-responsible for: maintaining forest integrity; hiring all project workers; and financing and implementing the project's socio-environmental actions.

To this extent, the project proponents collaborate on managing and executing the project's activities, in consultation and collaboration with communities, as fully described in section 2.4.

The project aims to continue to incorporate the communities into project decision-making throughout the life of the project, as has been done as of the design of the project actions (See section 2.3.7). Participatory

meetings will be held once a year to collect feedback regarding the project design and activities. For participation in project activities, community members will be consulted in order to verify those interested in participating in project actions.

Patrols will be initiated in the region for forest surveillance by field agents hired by the project proponent to ensure the conservation of the area, as presented in the Forest Integrity Plan (made available at audit). The creation of direct communication channels with the communities is also stipulated in the FIP. The project workers will be identified and hired according to an egalitarian and transparent process stipulated in Carbonext's OPs. The implementation of the project also relies on partnerships with various private sector and community actors, who in turn will liaise with public sector actors, to carry out the activities presented in the present document.

4.5.9 Smallholders/Community Members Capacity Development (GL2.9)

Local leaders and community members will be invited to actively participate in the IPOÁ project's design, implementation, and management. Community participation in all stages of the project is designed to be participatory, self-determined and inclusive, to address inequalities between genders, ages, and marginalized groups, promoting empowerment.

Where appropriate, outside private, and public collaborators will be procured to provide high-quality training, for instance in the community association creation and training process. From the complete list of activities described in Appendix 2, the five areas that involve community training and capacity-building are described below.

Table 129. IPOÁ project community training and capacity-building initiatives.

Activity	Description
Teacher and student education and training	Professional training of PZ teachers and extra lessons for students by a specialized service provider
Occupational health and safety	Training and workshops on health and safety in professional activities, including provision of personal protective equipment, and training on their use, for community workers
Sustainable Practices and Conservation Courses	Workers education on forest conservation and smallholder sustainable agricultural production

Employment	Support for creation of a community association and training by a specialized consultant
	Community forest monitor training
Fire brigade training (according to PIF)	Capacity building for community fire-fighting to combat fire effectively

5 BIODIVERSITY

5.1. Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (B1.1)

The initial study, described below, of the existing biodiversity conditions within the Project Area and its surroundings was based on literature review, the main sources of which were:

- The management plan of the Juma Sustainable Development Reserve (RDS)²¹⁸, a federal protected area located 13km from the Project Area.
- The fauna survey and inventory report carried out as part of the licensing process for forest suppression, conducted in the field for Maanaim and Campos Farms, was used to compose the list of threatened species.

Following the project's first verification, Carbonext will perform a Fauna and Flora Inventory – described in item 5.4 - within the IPOÁ Project Area.

The methodology used to complete the document for the fauna and flora biodiversity survey, from the JUMA RDS data, is described below.

Fauna biodiversity

To conduct the fauna survey, the Juma Reserve Management Plan utilized secondary data from studies of the main terrestrial vertebrate groups: non-flying mammals, birds, reptiles, and amphibians. The results

²¹⁸ https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf

showed that species diversity in the RDS is exceptionally high, even by Amazonian standards. A total of 70 species of non-flying mammals (55 genera, 28 families, and 10 orders) were recorded. As for avifauna, 398 were recorded within the UC boundaries and 214 were recorded in the Reserve's surroundings, a total value (612), which represents half of the avifauna of the Brazilian Amazon, one third of the avifauna of Brazil as a whole, and among the richest avifauna of any protected area in the world. The herpetofauna survey resulted in 43 species of reptiles and 27 of amphibians.

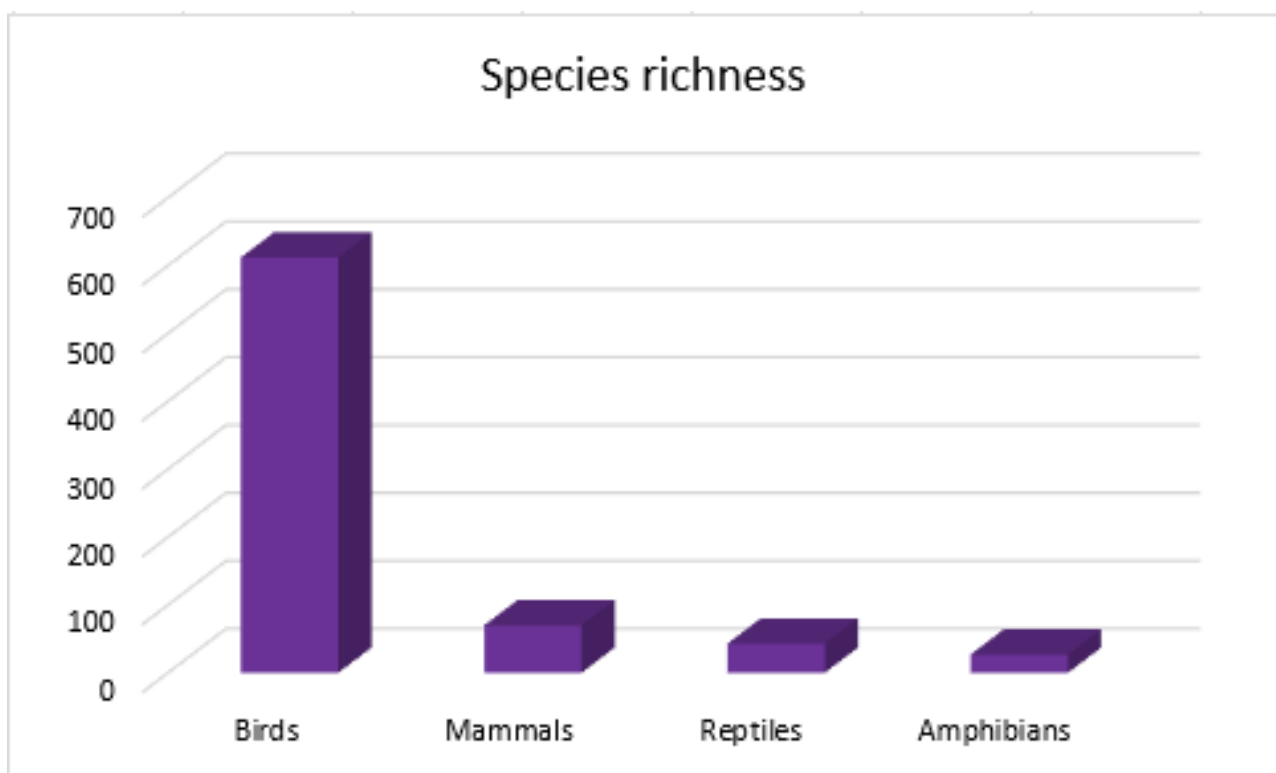


Figure 48. Species richness of fauna likely to occur in the project area, from literature

According to the Juma Reserve Management Plan, beyond this, new species for science from all groups of fauna are being described from this region at a rate above normal, suggesting that the region has not yet been adequately studied and that there is still more diversity to be discovered. This highlights the importance of the biodiversity inventory planned after the first verification, subsequent study, monitoring and conservation actions to be defined, as well as the conservation activity already implied by the implementation of the IPOÁ REDD+ Project as a whole, enabling more specific studies for the region and preservation of habitat for this fauna diversity.

The JUMA RDS contains 14 species classified at some level of threat of extinction ("VU", "EN", etc.), of which 8 are mammal species and three are species of the genus *chelonina*, as indicated by the following table. There are two bird species listed as vulnerable recorded in the RDS. Of the mammals, four are large species with a wide distribution in Brazil: *Tapirus terrestres* (tapir), *Trichechus inunguis* (manatee), *Priodontes maximus* (giant armadillo), *Pteronura brasiliensis* (giant river otter) which according to the JUMA RDS Management Plan are likely to have healthy populations at natural levels within the RDS. The largest monkey species (*Chiropotes albinasus*, *Ateles* sp., *Lagothrix* sp.) are classified as "endangered" and the tiny *Callibella humilis* is "Vulnerable." and the others are primates with restricted distributions. Furthermore, the fauna survey for the Maanaim and Campos Farms recorded, in addition to the *Tapirus terrestres* species other threatened species namely *Alouatta guariba clamitans*. The total number of species with some degree of threat, with potential presence in the project area is described below:

Table 130 - List of threatened and/or endemic birds, mammals, amphibians, and reptiles in the region of the Juma RDS and Fazenda Maanaim and Campos Fauna Report.

Taxon	Popular name	Threat (IUCN)		Endemism	Source
Podocnemidae					
<i>Podocnemis unifilis</i>	Yellow-spotted river turtle	VU		-	Vogt et al 2007
<i>Podocnemis sextuberculata</i>	Six-tubercled turtle	VU		-	Vogt et al 2007
Testudinidae					
<i>Chelonoidis denticulata</i>	Yellow-footed tortoise	VU		-	Vogt et al 2007
Tapiridae					
<i>Tapirus terrestres</i>	South American tapir	VU		-	Röhe 2007; 4
Trichechidae					
<i>Trichechus inunguis</i>	Amazonian manatee	VU		-	Röhe 2007
Dasypodidae					
<i>Priodontes maximus</i>	Giant armadillo	VU		-	Röhe 2007
Mustelidae					
<i>Pteronura brasiliensis</i>	Giant otter	EN		-	Röhe 2007
Pitheciidae					
<i>Chiropotes albinasus</i>	White-nosed saki	EN		END	Röhe 2007
Atelidae					
<i>Ateles</i> sp.,	Monkey	EN*		-	Röhe 2007
<i>Lagothrix</i> sp.	Monkey	EN		-	Röhe 2007
<i>Alouatta guariba clamitans</i>	Southern brown howler	VU		-	4
Callitrichidae					
<i>Callibella humilis</i>	Monkey	VU		EM	Röhe 2007
Psittacidae					
<i>Pyrrhura perlata</i>	-	VU		MD M	1,2,3
Thamnophilidae					
<i>Herpsilochmus</i> sp	-	-		M/AR	1,3

Taxon	Popular name	Threat (IUCN)		Endemism	Source
<i>Rhegmatorhina hoffmannsi</i>	White-breasted antbird	LC		M/T	1,2,3
<i>Skutchia borbae</i>	Pale-faced bare-eye	LC		M/T	1,2,3
Conopophagidae					
<i>Conopophaga melanogaster</i>	Black-bellied gnateater	LC		MD M	1,2,3
Dendrocolaptidae					
<i>Dendrocolaptes hoffmannsi</i>	Hoffmanns's woodcreeper	VU		M/T	1,2,3
Tyrannidae					
<i>Poecilatriccus senex</i>	Buff-cheeked tody-flycatcher	LC		M/T	1,2,3
Troglodytidae					
<i>Odontorchilus cinereus</i>	Tooth-billed wren	NT		MD M	1,3
Hylidae					
<i>Hypsiboas sp.</i>	-	NE		IND	Vogt et al 2007
<i>Osteocephalus sp</i>	-	NE		IND	Vogt et al 2007
Leptodactylidae					
<i>Leptodactylus sp. 1</i>	-	NE		IND	Vogt et al 2007
<i>Leptodactylus sp. 2</i>	-	NE		IND	Vogt et al 2007
Microhylidae					
<i>Chiasmocleis sp</i>	-	NE		IND	Vogt et al 2007
Strabomantidae					
<i>Pristimantis sp.</i>	-	NE		IND	Vogt et al 2007
Dipsadidae					
<i>Atractus sp</i>	-	NE		IND	Vogt et al 2007
<i>Oxyrhopus sp</i>	-	NE		IND	Vogt et al 2007

Threat level IUCN: VU – Vulnerable; EN – Endangered; NE – Not Evaluated; LC – Least Concern

Endemism: MD M - endemic to the right margin of the Madeira river; M/T - endemic to the Madeira and Tapajós rivers interfluvium; M/AR - endemic to the Madeira and Aripuanã river interfluvium; IND - Endemism analysis to be identified in future research (possible new and endemic species).

Sources: 1 – Cohn-Haft et al. 2007 2 – Banco de dados não publicados de M. Cohn-Haft 3 – Whittaker 2009. 4 – Fazenda Maanaim and Campos Fauna Report

The Juma RDS can be considered one of the most important state protected areas in the state of Amazonas from the point of view of terrestrial fauna²¹⁹ due to the presence of species such as *Callibella humilis*, where approximately half of the species' global distribution is contained within the RDS. Another important case

²¹⁹ https://pdfhost.io/v/edXfNIR6_Microsoft_Word_Plano_GestoRDS_Juma_FINAL_aprovada_Conselho_Deliberativo.pdf

is the *Mico acariensis*, whose entire distribution consists of an area surrounding the Reserve. Besides these, some other examples include *Mico manicorensis* (Manicoré - Aripuanã), *Mico chrysoleucus* (Aripuanã - Canumã/Acarí), and a new species of bird (not yet described) *Herpsilochmus* sp. (Machado/Jiparaná - Aripuanã).

Most of the endemic species in Aripuanã have no protection in other protected areas within their range²²⁴. Considering the distance between the RDS and the PA and the possible presence of these taxa within the Project Area, we emphasize the importance of REDD+ initiatives for the maintenance of habitat for such fauna biodiversity.



Figure 49. *Tapirus terrestris* (IUCN, 2022)



Figure 50. *Priodontes maximus* (IUCN, 2022)

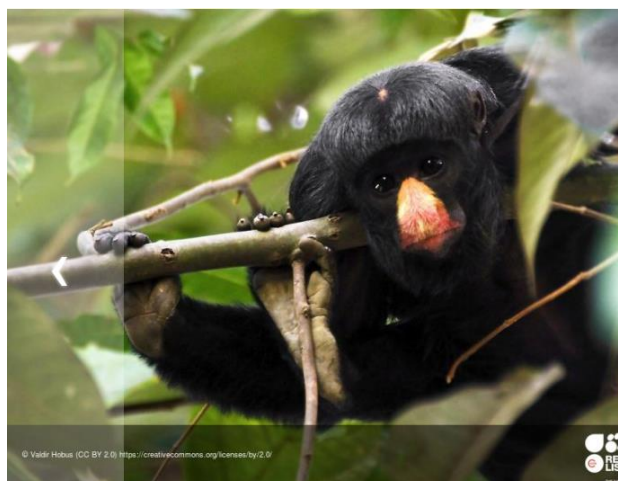


Figure 51. *Chiropotes albinus* (IUCN, 2022)



Figure 52. *Conopophaga melanogaster* (Cláudio Dias Timm; Avibase)²²⁰

Flora biodiversity

²²⁰ <https://avibase.bsc-eoc.org/species.jsp?avibaseid=C0F59C9F8A109FC9>

The JUMA RDS protected area's flora inventory was conducted in October 2019 in the region of the AM 174 highway (Higuchi et al., 2010). The study found 358 tree species, with the taxon *Protium* sp. (red rosewood) - Burseraceae being the most frequently recorded, followed by *Eschweilera* sp. (yellow matamata) - Lecythidaceae, a pattern commonly found throughout the Amazon region (Higuchi et al., 2010). The family with the highest number of species was Mimosoideae (29), followed by Sapotaceae (21), and Lauraceae and Papilionoideae (17), as observed by the following histogram:

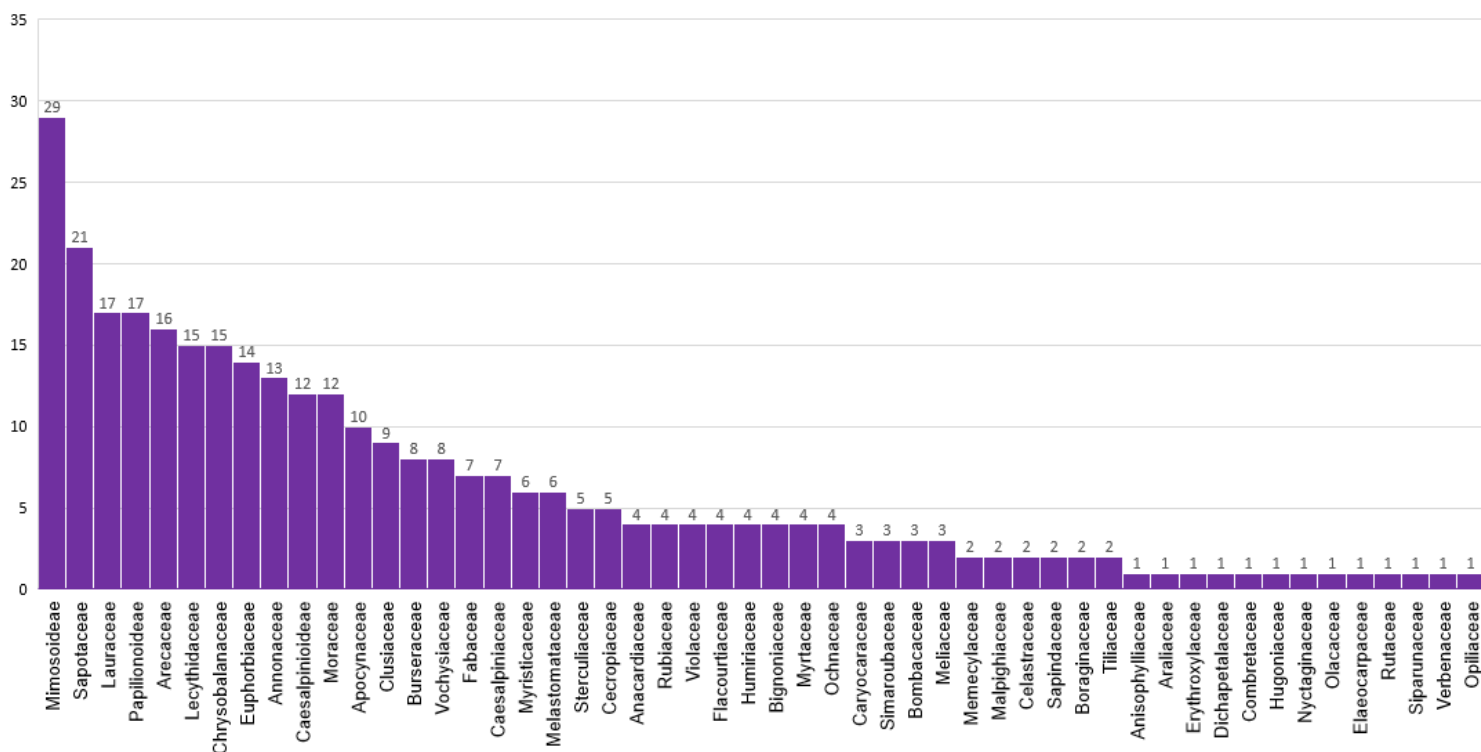


Figure 53. Distribution of number of species by inventoried botanical families of probable occurrence for the project area (HIGUCHI et al., 2010)

As for the occurrence of species, the presence of *Euterpe precatoria* (açai) and *Bertholletia excelsa* (Brazil nut tree) stands out in this study, both recorded in the literature as having significant contribution as a source of employment and income generation for thousands of rural and urban workers in the state of Amazonas.²²¹ The Brazil nut tree (*Bertholletia excelsa*), moreover, is classified as "Vulnerable" by both the

²²¹ <https://www.scielo.br/j/cerne/a/bcpmyb9fdlgfsgslqvnfnfs/?format=pdf&lang=pt>

IUCN and CNCFlora, as indicated by table above. As described in item 4, the Brazilian nut tree are possibly the same species reported by members of the communities, which use it for food and economic purposes. Thus, the importance of protecting these species and their natural habitats through the creation of preservation projects is highlighted.

From the list of inventoried species, available in the JUMA RDS Management Plan, we classified the taxa according to the level of extinction, based on the IUCN RED LIST (IUCN, 2018)²²² and the National Center for the Conservation of Flora (CNCFlora)²²³. According to the flora inventoried for the JUMA RDS, below are the species classified at some level of threat of extinction.

Table131. List of threatened and/or endemic flora in the region of the Juma RDS.

Taxon	Popular name	Threat (IUCN)	CNCFlora
Bignoniaceae			
<i>Tabebuia incana</i>	Yellow Arched Wood	VU	-
Bombacaceae			
<i>Catostema albuquerquei</i>	Mamãorana	em	-
Caesalpiniaceae			
<i>Aldina heterophylla</i>	Macucu de paca	VU	VU
<i>Vouacapoua americana</i>	Wacapou	CR	EN
Chrysobalanaceae			
<i>Couepia longipendula</i>	Egg nut	EN	
<i>Licania adolphoduckei</i>	Macucu fluffy	EN	
Lauraceae			
<i>Mezilaurus itauba</i>	Itaúba	VU	VU
<i>Aniba ferrea</i>	Louro chumbo	EN	EN
<i>Aniba cf. permollis</i>	Louro rosa	EN	
Lecythidaceae			
<i>Bertholletia excelsa</i>	Brazil nut	VU	VU
<i>Lecythis prancei</i>	Jarana nut	EN	
Meliaceae			
<i>Guarea convergens</i>	White gitó	VU	
<i>Trichilia areolata</i>	Red gitó	VU	
<i>Sorocea guilleminiana</i>	Wild jackfruit	VU	
Papilionoideae			
<i>Aldina heterophylla</i>	Macucu	VU	VU
Rutaceae			
<i>Hortia superba</i>		VU	VU
Sapotaceae			
<i>Pouteria petiolata</i>	Abiurana cutiti	VU	
<i>Pouteria freitasii</i>	Abiurana sabiá	VU	VU

Threat level IUCN and CNCFlora : VU – Vulnerable; EN – Endangered; CR - Critically Endangered

²²² <https://www.iucnredlist.org/>

²²³ <http://cncflora.jbrj.gov.br/portal/>



Figure 54. *Bertholletia excelsa*²²⁴



Figure 55. *Euterpe precatoria*²²⁹

Importance of IPOÁ REDD+ for biodiversity

It is noteworthy that the creation of the IPOÁ REDD+ Project directly impacts the preservation of fragments located 13km (Fazenda Campos) and 15km (Fazenda Maanaim) from the Juma RDS, thus it is possible to consider that several species recorded in the RDS have potential occurrence in the Project Area. Moreover, it is emphasized that other possible new mammal species are being studied by the researcher in the region, including jaguars, deer, tapirs, manatees, anteaters, and primates²²⁶. The biodiversity inventory study to be conducted by Carbonext after the first monitoring period has the potential to contribute to science with fauna and flora studies of the Aripuanã region. If the occurrence of new endemic and threatened species is recorded, new conservation decision strategies can be planned for the region.

In addition, the Juma Reserve Management Plan points out the low occurrence of protected areas in the surrounding region of the RDS. Thus, for some species, if they depended entirely on the RDS to guarantee their continued existence, they would qualify as threatened and possibly would not maintain sustainable populations. This also draws attention to the importance of pressures outside the RDS as well as the creation of protected areas. Noting that the Project Area of the IPOÁ REDD+ Project is located on the eastern frontier of the JUMA RDS, the conservation of these areas should contribute to the conservation of

²²⁴ <https://hablemosdeflores.com/bertholletia-excelsa/>

²²⁵ <https://puppy-party.info/euterpe-precatoria-54/>

²²⁶ https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf

species mainly in the Juma RDS located on the right bank of the Aripuanã River, which apparently do not present physical barriers to the use of the conserved habitats of the Maanaim and Campos properties. The creation of the IPOÁ REDD+ Project, especially the Campos property, may constitute a significant fauna refuge for species located in the Acari National Park.

Finally, the creation of the IPOÁ REDD+ project protects forest mosaics with important Amazon flora species, besides serving as habitat, an ecological springboard, and a refuge for different fauna species such as mammals, reptiles, and local avifauna.

5.1.2 High Conservation Values (B1.2)

For the Survey on High Conservation Areas, High Conservation Values (HCVs) were defined according to the "General Guide to Identifying High Conservation Values"²²⁷ relating to the region's biodiversity and communities. The definitions of HCVs consider a classified biological, ecological, social, or cultural value of exceptional or critical importance, which can be classified into different categories. In the project zone, the following beneficial changes to biodiversity are expected:

High Conservation Value	HCV 1: Species diversity - possible occurrence of threatened and endemic species in the Project zone
Qualifying Attribute	The Amazon Forest ecosystems comprise the largest and richest forest in the world in terms of biodiversity ²²⁸ . In the Amazon biome, about 183 species are threatened with extinction and among these 122 species are endemic ²²⁹ . According to studies conducted in the vicinity of the project area, endemic and endangered species are likely to occur in these regions, as noted in the tables under 5.1.1.
Focal Area	It is necessary to conserve and protect in totality the current vegetation in the project area, maintaining biodiversity monitoring

²²⁷ BROWN, E., N. Dudley, A. Lindhe, D.R. Muhtaman, C. Stewart, e T. Synnott (eds.). 2013 (outubro). Guia geral para identificação de Altos Valores de Conservação. HCV Resource Network.

<https://webcache.googleusercontent.com/search?q=cache:XmxEG_PJVjAJ:https://ic.fsc.org/download.guia-geral-para-identificacao-de-altos-valores-de-conservacao-portugues.a-3705.pdf&cd=1&hl=pt-BR&ct=clnk&gl=br>

²²⁸ <https://www.scielo.br/j/ea/a/yf6MQVWvKWGD33jN3jqw47n/?format=pdf&lang=pt>

²²⁹ https://www.researchgate.net/profile/Maria-Fabiola-Barros/publication/350531320_Reflexoes_em_Biologia_da_Conservacao_-_Volume_2/links/60650c1292851c91b194596e/Reflexoes-em-Biologia-da-Conservacao-Volume-2.pdf#page=255

	and carrying out educational activities for create awareness on the importance of biodiversity and conservation.
--	--

High Conservation Value	HCV 2: Ecosystems and mosaics at landscape level, and Intact Forest Landscapes - proximity of PA to protected areas
Qualifying Attribute	The PA of the REDD+ IPOÁ Project is near protect areas, as Juma Sustainable Development Reserve (RDS) (13km) and Acari National Park (25km), Aripuanã National Forest (50km) and Indigenous Land like Coata-Laranjal (30km) and Setemã (48km). The presence of protected areas in the vicinity, especially for the JUMA RDS, underscore the importance of protecting surrounding habitats. It is likely that these environments share similar species diversity, and perhaps animal populations that use the forest fragments as ecological steppingstones and other ecosystem services. Protecting areas near protected areas intensifies actions against deforestation in the Amazon
Focal Area	The IPOÁ REDD+ Project allows for the protection of the PA area, which serves as habitat for several species. Some species may belong to common populations between protected areas near the PA. In this way, conservation of the PA also contributes to the surrounding biodiversity.

High Conservation Value	HCV 5: Community needs - Brazil nut and copaíba extraction in the project zone and fishing in the Sucunduri River
Qualifying Attribute	The forest and aquatic ecosystems of the Amazon consist of a diverse source of food, nutritional and economic resources of importance to traditional communities. The main economic activities of the Boa Esperança community, located in the project zone, are the extraction of copaiba oil and Brazil nuts, cassava plantations for flour production, and vegetable production, the first two being directly dependent on forest resources. In addition, people from other communities have come to the Sucunduri River to engage in fishing activities.

Focal Area	The protection of the project area as well as the development of educational activities together with the surrounding communities impacts the quality of life and the basic needs of local communities in terms of livelihood, health, nutrition, water, among others.
------------	--

5.1.3 Without-project Scenario: Biodiversity (B1.3)

In the project's Reference Region the main causes for the deterioration of the environment are the conversion of forests to pasturelands as shown in section 2.1.6. Despite the fact that mining is the main economic activity in Novo Aripuanã, there are no known mines in the Reference Region.

The replacement of natural forest systems by pasture ecosystems and other anthropogenic uses is directly related to significant biodiversity loss in the Amazon. It is known that forests are considered the main ecosystems for terrestrial animals and the Amazon Forest has the largest biodiversity in the world²³⁰. The removal of vegetation cover, degradation, and deforestation is related to direct and indirect losses of thousands of species across the planet. In addition to resulting directly in the mortality of individual trees and other taxa of the plant kingdom, constituting a loss of forest habitat forces the migration of populations of several animal species, when not affected directly, putting species at risk of extinction.

The IPOÁ REDD+ Project prevents a predicted 4,200.43 hectares from being deforested over 30 years. In a without-project scenario, it is very likely that forest clearing for pasture conversion would reach the Project Area, negatively impacting biodiversity in the region. Additionally, without the project's biodiversity research, monitoring, and conservation actions, as well as social educational action foreseen in item 4.4.2, it would not be possible to expect biodiversity intervention or environmental awareness from the region's communities, being the main impacts causing positive change for biodiversity, enabled by the project.

5.2. Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (B2.1)

²³⁰ <https://www.scielo.br/j/ea/a/yf6MQVWvKWGD33jN3jqw47n/?format=pdf&lang=pt>

Biodiversity Element	Flora
Estimated Change	Positive - Biodiversity Conservation
Justification of Change	The REDD+ IPOÁ Project consists of preventing deforestation and degradation of Amazonian Forest ecosystems. In 2022 alone, 295 km² of deforestation were registered in the state of Amazonas²³¹. The project prevents deforestation and degradation by cattle-ranchers, land speculators and other actors, avoiding the threat to plant species by promoting conservation of suitable ecosystems for plant species. Furthermore, flora study, monitoring, and conservation initiatives to be defined by third-party project partners will promote flora biodiversity in ways to be defined.

Biodiversity Element	Fauna
Estimated Change	Positive - Biodiversity Conservation
Justification of Change	The Project activities aim to reduce deforestation and forest degradation, based on forest monitoring and conservation practices established in Section 2.1.11. This will be achieved through, for example, application of the project's Forest Integrity Plan. These actions aim to avoid the felling of forest, and associated destruction of habitat, harm and extinction of species, and complex systemic consequences for ecosystems and the environment. The maintenance of vegetation cover promotes adequate habitat for the species of Amazon birds, mammals, amphibians, and

²³¹ PRODES: http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/increments

	reptiles in different stages of threat and endemism, which are described in section 5.1.1, and will be further documented by future field inventories. Furthermore, fauna study, monitoring, and conservation initiatives to be defined by third-party project partners will promote fauna biodiversity in ways to be defined.
--	--

5.2.2 Mitigation Measures (B2.3)

The IPOÁ project is not predicted to have any negative impacts on biodiversity, given that the Project Area will be dedicated exclusively to the conservation of forest vegetation cover.

The pressures on biodiversity in the region are related to external factors and the presence of REDD+ IPOÁ aims to mitigate these impacts. Based on the study of the local socioeconomic scenario and the reality of the communities observed in the field so far, we identified two main current external factors of risk to biodiversity, specifically: a) Deforestation pressure for pastureland/land use change/disputes and changes in land rights or access/use rights; and b) Hunting habits in the communities

To mitigate negative impacts to biodiversity related to deforestation and forest degradation, the IPOÁ REDD+ Project will develop sustainable revenue streams for communities (see sections 4.2 and appendix 2), and carry out surveillance and monitoring activities as described in item 3.3.3. In case deforestation alerts are observed via satellite or in field monitoring, measures will be taken to address the deforestation as defined by the deforestation event OP, including the registration of a police report, actions in the field, and communications with relevant local authorities and other actors.

Furthermore, the environmental education work developed with the community, as described in 5.4, promotes awareness about the protection of local biodiversity, as well as sustainable practices that avoid damage to the environment, such as raising awareness about the impacts caused by hunting and small deforestation events for subsistence crop production.

The following table highlights the activities to be implemented to conserve the identified HCVs and avoid negative impacts on biodiversity.

Table 132 - Main biodiversity threats for the project region, mitigating measures, and related HCVs.

HCV Related	External threat to biodiversity	Possible impact agent	Mitigation Measure
HCV 1: Species diversity	Deforestation pressure for pastureland/land use change.	Farmers; external landowners; local residents.	Constant surveillance and monitoring (CarbonEye System). Encouraging farmers in the region to start REDD+ projects. Identify and conserve HCV species and areas, as well as those that have economic and sociocultural importance to the communities: fauna and flora inventory.
	Hunting.	Community.	Training: Sustainable Practices and Conservation Courses.
HCV 2: Ecosystems and mosaics at landscape level, and Intact Forest Landscapes - proximity of PA to protected areas	Deforestation pressure/Disputes and changes in land rights or access/use rights.	Farmers; external landowners; local residents.	Training: Disseminate sustainable practices in the use of forest, agricultural resources, and environmental education to combat degradation. Automatic satellite monitoring (Carbonext's CarbonEye System) to identify and act on deforestation. Hiring and training of forest monitors.
HCV 5: Community needs - Brazil nut and copaíba extraction in the project zone and fishing in the Sucunduri River.	Disputes and changes in land rights or access/use rights.	Farmers; external landowners.	Identify and development of educational activities with communities, establish a set of skills and training, identify suppliers, quote prices, hire suppliers. Identify species as those that have economic and sociocultural importance to the communities.

In addition, the survey and monitoring of fauna and flora biodiversity allows us to observe the stability of species and to demonstrate any possible changes in biodiversity. During the course of the project, with the participation of the external biodiversity experts, new biodiversity study, monitoring and conservation actions will be planned and implemented.

5.2.3 Net Positive Biodiversity Impacts (B2.2, GL1.4)

As outlined in Section 5, the baseline scenario will continue to compromise the integrity of Amazonian ecosystems with the advance of use for pasture, in addition to cattle and soy, as shown under the region, causing damage and putting biodiversity at risk. In addition, the activities to be undertaken described in sections 5.2.2 and 5.4 promote the continued presence of vegetation cover, biodiversity integrity and local ecosystem balance. As described in Section 5.2, the project will cause positive gains for biodiversity in the project area, as described below:

Table 133 - Fauna and flora conservation program

Project strategic area	Without-Project Biodiversity Scenario	Expected Biodiversity Impacts – summary	Possible Negative Biodiversity Impacts	Net Positive Biodiversity Impact
Fauna and flora inventory And Biodiversity Research, Monitoring, or conservation activities.	The main causes for the deterioration of the environment are the transformation of forest to pasture and other land-uses. Forest degradation, and deforestation is directly and indirectly related to loss of species across the Amazon biome in recent years. (Prist et al., 2012; Middleton, 2016; Brejão et al., 2018) In addition, the loss of forest habitat, increase of threat level, or extinction of several animal species (ICMbio, 2018; WWF, 2022).	Increase of information of local fauna and flora through inventory. Implementation of research, monitoring or conservation initiatives (to be determined) for fauna and flora biodiversity. Increase of local community knowledge and awareness through environmental education and training.	The project is not predicted to generate negative impacts on biodiversity.	The expectation is to promote both knowledge, through inventories, and subsequent research, monitoring or conservation of biodiversity through the activities summarized at left. This aims to effect significant biodiversity benefits. Given that no negative impacts are foreseen, it is concluded that the positive benefits outweigh the negative, and that the net impacts of the IPOÁ project on biodiversity are positive.

5.2.4 High Conservation Values Protected (B2.4)

The project activities aim to avoid deforestation and maintain forest cover. No environmental impact activities will be carried out, so no negative effects on the HCV are anticipated.

5.2.5 Species Used (B2.5)

This item is not applicable as no restoration activities are planned in the area.

5.2.6 Invasive Species (B2.5)

This item is not applicable as no restoration activities are planned in the area.

5.2.7 Impacts of Non-native Species (B2.6)

This item is not applicable as no restoration activities are planned in the area.

5.2.8 GMO Exclusion (B2.7)

This item is not applicable as no restoration activities are planned in the area.

5.2.9 Inputs Justification (B2.8)

This item is not applicable as no restoration activities are planned in the area.

5.2.10 Waste Products (B2.9)

Regarding the project activities in the areas of education, occupational health and safety and training, which involve training and lectures, will generate residual materials consisting of paper products. For the coffee-breaks, the team is prioritizing the use of reusable items in order to avoid the production of plastic waste. In addition, donations of basic hygiene kits, such as reusable ecological sanitary pads (no waste generation) and oral hygiene kits (contains plastic) are also planned. The waste generated (plastic and paper) are classified as class two B - non-hazardous and inert according to the ABNT NBR 10004 standard.

If not disposed of properly, waste material can harm the local aquatic and terrestrial biodiversity. Especially the populations of the three communities have the habit of burning all the garbage produced. Unfortunately, sending it for recycling ends up being an unviable alternative for the region's populations, due to the distance and logistics. Despite preventing this material from going into the forest or river, the burning of the

material releases toxic substances that not only contribute to air pollution but also have carcinogenic potential.

The further areas of project activities are not predicted to create residual materials which it is the project proponents' responsibility to dispose of.

Therefore, the project activities are being conducted in the most environmentally friendly way possible in order to avoid the generation of waste. As new activities are selected to compose the project, the identification, classification, and management of its waste products will be properly conducted. Offsite Biodiversity Impacts

5.3. Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2)

No negative offsite biodiversity impacts resulting from the implementation of the project's activities are expected. In addition, the proposed activities were designed to mitigate any possible waste or other negative impacts from the subsistence agriculture production practiced by local communities, which might affect offsite locations. The project activities are designed to enhance sustainable practices and promote environmental awareness, thus providing more harmony among man and the natural environment. For this reason, no mitigation measures were considered necessary.

5.3.2 Net Offsite Biodiversity Benefits (B3.3)

The net offsite biodiversity impacts of the Project are positive, since there are no mapped negative offsite biodiversity impacts resulting from the project's activities, and substantial benefits regarding fauna and flora conservation in the PA are foreseen. These biodiversity benefits will extend outside the project's boundaries due to factors including ecological effects, animal ranges, and dissemination of knowledge and awareness raised in local communities.

5.4. Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

The monitoring plan below identifies, in accordance with PD template guidelines, the communities, community groups and other stakeholders to be monitored, variables to be monitored, types of measurements and sampling methods, and the frequency of monitoring and reporting for each type and method.

Table 134. Monitoring Plan Biodiversity.

Actions Descriptions							
Scope	Activity	Reason	Period	Frequency	Area	Stakeholders	
Biodiversity	Fauna and flora inventory.	Fauna Inventory	Medium Term	Every 5 years	Project Area	Carbonext, Supplier, Communities	
		Flora Inventory		Every 5 years			
	Fauna and flora research, monitoring and conservation initiatives	Environmental awareness and education actions	Involvement of the local community and community environmental agents.	Medium Term	Every 5 years	Project Area	Carbonext, Supplier, Communities
		Post-Inventory Actions: Target Species Research, Monitoring or Conservation Program	Knowledge about biodiversity and ensure the perpetuation of species. The actions will be proposed by the expert.	Long Term	Every 5 years	Project Area	Carbonext, Supplier, Communities
	High Conservation Value areas	Survey of High Conservation Value areas	Identify and conserve areas that have economic and sociocultural importance for the communities	Medium Term	Every 5 years	Project Area	Carbonext, Supplier, Communities

5.4.2 Biodiversity Monitoring Plan Dissemination (B4.3)

The results of the monitoring carried out will be made public on the Internet and through the Carbonext website. All documents and information regarding the monitoring and verification results of this project will be published on the Verra platforms.

Regarding the local stakeholders, the results obtained from the monitored project will be shared during the periodic site visits, which will be arranged for events such as training, verification, or verification periods.

5.5. Optional Criterion: Exceptional Biodiversity Benefits

5.5.1 High Biodiversity Conservation Priority Status (GL3.1)

This item does not apply to the present project at this time. This study will be possible after the Fauna and Flora inventory foreseen in item 5.4, which will be carried out every five years, the first one being foreseen after receiving the first lot of credits.

To date, as described in 5.1.1, given previous studies conducted in the region and the state of conservation of the forests belonging to the properties, it is expected that more threatened and endemic species will be found in the Project Zone during the course of the fauna and flora inventories.

5.5.2 Trigger Species Population Trends (GL3.2, GL3.3)

This item does not apply to the present project at this time.

6 APPENDICES

Appendix 1: Project Activities

The activities of the IPOÁ REDD+ Project are presented in the table below, fulfilling the requirements of Section 2.1.11. They are designed based on the communities' necessities self-identified requirements, are collaborative, and will be undertaken throughout the project's lifetime.

Table 135. Project Activities

Activity Description						
Pillar/ Scope	Activity	Objective	Period	Frequency	Area	Stakeholders
Social	Healthcare	Building maintenance of basic health unit	Long-term	Annual	Project Zone	Carbonext, Project Proponent, Supplier
		Health transport programme		Annual		Carbonext, Project Proponent, Supplier, City Hall
	Education	School refurbishment	Long-term	Annual	Project Zone	Carbonext, Project Proponent, Supplier, City Hall

		Teacher support and training	Ensure the presence of qualified teachers and improve the quality and quantity of lessons in community school	Short-term			
Occupational Health and Safety		Training and workshop: Occupational health and safety	Provide safety equipment and improve knowledge of occupational health and safety	Medium-term	Annual	Project Zone	Carbonext, Project Proponent, Supplier
		Personal protective equipment for worker health and safety					Carbonext, Project Proponent
Communication and Electrification		Installation and maintenance of technology infrastructure for communication and internet access, and electricity	Improve internet access and electricity provision in the communities	Short-term	Annual	Project Zone	Carbonext, Project Proponent, Supplier, Communities
Employment		Hiring and training of forest monitors	Increase surveillance on the ground to combat trespassing and deforestation	Short-term	Annual	Project Zone	Project Proponent
Training		Sustainable Practices and Conservation Courses	Disseminate sustainable practices in the use of forest and agricultural resources	Medium-term	Annual	Project Zone	Carbonext, Communities Supplier,

	High Conservation Values	Survey of social High Conservation Value areas	Implement survey of High Conservation Value areas (HCVs 4, 5 and 6) ²³²	Long-term	3 years	Project Zone	Carbonext, Communities
Biodiversity	Fauna and flora study initiatives	Fauna Inventory	Implement fauna inventory activities	Medium-term	5 years	Project Zone	Carbonext, Communities Supplier,
		Flora Inventory	Implement flora inventory activities		5 years		
	Biodiversity Research, Monitoring, conservation activities or	Environmental awareness and education actions	Involvement of the local community and community environmental agents.	Medium Term	5 years	Project Area	Carbonext, Communities Supplier,
		Target Species Research, Monitoring or Conservation Program	Knowledge about biodiversity and ensure the perpetuation of species. The actions will be proposed by the expert.	Long Term	5 years	Project Area	Carbonext, Communities Supplier,
		Survey of High Conservation Value areas	Identify and conserve HCV species and areas, as well as those that have economic and	Medium-term	5 years	Project Zone	Carbonext, Communities Supplier,

²³² See section 4.1.3 for more details

			socio-cultural importance to the communities.				
	High Conservation Value areas	Remote Monitoring	Detect deforestation events in the in the Project Zone and its surroundings	Short Term	Weekly	Project Zone	Carbonext
Climate	Forest Conservation	Fire brigade training (according to PIF)	Community trained in firefighting	Short Term Medium-term	Annual	Project Zone	Carbonext, Project Proponent, Supplier
		Deforestation prevention and mitigation actions	Implement actions to combat deforestation and land-squatting		Annual	Project Zone	Carbonext, Project Proponent, Supplier
		Relationship-building actions with communities, such as festive, cultural, and sports activities.	Strengthen the relationship with and within the communities		Annual	Project Zone	Carbonext, Project Proponent
Other	Further Leisure and Cultural Activities	Unspent excess budget may be spent on unforeseen costs (e.g. maintenance) or allocated to any of the categories above, as required	N/A	N/A	N/A	Project Zone	Carbonext, Project Proponent

Appendix 2: Theory of Change

Table 135. Theory of Change – Social Pillar

Scope	Activity	Outputs (Short term)	Outcomes (Medium term)	Impacts (Long term)	Relevance to project's objectives
Healthcare	Basic Health Facility (BHF)	Basic Health Facility (BHF): Electricity and telecommunication infrastructure installed and functioning in the BHF. N° people (by gender/age) receiving ongoing tele-medicine medical care. N° of medicines and medical supplies procured/transported for the BHF. N° and effectiveness of meetings carried out to facilitate community- government PPP.	Increased range in medical diagnosis and attendance capacities through tele-medicine. Increase in supplies of medicines and medical supplies at the BHF. Reduction in cases and impact of a range of diseases and conditions in communities. Facilitated Public-Private relations between the community and the relevant public sector department.	Improved community access to healthcare services. A healthier community population. Contribution to good health and well-being (SDG 3).	The project's overall objectives of forest conservation, GHG reductions, sustainable development and biodiversity conservation are closely inter-related. The Social Theory of Change framework here presented emphasizes how socioeconomic initiatives contribute to a variety of the project's objectives.
	Support for health transport	N° equipment items and operational supplies purchased/ transported for water ambulance. N° of community training events relating to medical transport. N° people transported for medical using project supported water ambulance. N° and effectiveness of meetings carried out to facilitate community- government PPP	Increase in patients transported for medical conditions, accidents, births, etc. Increase in emergency cases treated in city medical facilities. Electricity and telecommunication infrastructure installed and functional for healthcare purposes. Facilitated Public-Private relations between the community and the relevant public sector department.	An increase in emergency cases treated in medical facilities. A reduction in deaths from emergency medical diseases, conditions, and accidents. A reduction in infant mortality. A reduction in mother mortality during childbirth.	

Education	School infrastructure and equipment	N° of electricity and telecommunication equipment installed and maintained within education facilities. N° of educational equipment and material purchased or transported. N° and effectiveness of meetings carried out to facilitate community- government PPP.	Presence of electricity and telecommunication infrastructure within education facilities. Good school equipment in the PZ. Facilitated Public-Private relations between the community and the relevant public sector department.	Contribute to quality education for communities. School facilities in PZ adequate and conducive to quality education.	
	Teacher and student education and training	N° enrolments in training and educational courses. Sum spent (R\$) on teacher and student training and courses. N° students (by gender/age) studying at PZ schools Progress in key subject areas (by gender/age) of students and teachers Student satisfaction survey²³³	Improved additional training for teachers. Improved quality of lessons for students. Educational service in the PZ brought above regional level.	Communities with an education level above the regional average. Communities with improved choices for their futures. Contribution to quality education within the PZ (SDG 4)	
Occupational Health and Safety	Training and workshop: Occupational health and safety	N° people (by gender/age) trained in best practices for occupational health and safety	Increase in people following occupational health and safety guidelines. Reduction in work-related accidents.	Population aware of occupational risks related to health and safety. Contribution to decent work and economic growth (SDG 8)	

²³³ Specific output suggested by specialist education service provider

	Personal protective equipment (PPE) distribution for worker health and safety.	N° people (by gender/age) given PPE equipment.			
Communication and Electrification	Installation and maintenance of infrastructure and equipment for electricity, telecommunication, and internet access	N° equipment items purchased and transported. N° equipment items installed and maintained in community buildings. N° of people (by gender/age) with access to electricity and communication technology. N° of telecommunication devices in the Project Zone. N° of community members (by gender/age) trained in: i) communication technology use and ii) maintenance. N° of domestic appliances in community households or buildings.	Functioning electricity, telecommunication, and internet infrastructure in PZ community residences. Population more informed on many topics from national current affairs to REDD+ and the Ipoá project itself.	Increase in community members with access to the internet and electricity, broadening their life opportunities. Population informed about key topics such as world and national affairs, REDD+ methodologies and the IPOÁ Project itself, increasing their influence and autonomy. Population able to communicate at the pace of the modern world. Contribution to ensuring access to affordable, reliable, sustainable and modern energy (SDG 7). Contribution to industry, innovation, and infrastructure (SDG 9). SDG 9.C "Significantly increase access to information and communications technology and strive to provide universal and affordable access to the Internet in least developed countries (...)"	

Employment	Community association creation	N° of participants in association training events, assemblies etc. Consultant identified/ hired for community association creation. Association constitution initiated, finalized, or maintained. N° of members and official positions in association. N° of women promoted to become members and official positions in association.	Strengthened community governance. Improved ability for communities to demand and obtain public services. Increase in community members (by gender) becoming members and occupying official positions in association.	Community entrepreneurship enabled. Improvement in income generation and the quality of life of employees. Reduction of poverty (SDG 1). SDG 8.3 Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services	
	Hiring and training monitors	N° community members identified (by gender/age) for forest monitoring work. N° people (by gender/age) trained in forest monitoring practices. N° of monthly surveillance rounds conducted. N° of reports, protocols, and alerts sent from monitors to PP. Sum spent (R\$) on monitor salaries.	Increase in people hired and carrying out surveillance rounds. Increase in number of community members employed in sustainable economy.	Contribution to sustainable communities (SDG 11)	
Training	Sustainable and Conservation Courses	N° people (by gender/age) educated in Sustainability and Conservation practices.	Increase in range of employment opportunities available to community members, especially in sustainability and bioeconomy. Increase in community knowledge and adoption of sustainability and conservation practices.	Contribution to creation of a new bioeconomy in the region. Population empowered with autonomy, leadership and innovation in sustainability and conservation. Contribution to sustainable communities (SDG 11).	
High Conservation Values	Survey of social High Conservation Value areas	HCV 4: Hectares of remaining forest in the PA at the end of each MR	Effective HCV study and monitoring programmes active in the PZ.	Tangible improvements in HCV 4 regarding ecosystem services for the community	

		Deforested areas cultivated by communities in the PZ (has)²³⁴ Areas disturbed by human activity in the PZ (has) HCV 5 and 6: N° of initiatives promoting community forest preservation, or other aspects of ecosystem services, in the PZ. N° or size of areas of importance for livelihoods or sociocultural value of communities mapped.	Effective HCV enhancement and conservation programmes active in the PZ. Active community engagement in HCV monitoring and conservation programmes.	Tangible improvements in HCVs 5 and 6 regarding community livelihoods and economic and sociocultural value to the communities	
--	--	---	--	--	--

Table 136. Theory of Change – Biodiversity Pillar

Scope	Activity	Outputs (Short term)	Outcomes (Medium term)	Impacts (Long term)	Relevance to project's objectives
-------	----------	----------------------	------------------------	---------------------	-----------------------------------

²³⁴ It will be key to measure this indicator for HCV 4 within and outside areas foreseen for expansion of community land-uses.

Fauna and flora study initiatives	Fauna Inventory	Nº of taxonomic groups studied: mammals, herpetofauna, insects, etc. Intensity or area of fauna sampling (Nº of plots or geographic area (ha). Nº of key indicator species identified: endemic, flagship, threatened, or other.	Presence of effective fauna: i) study and ii) conservation programmes in the PZ.	Improvement in knowledge of fauna within the PZ or greater region. Contribution to Life on Land (SDG 15).	The project's overall objectives of forest conservation, GHG reductions, sustainable development and biodiversity conservation are closely inter-related. The Biodiversity Theory of Change framework here presented emphasizes how biodiversity study and conservation initiatives contribute to a variety of the project's objectives.
	Flora Inventory	Intensity or area of flora sampling (Nº of plots or geographic area (ha). Nº of taxonomic groups studied: Order, family, genus, etc. Nº of key indicator species identified: endemic, flagship, threatened, or other.	Presence of effective fauna: i) study and ii) conservation programs in the PZ.	Improvement in knowledge of flora within the PZ or greater region. Contribution to Life on Land (SDG 15).	
Biodiversity Research, Monitoring, or conservation activities	Environmental education	Nº of environmental education events conducted. Nº of people educated in environmental awareness.	Understanding increased of community perceptions of biodiversity and their possible impacts. Community awareness of the importance of conserving the target species increased.	Alteration in community behavior harmful to target species. Community capacity created to autonomously monitor fauna, flora, manage threats, and autonomously promote biodiversity conservation.	Conservation of the area and monitoring of threats by the local community contribute to project's social development and biodiversity objectives.

	Biodiversity Research, Monitoring, conservation activities ²³⁵ or	Nº and effectiveness of fauna research, monitoring, or conservation initiatives implemented. Further biodiversity indicators to be determined by consultants during the inventory stage ²³⁸	Increased understanding of current populations in the PZ through monitoring. Increased knowledge of biodiversity indicators within the PZ through research. Structuring and implementation of targeted actions directed at species within the PZ ²³⁸ .	Improvement in conservation status of biodiversity within the PZ and greater region. Contributions to scientific understanding and literature.	The project's biodiversity research, monitoring and conservation are central to the its biodiversity objectives.
High Conservation Values	Survey of High Conservation Value areas and species	Nº of taxonomic groups studied: Order, family, genus, etc. Intensity or area of HCVs sampling (Nº of plots or geographic area (ha).	Presence of effective HCV study and monitoring programmes in the PZ. Presence of effective HCV conservation initiatives in the PZ. Improvement in community participation in identification and conservation of HCVs.	Improvement in HCV population health and diversity within the PZ or greater region. Improvement in conservation status of HCVs within the PZ or greater region. Contribution to Life on Land (SDG 15).	Increased knowledge and improved conservation status of target HCV species.

²³⁵ Specific activities and indicators to be determined by biodiversity consultant following fauna and flora inventories

Table 137. Theory of Change – Climate Pillar

Scope	Activity	Outputs (Short term)	Outcomes (Medium term)	Impacts (Long term)	Relevance to project's objectives
Forest Conservation	Remote Monitoring	Nº of deforestation events detected remotely.º	Reduction in deforestation within the PZ.	Conservation of 4,200.43 ha of forest, corresponding to avoided net emissions of 2,522,057 tCO2e over 30 years. Contribution to Life on Land (SDG 15). Contribution to Peace, Justice, and Strong Institution (SDG 16).	The project's overall objectives of forest conservation, GHG reductions, sustainable development and biodiversity conservation are closely inter-related. The Climate Action Theory of Change here presented emphasizes how project climate actions enhance and contribute to a variety of the project's objectives.
	Fire brigade training (according to PIF)	Nº of community firefighters identified and trained. Nº of fire-fighting training events held.	Community readiness for firefighting increased. Reduction in deforestation and fire events within the PZ.		
	Deforestation prevention and mitigation actions	Nº of deforestation events visited in the field. Nº and effectiveness of meetings carried out to combat deforestation. Nº of infrastructure items constructed for deforestation combat. Nº of equipment items and vehicles purchased for deforestation combat. Nº of protocol applications (deforestation event protocol; forest integrity protocol, etc.)	Reduction in deforestation within the PZ. Reduction in illegal trespassing and land-squatting attempts within the PZ.		

Table 138. Theory Change – Other

Scope	Activity	Outputs (Short term)	Outcomes (Medium term)	Impacts (Long term)	Relevance to project's objectives
Further Leisure, Cultural and Sporting Activities	Relationship-building actions with communities such as festive, cultural, and sports activities.	N° of actions carried out with communities. N° of participants (by gender/age) in activities N° and effectiveness of meetings carried out to promote such activities.	Increase in number of community members with access to leisure and cultural activities. Improved intra-community relations. Improved inter-community relations between the PZ communities. Improved relations between communities and project proponents and developer.	Improved quality of life for community members. Contribution to community Good Health and Wellbeing (SDG 3).	The project's overall objectives of forest conservation, GHG reductions, sustainable development and biodiversity conservation are closely inter-related. The Theory of Change regarding the project's other actions, here presented, emphasizes how leisure, cultural and sporting activities contribute to a variety of the project's objectives.

Appendix 3: Total VCUs
Table 139. Ipoá REDD+ Project – Total VCUs

REDD Benefits			
Year	REDD Benefits (buffer included) (tCO ₂)	Buffer discount (10.0%)	Annual Net tradeable VCUs (tCO ₂)
2022	246,706	24,295	222,410
2023	463,954	45,704	418,250
2024	483,341	47,636	435,705
2025	498,194	49,124	449,070
2026	164,407	16,277	148,130
2027	85,314	8,998	76,316
2028	79,726	8,363	71,362
2029	56,353	5,660	50,693
2030	56,353	5,660	50,693
2031	56,322	5,660	50,662
2032	61,237	6,321	54,916
2033	47,115	4,905	42,210
2034	53,940	5,927	48,013
2035	10,868	1,153	9,715
2036	6,476	699	5,777
2037	18,598	2,075	16,523
2038	7,310	806	6,504
2039	20,051	2,273	17,778
2040	41,527	4,764	36,763
2041	14,952	1,704	13,247
2042	7,102	777	6,325
2043	6,607	739	5,868
2044	3,511	391	3,120

CCB Version 3, VCS Version 3

2045	8,805	1,037	7,768
2046	2,688	304	2,385
2047	2,773	320	2,453
2048	2,131	242	1,889
2049	1,703	194	1,509
2050	11,861	1,371	10,489
2051	825	97	728
2052	1,307	153	1,154
Total	2,522,057	253,632	2,268,424

7 BIBLIOGRAPHIC REFERENCES

AALDE, H. et al. Generic methodologies applicable to multiple landuse categories. In: IPCC Guidelines for National Greenhouse Gas Inventories. IPCC, 2006. <https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_02_Ch2_Generic.pdf>. Accessed on 22/09/2022.

AGÊNCIA EMBRAPA DE INFORMAÇÃO TECNOLÓGICA - Ageitec. <<https://www.agencia.cnptia.embrapa.br/solostropicais/arvore/l.html>>. Accessed on 21/09/2022.

AGÊNCIA NACIONAL DE TELECOMUNICAÇÕES – ANATEL. Acessos Telefonia Móvel. <<https://informacoes.anatel.gov.br/paineis/meu-municipio/acessos-telefonias-movel>>. Accessed on 22/09/2022.

ALVES, V. F. Composição florística e fitossociologia de uma área com exploração florestal no município de Novo Aripuanã, Amazonas. Trabalho de conclusão de curso (Graduação em Engenharia Florestal) Universidade Federal Rural da Amazônia, Campus Paragominas, 2019. Paragominas, PA. 2019. <<http://www.bdta.ufra.edu.br/jspui/bitstream/l.pdf>>. Accessed on 21/09/2022.

AMAZONAS. Decreto n. 26.010, de 03 de Julho de 2006. Cria a Reserva de Desenvolvimento Sustentável do Juma. Diário Oficial do Estado. 2006.

AMAZONAS. Decreto n.º 15.780, de 05 de Janeiro de 1994. ALTERA o Art. 57 do Decreto Estadual N.º 10.028, de 04 de fevereiro de 1987, que regulamentou a Lei n.º 1.532, de 06.07.82, que dispõe sobre o Sistema Estadual de Licenciamento de Atividades com Potencial de Impacto ao Meio Ambiente e aplicação de penalidades. <https://www.mpam.mp.br/images/stories/decreto_15780_94.pdf> Accessed on 21/09/2022.

AMAZONAS. Decreto n.º 15.842, de 09 de Fevereiro de 1994. Altera o Art. 44 do Decreto Estadual n.º 10.028, de 04 de fevereiro de 1987, que regulamentou a Lei n.º 1.532 de 06.07.82, que dispõe sobre o Sistema Estadual de Licenciamento de Atividades com Potencial de Impacto ao Meio Ambiente e aplicação de penalidades. <https://www.mpam.mp.br/images/stories/decreto_15842_94.pdf>. Accessed on 21/09/2022.

AMAZONAS. Decreto nº 44.716, de 25 de outubro de 2021. Formaliza a adesão do Estado do Amazonas às campanhas "Race to Zero" e "Under2 Coalition", no âmbito da Convenção-Quadro das Nações Unidas sobre Mudança do Clima (CQNUMC/UNFCCC), e dá outras providências. <<https://leisestaduais.com.br/am/decreto-n-44716-2021>>. Accessed on 21/09/2022.

AMAZONAS. Lei n.º 3.135, de 05 de junho de 2007. Política Estadual sobre Mudanças Climáticas, Conservação Ambiental e Desenvolvimento Sustentável do Amazonas. <https://legisla.imprensaoficial.am.gov.br/diario_am/12/2007/6/4939?q=3.135>. Accessed on 21/09/2022.

AMAZONAS. Lei n.º 3.785, de 24 de julho de 2012. Dispõe sobre o licenciamento ambiental no Estado do Amazonas, revoga a Lei nº 3.219, de 28 de dezembro de 2007, e dá outras providências. <https://legisla.imprensaoficial.am.gov.br/diario_am/12/2012/7/2494?q=3.785>. Accessed on 21/09/2022.

AMAZONAS. Lei nº 2.416, de 22 de agosto de 1996. Disciplina a Política Estadual da Prevenção e Controle da Poluição, Melhoria e Recuperação do Meio Ambiente e da Proteção aos Recursos Naturais, e dá outras providências. <https://www.mpam.mp.br/images/stories/lei_2416_96.pdf>. Accessed on 21/09/2022.

AMAZONAS. Lei nº 4.406, de 28 de dezembro de 2016. Estabelece a Política Estadual de Regularização Ambiental, dispõe sobre o Cadastro Ambiental Rural - CAR, o Sistema de Cadastro Ambiental Rural - SICAR-AM, o Programa de Regularização Ambiental - PRA, no Estado do Amazonas. <https://legisla.imprensaoficial.am.gov.br/diario_am/12/2016/12/1433?q=4.406>. Accessed on 21/09/2022.

AMAZONAS. Lei nº 1.532 de 06 de julho de 1982. <<http://www.ipaam.am.gov.br/wp-content/uploads/2021/01/LDE-1.532-82-Pol%C3%ADtica-Estadual-da-Preven%C3%A7%C3%A3o-e-Controle-da-Polui%C3%A7%C3%A3o.pdf>>. Accessed on 21/09/2022.

AMAZONAS. Unidades de Conservação. SEMA – Secretaria de Estado do Meio Ambiente. 2023. <<https://meioambiente.am.gov.br/unidade-de-conservacao/>>. Accessed on 22/09/2022.

AVIBASE. Conopophaga melanogaster (Black-bellied Gnatcatcher). Avibase, Available in: <https://avibase.bsc-eoc.org/species.jsp?avibaseid=C0F59C9F8A109FC9>. Accessed on 22/09/2022.

AZEVEDO-RAMOS, C. et al. Lawless land in no man's land: The undesignated public forests in the Brazilian Amazon. Land Use Policy. Vol. 99, December 2020, 104863. <https://ipam.org.br/wp-content/uploads/2020/06/lawless-land-in-no-man-s-land.pdf>. Accessed on 21/09/2022.

BANCO NACIONAL DE DESENVOLVIMENTO ECONÔMICO E SOCIAL – BNDES. Responsável por cerca de 20% do PIB e principal geradora de superávits comerciais, a agropecuária conta com diversos instrumentos de apoio do BNDES. <<https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos/agropecuaria>>. Accessed on 21/09/2022.

BNC AMAZONAS. <<https://bncamazonas.com.br/>>. Accessed on 04/09/22.

BNDES – O Banco Nacional do Desenvolvimento. Onde atuamos. <<https://www.bndes.gov.br/wps/portal/site/home/onde-atuamos>>. Accessed on 22/09/2022.

BORGES, T.; BRANFORD, S. Ao afrouxar leis de exportação, Brasil permite saída de madeira ilegal da Amazônia. Mongabay. 2020. <<https://brasil.mongabay.com/2020/04/ao-afrouxar-leis-de-exportacao-brasil-permite-saida-de-madeira-ilegal-da-amazonia/>>. Accessed on 21/09/2022.

BRASIL INDICADORES. Valor atual e histórico do Salário-Mínimo. Brasil Indicadores, Available in: <https://brasilindicadores.com.br/salario-minimo/>. Accessed on 22/09/2022.

BRASIL. Decreto de 11 de Maio de 2016. Cria o Parque Nacional do Acari. Diário Oficial da União. 2016. <https://www.gov.br/icmbio/pt-br/assuntos/biodiversidade/unidade-de-conservacao/unidades-de-biomas/amazonia/lista-de-ucs/parna-do-acari/arquivos/dcom_decreto_criacao_parna_do_acari.pdf> Accessed at 14/03/2022.

BRASIL. Decreto nº 9.578, de 22 de novembro de 2018. Dispõe sobre diretrizes para as negociações coletivas de trabalho, das entidades estatais submetidas ao regime jurídico das empresas privadas. <http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/decreto/d9578.htm>. Accessed on 21/09/2022.

BRASIL. Decreto-lei nº 5.452, de 1º de maio de 1943. Aprova a Consolidação das Leis do Trabalho <https://www.planalto.gov.br/ccivil_03/decreto-lei/Del5452.htm>. Accessed on 21/09/2022.

BRASIL. Constituição da República Federativa do Brasil de 1988. <http://www.planalto.gov.br/ccivil_03/constituicao/constituicao.htm>. Accessed on 21/09/2022.

BRASIL. Crédito Rural, s/n. Banco Central do Brasil. <<https://www.bcb.gov.br/estabilidade/financeira/creditorural>>. Accessed on 22/09/2022.

BRASIL. Decreto nº 5746/2006. Regulamenta o art. 21 da Lei no 9.985, de 18 de julho de 2000, que dispõe sobre o Sistema Nacional de Unidades de Conservação da Natureza. <http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm>. Accessed on 22/09/2022.

BRASIL. Decreto Nº 5.092, de 21 de maio de 2004. Define regras para identificação de áreas prioritárias para a conservação, utilização sustentável e repartição dos benefícios da biodiversidade, no âmbito das atribuições do Ministério do Meio Ambiente. 2004. <https://www.planalto.gov.br/ccivil_03/_Ato2004-2006/2004/Decreto/D5092.htm>. Accessed on 21/09/2022.

BRASIL. **Decreto nº 5.746**, de 5 de abril de 2006. Regulamenta o art. 21 da Lei no 9.985, de 18 de Julho de 2000, que dispõe sobre o Sistema Nacional de Unidades de Conservação da Natureza <http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2006/decreto/d5746.htm>. Accessed on 21/09/2022.

BRASIL. **Decreto nº 7.378**, de 1º de dezembro de 2010. Aprova o macrozoneamento ecológico-econômico da Amazônia legal - MacroZEE da Amazônia legal e altera o decreto nº 4.297, de 10 de julho de 2002. 2010. <http://abce.org.br/downloads/Decreto_7_378_2_.pdf>. Accessed on 21/09/2022.

BRASIL. **Decreto nº 9.579**, de 22 de novembro de 2018. Consolida atos normativos editados pelo Poder Executivo federal que dispõem sobre a temática do lactente, da criança e do adolescente e do aprendiz, e sobre o Conselho Nacional dos Direitos da Criança e do Adolescente, o Fundo Nacional para a Criança e o Adolescente e os programas federais da criança e do adolescente, e dá outras

providências.<http://www.planalto.gov.br/ccivil_03/_ato2015-2018/2018/decreto/D9579.htm>. Accessed on 21/09/2022.

BRASIL. E-Social – Novo salário-mínimo 2022: Veja como registrar o reajuste no e-Social doméstico.<<https://www.gov.br/esocial/pt-br/noticias/novo-salario-minimo-2022-veja-como-registrar-o-reajuste-no-esocial-domestico>>. Accessed on 22/09/2022.

BRASIL. **Lei nº 10.406**, de 10 de Janeiro de 2002. Institui o Código Civil. <<http://www.planalto.gov.br/ccivil03/lei/2002/l10406compilada.htm>>. Accessed on 21/09/2022.

BRASIL. **Lei nº 12.305**, de 2 de agosto de 2010. <http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2010/lei/l12305.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 12.651**, de 25 de Maio de 2012. Dispõe sobre a proteção da vegetação nativa; altera as Leis nºs 6.938, de 31 de agosto de 1981, 9.393, de 19 de dezembro de 1996, e 11.428, de 22 de dezembro de 2006; revoga as Leis nºs 4.771, de 15 de setembro de 1965, e 7.754, de 14 de abril de 1989, e a Medida Provisória nº 2.166-67, de 24 de agosto de 2001; e dá outras providências. <http://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 13.105**, de 16 de Março de 2015. Código de Processo Civil. <>. Accessed on 21/09/2022.

BRASIL. **Lei nº 6.015**, de 31 de Dezembro de 1973. Dispõe sobre os registros públicos, e dá outras providências <http://www.planalto.gov.br/ccivil_03/leis/l6015compilada.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 8.069**, de 13 de julho de 1990. Estatuto da Criança e do Adolescente. Brasília, DF, 1990.

BRASIL. **Lei nº 9.433**, de 8 de janeiro de 1997. Institui a Política Nacional de Recursos Hídricos, cria o Sistema Nacional de Gerenciamento de Recursos Hídricos, regulamenta o inciso XIX do art. 21 da Constituição Federal, e altera o art. 1º da Lei nº 8.001, de 13 de março de 1990, que modificou a Lei nº 7.990, de 28 de dezembro de 1989. <http://www.planalto.gov.br/ccivil_03/leis/l9433.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 12.187**, de 29 de dezembro de 2009. Institui a Política Nacional de Resíduos Sólidos; altera a Lei no 9.605, de 12 de fevereiro de 1998; e dá outras providências. <http://www.planalto.gov.br/ccivil_03/_ato2007-2010/2009/lei/l12187.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 4.504**, de 30 de Novembro de 1964. Dispõe sobre o Estatuto da Terra, e dá outras providências. <http://www.planalto.gov.br/ccivil_03/leis/l4504.htm>. Accessed on 21/09/2022.

BRASIL. **Lei nº 6.938**, de 31 de agosto de 1981. Dispõe sobre a Política Nacional do Meio Ambiente, seus fins e mecanismos de formulação e aplicação, e dá outras providências. <https://www.planalto.gov.br/ccivil_03/Leis/L6938.htm>. Accessed on 21/09/2022.

BRASIL. Lei no 9.985, de 18 de julho de 2000. Regulamenta o art. 225, § 1o, incisos I, II, III e VII da Constituição Federal, institui o Sistema Nacional de Unidades de Conservação da Natureza e dá outras providências. <http://www.planalto.gov.br/ccivil_03/leis/l9985.htm> Accessed on 21/09/2022.

BRASIL. Ministério do Meio Ambiente. Resolução CONAMA Nº 001, de 23 de janeiro de 1986 <<http://www.ibama.gov.br/sophia/cnia/legislacao/MMA/RE0001-230186.PDF>>. Accessed on 21/09/2022.

BRASIL. Site do eSocial, 2022. Novo salário mínimo 2022: veja como registrar o reajuste no e-Social Doméstico. Available in: <<https://www.gov.br/esocial/pt-br/noticias/novo-salario-minimo-2022-veja-como-registrar-o-reajuste-no-esocial-domestico>>. Accessed on 22/09/2022.

BRASIL. Site do Ministério do Desenvolvimento e Assistência Social, Família e Combate à Fome: Auxílio Brasil, 2022. Available in: <<https://www.gov.br/cidadania/pt-br/auxilio-brasil#oque>>. Accessed on 22/09/2022.

BREJÃO G. L. et al. Threshold responses of Amazonian stream fishes to timing and extent of deforestation, Conservation Biology, n 32, Issue 4, p. 860–871. 2018.

BRONZE, G. Amazônia Legal tem recorde de alertas de desmatamento no 1o trimestre de 2022. CNN Brasil, 2022. Available in: <<https://www.cnnbrasil.com.br/nacional/amazonia-legal-tem-recorde-de-alertas-de-desmatamento-no-1o-trimestre-de-2022/>>. Accessed at 22/09/2022.

BROWN, E. et al. Common Guidance for Identification of High Conservation Values. Proforest, October, 2013. Available in: <https://www.proforest.net/fileadmin/uploads/proforest/Documents/Publications/common-guidance-for-the-identification-of-high-conservation-values.pdf>. Accessed on 13/09/2022.

BUSTAMANTE M. et al. Terceiro inventário brasileiro de emissões e remoções antrópicas de gases de efeito estufa. Relatórios de Referência. Setor uso da terra, mudança do uso da terra e florestas. Ministério da Ciência, Tecnologia e Inovação 2015. <http://redd.mma.gov.br/images/FREL/RR_LULUCF_Mudana-de-Uso-e-Floresta.pdf>. Accessed on 22/09/2022.

CAMPANHA, J. S.; BASTOS, C. P. Quem é o Brasil da COP 26? Fundação Getúlio Vargas - FGV <<https://portal.fgv.br/artigos/quem-e-brasil-cop-26>>. Accessed on 12/05/2022.

CARBONEXT. Anti-corruption policy and code of ethics and organizational conduct in force. 2022. <<https://carbonext.com.br/governanca-corporativa/>>. Accessed on 21/09/2022.

CARRANÇA. T. Por que o consumo de carne bovina no Brasil deve voltar em 2021 ao patamar de décadas atrás. BBC. São Paulo. 2021. <<https://www.bbc.com/portuguese/brasil-55664305>>. Accessed on 21/09/2022.

CARRERO, C. C. et al. A Cadeia Produtiva de Carne Bovina no Amazonas. IDESAM. Manaus, 2015. <http://www.idesam.org.br/publicacao/cadeia-produtiva-corte-amazonas.pdf> . Accessed on 21/09/2022.

CAVALCANTE, K. V. Vulnerabilidade socioambiental como reverso da sustentabilidade nos municípios das microrregiões Madeira e Parintins, no Estado do Amazonas. 2013. Tese (doutorado). Universidade de Brasília. Brasília. 2013. <<https://1library.org/article/microrregi%C3%A3o-madeira-no-estado-do-amazonas.y816jjwz>>. Accessed on 21/09/2022.

CENAMO, M. C. et al. REDD+ Estudo de Oportunidades para a Região Sul do Amazonas. Instituto de Conservação e Desenvolvimento Sustentável - IDESAM. Série Relatórios Técnicos, nº 1. 2011. <[https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas](https://idesam.org/publicacao/REDD_Estudo_de_Oportunidades_Sul_Amazonas.pdf)>.pdf>. Accessed on 13/08/21

CENTRO NACIONAL DE CONSERVAÇÃO DA FLORA. CNC Flora. Referência nacional em geração, coordenação e difusão de informação sobre biodiversidade e conservação da flora brasileira ameaçada de extinção. Available in: <<http://cncflora.jbrj.gov.br/portal/>>. Accessed on 22/09/2022.

CORREIO BRASILIENSE. Câmara pode aprovar hoje licenciamento ambiental automático. 2020. <<https://www.correiobraziliense.com.br/app/noticia/brasil/2020/04/29/interna-brasil,849652/camara-pode-aprovar-hoje-licenciamento-ambiental-automatico.shtml>> . Accessed on 21/09/2022.

CUNHA, J. M. Atributos do solo e emissão de CO₂ em Terra Preta Arqueológica preservada e sob cultivo em Novo Aripuanã, Amazonas, Brasil. 2016. 80 f. Tese (Doutorado em Física Ambiental) - Universidade Federal de Mato Grosso. Cuiabá, 2016. <https://ri.ufmt.br/bitstream/1/2160/1/TESE_2016_20Maur%c3%adcio%20da%20Cunha.pdf>. Accessed on 21/09/2022.

DATASUS - TECNOLOGIA DA INFORMAÇÃO A SERVIÇO DO SUS. TabNet Win32 3.0: Nascidos vivos - Amazonas. <<http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinasc/cnv/nvam.def>>. Accessed on 22/09/2022.

ECOLÓGICA ASSESSORIA. Monitoring Report Florestal Santa Maria Project, 2021. <https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqo wrnoiuomnckjashoufifmln902309ksdfiku098l71896923>. Accessed on 20/09/2022.

EMPRESA BRASILEIRA DE PESQUISA AGROPECUÁRIA. Sistema brasileiro de classificação de solos. SiBCS. 6ª ed. EMBRAPA. Brasília. 2006. <https://www.embrapa.br/en/busca-de-publicacoes/-/publicacao/1094003/sistema-brasileiro-de-classificacao-de-solos>. Accessed on 21/09/2022.

FEARNSIDE, P. M. Human carrying capacity of the Brazilian rainforest. New York, Columbia University Press, 1986.

FELLET, J. Como a 'MP da grilagem' pode mudar o mapa de regiões da Amazônia. Amazônia – Notícia e Informação. 2020. < <https://amazonia.org.br/como-a-mp-da-grilagem-pode-mudar-o-mapa-de-regioes-da-amazonia/>>. Accessed on 21/09/2022.

FUNDAÇÃO AMAZONAS SUSTENTÁVEL. Plano de gestão da reserva de desenvolvimento sustentável do JUMA: Versão Consulta Pública. Novo Aripuanã: Fundação Amazonas Sustentável, 2010. Available in: https://documentacao.socioambiental.org/noticias/anexo_noticia/9857_20100312_102543.pdf.

Accessed on 22/09/2022.

FUNDAÇÃO REDE AMAZÔNICA. Portal Amazônia - Novo Aripuanã. 2021 < <https://portalamazonia.com/amazonia-az/letra-n/novo-aripuana>>. Accessed on 21/09/2022.

GALUCH, V. M.; MENEZES, T. C. G. Da reforma agrária ao agronegócio: Notas sobre dinâmicas territoriais na fronteira agropecuária amazônica a partir do município de Apuí (Sul do Amazonas). Estudos Sociedade e Agricultura. 28 (2), 388-412, jun. a set. 2020. <<https://www.redalyc.org/journal/5999/599963212009/599963212009.pdf>>. Accessed on 21/09/2022.

GLOBO. Documentos mostram que Ibama facilitou exportação de madeira extraída ilegalmente. 2020 <<https://g1.globo.com/natureza/noticia/2020/11/17/documentos-mostram-que-ibama-facilitou-exportacao-de-madeira-extraida-ilegalmente.ghtml>>. Accessed on 21/09/2022.

GLOBO. Queimadas e desmatamento estão relacionados na Amazônia; entenda. 2019. <<https://g1.globo.com/natureza/noticia/2019/08/23/queimadas-e-desmatamento-estao-relacionados-na-amazonia-entenda.ghtml>>. Accessed on 06/09/2022.

GONÇALVES, A. Nova fronteira do desmatamento, sul do Amazonas tem capitais das queimadas e garimpo e ausência do Estado. O Globo. 2021. <<https://oglobo.globo.com/brasil/nova-fronteira-do-desmatamento-sul-do-amazonas-tem-capitais-das-queimadas-garimpo-ausencia-do-estado-25314282>>. Accessed on 21/09/2022.

HABLEMOS DE FLORES. Site do Hablemos de Flores, 2015. Descubra tudo acerca de la Bertholletia Excelsa aqui. Available in: <<https://hablemosdeflores.com/bertholletia-excelsa/>>. Accessed on 22/09/2022.

IMAZON. Desmatamento na Amazônia cresce quase 70 e atinge pior fevereiro em 10 anos. 2020. <<https://imazon.org.br/imprensa/desmatamento-na-amazonia-cresce-quase-70-e-atinge-pior-fevereiro-em-10-anos/>>. Accessed on 21/09/2022.

INFORME MANAUS. Governo do Amazonas adere à campanha global “Race to Zero” para zerar emissões até 2050. Informe Manaus. 2021. <<https://informemanaus.com/2021/governo-do-amazonas-adere-a-campanha-global-race-to-zero-para-zerar-emissoes-ate-2050/>>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA – IBGE. Censo Agropecuário. 2017. <<https://cidades.ibge.gov.br/brasil/pesquisa/24/76693>>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA – IBGE. Estimativas da População. 2021 <https://www.ibge.gov.br/estatisticas/sociais/populacao/9103-estimativas-de-populacao.html>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA - IBGE. Novo Aripuanã. <<https://ibge.gov.br/cidades-e-estados/am/novo-aripuan.html>>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA – IBGE. Síntese de Indicadores Sociais: Uma análise das condições de vida da população brasileira. Rio de Janeiro, 2020.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA. IBGE. 2016. Mapa das divisões hidrográficas do Brasil. <https://geofp.ibge.gov.br/informacoes_ambientais/estudos_ambientais/bacias_e_divisoes_hidrograficas_do_brasil/2021/Divisao_Hidrografica_Nacional_DHN250/mapas/mapa_das_divisoes_hidrograficas_do_brasil_2021.pdf>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA. Manual técnico da vegetação brasileira. Departamento de Recursos Naturais e Estudos Ambientais Rio de Janeiro: IBGE, 1991. <<https://biblioteca.ibge.gov.br/visualizacao/livros/liv23267.pdf>>. Accessed on 16/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA. Sistema IBGE de Recuperação Automática – SIDRA. Pesquisa da Pecuária Municipal. <<https://sidra.ibge.gov.br/tabela/3939>>. Accessed on 21/09/2022.

INSTITUTO BRASILEIRO DE GEOGRAFIA E ESTATÍSTICA. Site do IBGE cidades, 2017. Available in: <<https://cidades.ibge.gov.br/brasil/am/novo-aripuan/panorama>>. Accessed on 22/09/2022.

INSTITUTO BRASILEIRO DO MEIO AMBIENTE E DOS RECURSOS NATURAIS RENOVÁVEIS - IBAMA. Manejo sustentável autorizado pelo Ibama em 2019 totalizou 39 mil hectares. Brasília, 2020. <<http://www.ibama.gov.br/ultimas/2142-manejo-sustentavel-autorizado-pelo-ibama-em2019totalizou39milhectares#:~:text=Manejo%20sustent%C3%A1vel%20autorizado%20pelo%20Ibama%20em%202019%20totalizou%2039%20mil%20hectares>>. Accessed on 22/09/2022.

INSTITUTO CHICO MENDES DE CONSERVAÇÃO DA BIODIVERSIDADE - ICMBIO. Painel Dinâmico de Informações, 2021. <https://qv.icmbio.gov.br/QvAJAXZfc/opendoc2.htm?document=painel_corporativo_6476.qvw&host=Local&anonymous=true>. Accessed on 22/09/2022.

INSTITUTO CHICO MENDES DE CONSERVAÇÃO PARA A BIODIVERSIDADE- ICMBIO. Livro Vermelho da Fauna Brasileira Ameaçada de Extinção. Brasília, DF, 2018.

INSTITUTO DE CONSERVAÇÃO AMBIENTAL E DESENVOLVIMENTO SUSTENTÁVEL DA AMAZÔNIA – IDESAM. COP-26: entre erros e acertos. 2021. <<https://idesam.org/opinioao/cop26-entre-erros-e-acertos/>>. Accessed on 21/09/2022.

INSTITUTO DE PESQUISA AMBIENTAL DA AMAZÔNIA - IPAM. 35% do desmatamento na Amazônia é grilagem, indica análise do IPAM. 2019. < <https://ipam.org.br/35-do-desmatamento-na-amazonia-e-grilagem-indica-analise-do-ipam/>>. Accessed on 21/09/2022.

INSTITUTO DE PESQUISA ECONÔMICA APLICADA - IPEA. Metas Nacionais dos Objetivos de Desenvolvimento Sustentável – ODS. Brasília. 2018. <<https://portalods.com.br/publicacoes/ods-metas-nacionais-dos-objetivos-de-desenvolvimento-sustentavel/>>. Accessed on 04/05/2022.

INSTITUTO DO HOMEM E MEIO AMBIENTE DA AMAZÔNICA - IMAZON. Desmatamento na Amazônia cresce quase 70% e atinge pior fevereiro em 15 anos, 2022. <<https://imazon.org.br/imprensa/desmatamento-na-amazonia-cresce-quase-70-e-atinge-pior-fevereiro-em-10-anos/>>. Accessed on 22/09/2022.

INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS - INPE. Coordenação geral de observação da terra. Programa de monitoramento da amazônia e demais biomas. Desmatamento – Amazônia Legal. <<http://terrabrasilis.dpi.inpe.br/downloads/>> Accessed at 05/01/2019.

INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS - INPE. Mapa Índice TOPODATA. <webmapit.com.br>. Accessed on 21/09/2022.

INSTITUTO NACIONAL DE PESQUISAS ESPACIAIS. Programa de monitoramento da Amazônia e demais biomas Coordenação geral de observação da terra.. <http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/increments>. Accessed on 20/09/2022.

IUCN. Red List of Threatened Species. V. 2022-1. <https://www.iucnredlist.org>. Accessed on 21/09/2022.

MAHAR, D. J. Desenvolvimento econômico da Amazônia: uma análise das políticas governamentais. Rio de Janeiro: IPEA/INPES, 1978.

MAPBIOMAS BRASIL. <<https://mapbiomas.org/>>. Accessed on 22/09/2022.

MARTINS, M. B. et al. Reflexões em Biologia da Conservação- Volume 2. Belém: Museu Paraense Emílio Goeldi, 2020.

MEDEIROS, C. Pecuária responde por 75% do desmatamento em terras públicas da Amazônia. UOL. 2021. <<https://noticias.uol.com.br/meio-ambiente/ultimas-noticias/redacao/2021/10/27/amazonia-87-do-desmate-em-terras-publicas-ocorreu-em-areas-nao-destinadas.htm>>. Accessed on 21/09/2022.

MIDDLETON, T. C. P. Declínio de aves no arco do desmatamento amazônico. 2016. Tese (Doutorado em Ciências Biológicas). Universidade de São Paulo, São Paulo, 2016.

MILES, L. et al. The impact of global climate change on tropical forest biodiversity in Amazonia. *Global Ecology and Biogeography*, v. 13, n. 6, p. 553–565, nov. 2004. <<https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1466-822X.2004.00105.x>>. Accessed at 07/09/22.

MINISTÉRIO DO MEIO AMBIENTE – MMA. Áreas Prioritárias para Biodiversidade – Metodologia. 2021. <<https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias/metodologia-1>>. Accessed on 21/09/2022.

MINISTÉRIO DO MEIO AMBIENTE – MMA. 2ª Atualização das Áreas Prioritárias para Conservação da Biodiversidade. 2021. <<https://www.gov.br/mma/pt-br/assuntos/servicosambientais/ecossistemas-1/conservacao-1/areas-prioritarias/2a-atualizacao-das-areas-prioritarias-para-conservacao-da-biodiversidade-2018>> . Accessed on 21/09/2022.

MINISTÉRIO DO MEIO AMBIENTE - MMA. Áreas Prioritárias para a Conservação, Utilização Sustentável e Repartição dos Benefícios da Biodiversidade Brasileira. <<https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/conservacao-1/areas-prioritarias>> . Accessed on 21/09/2022.

MINISTÉRIO DO MEIO AMBIENTE. – MMA. A grilagem de terras publica na Amazonia brasileira. Instituto de Pesquisa Ambiental da Amazônia. Série Estudos 8. Brasília. 2006. <https://antigo.mma.gov.br/estruturas/225/_arquivos/9_a_grilagem_de_terras_publicas_na_amazonia_brasileira_225.pdf>. Accessed on 21/09/2022.

MINISTÉRIO DO MEIO AMBIENTE. Plano de Ação para Prevenção e Controle do Desmatamento na Amazônia Legal – PPCDAm. 2016. <PPCDAm (mma.gov.br)>. Accessed on 12/05/2022.

MINISTÉRIO DO MEIO AMBIENTE. REDD+ e a NDC do Brasil. 2016 < <http://redd.mma.gov.br/pt/redd-e-a-indc-brasileira>>. Accessed on 21/09/2022.

MINISTÉRIO DO MEIO AMBIENTE. Sumário de informações sobre como as salvaguardas de Cancun foram abordadas e respeitadas pelo Brasil durante a implementação de ações de redução de emissão provenientes do desmatamento no bioma Amazônia entre 2006 e 2010. Painel Técnico de Especialistas Nacionais em Salvaguardas REDD+. Brasília. 2015. <http://redd.mma.gov.br/images/publicacoes/salvaguardas_1sumario.pdf>. Accessed on 21/09/2022.

MINISTÉRIO DO TRABALHO E PREVIDÊNCIA. Normas regulamentadoras. 2022. <<https://www.gov.br/trabalho-e-previdencia/pt-br/composicao/orgaos-especificos/secretaria-de-trabalho/inspecao/seguranca-e-saude-no-trabalho/ctpp-nrs/normas-regulamentadoras-nrs>> . Accessed on 21/09/2022.

MOORE, A. Amazon Rainforest Fires: Everything You Need to Know. NC State University. College of Natural Resources News. 2019. < <https://cnr.ncsu.edu/news/2019/09/amazon-rainforest-fires-everything-you-need-to-know/>>. Accessed on 20/09/2022.

MÜLLER-HANSEN, F. et al. Can Intensification of Cattle Ranching Reduce Deforestation in the Amazon? Insights From an Agent-based Social-Ecological Model. *Ecological Economics*. Vol. 159, May 2019, Pages 198-211. 2019 <<https://www.sciencedirect.com/science/article/abs/pii/S0921800918304130>>. Accessed on 21/09/2022.

MURER, B. M.; FUTADA, S. M. Painel de Dados - Unidades de Conservação no Brasil. Instituto Socioambiental – ISA. 2022. <<https://uc.socioambiental.org/pt-br/paineldedados>>. Accessed on 22/09/2022.

NASCIMENTO, G. Meat exports in Brazil will be an important differential in 2021 Safras e Mercado, 2021. <<https://www2.safras.com.br/eng/2020/09/23/meat-exports-in-brazil-will-be-an-important-differential-in-2021/>>. Accessed on 21/09/2022.

ODS – OBJETIVOS DO DESENVOLVIMENTO SUSTENTÁVEL. Agenda 2030. <<https://www.pactoglobal.org.br/ods>>.

OLIVEIRA, B. F. A. et al. Deforestation and climate change are projected to increase heat stress risk in the Brazilian Amazon. *Communications Earth & Environment*, v. 2, n. 1, p. 1–8, 1 out. 2021. <https://www.nature.com/articles/s43247-021-00275-8.epdf?sharing_token=uFI7d8ofpF7MzxbDMdjBWtRgN0jAjWel9jnR3ZoTv0N__2v5vta8OWNgXc7jU-SMJowczw5uKk1DYe9rea6awE4cMAAKg9hEatBKF8y_SlaiM0m_O2O1BVGJ6T6J_QdmiS1dZ_M3zv8dt rU9mSiyLJO19GldF4OEZ8n5C9HabVA%3D>. Accessed at 04/09/22.

PEREIRA, C. F. et. al., Comunidades tradicionais e conflitos socioambientais: o caso da flona Aripuanã, Amazonas. Terceira Margem Amazônia. 2019. <<https://www.revistaterceiramargem.com/index.php/terceiramargem/article/view/318/225>>. Accessed on 21/09/2022.

PIEROLLA, G. P.; MEIRELLES, C. R. M. Estudos bioclimáticos da habitação ribeirinha amazônica: Análise dos sistemas de fechamento verticais e as aberturas. XVI Jornada de Iniciação Científica e X Mostra de Iniciação Tecnológica. Universidade Presbiteriana Mackenzie. 2020. <<http://eventoscopq.mackenzie.br/index.php/jornada/xvijornada/paper/view/2159/1295>>. Accessed on 21/09/2022.

PRIST, P. R. et al. How deforestation pattern in the Amazon influences vertebrate richness and community composition, *Landscape Ecology*, n. 27, p. 799–812, 2012.

PRODES. Monitoramento do Desmatamento da Floresta Amazônica Brasileira por Satélite. INPE. 2022. <<http://www.obt.inpe.br/OBT/assuntos/programas/amazonia/prodes>>. Accessed on 22/09/2022.

RELATÓRIO ANUAL DE DESMATAMENTO, 2021. São Paulo, Brasil MapBiomas, 2022. <<http://alerta.mapbiomas.org>>. Accessed on 04/09/22.

REVISTA ECO 21. Desmatamento, perda de biodiversidade e pobreza. Ano XIII, Edição 80, julho 2003. <https://ambientes.ambientebrasil.com.br/natural/artigos/desmatamento_perda_de_biodiversidade_e_pobreza.html>. Accessed at 23/09/22

RICHARDS, M.; PANFIL, S. N. Social and Biodiversity Impact Assessment (SBIA) - Manual for REDD+ Projects: Part 1 – Core Guidance for Project Proponents. Climate, Community & Biodiversity Alliance, Forest Trends, Fauna & Flora International, and Rainforest Alliance. Washington, DC. 2011. <SBIA_Part_1.pdf (verra.org)>. Accessed on 21/09/2022.

RICHARDS, M. 2011. Social and Biodiversity Impact Assessment (SBIA) Manual for REDD+ Projects: Part 2 – Social Impact Assessment Toolbox. Climate, Community & Biodiversity Alliance, and Forest Trends with Rainforest Alliance and Fauna & Flora International. Washington, DC. Accessed at 08/12/2022.

PITMAN, N. 2011. Social and Biodiversity Impact Assessment Manual for REDD+ Projects: Part 3 – Biodiversity Impact Assessment Toolbox. Forest Trends, Climate, Community & Biodiversity Alliance, Rainforest Alliance and Fauna & Flora International. Washington, DC. Accessed at 08/12/2022.

RODRIGUES, D.C.B. et al. Organização e trabalho das mulheres ribeirinhas amazônicas: um estudo nas comunidades de santa luzia e são lazaro no grande lago de Manacapuru/AM, Revista Retratos de Assentamentos, v.18, n.1, p. 113-134, 2015. Available in: <https://core.ac.uk/reader/228824080>. Accessed on 22/09/2022.

SALOMÃO, R. P. Density, structure, and spatial distribution of Brazilnut trees (*Bertholletia excelsa* H. & B.) on two plateaus of moist evergreen forest in the northern Brazilian Amazon. Ciências Naturais, Belém, v. 4, n. 1, p. 11-25. 2009. <<http://scielo.iec.gov.br/pdf/bmpegn/v4n1/v4n1a02.pdf>>. Accessed on 20/09/2022.

SANTANA, A. C. et al. Valoração e sustentabilidade da castanha-do-brasil na Amazônia. Revista de Ciências Agrárias, 60, 77-89, 2017. <<http://periodicos.ufra.edu.br/index.php/ajaes/article/view/2690>>. Accessed on 22/09/2022.

SANTANA, A. C. et al. Valoração e sustentabilidade da castanha-do-brasil na Amazônia. Revista de Ciências Agrárias - Amazon Journal of Agricultural and Environmental Sciences, n 60, p. 77-89, 2017.

SANTOS, H. G. et al. O novo mapa de solos do Brasil: Legenda atualizada. EMBRAPA Solos. Rio de Janeiro. 2011. <http://geoinfo.cnps.embrapa.br/layers/geonode%3Abrasil_solos_5m_20201104>. Accessed on 21/09/2022.

SERVIÇO FLORESTAL BRASILEIRO. Florestas Naturais. Sistema Nacional de Informações Florestais (SNIF). <<https://snif.florestal.gov.br/pt-br/os-biomas-e-suas-florestas>>. Accessed on 21/09/2022.

SERVIÇO FLORESTAL BRASILEIRO. Tipologias Florestais. Sistema Nacional De Informações Florestais. 2020. <<https://snif.florestal.gov.br/pt-br/conhecendo-sobre-florestas/168-tipologias-florestais>>. Accessed on 21/09/2022.

SOARES, P. G. Trajetória da fronteira agropecuária no projeto de assentamento Rio Juma – Apuí - AM. Dissertação (Mestrado). Universidade de São Paulo. Piracicaba. 2022. <https://www.teses.usp.br/teses/disponiveis/91/91131/tde-11022022-160839/_revisada.pdf>. Accessed on 21/09/2022.

SPRING, J. Brazil may be exporting illegally deforested wood. Business Day. 2020 <<https://www.businesslive.co.za/bd/world/americas/2020-03-04-brazil-may-be-exporting-illegally-deforested-wood/>> Accessed on 21/09/2022.

STRAND, J. et al. Spatially explicit valuation of the Brazilian Amazon Forest's Ecosystem Services. Nature Sustainability 1, 657–664, 2018. <<https://doi.org/10.1038/s41893-018-0175-0>>. Accessed on 05/09/22.

STRAND, J. et al. Spatially explicit valuation of the Brazilian Amazon Forest's Ecosystem Services. Nature Sustainability, v. 1, n. 11, p. 657–664, nov. 2018. <<https://www.nature.com/articles/s41893-018-0175-0>>. Accessed at 04/09/22.

UNITED NATIONS. Calculation of the number of sample plots for measurements within A/R CDM project activities. Clean Development Mechanism – CDM. United Nations Climate Change Conference - UNFCCC. <<https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf>>. Accessed on 13/08/21.

UNITED NATIONS. Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities. Clean Development Mechanism – CDM. United Nations Climate Change Conference - UNFCCC. <<https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-02-v1.pdf>>. Accessed on 13/06/22.

UNITED NATIONS. Decision 1/CP.16, Annex I, paragraph 2. Framework Convention on Climate Change <<https://www.un-redd.org/glossary/cancun-safeguards>>. Accessed on 21/09/2022.

UNITED NATIONS. The Cancun Agreements: Outcome of the work of the Ad Hoc Working Group on Long-term Cooperative Action under the Convention. Framework Convention On Climate Change. 2011. <<https://unfccc.int/documents/6527>>. Accessed on 05/09/2022.

VALERIANO D. M. Definição de floresta utilizada no FREL-Cerrado. Programa de Monitoramento da Amazônia e Demais Biomas Brasileiros. VII Reunião do GTREDD+. Brasília. 2016. <http://redd.mma.gov.br/images/gttredd/daltonvaleriano_definicaofloresta.pdf>. Accessed on 21/09/2022.

VELOSO, H. P. et al. Classificação da vegetação brasileira adaptada a um sistema universal. Instituto Brasileiro de Geografia e Estatística (IBGE). Rio de Janeiro, 1991. <https://ambientes.ambientebrasil.com.br/natural/regioes_fitoeologicas/regioes_fitoeologicas_-_floresta_ombrofila_densa.html>. Accessed on 21/09/2022.

VERRA. Florestal Santa Maria Project (FSM-REDD Project). Monitoring Report. 2021. <https://registry.terra.org/mymodule/ProjectDoc/Project_ViewFile.asp?FileID=52137&IDKEY=9lksjoiuwqo wrnoiuomnckjashoufifmln902309ksdfiku098l71896923> Accessed on 20/09/2022.

VERRA. VCS Module VMD0001 - Estimation of carbon stocks in the above-and below ground biomass in live tree and non-tree pools (CP-AB). V. 1.1, 2013. <<https://terra.org/wp-content/uploads/2017/11/VMD0001v1.1.pdf>>. Accessed on 22/09/2022.

VERRA. VCS Module VMD0005 - Estimation of carbon stocks in the long-term wood products pool (cp-w), Version 1.1. 2012. <<https://terra.org/wp-content/uploads/2018/03/VMD0005-CP-W-v1.1.pdf>>. Accessed on 22/09/2022.

VERRA. VCS Module VMD0011 - Estimation of emissions from market-effects (LK-ME). V. 1.1, 2015. <<https://terra.org/wp-content/uploads/2018/03/VMD0011-LK-ME-v1.1.pdf>>. Accessed on 22/09/2022.

VEZZANI, F. M. Solos e os serviços ecossistêmicos (Soils and the Ecosystem Services), Revista Brasileira de Geografia Física, v.8, p. 673-684, 2015.

VIEIRA, I. C. G. et al. Estratégias para evitar a perda de biodiversidade na Amazônia. Estudos Avançados, [S. l.], v. 19, n. 54, p. 153-164, 2005. Available in: <https://www.revistas.usp.br/eav/article/view/10075>. Accessed on: 22/09/2022.

WWF. Relatório Amazônia Viva. Quito, 2022.